

ECONOMICS

Senior Five



Student's Book

Experimental version

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FOREWORD

Dear Student,

Rwanda Basic Education Board is honoured to present to you Economics book for Senior 5 which serves as a guide to competence-based teaching and learning to ensure consistency and coherence in the learning of economics subject. The Rwandan educational philosophy is to ensure that you achieve full potential at every level of education which will prepare you to be well integrated in society and exploit employment opportunities.

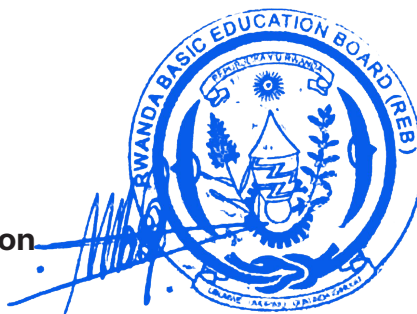
The government of Rwanda emphasizes the importance of aligning teaching and learning materials with the syllabus to facilitate your learning process. Many factors influence what you learn, how well you learn and the competence you acquire. Those factors include quality instructional materials available, assessment strategies for the students among others. Special attention was paid to activities that facilitate learning process, develop your ideas and make new discoveries during concrete activities carried out individually or with peers. In Competence-Based Curriculum (CBC), learning is considered as a process of active building and developing knowledge and meanings by the student where concepts are mainly introduced by an activity, a situation or a scenario that helps the student to construct knowledge, develop skills and acquire positive attitudes and values. For effective use of this textbook, your role is to:

- Work on given activities which lead to the development of skills;
- Share relevant information with other students through presentations, discussions, group work and other active learning techniques such as role play, case studies, investigation and research in the library, from the internet or from your community;
- Participate and take responsibility for your own learning; and
- Draw conclusions based on the findings from the learning activities.

To facilitate you in doing activities, the content of this book is designed to be self-explanatory so that you can use it independently, acquire knowledge, and assess your competences. The book is organized into units. Each unit includes the key unit competence, followed by an introductory activity before the development of Economic concepts that are connected to real world situations. The number of lessons in each unit depends on its length. Each lesson begins with a learning activity, and each unit ends with an end-of-unit assessment.

I wish to sincerely extend my appreciation to REB staff who organized the editing process of this textbook. Special gratitude also goes to lecturers, teachers, illustrators and designers who supported the exercise throughout. Any comment or contribution would be welcome to the improvement of this textbook for the next edition.

Dr. MBARUSHIMANA Nélon
Director General, REB



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I wish to express my appreciation to all the people who played a major role in the editing process of this Economics textbook for senior 5. It would not have been successful without their active participation.

Special thanks are given to those who gave their time to read and refine this textbook to meet the needs of Competence-Based Curriculum. I owe gratitude to different Universities and schools in Rwanda that allowed their staff to work with REB to edit this textbook.

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Finally, I would like to express my gratitude to the Rwanda Basic Education Board (REB) staff, particularly those from Curriculum, Teaching and Learning Resources Department, who were involved in the entire editorial process



Joan MURUNGI,

Head of Curriculum, Teaching and Learning Resources Department

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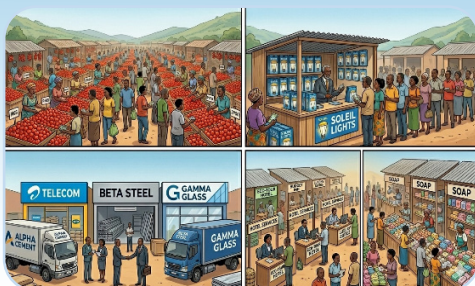
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UNIT 1

MARKET STRUCTURE



Key Unit Competence:

Analyze the operation of firms under different market structures

Learning objectives

Knowledge and understanding

- 1
 - Explain market structure and its importance in economics.
 - Identify the characteristics of each market structure.
 - Explain the process of profit maximization of firms under different markets in both the short run and long run.
 - Explain the advantages and disadvantages of firms under different market structures.
 - Explain why firms continue to produce
 - Explain the conditions for successful price discrimination under monopoly conditions
 - Identify the different forms of product differentiation under monopolistic competition.
 - Explain how prices are determined under oligopoly
 - Explain the concept of barriers to entry and their impact on competition.
 - Explain the effects of market barriers on competition
 - Compare the advantages and disadvantages of each structure to consumers and producers.

Skills

- 2
 - Analyze the conditions under which different market structures operate.
 - Analyze the operation of firms under different markets in both the short run and long run.
 - Illustrate the process of profit maximization of firms under different markets in both the short run and long run.
 - Illustrate the break-even and shut down points under perfect competition.

- Analyze how firms under oligopoly win markets without price wars.
- Illustrate the kinked demand curve under an oligopoly.
- Assess the role of government intervention in promoting fair competition and protecting consumers.
- Analyze how specific barriers affect market entry and competition.
- Evaluate which market structures are most efficient or beneficial to society.

Attitudes and values

3

- Appreciate the conditions under which market structures operate and choose the appropriate one.
- Appreciate the importance of fair competition in promoting efficiency and consumer welfare.



Introductory activity

Scenario: During a class visit to a local market in Kigali, students observe that many sellers are selling tomatoes, a few companies are selling soft drinks, and only one provider supplies electricity to the entire city.

Questions:

- List other local examples of goods or services in Rwanda that fit each of the above situations
- Discuss how prices and the level of competition differ across the three situations above.

Every market we encounter is organized differently, depending on how many firms operate within it, how they compete, and how much control they have over prices. This unit introduces the concept of market structure, the framework economists use to classify and analyze these differences, and equips you to identify the main types of market structures, explain their key features, and evaluate how each affects pricing, competition, and consumer welfare

1.1 Meaning and features of market structures

Learning activity 1.1



1. Think of two different products you or your family buy regularly. For each, how many sellers do you think are involved in providing that product in your local area?
2. In your experience, does the number of sellers of a product affect how much control they have over its price? Explain using an example from your own community.
3. Have you noticed any product or service in Rwanda that is supplied by only one company? What do you think this means for customers' choices and the prices they pay?

4. Based on your observations so far, what do you think the term "market structure" might mean in economics
5. Have you noticed any product or service in Rwanda that is supplied by only one company? What do you think this means for customers' choices and the prices they pay?
6. Based on your observations so far, what do you think the term "market structure" might mean in economics

i. Meaning of market structure

A market is any system or arrangement that connects buyers and sellers, enabling them to exchange goods, services, and price information, whether in person, by phone, or online. However, not all markets operate in the same way; some have many competing sellers, while others are dominated by just one or a few firms. This is where the concept of market structure becomes important. Market structure refers to the characteristics and organization of a market that influence the behaviour of buyers and sellers. It describes the nature and level of competition among firms producing similar goods or services in an economy, and it determines how prices are set, how much firms produce, and the degree of competition within a market.

ii. Features used to classify market structures

Market structures are classified according to the following major features:

1. Number of Firms in the Market

This refers to the number of sellers or producers operating in a particular market. The number of firms determines the degree of competition.

- In some markets, many small firms are competing with each other.
- In others, there may be only one or a few large firms controlling the market.

For example:

- Agricultural markets often have many producers.
- Electricity supply in some countries may be controlled by one dominant producer.

The greater the number of firms, the higher the level of competition.

2. Nature of the product

Markets are also classified according to the type of products firms sell.

Homogeneous products

These are identical products with no significant differences in quality, design, or branding. Consumers view them as perfect substitutes.

Example: Sugar, Maize, Salt etc

Differentiated products

These are products that are similar but distinguished by branding, packaging, quality, or advertising.

Example: Soft drinks, Mobile phones, Clothing brands, etc

Product differentiation increases consumer choice and affects competition among firms.

3. Ease of entry and exit

This refers to how easy or difficult it is for new firms to join or leave a market.

Easy entry and exit

When firms can easily start or close businesses with few restrictions, competition tends to increase.

Barriers to entry

Some markets have obstacles that make entry difficult, such as:

- High capital requirements
- Government regulations and licensing
- Advanced technology
- Strong brand loyalty
- Patents and legal protections

Markets with high barriers to entry are usually dominated by a few large firms.

4. Degree of price control

This shows the extent to which firms can influence or control market prices.

Price takers

Some firms have no control over prices and must accept prices determined by market forces of demand and supply.

Price makers

Other firms can influence prices because they control a large share of the market or produce unique products.

The level of price control depends on the market structure.

5. Method of price determination

This explains how prices are established in the market.

- In highly competitive markets, prices are determined mainly by demand and supply forces.
- In less competitive markets, dominant firms may influence or fix prices.
- Price determination affects consumers, producers, and the allocation of resources in the economy.

6. Availability of market information

This refers to the amount of information buyers and sellers possess about:

- Prices
- Product quality

- Market conditions
- Availability of substitutes

Perfect information

In some markets, buyers and sellers have full knowledge of market conditions.

Imperfect information

In other markets, information may be limited or unequal, giving some firms an advantage over others.

Availability of information affects decision-making and market efficiency.

i. Importance of market structure

Market structure is important because it influences the operation and performance of an economy in several ways.

1. Determines price and output

Market structure affects how firms decide the prices of goods and services and the quantity to produce.

For example:

- Competitive markets usually lead to lower prices.
- Monopolies may charge higher prices.

2. Influences the level of competition

It determines whether competition is:

High, Moderate or Limited.

A highly competitive market encourages firms to improve efficiency and reduce costs.

3. Affects resource allocation and efficiency

Market structure influences how efficiently resources are used in the production process.

Efficient markets help:

- Reduce wastage,
- Improve productivity, and
- Increase economic growth.

4. Impacts consumer welfare

The type of market structure affects:

- Prices paid by consumers,
- Quality of goods and services,
- Availability of products, and
- Consumer choice.
- Competitive markets usually provide better services and more variety to consumers.

5. Encourages innovation and investment

Some market structures encourage firms to invest in:

- Research,
- Technology, and
- Product improvement

6. Guides Government policy and regulation

Understanding market structures helps governments:

- Formulate economic policies,
- Regulate unfair competition,
- Protect consumers, and
- Promote economic stability.

Governments may intervene in markets where firms abuse market power.

ii. Categories of market structures

Market structures are broadly classified into two major categories:

A. Perfect markets

Perfect markets are markets where competition is very high and no single firm has control over prices. Under perfect markets, there is basically one market situation that is known as perfect competition.

Perfect Competition

Perfect competition is the main example of a perfect market. It is characterized by:

Many buyers and sellers,

- Homogeneous products,
- Free entry and exit,
- Perfect market information, and
- Firms being price takers.

In this market, prices are determined entirely by demand and supply forces.

B. Imperfect markets

Imperfect markets are markets where firms have some degree of control over prices and competition is limited.

Types of imperfect markets include:

- **Monopoly:** A market dominated by one seller producing a product with no close substitutes.
- **Oligopoly:** A market controlled by a few large firms that dominate production and pricing.

- **Monopolistic Competition:** A market with many firms selling differentiated products and competing through advertising and branding.

Imperfect markets are common in real-world economies because firms often differentiate products and exercise some control over prices.

Application activity 1.1



1. In your own words, what does the term "market structure" mean?
2. List two main features that help distinguish one market structure from another.
3. Using your answers from question 2, give a Rwandan example for each market structure and explain why it fits

1.2 Perfect market (Perfect competition)

Learning activity 1.2



Market Day Scenario

In a rural village, farmers gather every Saturday to sell tomatoes. There are many farmers selling tomatoes, and many buyers coming to purchase them. All the tomatoes look almost the same in size and quality. Buyers move freely from one seller to another comparing prices.

If one farmer tries to sell at a higher price, buyers simply go to another seller. All sellers end up charging nearly the same price.

Questions for Learners:

1. What do you notice about the number of sellers and buyers in this market?
2. Are the tomatoes different or similar?
3. Can one seller decide the price alone? Why or why not?
4. What happens when a seller increases the price?

1.2.1 Meaning and characteristics of perfect competition market

Meaning of Perfect Competition



Perfect competition is a market structure where there are many buyers and sellers dealing in identical products, and no single firm has the power to influence the market price.

Prices are determined purely by the forces of demand and supply, and all firms operate under the same market conditions.

In this type of market, firms are known as price takers, meaning they accept the prevailing market price and cannot influence it.

Characteristics of Perfect Competition

1. Large Number of Buyers and Sellers

There are many buyers and sellers in the market, each operating on a small scale. Because of this, no single buyer or seller can influence the market price or output.

2. Homogeneous Products

All firms produce and sell identical or very similar products. There is no difference in quality, design, or branding.

- Consumers have no preference for a specific seller.
- Products are perfect substitutes.

Example: agricultural products like maize, beans, or sugar.

3. Free Entry and Exit of Firms

Firms are free to enter or leave the market without restrictions.

- No barriers such as high capital requirements, licensing restrictions, or legal obstacles.
- This ensures long-run normal profits in the industry.

4. Perfect Knowledge (Perfect Information)

Both buyers and sellers have full and accurate information about:

- Prices,
- Quality of goods, and
- Market conditions.

This prevents exploitation and ensures rational decision-making.

5. Firms are Price Takers

Individual firms cannot influence the market price.

- The price is determined by overall market demand and supply.

- Each firm accepts the prevailing price.

6. Perfect Mobility of Factors of Production

Resources such as labour and capital can move freely between industries and firms.

- Workers can easily change jobs.
- Capital can shift to more profitable industries.

This improves efficiency in resource allocation.

7. Many Firms with No Market Control

Although there are many firms, none of them has control over the market.

- No monopoly or dominance by a single firm.
- Competition is very high.

8. The demand curve is perfectly elastic

Each individual firm faces a perfectly elastic demand curve.

- This means the firm can sell any quantity at the market price.
- If it increases its price, demand falls to zero because consumers can buy from other firms.
- Graphically, the demand curve is a horizontal straight line.

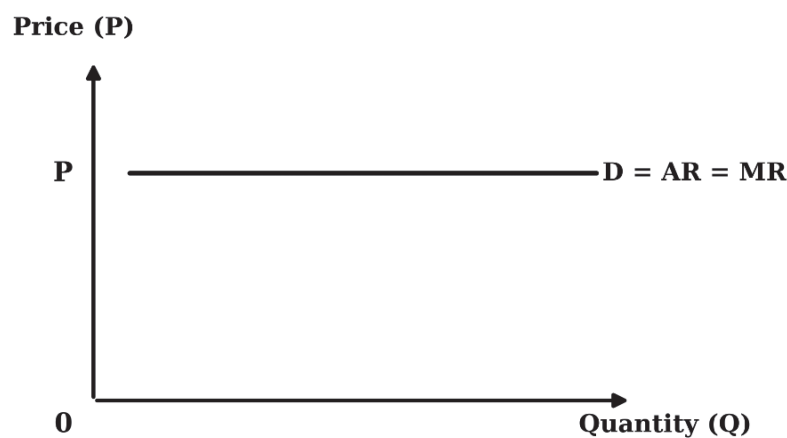


Figure 1: The demand curve under perfect competition

The figure shows that under perfect competition; the firm's demand curve (D) is perfectly elastic and coincides with the average revenue (AR) and marginal revenue (MR) curves. As a price taker, the firm can sell any quantity of output at the prevailing market price (P), resulting in $D = AR = MR$ throughout the curve.

Examples of perfect competition market Rwanda



- Vegetable sellers in local markets such as tomatoes, beans, and onions.
- Irish Potato Trade (Musanze/Nyabihu)

While no real market perfectly satisfies all the conditions of perfect competition, small-scale agricultural markets in Rwanda, such as vegetable, potato, and grain trading, closely approximate the model due to their many sellers, largely standardized products, and relatively free entry.

These real-world examples help illustrate the idea of highly competitive markets, but economists actually distinguish between two related terms, pure competition and perfect competition, each defined by a specific set of conditions outlined below.

Pure competition is characterised by: Many firms and buyers, product homogeneity, free entry and exit, profit-maximisation, no government interference, no transport costs, no persuasive advertising.

Perfect competition, on the other hand, requires the fulfillment of two additional conditions: Perfect mobility of factors of production and perfect knowledge of market conditions.

Therefore, perfect competition is a wider term than pure competition. In other words, pure competition is a subset of perfect competition.

Effects of Competitive Market

1. Lower prices for consumers

Many sellers compete, so prices tend to be low and affordable.

2. Better quality products

Firms improve the quality of goods and services to attract more customers.

3. Efficient use of resources

Businesses try to reduce costs and avoid waste in order to compete.

4. More choices for consumers

Buyers can choose from many sellers and products.

5. Encourages innovation and improvement

Firms introduce new ideas, technology, and better services to stay competitive.

6. Firms that are inefficient may leave the market

Businesses that cannot compete **may close or exit the market.**

7. Uniform Price in the Market

All firms sell their products at a single market price determined by demand and supply.

- No price discrimination is possible.
- All consumers pay the same price.

8. Absence of Selling Costs

In its pure form, perfect competition assumes there are no advertising or promotional costs because products are identical and buyers already have full knowledge.

Perfect competition represents an ideal market structure where competition is at its highest level, ensuring efficient allocation of resources, fair pricing, and maximum consumer welfare. However, in reality, it is rare and mainly used as a theoretical model for comparing other market structures.

1.2.2 The Cost curves for a firm under perfect competition

In a perfectly competitive market, firms are price takers and must operate efficiently to maximize profits and remain competitive. One of the most important tools for analyzing a firm's production decisions is the cost curve, which illustrates how production costs change as the level of output varies. Cost curves help identify the most efficient level of production and explain the relationship between cost and output. They also provide valuable insights into firm behavior in both the short run and long run. Under perfect competition, firms strive to produce at the lowest possible cost in order to remain competitive, maximize profits, and achieve efficient resource allocation

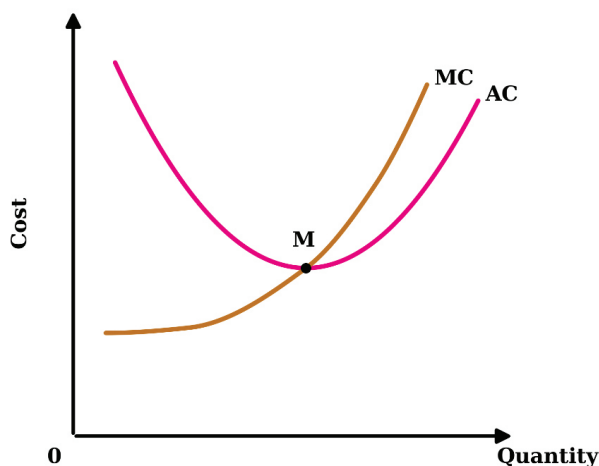


Figure 2: The Cost curves for a firm under Perfect Competition

The Average Cost (AC) and Marginal Cost (MC) curves are generally U-shaped due to the effects of economies and diseconomies of scale in the long run and the law of diminishing returns in the short run. The MC curve intersects the AC curve at point M, which is the minimum point of the AC curve. When MC is below AC, average cost falls; when MC is above AC, average cost rises. Therefore, point M represents the firm's optimum scale or minimum-cost level of production, where average cost is at its lowest.

1.2.3 Relationship between Average Revenue (AR), Marginal Revenue (MR), Price, and Demand under Perfect Competition

Under perfect competition, the Average Revenue (AR) curve coincides with the Marginal Revenue (MR) curve because the market price remains constant for every unit sold. As a price taker, the firm can sell additional units of output without affecting the market price. Therefore, each extra unit sold adds the same amount to total revenue, making Marginal Revenue equal to Average Revenue, Price, and Demand. Thus, under perfect competition: $AR = MR = P = D$. This relationship is illustrated by a horizontal demand curve, indicating that the firm can sell any quantity of output at the prevailing market price.

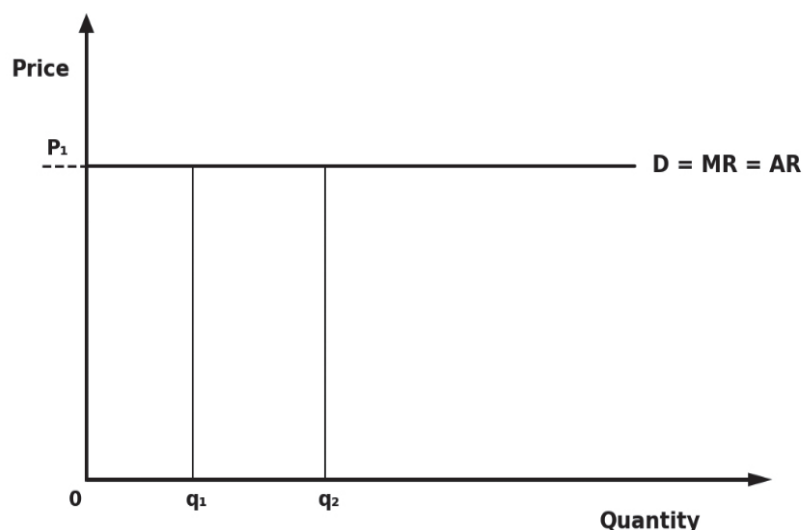


Figure 3: Demand curve representing relationship between AR and MR

As shown in the figure, under perfect competition the firm's demand curve is perfectly elastic and is represented by a horizontal line at the prevailing market price (P_1). Since the firm is a price taker, it can sell different quantities of output (q_1 and q_2) without affecting the market price. Consequently, Price (P), Average Revenue (AR), and Marginal Revenue (MR) are equal at all levels of output, causing the demand curve to coincide with the AR and MR curves ($D = MR = AR$). This implies that the firm can increase its sales while receiving the same price per unit of output.

1.2.4 Short-Run and Long-Run Equilibrium Position of a Firm under Perfect Competition

In both the short run and the long run, a perfectly competitive firm is in equilibrium when it has no incentive to change its level of output because profits are maximized. The necessary condition for equilibrium is that Marginal Cost (MC) equals Marginal Revenue (MR) i.e., $MC = MR$. However, this condition alone is not sufficient. The sufficient condition requires that the MC curve cuts the MR curve from below and is rising at the point of intersection, ensuring a stable profit-maximizing equilibrium. Since the firm is a price taker, Price (P), Average Revenue (AR), and Marginal Revenue (MR) are equal, such that $P = AR = MR$. Therefore, equilibrium occurs where $MC = MR = AR = P$.

While the equilibrium conditions are the same in both periods, the outcomes may differ. In the short run, a firm may earn abnormal profits, normal profits, or incur losses, depending on its cost structure and market price. In the long run, however, the entry and exit of firms eliminate abnormal profits and losses, leaving firms to earn only normal (zero economic) profits. Consequently, long-run equilibrium is achieved where $P = AR = MR = MC = AC$, and the firm operates at the most efficient scale of production

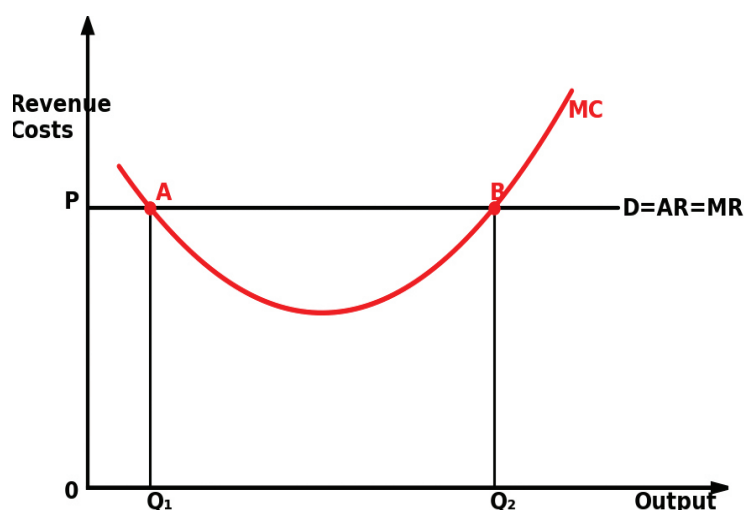


Figure 4: Equilibrium position of a firm under perfect competition.

As shown in the figure above, both points A (OQ_1) and B (OQ_2) satisfy the condition $MC = MR$. However, at point A, the MC curve is falling, so equilibrium is unstable. The firm therefore increases output to OQ_2 . At point B, MC equals MR and is rising, satisfying both the necessary and sufficient conditions for profit maximization. Hence, the firm's equilibrium occurs at B. At this equilibrium, the firm may earn abnormal profits, normal profits, or incur losses depending on its average total cost (ATC).



Note: In perfect competition, Price (P) = Average Revenue (AR) = Marginal Revenue (MR). This is represented by a horizontal line.

1.2.5 Price and output determining of a firm under perfect competition in the short-run

In a perfectly competitive market, firms are price takers and must accept the market price determined by demand and supply forces. In the short run, a firm chooses the level of output that maximizes profit or minimizes loss by producing where Marginal Cost (MC) equals Marginal Revenue (MR) and the MC curve is rising. Since the firm cannot influence price, Price (P) equals Average Revenue (AR) and Marginal Revenue (MR). Depending on its cost structure, the firm may earn abnormal (supernormal) profits when $P > ATC$, normal profits when $P = ATC$, or losses when $P < ATC$ but remains above AVC. Even when incurring losses, the firm may continue operating in the short run provided it can cover its variable costs, while seeking to improve efficiency and reduce production costs.

1.2.6 Abnormal Profits and Loss-Making Firm in the Short Run under Perfect Competition

In a perfectly competitive market, firms can earn different levels of profits in the short run, depending on their costs and the market price. Some firms may make abnormal profits, while others might incur losses. Abnormal profits occur when a firm's total revenue exceeds its total costs, attracting new firms into the market. Conversely, loss-making firms face total costs higher than their total revenue, which may lead to a reduction in output or exit from the market. Understanding how firms earn abnormal profits or incur losses in the short run helps to explain the dynamics of competition and market adjustments.

a) Abnormal Profits in the Short Run under Perfect Competition

A firm under perfect competition earns **abnormal (supernormal) profits** in the short run when its total revenue is greater than its total cost of production.

The firm reaches equilibrium at the output level where: $MC=MR$ and the marginal cost (MC) curve is rising.

Conditions for Abnormal Profits

A perfectly competitive firm earns abnormal (supernormal) profits when:

- Price (P), which is equal to Average Revenue (AR), exceeds Average Total Cost (ATC) at the equilibrium level of output.
- The firm produces at the profit-maximizing output where Marginal Cost (MC) equals Marginal Revenue (MR), and the MC curve cuts the MR curve from below.
- The difference between Price (or AR) and ATC represents the abnormal profit earned per unit of output.

Thus:

$$P=AR>ATC$$

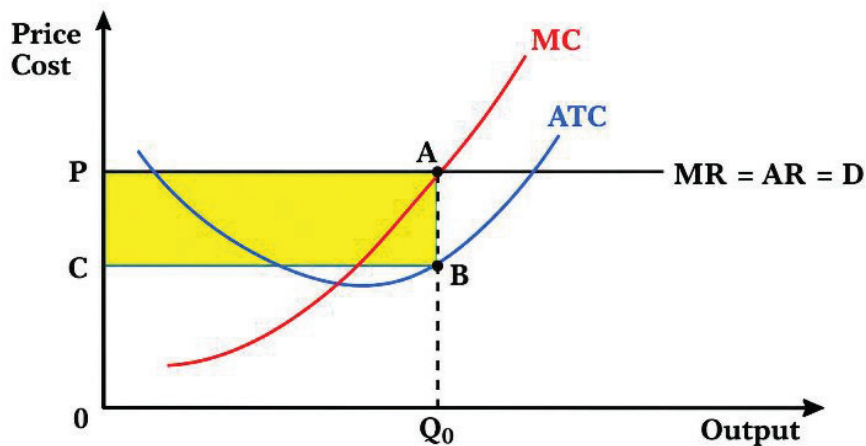


Figure 5: Abnormal profits in the short run under perfect competition

From the figure above, the firm is in equilibrium at point A, where $MC = MR$, at the output level OQ_0 (equilibrium output). This output (OQ_0) is produced at a total cost of $OCBQ_0$ and sold at a price of OP , generating a total revenue of $OPAQ_0$, a price that has been predetermined by the forces of demand and supply in the market. This results in the firm earning abnormal (supernormal) profits, shown by the shaded area $CPAB$.

Profit = Total Revenue – Total Cost

From the figure: Profit = $OPAQ_0 - OCBQ_0 = CPAB$

Since the firm produces at output OQ_0 , which lies beyond the point of minimum average total cost (the lowest point on the ATC curve), the firm is operating at overcapacity in the short run.

b) Losses under Perfect Competition in the Short Run

A firm under perfect competition incurs **losses (sub-normal profits)** in the short run when its total cost exceeds total revenue. This happens when the market price is lower than the average cost of production.

The firm is still in equilibrium at the output level where:

$MC=MR$ and the marginal cost (MC) curve is rising.

A perfectly competitive firm makes losses under the following conditions:

- Price (P), which is equal to Average Revenue (AR), is less than Average Total Cost (ATC) at the equilibrium level of output.
- The firm produces at the equilibrium output where Marginal Cost (MC) equals Marginal Revenue (MR), and the MC curve cuts the MR curve from below.
- The difference between ATC and Price (or AR) represents the loss per unit of output.

Thus: $P=AR < ATC$

Causes of Losses in the Short Run

A firm may incur losses because:

- production costs are high,
- demand for the product is low,
- the firm is inefficient,
- it is newly established, or
- market prices have fallen.

However, the firm may continue operating in the short run if total revenue can cover variable costs, hoping to improve efficiency or market conditions in the future.

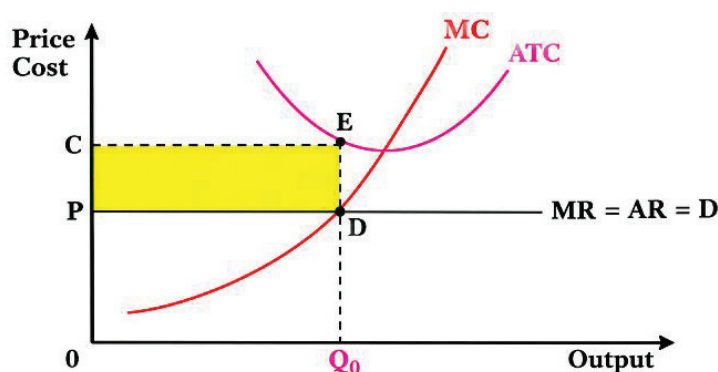


Figure 6: Losses under Perfect Competition in the Short Run

As shown in the figure above, the firm produces output OQ_0 at total cost $OCEQ_0$ and sells it at price OP , getting total revenue $OPDQ_0$, hence making losses $PCED$ because the AC is greater than the AR.

Losses = $TC - TR$. Thus, from the above curve, losses = $OCEQ_0 - OPDQ_0 = PCED$. Since the firm is producing output Q_0 , which is less than the optimum output (where $MC = ATC$), the firm is producing at excess capacity.

1.2.7. Price and output determination under perfect competition in the long run.

Abnormal (supernormal) profits earned in the short run attract new firms into the industry because there are no barriers to entry under perfect competition. As more firms enter the market, industry supply increases, causing the market price to fall. Consequently, the abnormal profits of existing firms are gradually reduced. This process continues until all firms earn only normal (zero economic) profits. Long-run equilibrium is therefore achieved when firms have no incentive to enter or leave the industry and each firm produces at the profit-maximizing output where $MC = MR$.

In the long run, a perfectly competitive firm operates under the following conditions:

- Average Revenue (AR), which is equal to Price (P), is equal to Average Cost (AC or ATC), implying that the firm earns only normal profits.
- The AR (Price) curve is tangent to the AC curve at the equilibrium output, indicating that the firm is covering all its costs, including normal profit.
- Marginal Cost (MC) equals Marginal Revenue (MR), and the MC curve cuts the MR curve from below at the equilibrium point.

- The MC curve intersects the AC curve at its minimum point, showing that the firm is operating at its optimum scale and producing at the lowest possible average cost.
- At long-run equilibrium, **$P = AR = MR = MC = AC$** , and neither entry nor exit of firms occurs since all firms earn normal profits.

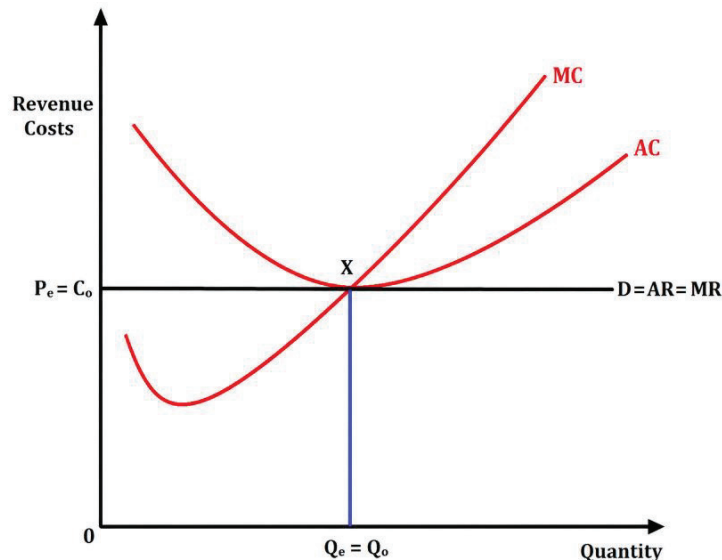


Fig 7: Normal profits under perfect competition

As illustrated in the figure, the firm attains long-run equilibrium at point X, where the Marginal Cost (MC) curve intersects the Marginal Revenue (MR) curve and the equilibrium output is $0Q_e$ ($0Q_o$). At this point, the firm satisfies the conditions for equilibrium, such that **$MC = MR = AR = P = AC$** . The firm produces $0Q_e$ units of output at an average cost of $0C_o$ and sells its output at the market price $0P_e$, where $P_e = C_o$. Since Average Revenue (AR) equals Average Cost (AC), the firm earns normal (zero economic) profit, covering all production costs, including normal returns to the entrepreneur. Furthermore, the MC curve intersects the AC curve at its minimum point, indicating that the firm is operating at its optimum scale (**$Q_e = Q_o$**) and producing at the lowest possible average cost.

1.2.8 Break-even and Shutdown Points of a Firm under Perfect Competition

Under perfect competition, a firm may experience different profit positions depending on the relationship between market price and its costs of production. In the long run, abnormal (supernormal) profits attract new firms into the industry because entry is unrestricted. As more firms enter, industry supply increases, causing market prices to fall until abnormal profits are eliminated. The firm eventually reaches the break-even point, where Price (P) or Average Revenue (AR) is equal to Average Total Cost (ATC), that is, **$P = AR = ATC$** . At this point, the firm covers all its production costs, including normal returns to the entrepreneur, and earns only normal (zero economic) profit. The break-even point is therefore defined as the point at which a firm neither earns abnormal profits nor incurs losses because total revenue is equal to total cost.

If market supply continues to increase and prices fall below the break-even level, the firm begins to incur losses. However, it may continue operating in the short run provided it can cover its Average Variable Costs (AVC). The firm reaches the shutdown point when Price (P) or Average Revenue (AR) is equal to Average Variable Cost (AVC),

that is, $P = AR = AVC$. At this point, the firm is only able to cover its variable costs and makes no contribution towards fixed costs. The shutdown point is therefore defined as the point below which a firm cannot continue operating because it is unable to cover its variable costs. If price falls below AVC , the firm is better off temporarily ceasing production in the short run. Thus, while the break-even point represents the minimum price required to cover all costs ($P = ATC$), the shutdown point represents the minimum price required for the firm to remain in operation in the short run ($P = AVC$).

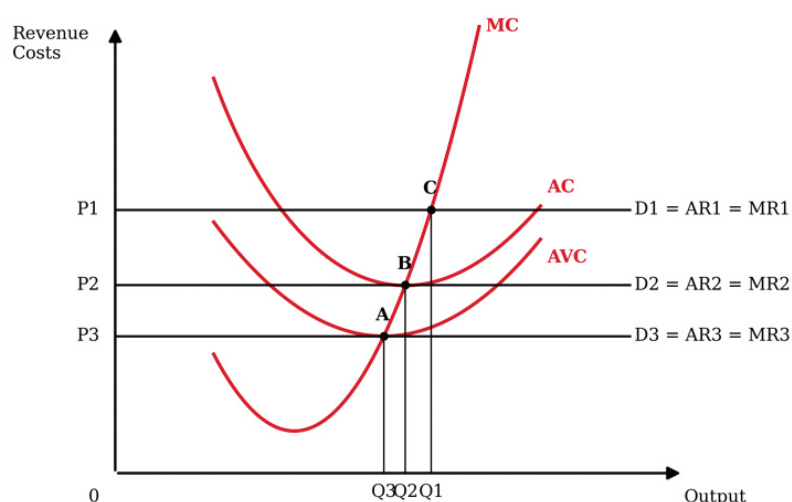


Figure 8: Breakeven and Shutdown points of a firm under perfect competition.

As illustrated in the figure, the firm initially earns abnormal profits at point C in the short run, attracting new firms into the industry. As market supply increases, price falls until firms earn only normal (zero economic) profit at point B, where price equals average cost ($P = AC$). Therefore, B is the break-even point, where the firm covers all production costs but earns no abnormal profit.

If price falls further to point A, the firm can only cover its average variable costs (AVC). Thus, A is the shutdown point, where the firm is indifferent between producing and ceasing operations. Below this point, the firm cannot cover its variable costs and will shut down in the short run.

Importance of the Break-even and Shutdown Points

The break-even and shutdown points are important decision-making tools for firms operating under perfect competition because they help managers evaluate the viability and profitability of production. Specifically, these points enable firms to:

- Determine whether to continue production or temporarily cease operations based on prevailing market prices and production costs.
- Assess profitability and operational efficiency by comparing revenue with total and variable costs.
- Minimize losses by identifying the lowest price at which production remains economically justifiable.
- Make informed output and pricing decisions in highly competitive markets.

- Identify the minimum price required to cover all costs (break-even point) and the minimum price necessary to remain in operation in the short run (shutdown point).
- Guide long-run planning and resource allocation by indicating whether firms should remain in, enter, or exit an industry.

These concepts are therefore essential for ensuring efficient production, profit maximization, and the long-term survival of firms in competitive markets.

1.2.9 Why Firms Continue Operating Despite Losses

Firms may choose to remain in business even when they are not covering all their costs. The reasons include:

- **Access to external financing:** Firms may anticipate loans, subsidies, or bailouts from banks, investors, or government institutions.
- **Government ownership or support:** Public enterprises often prioritize social objectives such as employment creation, provision of essential services (e.g., water, electricity, newspapers), or stabilizing markets rather than profit.
- **Infant industry protection:** New firms may endure short-term losses with the expectation of overcoming startup challenges and achieving economies of scale in the long run.
- **Merger or acquisition prospects:** Firms may continue operating while awaiting integration with stronger companies.
- **Market presence:** Closing down risks losing market share and customer loyalty, which may be difficult to regain later.
- **Subsidiary support:** Loss-making branches may be sustained by profits from other divisions of the parent company.
- **Retention of skilled labor:** Firms may prefer to absorb losses rather than lose highly trained employees, whose replacement would be costly.
- **Welfare orientation:** Some firms are established to provide essential goods and services at affordable prices, regardless of profitability.
- **Management restructuring:** Losses may be temporary due to poor administration, with hopes of improvement under new leadership.
- **Reputation and credibility:** Firms may continue to operate to maintain goodwill, brand image, and trust among stakeholders.
- **Technological adaptation:** Entrepreneurs may sustain operations while adopting new production methods or innovative techniques.
- **Cost reduction strategies:** Owners may anticipate solving inefficiencies, such as reducing excess manpower or improving supply chains.
- **Marketing and pricing improvements:** Firms may expect better promotional strategies or favourable market conditions in the future.
- **Short-run vs. long-run considerations:** In microeconomics, firms may continue operating in the short run if they can cover variable costs, even if they cannot cover fixed costs, because shutting down immediately would result in greater losses.

1.2.10 Advantages and Disadvantages of Perfect Competition

Advantages

Perfect competition is a market structure characterized by many buyers and sellers, free entry and exit of firms, perfect information, and homogeneous products. These features influence how firms operate and how resources are allocated within the economy, leading to several important outcomes for producers, consumers, and society as a whole. The following are the major advantages associated with a perfectly competitive market structure:

- **Lowest possible average cost:** In the long run, firms operate at the most efficient scale.
- **No advertising costs:** Homogeneous products eliminate the need for costly promotions.
- **High output levels:** Free entry and exit encourage maximum production.
- **Efficient resource allocation:** Resources flow to the most productive firms.
- **Quality improvements:** Competition drives firms to enhance product standards.
- **Consumer welfare:** Abundant supply of affordable, high-quality goods raises living standards.
- **No resource wastage:** Inefficient firms exit, leaving resources in the hands of efficient producers.
- **Consumer sovereignty:** Buyers influence production decisions through demand.

Disadvantages

Although perfect competition promotes efficiency, it also has several limitations that may affect producers, consumers, and the economy as a whole. The intense competition and market conditions associated with this market structure can influence firms' profitability, innovation, product quality, and long-term growth. The following are the major disadvantages associated with a perfectly competitive market structure.

- **Only normal profits in the long run:** Firms cannot earn supernormal profits, limiting reinvestment and expansion.
- **Limited research and innovation:** Lack of excess profits discourages investment in R&D.
- **Unemployment risks:** Inefficient firms are forced out, leading to job losses.
- **Unrealistic assumptions:** Perfect competition rarely exists in reality (e.g., perfect knowledge, homogeneous products).
- **Resource exhaustion:** Free entry may lead to overuse of scarce resources.
- **Limited consumer choice:** Products are identical, reducing variety.

- **No abnormal profits:** Firms struggle to grow beyond their current scale.
- **Difficulty in long-term sustainability:** Without innovation and differentiation, firms may stagnate.

Application activity 1.2



1. Imagine you are a tomato seller in a market with many other sellers selling identical tomatoes.
 - a) What type of market are you operating in?
 - b) Would you have control over the price? Explain.
 - c) What would happen if you increased your price above others?
2. Explain how a market can exist without a physical location. Give an example.
3. Identify any four factors that determine the type of market structure.
4. In a market where all sellers sell identical products, how do buyers choose from whom to buy?
5. Explain why firms in perfect competition are called price takers.
 - a) A firm produces 100 units where its marginal cost (MC) equals marginal revenue (MR). What does this indicate?
 - b) What would happen if a firm produces at a level where MC is greater than MR?
 - c) Why is the condition $MC = MR$ important for profit maximization?
 - d) A firm is making losses but still covers its variable costs. Should it continue operating? Explain.
 - e) What happens if a firm cannot cover its variable costs?
 - f) Explain how perfect competition may lead to low profits.
 - g) Why is perfect competition considered unrealistic in real life?
6. A market has many sellers, no barriers to entry, and all firms earn normal profits in the long run.
 - a) Identify the type of market.
 - b) Explain why firms earn only normal profits in the long run.
 - c) What would happen if new firms enter this market?

1.3 Monopoly

Learning activity 1.3



Answer the following questions:

1. What type of market structure exists when a single seller dominates, and what are two defining features of this structure?
2. Identify and explain two major factors that can give rise to such market dominance (e.g., patents, control of raw materials).
3. At what point does a firm in this market structure maximize profit in the short run, and what happens if Average Revenue (AR) is less than Average Cost (AC)?
4. What are two measures governments can use to regulate or control firms with excessive market power, and why are they necessary?
5. Give one benefit and one drawback of this market structure for society, and briefly explain how charging different prices to different groups of consumers can affect both buyers and sellers.

Imagine a market where there is only one seller and consumers have no alternative source of the product or service. How does such a situation arise, and what effects does it have on prices, consumers, and the economy? This lesson takes you into an interesting area of economics where business power, consumer interests, and government action come together. As you read on, you will discover why some firms become so dominant, how they influence the markets they operate in, and why this type of market structure is known as a monopoly.

1.3.1 Meaning, forms and features of monopoly

Meaning of monopoly

A monopoly is a market structure in which there is only one producer or seller of a commodity or service with no close substitutes. Entry of other firms into the market is highly restricted, giving the firm significant control over price and supply.

Forms of Monopoly

- **Pure (Absolute) Monopoly:** A pure or absolute monopoly is a market situation where a single firm is the only producer and seller of a product that has no close substitutes, giving it complete control over the supply of that product. Because consumers cannot obtain the product from another source, the firm enjoys significant market power over price and output. However, pure monopolies are extremely rare in practice because consumers or other firms can usually develop alternative products or substitutes over time.
- **Monopsony:** A monopsony is a market situation where there is only one buyer of a product or factor of production, giving that buyer significant bargaining power over sellers. Since producers have only one major customer, they often have limited ability to negotiate prices and may be forced to accept the

terms offered. For example, if a government is the sole purchaser of military equipment from a specialized defense contractor, the contractor has only one buyer for its product, giving the government considerable influence over pricing and contract conditions.

- **Bilateral Monopoly:** A bilateral monopoly is a market structure where one seller deals with one buyer in a particular product or service. Since both parties have market power, neither can determine the price alone, so the final price is usually reached through negotiation. For example, a labour union (the sole seller of labour) negotiating with a single large employer (the sole buyer of labour) creates a bilateral monopoly situation, with wages and working conditions determined through bargaining.
- **Discriminatory Monopoly:** A discriminatory monopoly occurs when a monopolist charges different prices to different customers for the same product or service, even though the cost of providing it is essentially the same. The monopolist is able to separate customers according to their willingness or ability to pay and uses this power to maximize profits. For example, RwandAir may charge different fares to passengers travelling in the same cabin and receiving the same service, depending on factors such as the booking date, ticket conditions, or length of stay. Thus, customers purchase the same service but at different prices.
- **Collective Monopoly:** A collective monopoly occurs when two or more firms agree to act together as if they were one seller, rather than competing independently. These firms may coordinate prices, output levels, markets, or other business decisions, giving them greater control over the market and reducing competition. For example, if several cement manufacturers agreed to charge the same price and limit the quantity of cement supplied to the market, consumers would face conditions similar to those of a monopoly even though several firms exist.
- **Natural Monopoly:** A natural monopoly arises when one firm can supply the entire market at a lower average cost than several competing firms due to very large economies of scale or high infrastructure costs. In such industries, the market demand is usually only sufficient for one efficient producer, and having multiple firms would lead to unnecessary duplication of facilities and increased costs. A good example is electricity distribution, where building separate power lines and networks for different companies would be extremely costly and inefficient compared to having one provider serve the entire market.
- **Statutory Monopoly:** A statutory monopoly is a monopoly that exists because the government grants exclusive legal rights to a particular firm or organization to provide a specific product or service. Unlike other monopolies that arise from market conditions, a statutory monopoly is created and protected by law, preventing competitors from entering the market. For example, only the National Bank of Rwanda (BNR) has the legal authority to issue Rwanda's currency; no private institution is permitted to perform this function.

- **Spatial (Local) Monopoly:** A spatial or local monopoly occurs when a firm becomes the only or dominant supplier in a particular geographical area because competing suppliers are too far away or transport costs make competition difficult. As a result, consumers in that location have limited alternatives and must rely on the local supplier. For example, a pharmacy or retail shop in a remote village in Rwanda may enjoy monopoly power because residents would have to travel long distances and incur additional costs to access similar services elsewhere.

Features/Characteristics of monopoly

- There is only one seller in the market.
- The product has no close substitutes.
- Entry of new firms is blocked by barriers such as government regulations, patents, or high capital requirements.
- The monopolist is a **price maker** and has control over output and price.
- The main objective of the firm is profit maximization, where output is produced at the point where:
MC = MR
- The demand curve facing the monopolist is downward sloping from left to right.

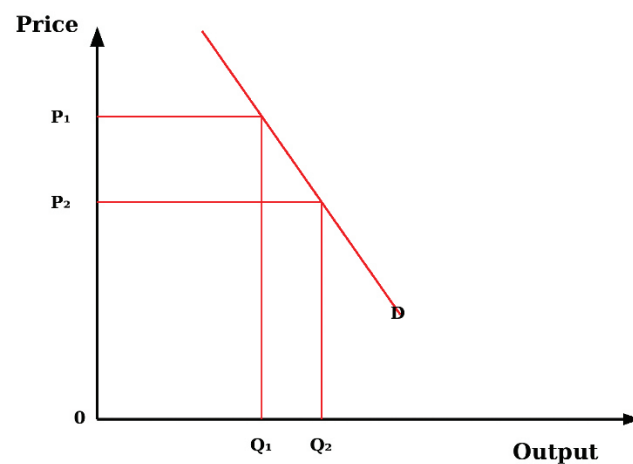


Figure 9: The demand curve under monopoly

The diagram shows the monopolist's downward-sloping demand curve. As price falls from P_1 to P_2 , quantity demanded increases from Q_1 to Q_2 . Demand is relatively inelastic because consumers have few or no close substitutes for the monopolist's product, so changes in price cause only a relatively small change in quantity demanded.

- The firm may operate with excess capacity in both the short run and long run.
- Consumers usually have limited choice under monopoly markets.
- Marginal revenue curve is below the Average revenue curve because for monopoly firm to increase revenue, it has to lower the price.

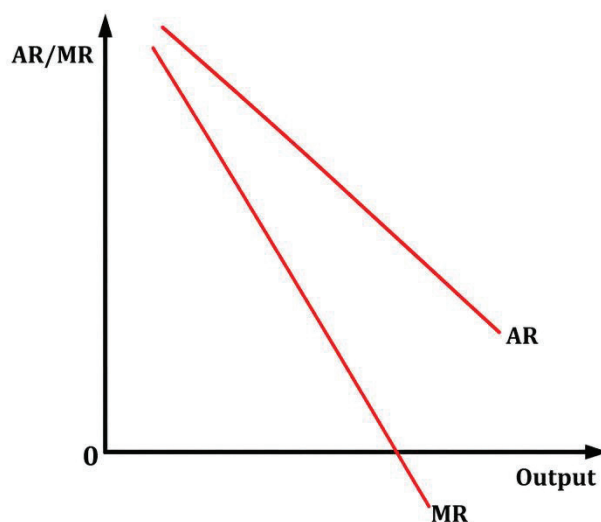


Figure 10: Average Revenue and Marginal Revenue curves under monopoly.

The MR curve follows the same downward-sloping shape as the AR curve but declines more rapidly and lies below it. This is because a monopolist must reduce the price to sell additional units, causing marginal revenue to be less than average revenue. Therefore, both AR and MR fall as output increases, but MR falls at a faster rate than AR.

Examples of Monopoly in Rwanda

- Water and Sanitation Corporation (WASAC)
- Rwanda Energy Group (REG)
- National Bank of Rwanda (NBR)

1.3.2 Sources/bases of monopoly Power

Monopoly power arises when a firm gains exclusive control over the production or supply of a good or service, making competition difficult or impossible. The main sources of monopoly include:

- **Patent Rights and Copyrights:** Governments may grant legal protection to inventors, authors, and companies, giving them exclusive rights to produce or sell a product for a certain period.
- **Ownership of Key Raw Materials:** A firm that controls important natural resources or raw materials can prevent other firms from producing similar products.
- **Advanced Technology and Special Skills:** Some firms develop unique technology, production methods, or technical knowledge that competitors cannot easily copy.
- **Economies of Scale:** Large firms can produce at lower costs than small firms, making it difficult for new firms to compete effectively.
- **Government Policy and Licensing:** Governments may grant exclusive licenses or protect local firms through tariffs, quotas, and restrictions on foreign competition.

- **Mergers and Takeovers:** When firms combine or acquire competitors, competition is reduced and monopoly power increases.
- **Geographical Isolation:** In remote areas, one firm may dominate the market because other firms find it expensive or difficult to operate there.
- **Heavy Capital Requirements:** Some industries require huge amounts of capital, discouraging new firms from entering the market.

1.3.3 Monopoly equilibrium and profit maximization in the short run

Equilibrium Position of a Monopoly Firm

A monopoly firm is in equilibrium when it produces the level of output that maximizes profit or minimizes loss. This occurs where marginal cost (MC) equals marginal revenue (MR) and the MC curve cuts the MR curve from below. At this point, the monopolist has no incentive to increase or reduce output because any change would lower profit or increase losses. In the short run, a monopolist may earn supernormal profits, normal profits, or even incur losses, depending on the relationship between the market price (Average Revenue) and the Average Total Cost (ATC). If price is above ATC, the firm earns profits; if price equals ATC, it earns normal profit; and if price is below ATC, it incurs losses.

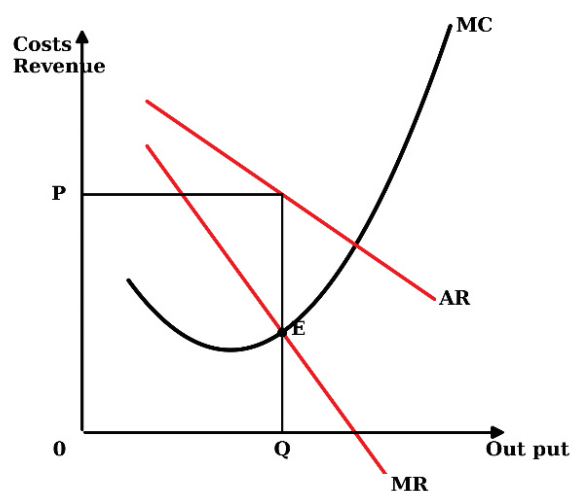


Figure 11: Equilibrium position of a firm under monopoly

The figure illustrates the equilibrium position of a monopoly firm at point E, where Marginal Cost (MC) equals Marginal Revenue (MR) and the MC curve cuts the MR curve from below. The monopolist produces output Q and charges price P, which is determined from the Average Revenue (AR) curve. At this point, the firm maximizes profit or minimizes loss, making it the monopoly equilibrium output and price.

Profit maximization of a firm under monopoly in the short-run

In the short run, a monopolist aims to maximize profits by producing the level of output where Marginal Cost (MC) equals Marginal Revenue (MR). When demand for the product increases, the monopolist can raise both the price and output by employing more variable inputs such as labour, raw materials, and working hours. Conversely, when demand falls, the monopolist can reduce output by cutting back on the use of

these variable inputs. As a price maker, the monopolist has the power to influence the market price and can charge a price above the Average Cost (AC) of production. When the price exceeds average cost, the firm earns supernormal (abnormal) profits; when price equals average cost, it earns normal profit; and when price falls below average cost, it incurs losses. Therefore, the monopolist maximizes profit by choosing the output where $MC = MR$ and then charging the highest price consumers are willing to pay for that output, as indicated by the demand (AR) curve.

Conditions for short run profit maximization under Monopoly

A monopolist earns supernormal (abnormal) profits when:

- Marginal Revenue (MR) = Marginal Cost (MC).
- The MC curve cuts the MR curve from below.
- Average Revenue (AR) is greater than Average Cost (AC) at the equilibrium output.

The equilibrium output is determined where $MR = MC$, while the equilibrium price is obtained from the AR (demand) curve at that output level. Therefore, a monopolist maximizes profit in the short run by producing the output where $MR = MC$ and earns supernormal profits when AR exceeds AC.

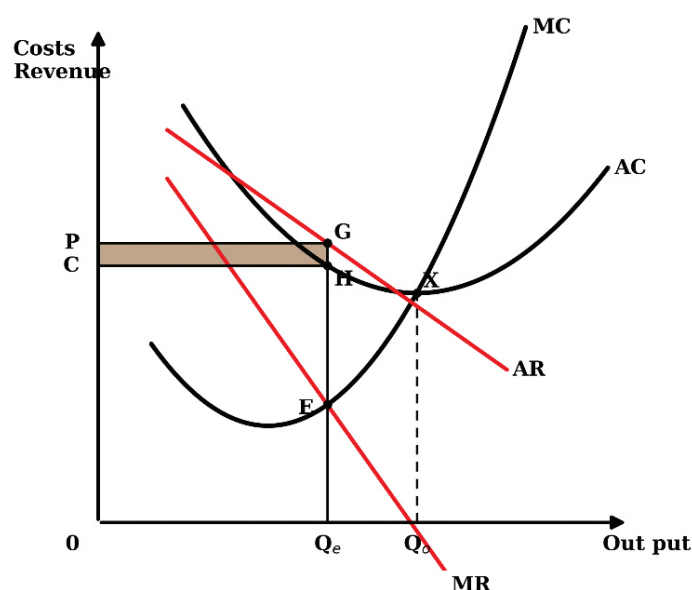


Figure 12: Abnormal profits of a firm under monopoly in the short-run

The diagram shows a monopolist maximizing profit in the short run at output OQ_e , where **Marginal Cost (MC) equals Marginal Revenue (MR)** and the **MC curve cuts the MR curve from below** at point E, satisfying the conditions for equilibrium. At this output, the firm produces at an **Average Cost (AC) of OC** and charges a price of **OP**, determined from the **Average Revenue (AR) or demand curve**. The firm's **Total Revenue** is represented by **OPGQ_e**, while **Total Cost** is represented by **OCHQ_e**. The difference between total revenue and total cost, shown by the area **CPGH**, represents the firm's **supernormal (abnormal) profit**. The equilibrium output OQ_e is less than the optimum output OQ_0 , where average cost is at its minimum, indicating that the monopolist operates with **excess capacity** while maximizing profits.

Losses under monopoly in the short-run

Losses under monopoly in the short run

A monopolist may make losses in the short run when the revenue earned from selling its output is less than the cost of producing that output. This occurs when the Average Revenue (AR) or price is lower than the Average Cost (AC) at the equilibrium output. Despite making losses, the monopolist may continue operating in the short run as long as it can cover its variable costs and minimize losses by producing at the equilibrium level.

Conditions for losses under monopoly

- Average Revenue (AR) is less than Average Cost (AC), that is, the AC curve lies above the AR curve at the equilibrium output.
- The MC curve cuts the MR curve from below, indicating equilibrium.
- The AR curve determines the selling price, while the AC curve determines the cost per unit of output.
- Since AC exceeds AR, total cost is greater than total revenue, resulting in losses.

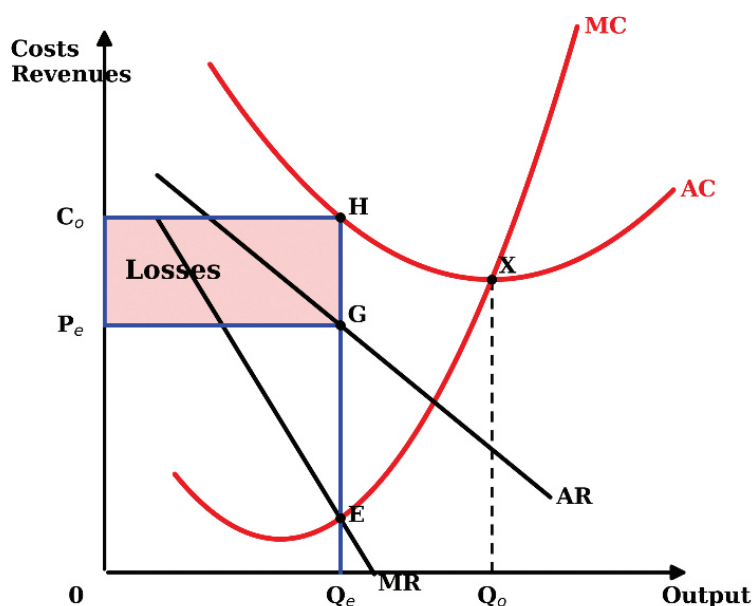


Figure 13: Losses of a firm under monopoly in the short-run

The diagram shows a monopolist minimizing losses in the short run by producing output OQ_e , where $MC = MR$ and the MC curve cuts the MR curve from below at point E , satisfying the equilibrium conditions. At this output, the firm charges price OP_e as determined by the AR (demand) curve, while its average cost is OC_o , resulting in a loss represented by the shaded area $P_e C_o H G$. The firm continues operating because producing at OQ_e minimizes losses in the short run. However, the equilibrium output OQ_e is lower than the optimum output OQ_o , where the AC curve reaches its minimum point at X . This indicates that the monopolist is operating with **excess capacity**, meaning it is producing below its full efficient scale while still minimizing losses.

Long-Run profit maximization and monopoly

In the long run, a monopolist can continue to earn **abnormal (supernormal) profits** because **barriers to entry** prevent new firms from entering the market and competing away profits. The monopolist is in equilibrium when it produces the output where **Long-Run Marginal Revenue (LMR) = Long-Run Marginal Cost (LMC)**, and sets price based on the **Average Revenue (AR) / demand curve**.

Conditions for Long-Run profit maximization

- **MR = MC (or LMR = LMC):** This condition determines the **profit-maximizing output (Q_e)**.
- **MC cuts MR from below:** Confirms that the equilibrium output yields **maximum profit**.
- **Price is determined from the AR (demand) curve:** At output Q_e , the monopolist reads the **equilibrium price (P_e)** from the AR curve.
- **AR > AC at Q_e :** Indicates the firm earns **abnormal profits** (revenue per unit exceeds cost per unit).
- **AC curve lies below AR at equilibrium output:** Shows a positive profit margin per unit of output.
- **MC curve intersects AC at its lowest point (minimum AC):** Reflects **long-run productive efficiency of cost structure**, even though output is not socially optimal.
- **Barriers to entry are essential**
These may include:
 - Legal protection (patents, licenses)
 - Control of key resources
 - High start-up costs
 - Brand loyalty and economies of scale

They prevent competition and allow abnormal profits to persist.

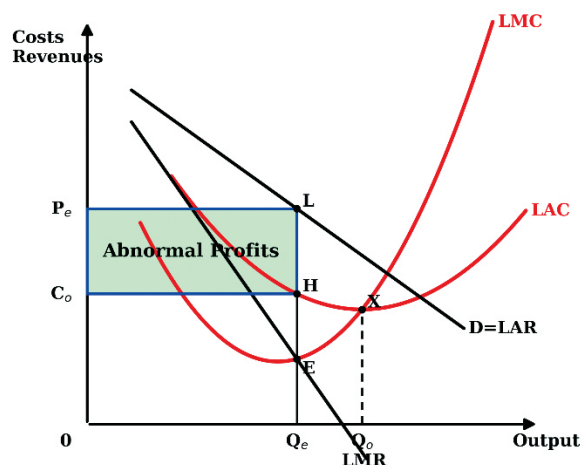


Figure 14: Long run profit maximization under monopoly

The figure above shows a monopolist earning **supernormal profits** in the long run. Equilibrium occurs at point E, where $LMC = LMR$. The firm produces OQ_e and charges price OP_e from the **LAR (demand) curve**, while its average cost is OC_o from the **LAC curve**. Since **price exceeds average cost ($AR > AC$)**, the monopolist earns **supernormal profits**, represented by the shaded area C_oP_eLH . The firm produces OQ_e , which is below the optimum output OQ_o , indicating **excess capacity**.

Key Points

- » The monopolist is a price maker, not a price taker.
- » Output is usually lower and price higher than in competitive markets.
- » There is no long-run supply curve for a monopoly because output and price depend on demand and cost conditions.
- » Monopolies are often allocative inefficient because $P > MC$, meaning resources are not optimally distributed.

1.3.4 Measures for controlling monopoly

Governments employ various strategies to regulate monopolies and protect consumers:

- **Price Legislation:** Setting maximum prices to prevent exploitation.
- **Anti-Monopoly (Anti-Trust) Laws:** Prohibiting monopolization, collusion, and unfair practices that restrict competition.
- **Nationalization:** Taking ownership of monopoly firms to ensure fair pricing and service delivery.
- **Subsidies for New Firms:** Supporting new entrants to compete effectively with established monopolies.
- **Resale Price Maintenance:** Producers may set fixed resale prices to prevent excessive markups.
- **Encouraging Imports:** Allowing foreign goods to compete with domestic monopolies.
- **Consumer Action:** Boycotts or demonstrations can pressure monopolists to lower prices.
- **Public Enterprises:** Governments may establish their own firms to compete directly with monopolies.
- **Removal of Protectionist Policies:** Reducing trade barriers and excessive taxation to foster competition.
- **Taxation:** Imposing taxes on monopolists to limit excessive profits and redistribute income. Such taxes may include:
 - a) Sur tax: This is an incentive tax that provokes monopolists to increase their scale of production.
 - b) Advalerem tax: This is a tax levied on the value of the commodity.

c) Specific tax: Tax charged per unit of out put.

d) Lump-sum tax: A tax levied on the total amount produced and it does not change even if output changes.

1.3.5 Advantages and disadvantages of monopoly

Monopolies are often described as a **double-edged sword** because they bring both benefits and drawbacks to the economy.

Advantages and disadvantages of monopoly

A. Advantages of Monopoly

- **Avoidance of duplication of services:** Monopoly helps to avoid unnecessary duplication of services and resources. For example, it may be uneconomical to establish several electricity or water supply systems in the same area.
- **Economies of scale:** Large-scale production enables monopolies to enjoy economies of scale, leading to lower average costs of production and increased efficiency.
- **Encourages research and innovation:** Monopoly firms often earn supernormal profits which can be used to finance research, technological improvement, and innovation. Patent rights and copyrights also encourage inventors and investors.
- **Possibility of price discrimination:** A monopolist may charge different prices to different groups of consumers according to their ability to pay. This can benefit low-income consumers in some cases, such as reduced transport or utility charges.
- **Stable supply of essential services:** Government monopolies in public utilities such as electricity, water, roads, and telecommunications can ensure reliable and coordinated service delivery.
- **Protection of infant industries:** New industries may be protected from foreign competition until they become strong enough to compete internationally.
- **Reduced advertising costs**
Since there is little or no competition, monopolies spend less on persuasive advertising, reducing wastage of resources.
- **Long-term investment**
Monopoly firms can undertake large and expensive projects because they have strong financial capacity and stable profits.

B. Disadvantages of monopoly

- **High prices to consumers:** Monopolists may charge high prices because consumers have few or no alternative suppliers.
- **Poor quality products and services:** Lack of competition may reduce the incentive to improve product quality or customer service.
- **Restricted consumer choice:** Consumers are denied a variety of products

Experimental version

since the monopolist may supply only one type of product or service.

- **Production at excess capacity:** Monopoly firms may deliberately produce less output to keep prices high, leading to under-utilization of resources.
- **Exploitation of consumers**
Consumers may be overcharged through unfair pricing policies and price discrimination.
- **Possibility of shortages**
If the monopolist stops production or experiences failure, the market may suffer severe shortages because there are no alternative suppliers.
- **Income inequality**
Monopoly profits are concentrated in the hands of a few owners, widening the gap between the rich and the poor.
- **Political influence and corruption**
Large monopoly firms may influence government policies and decision-making in their favor.
- **Barriers to new firms**
Monopoly discourages new entrants into the industry, limiting competition and entrepreneurship.
- **Economic inefficiency**
Lack of competition may result in waste, managerial inefficiency, and slow adoption of new methods of production.

1.3.6 Price discrimination under monopoly

Meaning

Price discrimination occurs when a monopolist sells the same product at different prices to different customers, regardless of production cost.

Forms of Price Discrimination

- **By Income:** Charging higher fees to wealthier clients, lower fees to poorer ones.
- **By Age:** Discounts for children, students, or seniors.
- **By Gender:** Different entry fees (e.g., ladies' nights).
- **By Geography:** Different prices in different regions.
- **By Time of Service:** Higher prices during peak demand, lower during off-peak.
- **By Product Nature:** Variations such as hardcover vs. Softcover books.
- **By Usage:** Different rates for industrial vs. Domestic electricity.

Conditions for success

- The seller must be a monopolist.
- Demand elasticity must differ across markets.
- Higher prices are charged where demand is less elastic.
- Lower prices are charged where demand is more elastic.

Advantages

- Provides affordable access for low-income groups.
- Increases total revenue and output.
- Promotes income redistribution.
- Boosts sales and consumption.
- Helps dispose of surplus or lower-quality goods.

Disadvantages

- May reduce service quality for lower-paying customers.
- Dumping can harm local industries.
- Poor-quality goods may be sold to disadvantaged groups.
- Can misallocate resources toward higher-paying markets.

Application activity 1.3



1. How does monopoly power influence market prices and overall resource allocation in an economy?
2. What happens to a firm's profitability when Average Revenue (AR) is less than Average Cost (AC), and under what conditions should the firm continue operating rather than shut down?
3. How do government interventions such as price controls or import promotion protect consumers from exploitation by dominant firms?

1.4 Monopolistic competition

Learning activity 1.4



Imagine walking through Kigali City Market looking for soap. You find many sellers offering soap, but no two products are exactly alike. Some soaps are scented, some are made for babies, and others differ in quality and price. As a result, customers can choose the product that best suits their preferences and budget.

Required:

1. Although all the vendors sell soap, what factors might influence a consumer to choose one type or brand of soap instead of simply buying the cheapest one available?
2. If the price of one brand of soap increased by 100 RWF, would all its customers immediately stop buying it and switch to another brand? Explain your answer.

Like monopoly, monopolistic competition is a market structure under imperfect competition and lies between perfect competition and monopoly. To understand this market structure, it is important to examine its features, identify examples from local communities, analyze the short-run and long-run operations of firms, and assess its effects on consumers, producers, and the wider community. This provides a basis for understanding the advantages and disadvantages of monopolistic competition.

1.4.1 Meaning and characteristics of firms under monopolistic competition

i. Meaning of monopolistic competition

Monopolistic competition is a market structure where many firms sell similar but differentiated products that are close substitutes for one another. Each firm tries to make its product appear different from competitors' products in order to attract customers.

Product differentiation may be through:

- Packaging,
- Branding,
- Quality,
- Design and style,
- Advertising,
- After-sales services,
- Giving credit facilities, and
- Location of the business.

Examples include restaurants, salons, clothing shops, soft drinks, and mobile phone brands.

ii. Characteristics of Firms under Monopolistic Competition

- **Many Buyers and Sellers:** The market consists of many firms and consumers, but no single firm dominates the entire market.
- **Product Differentiation:** Firms sell products that are slightly different in quality, packaging, design, or brand name, although they remain close substitutes.
- **Free Entry and Exit of Firms:** Firms can easily enter or leave the industry in the long run because barriers to entry are relatively low.
- **Advertising and Sales Promotion:** Firms engage in persuasive and informative advertising to attract customers and create brand loyalty.
- **Downward Sloping Demand Curve:** Each firm faces a downward-sloping and fairly elastic demand curve because consumers can switch to close substitutes if prices rise.
- **Limited Price Control:** Firms are price makers to a certain extent because of product differentiation, but they also face competition from rival firms.

- **Profit Maximization Objective:** Firms aim at maximizing profits by producing where: $MR = MC$
- **Excess Capacity Production:** Firms often produce below their full capacity to maintain higher prices and product uniqueness.
- **Brand Loyalty:** Consumers may develop a preference for a particular brand even when similar products are available.

1.4.2 Equilibrium position of a firm under monopolistic competition

A firm under monopolistic competition is in equilibrium when: $MC = MR$

At equilibrium, the firm chooses the output level where profits are maximized or losses are minimized.

Short-Run Equilibrium:

In the short run, a firm may experience:

- Supernormal (abnormal) profits when: $AR > AC$
- Normal profits when: $AR = AC$
- Losses when: $AR < AC$

Firms may increase advertising, improve product quality, or introduce new product features to attract more customers and increase profits. However, because many close substitutes exist, the power of each firm remains limited.

Long-Run equilibrium

In the long run, supernormal profits attract new firms into the industry because entry barriers are low. Increased competition reduces the demand faced by existing firms until they earn only normal profits.

Thus, in the long run:

- firms earn normal profits only,
- there is free entry and exit of firms, and
- firms continue producing at excess capacity.

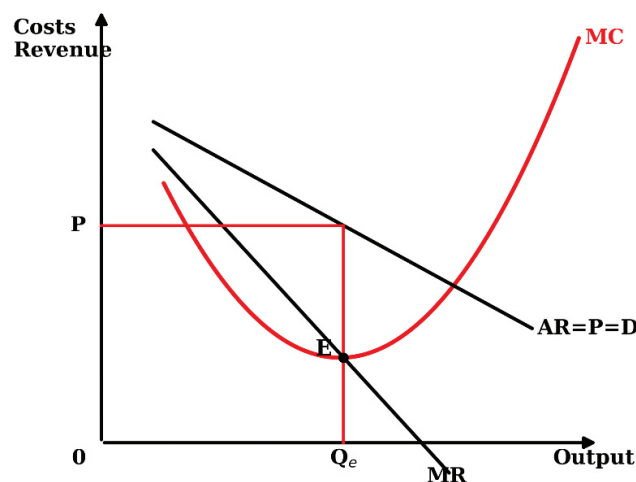


Figure 15: Equilibrium position of a firm under monopolistic competition

The firm is in equilibrium at point **E**, where **MC = MR**. At this point, the firm produces Q_e units of output. The corresponding price, P , is determined from the AR (Demand) curve directly above Q_e . Therefore, the firm maximizes profit by producing Q_e units and selling them at price P . In the short run, the firm may earn supernormal profits, but in the long run, the entry of new firms reduces these profits to normal profits.

1.4.3 Price and output determination under monopolistic competition in the short run

In monopolistic competition, many firms sell similar but differentiated products (e.g., toothpaste brands such as Colgate and Whitedent). Each firm has some degree of market power because of product differentiation, but it still faces competition from close substitutes. In the short run, firms can earn supernormal profits, normal profits, or incur losses depending on market demand and cost conditions. In essence, the short-run equilibrium allows firms to earn profits through differentiation, but competition ensures these profits don't last forever.

a) Demand curve facing a firm

In monopolistic competition, each firm faces a downward-sloping demand curve. This means that if the firm wants to sell more of its product, it must lower its price. Conversely, if it raises its price, the quantity demanded will decrease. The demand curve is relatively elastic because many other firms sell similar products, so consumers can easily switch to competitors if the price becomes too high.

The shape of the demand curve reflects the firm's ability to differentiate its product from others in the market. Since products are similar but not identical, consumers may be willing to pay a slightly higher price for a product they prefer, giving the firm some degree of market power. However, because close substitutes exist, the demand faced by the firm is not perfectly inelastic.

Each firm faces a **downward-sloping demand curve** because:

- Products are differentiated (branding, packaging, quality, advertising)
- Consumers have preferences for specific brands
- However, demand is relatively elastic due to availability of substitutes

This means the firm is a **price maker to some extent**, but not a pure monopoly.

b) Revenue conditions in the short run

Firms in monopolistic competition operate based on:

- **Average Revenue (AR)** = Price (P)
- **Marginal Revenue (MR)** < AR because the firm must reduce price to sell more units

This relationship influences the profit-maximizing output decision.

c) Profit-maximizing output condition

A firm maximizes profit where:

MR = MC (Marginal Revenue Equals Marginal Cost)

At this point:

- Output level is determined
- Price is set based on the demand (AR) curve at that output level

d) Possible equilibrium outcomes in the short run

Depending on cost and demand conditions, a firm may experience one of the following:

i. Supernormal Profits

Occurs when:

$AR > AC$ at the profit-maximizing output ($MR = MC$)

Features:

- Total revenue exceeds total cost
- Attracts new firms in the long run

ii. Normal Profits (Break-even)

Occurs when: $AR = AC$ at equilibrium output

Features:

- Firm covers all costs, including opportunity cost
- No incentive for entry or exit in the short run

iii. Losses

Occurs when: $AR < AC$ at equilibrium output

Features:

- Firm cannot cover total costs
- May continue operating if $AR \geq AVC$
- May shut down if $AR < AVC$ (shutdown point rule applies)

1.4.4 Abnormal profit and loss under a monopolistic competition firm

A firm is in **equilibrium** when **Marginal Cost (MC) = Marginal Revenue (MR)**. In the short run, firms may earn **abnormal profits** or incur **losses**, depending on the relationship between **Average Revenue (AR)** and **Average Cost (AC)**.

a) Abnormal Profits

A firm earns abnormal (supernormal) profits when:

- **AR > AC** at the equilibrium output.
- The **AR curve** (demand curve) determines the selling price.
- The **AC curve** lies below the AR curve at equilibrium.

- The **MC curve** cuts the AC curve at its lowest point.

This situation occurs when consumers are willing to pay higher prices due to brand loyalty or product differentiation.

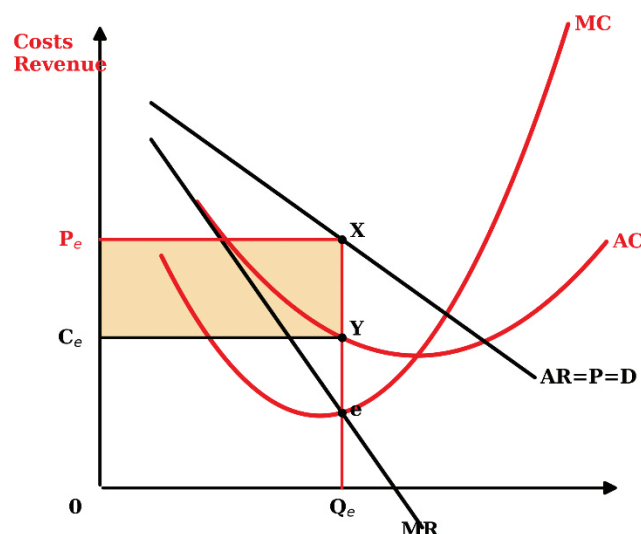


Figure 16: Abnormal profits under monopolistic competition in the short run

The figure above shows a firm in monopolistic competition earning **supernormal (abnormal) profits** in the short run. The firm is in equilibrium at point e, where **MC = MR**. At this point, it produces **OQ_e** units of output and charges a price of **OP_e**, determined from the **AR** (demand) curve. The firm's average cost is **OC_e**, obtained from the **AC** curve. Total Revenue (**TR**) is represented by **OP_eXQ_e**, while Total Cost (**TC**) is represented by **OC_eYQ_e**. Therefore, **Profit = TR - TC = OP_eXQ_e - OC_eYQ_e**, giving a supernormal profit represented by the shaded area **P_eXYC_e**. The firm produces **OQ_e**, which is less than its optimum output, indicating that it operates with excess capacity while maximizing profit in the short run.

b) Losses under monopolistic competition in the short run

A firm under monopolistic competition may incur losses in the short run when its total costs exceed total revenue. This usually happens when demand for the firm's product is low or production costs are high. However, the firm may continue operating in the short run if it can cover its variable costs.

A firm makes losses under the following conditions:

- Average Revenue (AR) is less than Average Cost (AC).
- $AR < AC$
- The AR curve (demand curve) determines the selling price of the product.
- The AC curve represents the average cost of production.
- At the equilibrium level of output, the AC curve lies above the AR curve, showing that the cost per unit is greater than the revenue earned per unit.

- The firm produces where marginal cost equals marginal revenue.
- $MC = MR$
- The Marginal Cost (MC) curve cuts the average cost (AC) curve at its minimum point.
- Since the selling price is lower than the average cost, the firm experiences short-run losses.

In the short run, firms may continue operating despite losses because they expect demand and profits to improve in the future. However, persistent losses may force some firms to leave the industry in the long run.

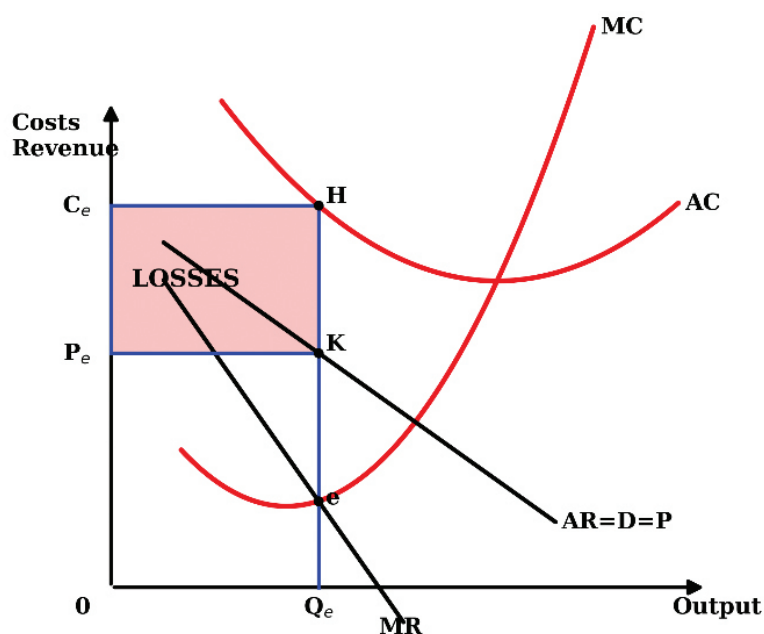


Figure 17: Short-Run losses under monopolistic competition

The figure shows a firm in monopolistic competition incurring losses in the short run. The firm is in equilibrium at point e , where $MC = MR$. At this point, it produces OQ_e units of output and sells them at price OP_e , determined by the AR (demand) curve. The firm's average cost is OC_e , obtained from the AC curve. Total Revenue (TR) is represented by $OP_e KQ_e$, while Total Cost (TC) is represented by $OC_e HQ_e$. Since AC exceeds AR (price), total cost is greater than total revenue. Therefore, $\text{Loss} = TC - TR = OC_e HQ_e - OP_e KQ_e$, represented by the shaded area $P_e C_e H K$. Despite making losses, the firm continues operating in the short run because it can still cover its variable costs and minimize losses by producing at the equilibrium output OQ_e .

1.4.5 Equilibrium of the firm under monopolistic competition in the long run

In the long run, firms operating under monopolistic competition earn only **normal profits**. This is because the supernormal (abnormal) profits earned in the short run attract new firms into the industry. These new firms introduce new brands and differentiated products, increasing competition within the market.

As more firms enter the industry:

- the market becomes more competitive,
- consumers enjoy a wider variety of products,
- each existing firm loses part of its market share,
- Demand for each firm's product decreases.

As a result, the firm's demand curve or Average Revenue (AR) curve gradually shifts to the left and becomes more elastic due to the availability of close substitutes. Firms that were making losses in the short run leave the industry, while profitable firms remain.

Entry and exit of firms continue until all firms earn only normal profits. At this point, the firm reaches long-run equilibrium where the Average Revenue (AR) curve becomes tangent to the Long-Run Average Cost (LAC) curve.

A firm in monopolistic competition is therefore said to be in long-run equilibrium when it produces the level of output where:

- Long-Run Marginal Cost (LMC) equals Long-Run Marginal Revenue (LMR), and
- Average Revenue (AR) is equal to Long-Run Average Cost (LAC).

At this equilibrium point:

- the firm covers all production costs,
- no supernormal profit or loss is earned,
- firms have no incentive to enter or leave the industry.

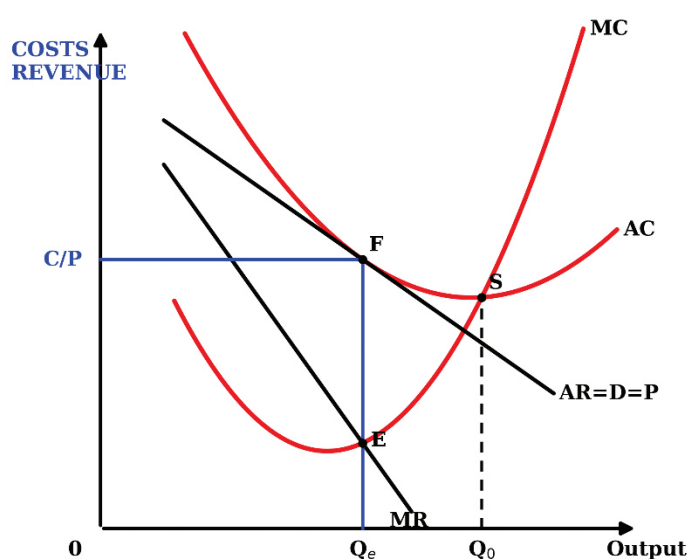


Figure 18: Normal profits under monopolistic competition long-run profits

The figure above shows a firm in long-run equilibrium under monopolistic competition at point E, where $MC = MR$. The firm produces OQ_e units of output and sells them at price OC/P , determined by the AR (demand) curve. At this output level, the AC curve is tangent to the AR curve at point F, meaning that Average Revenue (AR) equals Average Cost (AC). As a result, Total Revenue (OC/PFQ_e) equals Total Cost (OC/PFQ_e), and the firm earns normal (zero) profit, that is, $Profit = TR - TC = 0$. The firm produces OQ_e , which is less than the optimum output OQ_o where $LMC = LAC$ at point S. The difference Q_eQ_o represents excess capacity, indicating that the firm operates below its most efficient scale in the long run.

1.4.6 Product differentiation under monopolistic competition

Product differentiation refers to the process by which firms in a monopolistically competitive market make their products appear unique or distinct from those of competitors. Although many firms offer similar goods, each producer seeks to create a perception of difference to attract and retain customers.

In other words, it is a situation where a producer positions their product as **distinct in quality, design, or presentation**, even when the core function is similar to competing products.

Forms of product differentiation

Firms may differentiate their products through various strategies, including:

Form	Description
Packaging	Using attractive or functional packaging to appeal to consumers.
Design / Shape	Altering the physical appearance or ergonomics of the product.
Branding	Creating a recognizable brand name, logo, or identity.
Colour	Employing distinctive colors to stand out on shelves.
Scent / Flavor	Adding unique sensory features to enhance appeal.
Labelling	Providing informative or persuasive labels.
Salesmanship	Using personal selling or customer service to build loyalty.
Size / Quantity	Offering different sizes or quantities to suit various preferences.

Summary insight

Product differentiation is the cornerstone of monopolistic competition—it drives consumer choice and innovation but also introduces inefficiencies. In the short run, it allows firms to earn profits; in the long run, competition erodes these profits, leaving only normal returns while maintaining diversity in the market.

Advantages of monopolistic competition

- **Product differentiation** gives consumers a wide variety of choices, catering to different tastes, preferences, and income levels.
- Firms compete to **improve product quality**, design, and innovation, which benefits consumers.
- If one firm exits the market, **close substitutes** remain available, ensuring continuity of supply.
- **Prices are lower than monopoly prices** because competition among substitutes prevents firms from charging excessively high prices.
- **Freedom of entry and exit** allows new entrepreneurs to join the industry, creating employment opportunities and encouraging innovation.
- Firms gain **brand recognition and loyalty** through specialization in their own products.
- In the **short run**, abnormal (supernormal) profits can be reinvested to improve product quality, undertake research, and expand production capacity.
- Consumers benefit from **non-price competition** such as advertising, packaging, and customer service, which enhances product appeal.

Disadvantages of monopolistic competition

- **Under-utilization of resources** occurs both in the short run and long run, leading to **excess capacity** (firms do not produce at minimum cost).
- In the **long run**, firms earn only **normal profits**, limiting their ability to invest in research, innovation, or expansion to enjoy economies of scale.
- **Prices are higher than in perfect competition**, since firms have some market power due to product differentiation.
- Heavy reliance on **advertising and marketing** to maintain market share increases production costs, which may be passed on to consumers in the form of higher prices.
- Firms operate below optimal scale, resulting in **limited employment opportunities** compared to industries with full capacity utilization.
- **Output is lower than in perfect competition**, meaning society does not achieve maximum efficiency in resource allocation.
- Excessive product differentiation may lead to **duplication and waste**, as firms spend resources on superficial changes rather than genuine innovation.



Insight

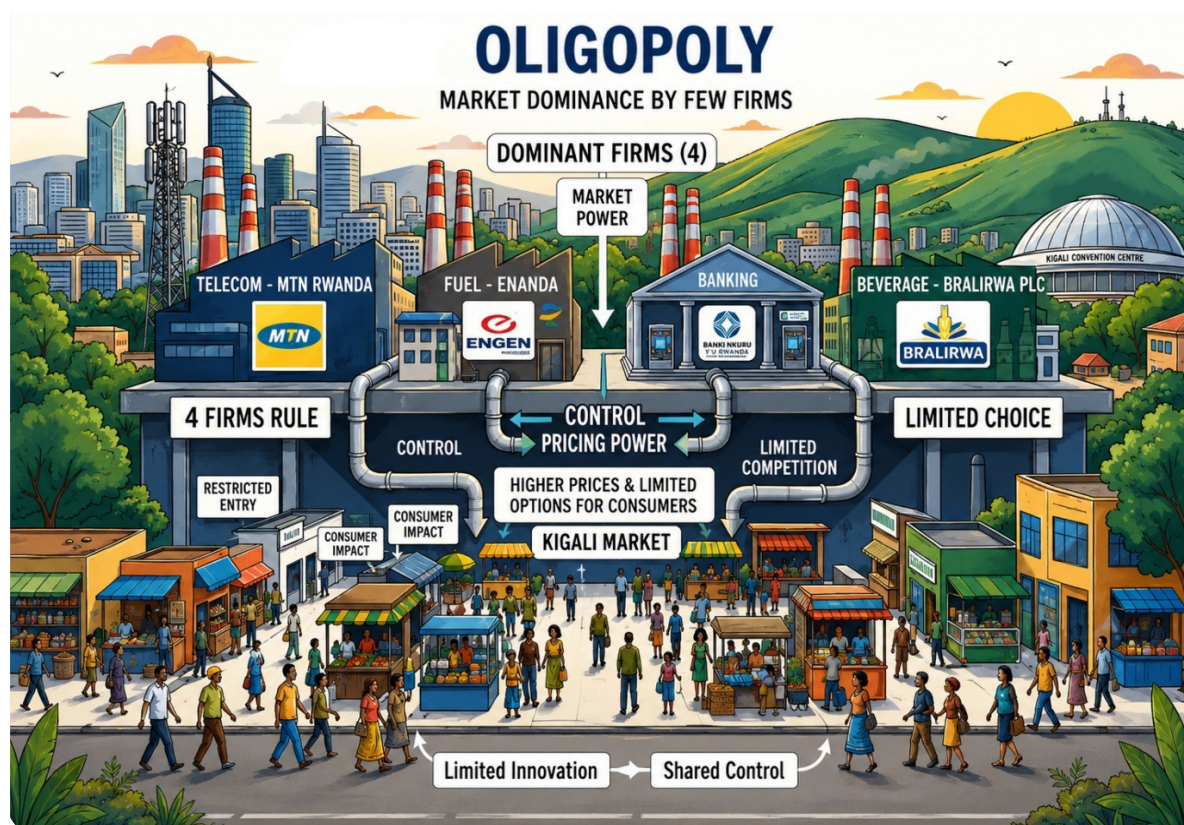
Monopolistic competition strikes a balance between **consumer choice and competitive pricing**, but it sacrifices efficiency. While consumers enjoy variety and innovation, firms face constant pressure to differentiate, often at the cost of higher prices and under-utilized resources.

Application activity 1.4



1. What is monopolistic competition, and how does it differ from perfect competition and monopoly?
2. Identify and explain the main characteristics of firms operating under monopolistic competition.
3. How does a firm under monopolistic competition achieve equilibrium in the market?
4. Using a diagram, explain how price and output are determined for a firm under monopolistic competition in the short run.
5. What conditions lead to abnormal profit or loss for a firm in monopolistic competition during the short run?
6. Describe the equilibrium position of a firm under monopolistic competition in the long run.
7. What is product differentiation under monopolistic competition, and what forms can it take?
8. Critically assess the advantages and disadvantages of monopolistic competition for consumers, producers, and the economy.

1.5 Oligopoly



Learning activity 1.5



Using the photographs represented above,

- How many firms do you think provide telecommunication services in Rwanda?
- How many firms do you think produce beer in Rwanda?
- What type of market structure is seen in the photographs and why?
- Identify characteristics of such a market structure.

Have you ever wondered why some industries in Rwanda and around the world are dominated by only a few large firms? Such firms are examples of businesses operating under imperfect competition. To understand how these firms and markets function, it is important to examine the features of oligopoly, how firms interact and make decisions, and the effects of their actions on consumers, producers, and the wider community. This also provides a basis for understanding the advantages and disadvantages of oligopoly in the economy.

1.5.1 Meaning, forms and features of oligopoly

i. Meaning of oligopoly

Oligopoly is a market structure characterized by the presence of **a few large sellers or producers and many buyers**, dealing in products that may either be similar (homogeneous) or differentiated. Because only a small number of firms dominate the market, each firm holds a significant share and has considerable influence over price, output, and market conditions.

The behaviour of one firm directly affects other firms in the industry. Therefore, firms under oligopoly must consider competitors' actions before making business decisions.

The term **oligopoly** is derived from two Greek words:

- *Oligos* meaning **few**
- *Polein* meaning to **sell**

Thus, oligopoly means **"market with few sellers."**

ii. Forms (Types) of oligopoly

Oligopoly can take different forms depending on the nature of products sold and the degree of cooperation among firms.

1. Perfect (Pure) oligopoly

- Firms produce and sell **homogeneous (identical) products**.
- Consumers view these products as close substitutes, with little or no differentiation.
- Competition is mainly through price and output decisions.

Examples: cement, steel, petroleum products, some agricultural processing industries.

Characteristics:

- Products are identical.
- Brand differences are minimal.
- Price competition dominates.

2. Imperfect (Differentiated) oligopoly

Firms sell **differentiated products** that perform similar functions but differ in quality, packaging, branding, design, or advertising.

Examples: mobile phone services, soft drinks, motor vehicles, consumer electronics.

Characteristics:

- Products are differentiated.
- Nonprice competition (advertising, branding, innovation) is common.

- Firms rely heavily on marketing strategies to attract customers.

3. Collusive Oligopoly

- Firms cooperate or collude to reduce competition and maximize joint profits.
- Collusion may take the form of:
 - Cartels (formal agreements on prices/output)
 - Price agreements
 - Output quotas

Market sharing arrangements

Example: OPEC (Organization of Petroleum Exporting Countries) coordinates production levels.

4. NonCollusive oligopoly

- Firms operate independently without formal agreements.
- Competition occurs through:
 - Advertising and promotion
 - Product quality improvement
 - Technological innovation
- Customer service strategies

5. Duopoly

- A special case of oligopoly where **only two firms dominate the market**.
- Each firm's decisions strongly affect the other.

Example: Airbus and Boeing in the aircraft manufacturing industry.

6. Duopsony

- A market with **two dominant buyers** purchasing from many sellers.
- Sellers depend heavily on the purchasing decisions of these two buyers.

7. Oligopsony

- A market with **few large buyers** and many sellers.
- Buyers have significant bargaining power over prices and terms.

Example: large supermarket chains purchasing from many small farmers.

Summary insight

Oligopoly can range from **pure competition with identical products** to **differentiated markets with heavy branding**. It may involve **collusion (cartels)** or **independent rivalry**, and in extreme cases, reduce to **duopoly** or **oligopsony**. The form depends on product nature, number of firms or buyers, and the level of cooperation among them.

Examples of Firms and Industries under Oligopoly

Examples of industries operating under oligopoly include:

Telecommunications Companies

Mobile telephone companies such as **MTN**, **Airtel**, and other telecom operators.

Petroleum Distribution Companies

Petroleum firms such as **Kobil**, **SP**, and **Mount Meru**.

Beverage Industries

Soft drink companies such as The Coca-Cola Company and PepsiCo.

Beer producers such as **Bralirwa** and **Skol Brewery**.

Media and Publishing Companies

Newspaper firms such as **The New Times**, **Umuseke**, and **Imvaho Nshya**.

Other Examples

- Airline industries
- Banking institutions
- Motor vehicle manufacturers
- Cement industries
- Internet service providers

iii. Features (Characteristics) of oligopoly

Oligopoly is a market structure with unique traits that distinguish it from perfect competition, monopoly, and monopolistic competition.

1. Few large firms, many buyers

- The market is dominated by a small number of large firms, each controlling a significant share.
- Consumers are numerous, but sellers are limited, giving each firm considerable influence.

2. Interdependence of firms

- Firms are mutually dependent: the actions of one (pricing, advertising, output) directly affect rivals.
- Example: if one firm lowers its price, competitors often follow to avoid losing market share.

3. NonPrice competition

Firms avoid price cuts to prevent profit erosion. Instead, they compete through:

- Advertising and promotion

- Product quality improvements
- Branding and packaging
- Customer service
- Innovation and technology

4. Product differentiation

- Products may be similar but are differentiated by brand, design, quality, or features.
- Examples: soft drinks, mobile networks, automobiles.

5. Market power and collusion

- Each firm has market power due to limited competition.
- Firms may collude (form cartels) to set prices, output, or market shares, behaving like a monopoly.

6. Uncertainty in decisionmaking

- Firms cannot predict rivals' reactions with certainty.
- Example: raising prices may cause loss of customers if competitors keep prices unchanged.

7. Barriers to entry

Entry of new firms is restricted by:

- High capital requirements
- Advanced technology needs
- Patents and legal restrictions
- Strong brand loyalty
- Economies of scale enjoyed by incumbents

8. Price rigidity

- Prices remain stable for long periods.
- Firms avoid frequent changes to prevent retaliatory moves and instability.

9. Kinked demand curve

- Demand curve has a "kink": Firms are faced with a kinked demand curve, i.e. a curve that has a bend (kink) and it is elastic above the kink and inelastic below the kink.
- Explains why prices are sticky.

10. Price wars

- Sometimes firms engage in aggressive price cutting to capture market share.
- Rival firms respond with further cuts, leading to reduced profits across the industry.

11.Importance of advertising and promotion

- Heavy spending on advertising, sponsorships, sales promotions, and online campaigns.
- Goal: attract customers, build brand loyalty, and compete without lowering prices.

12. Longterm strategic behaviour

Firms plan strategically because each decision influences rivals.

Expansion, investment, and pricing are carefully considered to maintain competitiveness.

Summary insight

Oligopoly is characterized by **few dominant firms, interdependence, and strategic competition**. While collusion can make it resemble monopoly, nonprice competition and product differentiation ensure consumer choice. However, barriers to entry, price rigidity, and excess advertising costs often reduce efficiency.

1.5.2 The demand curve of a firm under oligopoly

In oligopoly, firms face a **kinked demand curve** due to interdependence. Each firm considers rivals' reactions before changing prices.

This explains why prices remain relatively stable.

Features

1. Elastic above the kink

- If a firm raises prices, rivals don't follow.
- Customers switch to cheaper competitors.
- Demand falls sharply leading to the highly elastic.

2. Inelastic below the kink

- If a firm lowers prices, rivals match.
- Gains in customers are small.
- Demand rises slightly → relatively inelastic.

Implications

- Price rigidity and fear of retaliation.
- Firms avoid price changes.
- Competition shifts to non-price strategies:
- Advertising
- Branding & packaging
- Product quality
- Customer service & promotions

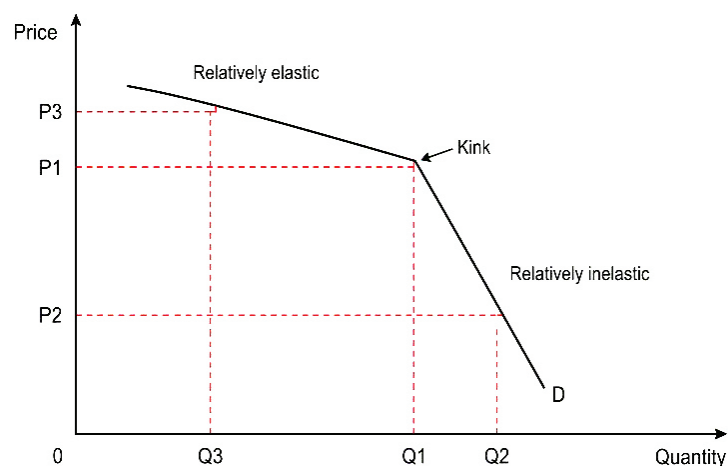


Fig 19: The kinked demand curve under Oligopoly

From Figure 19 above, P_1 is the market price or administered price. Should any firm increase its price above that price to price P_3 , it would lose its customers ($Q_1 - Q_3$) to other firms. If a firm decides to set the price below P_1 to P_2 , other firms will react by reducing their price even further or lower to win more customers; hence, the increase in quantity sold will be lower ($Q_1 - Q_2$) than the price reduction. Hence, the demand curve has a kink, meaning that the prices will remain rigid/stable for a long period of time.

The AR and MR Curves under Oligopoly

The average revenue (AR) curve under oligopoly is kinked. The kink is the sharp bend in the demand curve at the prevailing price and output. It arises because firms expect rivals to ignore price increases but match price decreases. Consequently, demand is relatively elastic above the kink and relatively inelastic below it. The marginal revenue (MR) curve derived from the kinked AR curve has three parts and is discontinuous at the kink, creating a vertical gap. This means that marginal cost can change within the gap without affecting the firm's profit-maximizing price and output. As a result, prices tend to remain stable or rigid in oligopolistic markets.

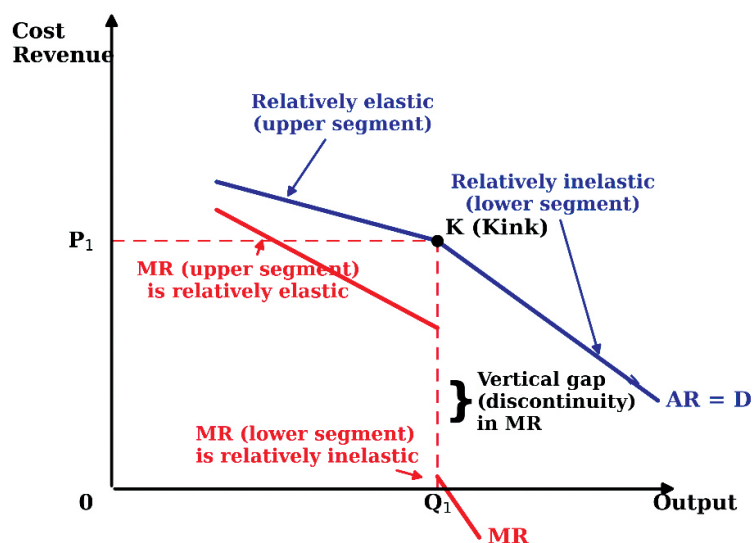


Figure 20: The AR and MR curves under Oligopoly

Figure 20 illustrates that the AR curve is relatively elastic above the kink point (K) because rival firms are unlikely to match price increases, and relatively inelastic below the kink because rivals tend to follow price reductions. Correspondingly, the MR curve has an upper elastic segment, a lower inelastic segment, and a vertical discontinuity at the kink output (Q_1). This discontinuity in the MR curve arises from the kink in the AR curve and explains price rigidity in oligopolistic markets, since marginal cost can fluctuate within the discontinuous gap without affecting the profit-maximizing price (P_1) or output level (Q_1).

1.5.3 Price determination under oligopoly (methods of price determination)

Under an oligopolistic market structure, there are only a few large firms, and each firm's pricing and output decisions affect the others. As a result, there is no single method of price determination under oligopoly. Prices may be determined through competition, cooperation, or the influence of dominant firms. The major methods of price determination under oligopoly include the following:

1. Independent pricing

Independent pricing occurs when each firm sets its own price and output level with the objective of maximizing profits without formally consulting rival firms. Although firms act independently, they must still consider the likely reactions of competitors because decisions made by one firm can significantly affect the market.

This method often leads to: intense competition, price instability, advertising battles, and price wars.

A price war arises when firms continuously reduce prices in an attempt to attract more customers than their rivals. Such competition may reduce profits for all firms in the industry.

2. Perfect collusion (Cartel pricing)

Perfect collusion occurs when firms in the same industry formally agree to cooperate in determining prices, output, and market shares. The firms form a cartel with the aim of reducing competition and maximizing joint profits, almost like a monopoly.

Under cartel agreements: firms agree on a common price, production quotas may be allocated among members, competition is minimized, and profits tend to increase.

A well-known example is Organization of the Petroleum Exporting Countries (OPEC), where oil-producing countries cooperate to influence oil prices and output levels in the world market.

However, cartels may collapse if:

- members cheat by secretly lowering prices,
- production costs differ among firms, or
- governments prohibit collusive practices through anti-monopoly laws.

3. Imperfect collusion (Price leadership)

Experimental version

Imperfect collusion exists where firms do not formally agree, but one dominant firm informally influences the pricing decisions of other firms. This method is commonly known as price leadership.

Under price leadership:

- the dominant or largest firm sets the price,
- smaller firms follow the price set by the leader,
- competition is reduced, and
- price stability may be maintained in the industry.

The dominant firm usually has advantages such as:

- lower production costs,
- a larger market share,
- superior technology, or
- stronger financial power.

If the price set by the leading firm is favorable, other firms follow it. If it is unfavorable, weaker firms may fail to compete and eventually leave the industry.

4. Predatory pricing

Predatory pricing occurs when a firm deliberately lowers prices below normal profit levels, or even below cost, in the short run with the intention of driving competitors out of the market.

The objective is to:

- eliminate rival firms,
- discourage new entrants, and
- gain monopoly power in the long run.

Once competitors are forced out, the dominant firm may later increase prices and recover earlier losses through higher profits.

Predatory pricing is often considered unfair competition and may be restricted by government regulations.

5. Limit pricing

Limit pricing refers to a situation where established firms intentionally keep prices low in order to discourage new firms from entering the industry.

This strategy is common in industries where:

- large-scale production leads to economies of scale,
- existing firms have cost advantages, and
- barriers to entry are high.

By charging relatively low prices:

- potential entrants may believe they cannot earn sufficient profits,
- entry into the market is discouraged, and
- existing firms maintain their market dominance.

In the long run, limit pricing helps established firms protect their market position and maintain control over the industry.

Here's a shortened version of your text, keeping the key points intact:

1.5.4 Non price competition under oligopoly

In oligopolistic markets, firms often avoid frequent price changes due to the risk of retaliation and price rigidity. Instead, they compete through **nonprice strategies** aimed at expanding market share, strengthening customer loyalty, and differentiating their products. This is a defining feature of oligopoly, including in Rwanda.

i. Forms of non- price competition in Rwanda

- **Advertising & promotion**
Persuasive campaigns via radio, TV, newspapers, billboards, and digital platforms to raise awareness and influence consumer preferences.
Examples: MTN's "Everywhere You Go," Airtel's "Express Yourself."
- **Branding & product differentiation**
Use of appealing brand names (e.g., Rwanda Tea), attractive packaging, and innovative designs to stand out. Firms also blend products to create unique varieties.
- **Credit facilities & incentives**
Offering instalment payments, credit sales, or loyalty rewards to encourage repeat purchases.
- **Free samples & gifts**
Petrol stations giving soap, or beverage companies offering promotional items to boost sales.
- **Branch expansion & accessibility**
Opening outlets across the country, mobile shops (bread vans, bicycles), and stopshopping centers at fuel stations to reach more customers.
- **Aftersales services**
Free delivery, warranties, spare parts, and customer care support to build trust and longterm relationships.
- **Sponsorships & social engagement**
Supporting sports, music, and cultural events (e.g., Primus Guma Guma Super Star) to strengthen brand visibility.
- **Promotions & raffle draws**
Campaigns like MTN Sharama or Tigo's "Millionaires Club" attract customers through prize draws.
- **Trade fairs & exhibitions**
Showcasing products to potential buyers, building networks, and educating consumers.

➤ **Customer experience enhancements**

Renovating premises for clients (telecoms, breweries), offering modern retail spaces, and improving service environments.

➤ **Quality improvement & innovation**

Continuous upgrades in product quality, introduction of new variants, and technological innovation to maintain competitiveness.

➤ **Slogans & emotional appeal**

Memorable phrases that connect emotionally with consumers, reinforcing brand identity.



Key insights

- » Nonprice competition is essential in oligopoly because price wars are costly and unstable.
- » Firms rely on differentiation, customer loyalty, and brand strength rather than undercutting prices.
- » In Rwanda, telecoms, breweries, and consumer goods companies are leading examples of firms using these strategies.

ii. Equilibrium position and profit maximisation of a firm under oligopoly

In oligopoly, a firm reaches **equilibrium** where **marginal cost (MC) equals marginal revenue (MR)**. At this point, the firm maximises profit. Due to price rigidity and interdependence among firms, oligopolists often earn **abnormal (supernormal) profits** both in the short run and long run. These profits are sustained because entry barriers (high capital requirements, brand loyalty, patents, government regulation) prevent new competitors from easily entering the market.

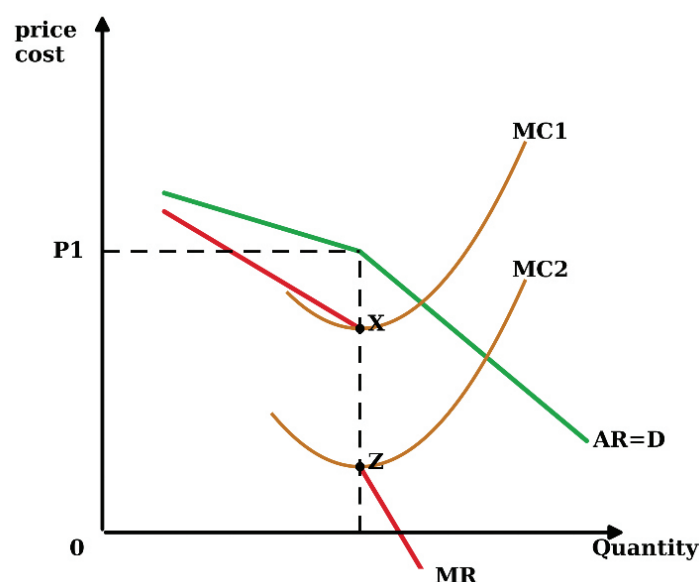


Fig 21: Equilibrium position of a firm under oligopoly

Figure 21 shows that the firm's equilibrium occurs within the discontinuous segment of the MR curve, between points X and Z. Any MC curve that intersects the MR curve within this gap satisfies the condition $MC = MR$, resulting in the same equilibrium price

(P_1) and output level. Therefore, even when marginal cost shifts from MC1 to MC2, equilibrium remains unchanged because both curves intersect the discontinuous section of MR. This explains price rigidity in oligopolistic markets, where moderate changes in production costs do not affect the firm's equilibrium price and output.

1.5.5 Equilibrium position and profit maximisation of a firm under oligopoly

Under an oligopoly market structure, a few large firms dominate the market and each firm's decisions regarding price, output, advertising, and product quality influence the actions of rival firms. Unlike perfect competition and monopoly, firms in an oligopoly are **interdependent**, meaning that the behaviour of one firm affects the performance and decisions of other firms in the industry.

A firm under oligopoly reaches equilibrium when it produces the level of output where profit is maximised and there is no incentive to change its price or output level. Similar to other market structures, profit maximisation generally occurs where:

$$MC=MR$$

However, equilibrium under oligopoly is more complex because firms must consider the possible reactions of competitors before making decisions. If one firm changes its price, other firms may respond by lowering or maintaining their prices, which affects demand and profits.

In the short run, oligopolistic firms may earn **abnormal (supernormal) profits, normal profits**, or even **incur losses**, depending on production costs, market demand, and competitors' actions. In the long run, firms may continue earning abnormal profits because barriers to entry such as large capital requirements, patents, brand loyalty, and economies of scale prevent new firms from entering the market easily.

Therefore, the equilibrium position of a firm under oligopoly depends not only on cost and revenue conditions but also on strategic interactions among firms. This makes oligopoly one of the most complex market structures in terms of price and output determination.

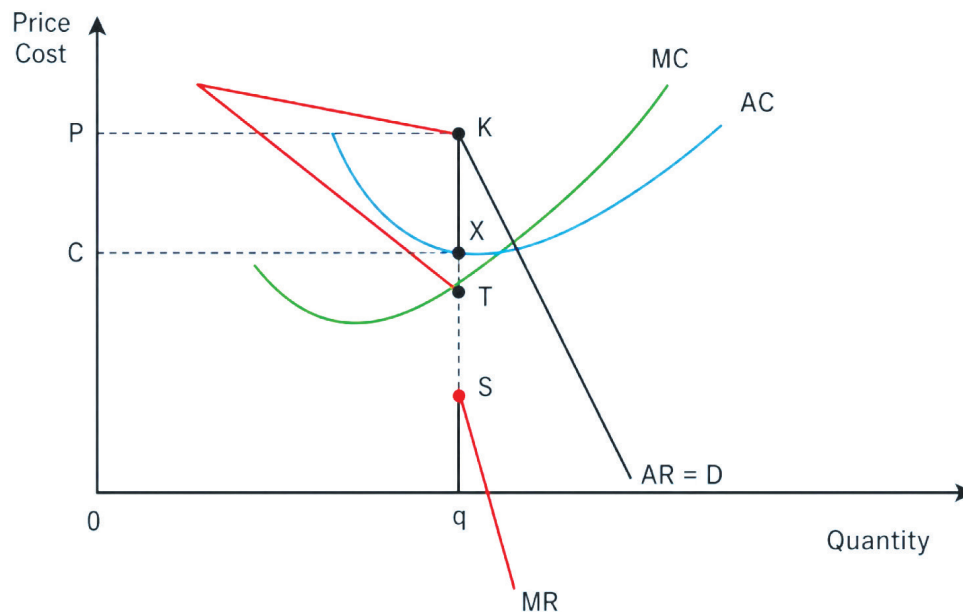


Figure 22: Equilibrium position and profit maximisation of a firm under oligopoly

From Figure 22 above, the firm produces output **Oq** at total cost

$OCxq$ and sells it at price OP getting total revenue $OPKq$. Profits = $TR - TC$. $OPKq - OCxq = CPKx$.

The marginal revenue curve is not continuous between sections **T** and **S** because of uncertainties. The firm is, therefore, in equilibrium at point **T** where **MC = MR** and earns abnormal profits shown by the area **CPKx**.

Advantages of oligopoly

- **Lower prices for consumers:** Fear of rival reactions and competitive pressure often keep prices relatively low.
- **Price stability:** Consumers benefit from stable prices, which makes budgeting easier.
- **Variety of products:** Branding and product differentiation (imperfect oligopoly) widen consumer choice.
- **Improved quality:** Competition encourages firms to produce highquality goods.
- **Innovation & R&D:** Abnormal profits are reinvested in research, leading to technological advancement and higher living standards.
- **Largescale production:** Firms operate on a large scale, enjoying economies of scale and increased output.
- **Consumer awareness:** Extensive advertising ensures customers are wellinformed.

- **Customer welfare:** Promotions, gifts, and loyalty rewards improve consumer satisfaction.
- **NonPrice competition:** Firms compete through branding, packaging, aftersales services, and sponsorships rather than frequent price changes.

Disadvantages of oligopoly

- **Consumer exploitation:** Collusion among firms may lead to overcharging.
- **Resource wastage:** Duplication of products and advertising campaigns can waste resources.
- **Small firm collapse:** Smaller firms often fail when outcompeted by dominant players.
- **Underutilisation of resources:** Firms may produce at excess capacity, leading to inefficiency.
- **Government pressure:** Large oligopolistic firms can influence government policy.
- **Distorted consumer choice:** Excessive advertising manipulates preferences rather than informing.
- **Income inequality:** High barriers to entry limit opportunities for new firms, concentrating wealth.
- **Price wars:** Rival firms sometimes engage in aggressive price cutting, which can reduce profits and destabilise the market.



Insights

- » Oligopolies balance between competition and collusion: while rivalry drives innovation, collusion can harm consumers.
- » Modern oligopolies (telecoms, airlines, tech giants) rely heavily on nonprice strategies such as digital marketing, customer loyalty programs, and technological upgrades.
- » Governments often regulate oligopolies to prevent anticompetitive practices and protect consumer welfare.

1.5.6 Comparison of the different market structures

In economics, **market structure** refers to the organizational and competitive characteristics of a market that influence the behavior of firms and the outcomes for consumers. The four main types of **perfect competition, monopoly, monopolistic competition, and oligopoly** differ in terms of the number of firms, product nature, entry barriers, pricing power, and level of competition.

Understanding these structures is crucial because they determine how prices are set, how resources are allocated, and how consumers benefit or suffer. For instance, perfect competition ensures efficiency and consumer welfare, while monopoly may lead to exploitation through high prices.

Monopolistic competition provides variety through product differentiation, whereas oligopoly is marked by interdependence among a few dominant firms, often leading to price rigidity and reliance on nonprice competition.

This comparison highlights the **key similarities and differences** across the four structures, showing how each affects profitability, innovation, consumer choice, and government regulation in both the short and long run.

Table showing the comparison of different market structures

Characteristic	Perfect Competition	Monopoly	Monopolistic Competition	Oligopoly
Number of sellers	Very many	One	Several	Few (dominant firms)
Number of buyers	Very many	Very many	Very many	Very many
Nature of product	Homogeneous (identical)	Single product, no close substitutes	Differentiated and heterogeneous	Homogeneous (pure oligopoly) or differentiated (imperfect oligopoly)
Demand Curve / AR Curve	Perfectly elastic (horizontal)	Inelastic (downward sloping)	Fairly elastic (downward sloping)	Kinked (elastic above kink, inelastic below)
Marginal revenue curve	Same as AR	Below AR, downward sloping	Below AR, downward sloping	Below AR, segmented due to kink
Entry of new firms	Free entry	Blocked (high barriers)	Relatively free	Limited entry (barriers such as capital, patents, brand loyalty)
Exit from industry	Free	Barriers to exit	Free	Limited exit
Profitability (short run)	Abnormal profits or losses	Abnormal profits	Abnormal profits or losses	Abnormal profits or losses
Profitability (long run)	Normal profits (due to free entry)	Abnormal profits (sustained)	Normal profits	Abnormal profits (due to barriers)
Role of Government	Minimal	May regulate to prevent abuse	Minimal	Often regulates to prevent collusion and protect consumers
Equilibrium condition	$MC = MR$	$MC = MR$	$MC = MR$	$MC = MR$

Advertising	None (products identical)	Informative only	High persuasive advertising	High persuasive advertising, brand wars
Price discrimination	None	Possible (first, second, third degree)	Rare	Possible in some cases (e.g., airlines, telecoms)
Market power	None (price taker)	Absolute (price maker)	Some (due to differentiation)	Significant (interdependence among few firms)
Efficiency	Allocative & productive efficiency	Often inefficient (deadweight loss)	Not fully efficient	May be efficient but prone to collusion
Examples	Agricultural markets (maize, wheat)	Utilities (water, electricity)	Restaurants, clothing brands	Telecoms, airlines, breweries, tech giants



Insights

- » Perfect competition is rare in reality; mostly theoretical, seen in agriculture.
- » Monopoly can lead to consumer exploitation but also economies of scale.
- » Monopolistic competition is the most common structure in retail and services, balancing variety with competition.
- » Oligopoly dominates modern economies (telecoms, oil, airlines, tech), where firms rely heavily on nonprice competition and sometimes collusion.

Application activity 1.5



Oligopoly

1. Define oligopoly and describe its main features. What are the different forms of oligopoly?
2. Using a diagram, illustrate the demand curve faced by a firm operating under oligopoly. Explain how this differs from demand curves in perfect competition and monopoly.
3. Discuss the ethical considerations and potential social impacts of firms engaging in non-price competition in an oligopolistic market. Why might firms prefer non-price competition over price wars?

4. Explain the methods of price determination in an oligopoly. How do firms decide on their pricing strategies?
5. Describe how firms under oligopoly achieve equilibrium and maximize profits. What factors influence their decision-making process?
6. Compare and contrast oligopoly with other market structures such as perfect competition and monopoly. What are the implications for consumers and society?

1.6 End Unit Assessment



Knowledge and Understanding

1. Define market structure and explain the meaning and features of perfect competition.
2. Identify the cost curves for a firm under perfect competition and describe the relationship between AR, MR, price, and demand.
3. Explain the short-run equilibrium position of a firm under perfect competition, including how price and output are determined.
4. Discuss why a firm may earn abnormal profits or incur losses in the short run under perfect competition and explain breakeven and shutdown points.

Skills

5. Given a firm in perfect competition, derive the short-run supply decision when price falls below average variable cost.
6. Compare long-run equilibrium under perfect competition with the short-run outcome, explaining entry and exit effects on profits.
7. Using a simple diagram, show how a monopolist maximizes profit in the short run and identify the areas of abnormal profit or loss.

Attitudes and Values

8. Evaluate the ethical considerations of price discrimination in a monopolistic market from a consumer perspective.

UNIT 2

NATIONAL INCOME



Key Unit Competence:

Assess national income concepts, measurement methods, and indicators, and their importance and limitations for economic planning and development.

Learning objectives

Knowledge and understanding

1

- Explain GDP, GNP, NNP, per capita income, and assess their relevance for Rwanda's development.
- Evaluate different measurement approaches (income, expenditure, output) and their suitability in policy contexts.
- Analyze determinants of national income (resources, technology, investment, human capital) and their impact on growth.
- Assess the importance and limitations of national income statistics for planning, focusing on data quality and comparability.
- Interpret equilibrium national income using AD-AS, inflationary/deflationary gaps, and leakages/injections.
- Examine phases of the business cycle and their implications for Rwanda's stability and growth.
- Evaluate income inequality using Lorenz curves and Gini coefficients, linking findings to social policy.
- Investigate causes of low per capita income in Rwanda and connect them to development challenges.

Skills

2

- Use national income ideas to explain real examples in Rwanda (like Vision 2050 or farming/coffee sector).
- Work out simple numbers for GDP, per capita income, and inequality using Rwanda data (BNR, NISR).

- Draw and read basic graphs such as AD–AS, circular flow of income, and Lorenz curves.
- Look at changes in national income and suggest simple ways the government can keep the economy stable.
- Compare how people live in Rwanda and other countries using per capita income or PPP.
- Write short reports that show how national income affects Rwanda's development goals.
- Take part in class debates and role-plays about "growth vs. equality," using facts to support your ideas.

Attitudes and values

3

- Appreciate the role of national income in guiding development planning and policy.
- Value equitable income distribution and fairness in economic systems.
- Show awareness of economic changes and their impacts on livelihoods.
- Recognize the government's role in stability, redistribution, and welfare improvement.
- Balance equity vs. growth trade-offs with a focus on inclusive development.
- Develop integrity in handling and interpreting economic data.
- Cultivate proactive problem-solving attitudes toward Rwanda's low income challenges.

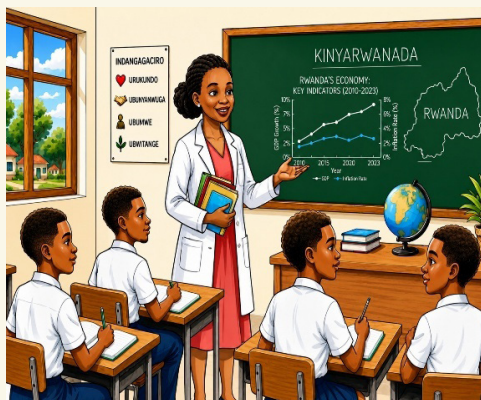
INTRODUCTION

National Income: National income refers to the **total value of all goods and services produced within a country during a specific period, usually one year**. It is an important measure because it shows the amount of wealth a nation creates and helps us understand the living standards of its people.

It includes **everything that is produced and consumed in the economy**, such as food, clothes, houses, transport, education, and health services. Economists use national income to measure the growth of the economy and to compare the performance of one country with another. For example, when Rwanda produces more tea, coffee, and services like tourism, the national income increases.



Introductory activity 2



1. What kinds of work are shown in the photographs, and how do they reflect everyday activities in Rwanda?
2. How do these activities help families meet their needs and improve their lives?
3. In what ways do farming, transport, and teaching contribute to the progress of the community and the country?
4. Who benefits when people take part in these activities, and why is their participation important?

2.1 Meaning of national income and related concepts

Learning activity 2.1



In a small village called **Nyagasambu**, TWAHIRWA is a farmer who grows premium organic coffee beans. Last year, TWAHIRWA sold his harvest to a local factory for **1,000,000 RWF**. The factory processed and packaged the beans, then sold them to tourists and shops for **2,500,000 RWF**. To keep the business running, the factory owner paid **500,000 RWF** in wages to workers and **200,000 RWF** in taxes. After covering all expenses, the owner had **800,000 RWF** left as profit, which he used to buy a bicycle and save for his children's school fees.

Questions

1. What was TWAHIRWA's income from selling coffee
2. If the price of bread in the village doubled this year, would TWAHIRWA's **real income** be higher or lower than his nominal income?
3. The factory owner had 800,000 RWF left after paying taxes and expenses. Is this his disposable income? Why or why not?
4. When calculating national income for Nyagasambu, should we count both TWAHIRWA's 1,000,000 RWF and the factory's 2,500,000 RWF?
5. What problem might occur if we count both?
6. How much extra value did the factory add to TWAHIRWA's raw coffee beans?

i. Definition of national income

National income is the total monetary value of all goods and services produced through the productive activities of a country within a given period, usually one year.

It also refers to the sum of all incomes earned by residents of a country during that period. However, incomes received as gifts or transfer payments from individuals, firms, or the government are not included, since they are not generated from productive economic activities.

Whenever any economic activity takes place, whether a farmer sells maize, a bakery sells bread, or a company pays wages to its workers, income is generated somewhere in the economy. National income is essentially the sum total of all these individual incomes earned across the entire country within a given period.

ii. Concepts used in national income

- **Gross Domestic Product (GDP):** Value of all goods and services produced **within the country's borders**, by both nationals and foreigners.

Example: Tea produced in Rwanda by Rwandan farmers and foreign investors is part of Rwanda's GDP. GDP is increasing when a country is producing more goods and services

Formula :

$$\text{i. } \mathbf{GDP = C + I + G + (X - M)}$$

Where;

C = Consumption by households: Total spending by households on goods and services.

I = Investment by firms: Total spending on capital goods that will be used for future production.

G = Government Spending: Total government expenditures on goods and services.

X = Exports: Total value of goods and services produced domestically and sold abroad.

M = Imports: Total value of goods and services produced abroad and purchased domestically

ii. GDP = GNP – Net Income from Abroad (X – M)

- **Gross National Product (GNP):** Value of goods and services produced by **nationals**, whether inside the country or abroad. Excludes income earned by foreigners living in Rwanda.

Formula:

$$\text{GNP} = \text{GDP} + \text{Net Factor Income from Abroad}$$

Where:

Net Factor Income from Abroad is the difference between income earned by nationals abroad and income earned by foreigners in Rwanda. Includes salaries, rent, dividends, profits, royalties.

- **Net National Product (NNP):** The net monetary value of all final goods and services produced by a country's citizens, both within and outside the country, during a given period, after deducting depreciation (wear and tear of capital assets). It reflects the economy's actual net production. GNP minus depreciation (loss in value of machines, buildings, and equipment).

Formula:

$$\text{NNP} = \text{GNP} - \text{Depreciation}$$

- **Net Domestic Product (NDP):** The net monetary value of all final goods and services produced within a country's borders by both nationals and non-nationals during a given period, after deducting depreciation (wear and tear or obsolescence of capital assets). It reflects the economy's actual net domestic production. GDP minus depreciation.

Formula:

$$\text{NDP} = \text{GDP} - \text{Depreciation}$$

Where:

Depreciation (also called consumption of fixed capital) is the loss in value of capital goods such as machinery, buildings, vehicles, and equipment due to wear and tear, aging, obsolescence, or accidental damage during the production process.

- **Per Capita Income:** Average income per person.

Formula:

$$\text{Per Capita Income} = \frac{\text{National income}}{\text{Total population}}$$

It Shows living standards in a country.

- **Personal Income:** Total income received by individuals from productive activities (wages, salaries) and non-productive sources (gifts, transfers).
- **Disposable Income:** Income left for spending or saving after paying taxes.

Formula:

i. **Disposable Income (DI)** = Personal Income (PI) - Personal Direct Taxes"

or simply, "DI" = "PI" - "Taxes"

Where:

Personal Income (PI) = Income received by individuals from all sources.

Personal Direct Taxes = Taxes paid directly by individuals, such as income tax.

Disposable income is the income that remains available to individuals after paying personal taxes. It can be used for consumption or saving.

ii. **Disposable Income** = Consumption + Savings

$DI = C + S$

Where:

C = Consumption expenditure

S = Savings

Example: If a teacher earns 200,000 RWF and pays 20,000 RWF in taxes, disposable income = 180,000 RWF.

- **Nominal vs. Real Income**

Nominal income: Income expressed in money terms (francs, dollars).

Real income: The goods and services that money can buy. If prices rise, real income may fall even if nominal income stays the same.

- **National Income at Market Price (NNPmp):** Value of goods and services at market prices.

Formula:

$NNPmp = NNPfc + \text{Indirect Taxes} - \text{Subsidies}$

- **National Income at Factor Cost (NNPfc):** Value of goods and services measured by payments to factors of production (wages, rent, interest, profit).

Formula:

$NNPfc = NNPmp + \text{Subsidies} - \text{Indirect Taxes}$

Application activity 2.1



- 1. Question:** If Rwanda produces goods and services worth 12 trillion RWF in one year, what do we call this total value?
- 2. Question:** A Rwandan company earns profits from selling tea in Kenya. Should this income be included in Rwanda's GDP or GNP? Why?
- 3. Question:** If Rwanda's national income is 13 trillion RWF and the population is 13 million, what is the per capita income, and what does it show?

2.2 Methods/Approaches of measuring national income

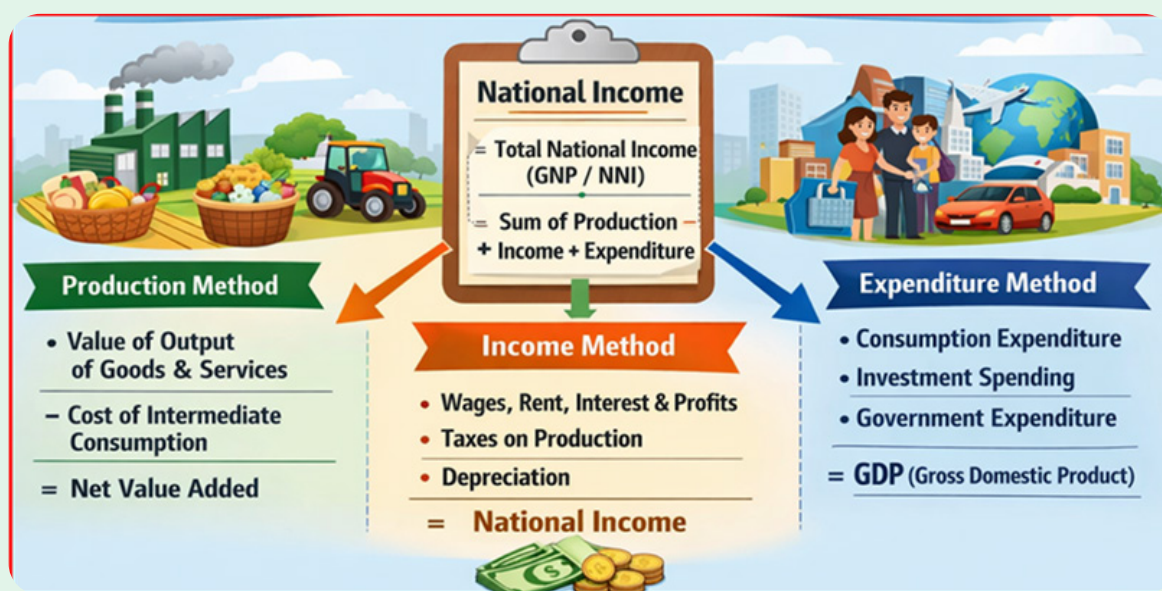
Learning activity 2.2



Using the illustrations in figure below:

Describe the activities shown in Figure below

1. Explain how each of the activities in the figure contributes to the measurement of national income.
2. Based on the activities shown in the figure, describe the approaches used to measure national income.
3. Identify the challenges or problems that may arise when using these different approaches to measure national income.
4. Explain why the three approaches to measuring national income should give the same final result.



Measuring national income is essential for understanding the economic performance of a country. It provides insights into the overall economic activity, living standards, and economic growth. To accurately assess a nation's economic health, economists have developed various methods or approaches to calculate national income.

These methods help in capturing the total value of goods and services produced within the country over a specific period, offering a comprehensive picture of economic development. The main approaches include the Production (or Output) Method, Income Method, and Expenditure Method, each providing a different perspective but collectively contributing to a complete understanding of a country's economic activity.

There are three main methods of measuring national income:

1. Income approach

2. Expenditure approach

3. Product (Output) approach

1. Income approach

This method adds up all the incomes received by individuals, firms, and government from productive activities. These include:

- **Wages and salaries** (payment to labour)
- **Rent** (payment for land and property)
- **Profits** (earnings of entrepreneurs)
- **Interest** (payment for capital)

Formula:

$$NY=W+R+P+I$$

Where:

- NY = National Income
- W = Wages
- R = Rent
- P = Profits
- I = Interest

Important: Only productive activities are included. Illegal activities (smuggling, black markets, prostitution) and transfer payments (gifts, grants, pocket money, famine relief, pensions) are excluded because they do not reflect production.

National income measured by this method is called National Income at Factor Cost, since it considers payments to the factors of production.

Example:

How Mining Contributes to National Income Mining is one of Rwanda's important economic activities. Minerals such as tin, tantalum, and tungsten are extracted and exported. The production process passes through several stages before the minerals reach international buyers. At each stage, value is added, and this is what economists use to measure national income.

Stages of production

Stage	Activity	Selling Price	Value Added
1	Miner extracts ore from the ground	Rwf40	Rwf40
2	Processing plant refines and packages the minerals	Rwf70	Rwf30
3	Export company sells the refined minerals abroad	Rwf100	Rwf30

Step-by-step explanation miner:

A miner: extracts ore and sells it for Rwf40. This is the miner's income, and the value added at this stage is Rwf40.

Processing plant: The ore is refined and packaged, then sold for Rwf70. The value added here is $Rwf70 - Rwf40 = Rwf30$.

Exporter: The export company sells the refined minerals abroad for Rwf100. The value added here is $Rwf100 - Rwf70 = Rwf30$.

Total Value Added Total Value Added: $40 + 30 + 30 = 100$

This equals the final selling price of the minerals (Rwf100).

Why Economists Use Value Added (Not Total Selling Prices) Economists do not simply add all selling prices together ($40 + 70 + 100 = 210$) because that would double count the same product at different stages. Instead, they calculate the value added at each stage to avoid exaggerating national income.



Key Takeaway

Mining contributes to Rwanda's national income through extraction, processing, and exporting.

The value added method ensures that only the new wealth created at each stage is counted.

This example shows how natural resources, when processed and exported, play a vital role in Rwanda's economy.

2. Expenditure approach

This method measures national income by adding up all expenditures on **final goods and services**. Only final goods are counted to avoid **double counting**. For example, we count expenditure on bread but not on wheat used to make bread.

Formula: $NY = C + I + G + (X - M)$

Where:

- C = Household consumption expenditure
- I = Investment expenditure (capital goods)

- G = Government expenditure on goods and services
- X-M = Net exports (exports minus imports)

National income measured by this method is called **National income at market price**, since it considers the actual prices paid for goods and services in the market.

Example:

Real-Life Scenario: Measuring GDP in Rwanda

In Rwanda, different groups spend money on goods and services throughout the year. These spending activities help economists measure the country's **Gross Domestic Product (GDP)** using the **expenditure approach**.

Scenario

- 1. Households (C)** Families in Kigali and other towns buy food, clothes, pay for transport, and use services such as salons and mobile money. These everyday purchases represent **consumption expenditure**.
- 2. Businesses (I)** A coffee company invests in new machines to process beans for export. This spending on equipment and production facilities represents **investment expenditure**.
- 3. Government (G)** The government constructs a new public school, pays teachers' salaries, and funds health centres. These are examples of **government expenditure** on goods and services.

4. Foreign Trade (X – M)

- Rwanda **exports** coffee and tea to other countries, earning foreign income.
- Rwanda **imports** cars, electronics, and machinery from abroad, which represents money leaving the country. The difference between exports and imports gives **net exports (X – M)**.

➤ Example Calculation

If in one-year period:

- Household spending (C) = **Rwf500 million**,
- Investment (I) = **Rwf200 million**,
- Government spending (G) = **Rwf150 million**,
- Exports (X) = **Rwf100 million**,
- Imports (M) = **Rwf50 million**

Then GDP is calculated as:

GDP=C+I+G+(X-M). GDP=500+200+150+(100-50). **GDP=900 million dollars**

Interpretation

This means Rwanda's total spending on goods and services during the year amounts to **Rwf900 million**. It reflects the country's overall economic activity — how households, businesses, government, and foreign trade contribute to national income.

3. Product (Output) approach

This is the most direct method. It measures national income by adding up the **value added** at each stage of production across all sectors of the economy.

- **Value added** = Value of final product – Value of intermediate inputs
- Only final goods and services are included. Intermediate goods are excluded to avoid double counting.

Example (Cotton to dress):

- Farmer sells cotton to ginnery: 10,000 RWF (value added = 10,000)
- Ginnery sells cotton to spinner: 15,000 RWF (value added = 5,000)
- Spinner sells thread to textile factory: 25,000 RWF (value added = 10,000)
- Textile factory sells dress to consumer: 40,000 RWF (value added = 15,000)

Total value added = 10,000 + 5,000 + 10,000 + 15,000 = **40,000 RWF**. This equals the final value of the dress, showing how value is added step by step.



Key Takeaway

- » Income Approach: Focuses on payments to factors of production.
- » Expenditure Approach: Focuses on spending on final goods and services.
- » Product Approach: Focuses on value added during production.

All three methods should, in theory, give the same measure of national income if calculated correctly.

Reasons why developing countries use the output approach to measure national income

Developing countries, including Rwanda, often rely on the **output (product) approach** when estimating national income. This method is considered more practical and reliable for their economic conditions. The main reasons include:

1. Limited and unreliable income data

- Many people in developing countries work in the **informal sector** (such as small traders, artisans, and farmers).
- Their earnings are not officially recorded, making the **income approach** difficult to apply accurately.

2. Incomplete information on private consumption

- Governments often lack detailed data on household spending patterns, especially in rural areas.
- This makes the expenditure approach less dependable for measuring total national expenditure.

3. Risk of double counting

- Both the income and expenditure approaches may count the same transactions more than once.
- The output approach helps avoid this by focusing only on the value added at each stage of production.

4. Difficulty in separating intermediate and final goods

- In developing economies, many goods are partly processed or traded informally.
- The expenditure approach may mistakenly include intermediate goods, inflating national income figures.

5. Ease of sectoral measurement

- The output approach allows governments to measure production by sector (agriculture, industry, services).
- This helps identify which sectors contribute most to economic growth and where improvement is needed.

6. Challenges in estimating government and foreign expenditure

- Data on government spending abroad or foreign investment flows are often incomplete.
- The output approach focuses on domestic production, making it simpler and more accurate for national planning.

7. Better alignment with available data sources

- Developing countries usually have more reliable records of **production volumes** (e.g., tons of tea, coffee, cement) than of income or expenditure.
- These production statistics make the output approach more practical and transparent.

Identical results of the three approaches

In theory, the **income**, **expenditure**, and **output** approaches should give the same national income figure if all data are correctly recorded and calculated. However, in practice, small differences may occur due to **errors in data collection, estimation, or valuation**.

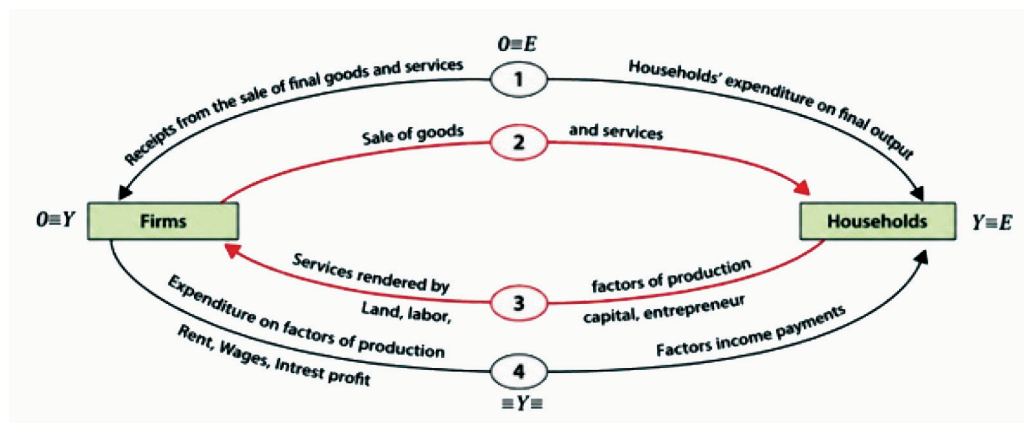


Figure 2.1.: Identical results of the approaches to national income

From Figure above

Spending on goods and services in the market becomes income for the factors of production as rewards for their contribution, such as wages, interest, rent, and profit. For this reason, the **income approach** gives the same result as the **expenditure approach** ($Y \equiv E$).

The value of goods and services produced by firms is reflected in the prices paid for them in the market. Therefore, the **output approach** is equal to the **expenditure approach** ($O \equiv E$).

At the same time, the value of production is distributed as income to the factors of production. This means the **output approach** is also equal to the **income approach** ($O \equiv Y$).

Hence, all three approaches are equal: $O \equiv Y \equiv E$.

Application activity 2.2



i. Calculating national income

Country X provides the following data (in million RWF):

- Production (Value added): Agriculture = 300, Industry = 500, Services = 700
- Income Method: Wages = 900, Rent = 100, Interest = 150, Profits = 350
- Expenditure Method: Consumption (C) = 1,000, Investment (I) = 300, Government Spending (G) = 250, Exports (X) = 200, Imports (M) = 250

Question: Calculate the national income (GDP) using the production, income, and expenditure methods.

ii. Importance of national income statistics

Question: Explain why national income statistics are important for understanding the economic performance of a country.

iii. Final Goods vs. intermediate goods

In Country X:

- Value of intermediate goods = 250 billion RWF
- Value of final goods = 1,500 billion RWF

Question: Why are intermediate goods excluded from national income calculations, while only final goods are included?

iv. Data collection and use of national income

Country X faces challenges such as:

- Many people working in the informal sector
- Delays in reporting data
- GDP figures changing by $\pm 5\%$ after revisions

Question:

- a) State the problems faced when collecting national income data.
- b) Explain how national income figures help the government in planning and decision-making.

2.3 Importance and difficulties of measuring national income in developing countries

Learner's activity 2.3



1. Why is measuring national income considered important for a country's economic planning and development?
2. How does national income measurement help the government of Rwanda in making policies for poverty reduction?
3. What role does national income data play in comparing Rwanda's economic performance with other countries?
4. Identify two major difficulties faced by developing countries like Rwanda in accurately measuring national income.
5. How do informal sector activities (such as small roadside businesses or subsistence farming) create challenges in measuring national income?
6. Suggest one way Rwanda could improve the accuracy of its national income statistics despite these difficulties.

i. Importance of national income statistics

National income statistics are vital for understanding the overall performance of an economy. They provide information about the **aggregate income, output, and expenditure** of a nation, showing how different sectors (households, businesses, government, and foreign trade) contribute to economic growth. Compiling and analysing these figures is important for several reasons:

1. National policy and planning

Governments use national income data to design policies on employment, investment, taxation, and poverty reduction.

For example, if output is low, policies may focus on stimulating production or supporting key industries.

2. Research and analysis

Economists and social scientists rely on these statistics to study savings, investment, consumption patterns, and income distribution.

Data helps in identifying trends and making evidence-based recommendations.

3. Measuring living standards

Per capita income is a key indicator of whether living standards are improving or declining.

It helps compare welfare across regions and over time.

4. Income distribution

Statistics show how income is shared among households, businesses, and government.

This guides tax policy, wage regulation, and government expenditure planning.

5. International transactions

National income data reveals the extent of imports and exports, showing how dependent a country is on foreign economies.

It helps in managing trade balances and foreign exchange reserves.

6. Budget preparation

Figures on private and public expenditure guide the preparation of the national budget.

They help estimate revenue needs and spending priorities.

7. Regional development

Comparing incomes across regions highlights disparities.

This supports targeted interventions to uplift less developed areas.

8. International comparisons

National income statistics allow comparisons between countries, helping measure progress and competitiveness globally.

They are also used by organizations like the World Bank and IMF to classify economies.

9. Tracking economic growth over time

Comparing national income figures across years shows whether the economy is expanding or contracting.

This helps in evaluating the success of policies and reforms.

10. Resource utilization

Rising national income often reflects better use of natural, human, and capital resources.

It shows whether resources are being used productively or wasted.

11. Sectoral analysis

National income data measures the size of different sectors (agriculture, industry, services).

This helps trace sources of growth and allocate resources effectively.

12. Attracting investment

Reliable statistics give confidence to investors and development partners.

They show the strength of the economy and areas with growth potential.

Challenges in data collection and accuracy

Specific difficulties in Rwanda and other developing countries

ii. Difficulties of measuring national income in developing countries

Challenges in data collection and accuracy: Measuring national income requires reliable statistics, but many countries especially developing ones face serious obstacles. These challenges reduce the accuracy of national income figures and make planning more difficult.

a) General difficulties of measuring national income in developing countries

- **Inadequate information:** Many economic activities, especially in the informal sector, are not recorded.
- **Double counting:** Intermediate goods may be counted along with final goods, inflating figures.
- **Non-monetary output:** Subsistence farming, household services, and barter trade are difficult to value in money terms.
- **Price changes and inflation:** Rising or falling prices distort national income figures even when production levels remain unchanged.
- **Timing of production:** Agricultural products may take more than one year to mature, making it hard to assign them to a specific year.
- **Transfer payments:** Benefits, gifts, and grants are difficult to classify, leading to confusion in data.
- **Omissions:** Illegal activities (smuggling, prostitution) generate income but are excluded from official statistics.
- **Shortage of facilities and skilled personnel:** Limited access to modern technology and trained statisticians leads to errors in data collection and processing.

b) Specific Difficulties in Rwanda and Other Developing Countries

1. Large informal sector

Many Rwandans earn income from small-scale farming, street vending, Motorcycle transport, and informal services. These activities are rarely documented, leading to underestimation of national income.

2. Subsistence agriculture

A significant portion of Rwanda's population grows crops mainly for home consumption. Since these outputs are not sold in markets, they are excluded from GDP, even though they contribute to welfare.

3. Limited statistical infrastructure

- National institutions like NISR (National Institute of Statistics of Rwanda) face resource constraints.
- Lack of advanced tools and trained personnel makes data collection slow and sometimes inaccurate.

4. Difficulty in tracking external transactions

- Rwanda's exports (coffee, tea, minerals) and imports (machinery, electronics) involve many small traders.
- Smuggling and informal cross-border trade with Uganda, DRC, and Burundi make it hard to capture accurate figures.

5. Price instability

- Fluctuations in food prices, fuel costs, and inflation distort national income statistics.
- For example, a rise in tea prices may show higher income even if production levels remain constant.

6. Regional disparities

- Rural areas often lack proper data collection systems compared to Kigali.
- This makes it difficult to measure regional contributions to national income accurately.

7. Dependence on donor aid and transfers

- Rwanda receives significant foreign aid and remittances.
- Distinguishing between productive income and transfer payments is complex.



Key Takeaway

National income statistics are essential for Rwanda's development planning, but challenges in **data collection, informal activities, subsistence farming, smuggling, and inflation** make accurate measurement difficult. Addressing these issues requires stronger statistical institutions, better technology, and improved monitoring of informal and cross-border trade.

Application activity 2.3



1. Why are national income statistics important for Rwanda's economic planning and policy making?
2. What challenges are faced in measuring national income in developing countries like Rwanda, especially due to double counting, non-monetary output, informal activities, subsistence production, and smuggling?
3. How do agriculture, industry, services, trade, households, businesses, government, and foreign trade contribute to Rwanda's national income?
4. Why is it difficult to measure the contribution of rural workers, informal sector activities, and subsistence production to national income?
5. What practical measures can Rwanda use to improve the accuracy of national income data collection, such as surveys, digital records, and better reporting systems?
6. Why are transparency, accountability, and teamwork important in improving national income measurement and solving related economic challenges?

2.4 Circular flow of income

Learner's activity 2.4



1. Who are the main actors in the circular flow of income, and how do households and firms interact in a basic two-sector model?
2. How do wages, rent, interest, and profits move from firms to households, and how do households, in turn, spend their income on goods and services?
3. What impact do injections (like government spending, investment, and exports) and leakages (such as taxes, savings, and imports) have on the circular flow?
4. How can saving and investment influence the circular flow in the short term? Why might higher savings reduce current consumption?
5. In what ways do taxation and government expenditure affect the overall size and activity of the economy within the circular flow framework?
6. What happens to the circular flow if exports decline? How does this change affect national income?
7. How do injections and leakages balance out to determine equilibrium national income in both closed and open economies?

The **circular flow of income** is a model that shows how resources, goods, services, expenditure, and incomes move between different sectors of the economy.

- **Real flow:** Movement of goods, services, and factors of production.
- **Money flow:** Movement of payments such as wages, rent, profits, interest, and consumer expenditure.

This model helps us understand how households, firms, the government, and the foreign sector interact in generating and spending income.

2.4.1 Circular flow of income in a closed economy

In a **closed economy** (no foreign trade, no government intervention), only **households and firms** interact.

- **Households** provide factors of production (land, labor, capital, entrepreneurship) to firms.
- **Firms** pay households in the form of wages, rent, interest, and profits.
- Firms use these factors to produce goods and services, which households buy.
- Households spend their income on goods and services, completing the cycle.

Key relationships:

In an economy, there is a continuous movement of goods, services, resources, and money between households and firms. This movement is explained through **real flows** and **money flows**.

Real flows

Real flows refer to the movement of actual goods, services, and factors of production in the economy. Households provide factors of production such as land, labour, capital, and entrepreneurship to firms. Firms then use these resources to produce goods and services, which are supplied to households for consumption.

Illustration:

Factors of production → Firms → Goods and services → Households

Money flows: Money flows refer to the movement of money between households and firms. Firms pay households rewards for providing factors of production in the form of wages, rent, interest, and profits. In return, households spend money on goods and services produced by firms through consumer expenditure.

Illustration: Wages, rent, interest, profits → Households → Consumer expenditure → Firms

These flows are interconnected because the spending of one group becomes the income of another in the economy.

- The three approaches (Income, expenditure, output) are equal: $O \equiv Y \equiv E$

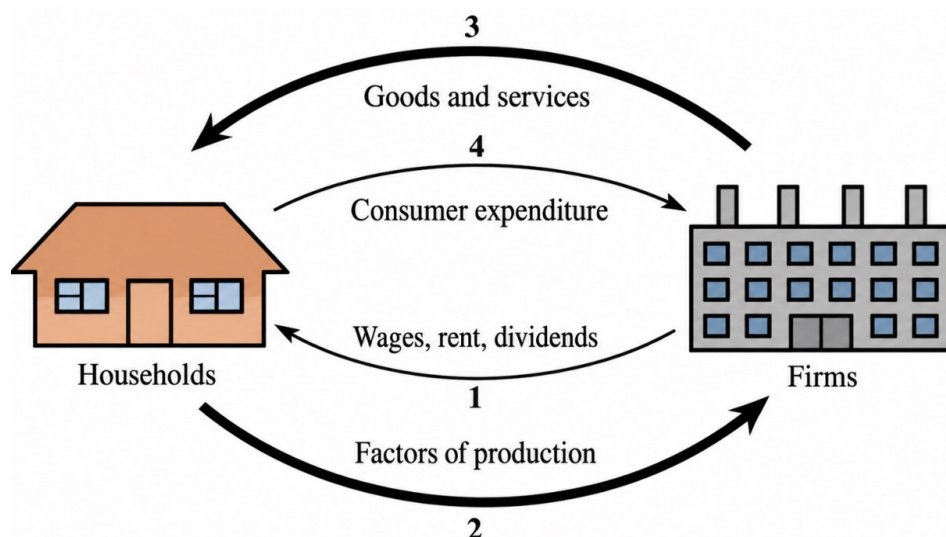


Figure 2.2.: Circular Flow in a Closed Economy

2.4.2 Circular flow of income in an open economy

In an **open economy**, households and firms interact with the **government, financial institutions, and the foreign sector**.

- **Households** earn income from firms and spend part on consumption. They also save through financial institutions.
- **Firms** produce goods and services, sell them domestically, and export abroad. They borrow from financial institutions for investment.
- **Government** collects taxes and spends on public services (education, health, infrastructure).
- **Financial institutions** channel savings into investments.
- **Foreign sector** buys exports and sells imports.

Key relationships:

- National income is measured as: $NY = C + I + G + (X - M)$
- This shows how consumption, investment, government spending, and net exports contribute to GDP.

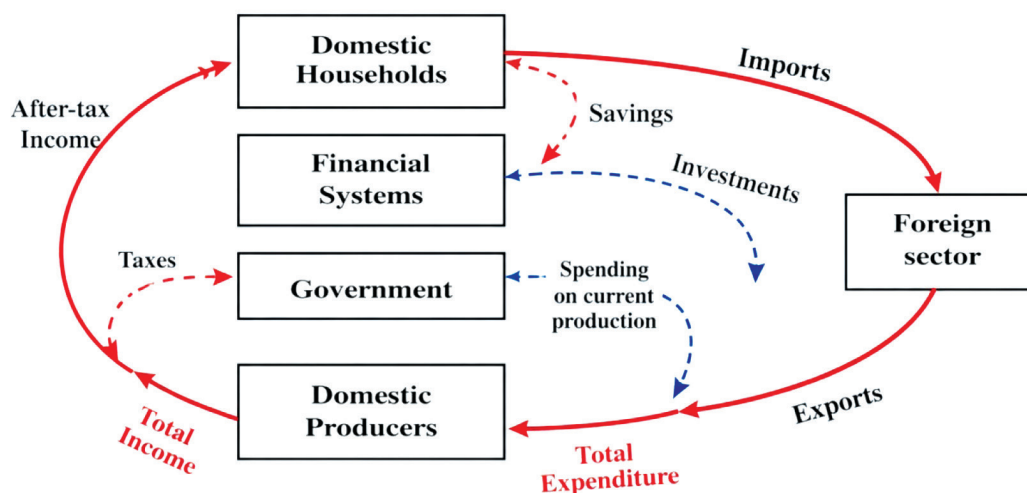


Figure 2.3.: Circular Flow in an Open Economy

From Figure 18 above, it can be noted that:

- Households own the factors of production and provide them to firms. In return, they receive income in the form of wages, rent, interest, and profits. This income is used to buy goods and services, while the remaining part is saved through financial institutions.
- Firms employ factors of production from households to produce goods and services for local consumption and export. They may borrow funds from financial institutions to finance production and investment.
- Firms pay taxes to the government from their profits.
- The government uses tax revenue to provide public and social services to the population.
- The foreign sector provides markets for exports and supplies imports to domestic consumers and firms.
- Financial institutions act as intermediaries between savers and borrowers by collecting household savings, paying interest to depositors, and providing loans to firms for investment. Their profit comes from the difference between interest paid and interest charged.

Application activity 2.4



1. In a simple closed economy, how do households and firms exchange goods, services, and factor incomes?
2. Give an example of how wages paid by firms become household spending in the circular flow.
3. Why does a closed economy exclude foreign trade, and how does this affect the flow of income?
4. How do exports and imports change the circular flow of income compared to a closed economy?

5. Explain how taxation and government spending act as leakages and injections in the circular flow.
6. What happens to national income if leakages (savings, taxes, imports) are greater than injections (investment, government spending, exports)?

2.5 Determinants of national income and limitations of national income figures

Learning Activity 2.5



1. What are the main factors that determine a country's national income? List at least three.
2. How does the level of investment in a country influence its national income? Explain briefly.
3. In what ways can government policies affect the national income of a country? Give two examples.
4. Why might the national income figures not accurately reflect the actual economic well-being of all citizens?
5. What are some limitations of using Gross Domestic Product (GDP) as a measure of a country's economic performance?
6. Can two countries with similar national income figures have very different standards of living? Why or why not?

National income is the total value of all goods and services produced within a country over a specific period. It is an important measure used to understand the economic performance of a nation. Several factors, known as determinants, influence the level of a country's national income. However, while national income figures provide useful information, they also have limitations and may not always give a complete picture of a country's economic well-being. In this lesson, we will explore what determines national income and discuss some of the challenges in using national income figures to assess economic progress.

2.5.1 Determinants of a country's national income

National income is a key indicator of a country's economic performance and is influenced by several determinants such as the level of production, technological advancement, labour force quality, capital investment, and resource availability. These factors collectively impact the overall income generated within a nation.

However, while national income figures provide valuable insights, they also have notable limitations. These figures often fail to account for income distribution, non-monetized activities, and environmental factors. Additionally, they may not accurately reflect the standard of living or the well-being of the population, as they focus solely on monetary transactions and output.

i. Key factors influencing national income

Stock of natural resources: Abundant land, fertile soils, minerals, forests, and water bodies raise production and income when properly exploited. Scarcity or poor use of resources leads to low national income.

Availability of capital: Capital in the form of machines, tools, and financial resources boosts productivity. A shortage of capital limits investment and reduces income.

Technological progress: Modern technology increases efficiency, output, and competitiveness. Outdated technology slows production and lowers income.

Human resources: A skilled, healthy, and motivated labor force with entrepreneurial ability raises productivity. Low-quality labor reduces national income.

Political situation: Stability encourages investment and production, while instability causes uncertainty, low output, and reduced income.

Market size: A large domestic and international market motivates firms to produce more, raising income. A small market discourages production.

Infrastructure development: Good roads, electricity, communication, and transport systems attract investment and facilitate trade. Poor infrastructure limits growth.

Government policy: Supportive policies (subsidies, fair taxation, investment incentives) encourage production. Excessive taxation or restrictive policies discourage investment and reduce income.

Organization of production factors: Efficient coordination of land, labor, capital, and entrepreneurship increases output. Poor organization wastes resources and lowers income.

Institutional and cultural factors: Positive attitudes toward work, supportive cultural values, and strong institutions raise productivity. Negative attitudes or weak institutions reduce income.

Investment levels: Higher investment in industries, education, and infrastructure increases future national income.

Consumption patterns: Strong consumer demand encourages production and income growth. Weak demand slows economic activity.

Foreign trade: Exports bring income into the country, while excessive imports can drain resources.

Natural disasters and climate: Floods, droughts, or epidemics reduce production and income. Favorable climate supports agriculture and growth.

Global economic conditions: International prices, global recessions, or booms affect national income through trade and investment flows.

ii. Application of factors influencing national income in Rwanda

The factors influencing national income are applied differently in Rwanda depending on the structure of the economy, government policies, and the country's development priorities. The following explains how each factor specifically applies to Rwanda.

1. Natural resources and national income in Rwanda

Rwanda utilizes its natural resources to promote agriculture, mining, tourism, and energy generation, which contribute significantly to national income.

Experimental version

Examples

- Fertile soils support the cultivation of crops such as coffee, tea, maize, beans, and bananas.
- Mineral resources, including coltan, cassiterite, wolfram, and gold, generate export revenue.
- National parks, mountains, and wildlife attract tourists and earn foreign exchange.
- Lakes and rivers support fishing activities and hydroelectric power production.

Challenges

- Limited land due to high population density.
- Soil erosion and environmental degradation.
- Climate change affecting agricultural output.

2. Human resources and national income in Rwanda

Rwanda places strong emphasis on developing skilled, healthy, and productive human resources to enhance economic growth.

Examples

- Expansion of free basic education and Technical and Vocational Education and Training (TVET).
- Promotion of ICT knowledge and digital literacy.
- Improvement of healthcare services through community-based health insurance.
- Youth employment and entrepreneurship support programs.

Contribution to national income

A skilled and healthy labour force increases productivity, innovation, efficiency, and business development.

3. Capital formation and national income in Rwanda

Rwanda encourages investment in both physical and financial capital to expand production and economic activities.

Examples

- Construction of roads, airports, factories, and industrial parks.
- Expansion of electricity and water supply networks.
- Growth of financial institutions and access to credit facilities.
- Promotion of savings and investment through banks and SACCOs.

Challenges

- Low domestic savings.
- Dependence on foreign investment and external aid.

4. Technology and innovation in Rwanda

Technology and innovation are central to Rwanda's economic transformation and modernization efforts.

Examples

- Use of e-government platforms such as Irembo.
- Expansion of mobile money and digital banking services.
- Adoption of modern farming technologies.
- Development of Kigali Innovation City and ICT innovation hubs.

Contribution to national income

Technology improves productivity, communication, service delivery, and efficiency in production.

5. Political stability and national income in Rwanda

Political stability has provided a conducive environment for investment, production, and economic development.

Examples

- Peace and security attract both local and foreign investors.
- Stable leadership supports long-term development planning.
- Effective public institutions improve investor confidence.

Contribution to National Income

Political stability promotes investment, employment creation, and expansion of economic activities.

6. Infrastructure development and national income in Rwanda

Infrastructure development plays a major role in stimulating economic growth and increasing national income.

Examples

- Construction and rehabilitation of roads linking different districts.
- Expansion of electricity supply in both rural and urban areas.
- Improvement of internet services and communication systems.
- Development of Kigali International Airport and the Bugesera International Airport project.

Contribution to national income

Efficient infrastructure reduces transport costs, facilitates trade, and attracts investment.

7. Market size and national income in Rwanda

Rwanda expands its market opportunities through regional and international economic integration.

Examples

- Membership in the East African Community (EAC).
- Participation in the African Continental Free Trade Area (AfCFTA).
- Promotion of exports to neighboring and international markets.

Challenges

The domestic market remains relatively small due to low purchasing power among some consumers.

8. Government policy and national income in Rwanda

Government policies in Rwanda strongly support economic growth, production, and investment.

Examples

- Provision of investment incentives to local and foreign investors.
- Policies encouraging industrialization and export diversification.
- Support for Small and Medium Enterprises (Small and Medium-sized Enterprises (SMEs)).
- Tax reforms and programs improving the ease of doing business.

Contribution to national income

Supportive government policies stimulate entrepreneurship, production, and employment creation.

9. Agriculture and national income in Rwanda

Agriculture remains one of the leading sectors contributing to Rwanda's national income.

Examples

- Crop intensification and land consolidation programs.
- Irrigation projects aimed at reducing reliance on rainfall.
- Exportation of coffee and tea.
- Livestock support initiatives such as the Girinka program.

Challenges

- Dependence on rain-fed agriculture.
- Small and fragmented land holdings.
- Climate-related shocks such as droughts and floods.

10. Industrial development and national income in Rwanda

Rwanda promotes industrialization to reduce dependence on agriculture and increase value addition.

Examples

- Development of the Kigali Special Economic Zone.
- Expansion of agro-processing industries.
- Growth of manufacturing and construction sectors.
- Promotion of "Made in Rwanda" products.

Contribution to national income

Industries generate employment opportunities, promote exports, and increase value addition to raw materials.

11. Service sector and national income in Rwanda

The service sector is among the largest contributors to Rwanda's Gross Domestic Product (GDP).

Examples

- Growth of tourism through conferences, hotels, and national parks.
- Expansion of banking and insurance services.
- Development of transport and communication services.
- Rapid growth of ICT and digital services.

Contribution to National Income

The service sector generates foreign exchange, employment, and government revenue.

12. Foreign trade and national income in Rwanda

Rwanda relies on international trade to earn foreign exchange and obtain goods and services not produced locally.

Main Exports: Coffee, Tea, Minerals and Tourism services

Main Imports: Fuel, Machinery, Manufactured goods and Vehicles

Challenges: Persistent trade deficits caused by high import expenditure.

13. Foreign aid and remittances in Rwanda

Foreign aid and remittances play an important role in supporting development and household welfare.

Examples

- Infrastructure and social projects financed by development partners.
- Money sent home by Rwandans working abroad.
- External support for healthcare, education, and public services.

Contribution to national income

These increase household income, investment, and economic development.

14. Environmental and climate factors in Rwanda

Environmental sustainability is an important part of Rwanda's development strategy.

Examples

- Environmental conservation programs.
- Afforestation and terracing activities.
- Ban on plastic bags to reduce environmental pollution.
- Promotion of green growth and climate resilience strategies.

Challenges

- Floods and landslides damaging infrastructure and crops.
- Climate variability reducing agricultural productivity.

15. Demographic factors and national income in Rwanda

Population changes influence labour supply, production, and market demand.

Examples

- A large youth population provides potential labour force.
- Urbanization increases demand for housing, transport, and social services.
- Population growth increases pressure on land and employment opportunities.

Contribution to national income

A productive and employed working-age population contributes positively to national income growth.

Summary

In Rwanda, national income is influenced by several interrelated factors such as:

- | | |
|---------------------------------|--|
| » Effective government policies | » Expansion of the service sector |
| » Infrastructure development | » Political stability |
| » Technological advancement | » Regional and international integration |
| » Agricultural transformation | |

However, the progress made, challenges including climate change, unemployment, limited capital, trade imbalances, and dependence on agriculture still affect national income growth. Continued investment in education, industrialization, technology, and infrastructure is essential for achieving sustainable economic growth and improving the standard of living in Rwanda.

2.5.2 Limitations/ Shortcomings of National Income Figures

Shortcomings of using national income figures

Although national income statistics are highly useful for policy, planning, and analysis, they have several limitations. These shortcomings mean that the figures must be interpreted with caution.

1. International comparisons

- Per capita income comparisons between countries can be misleading.
- Differences in culture, diet, climate, and consumption patterns are not reflected in money values.
- Example: A country where rice and fish are the staple diet cannot be directly compared with Rwanda, where diets are more varied.

2. Changes in capital stock

- National income figures often fail to account for changes in the stock of capital equipment.
- Depreciation estimates are usually inaccurate, making it difficult to measure the true productive capacity of an economy.

3. Ignoring quality improvements

- Statistics focus on quantity rather than quality.
- Improvements in the quality of goods and services (e.g., better healthcare, education, or technology) are not reflected in GNP figures.

4. Exclusion of non-market activities

- National income data only includes goods and services sold in the market.
- In many developing countries, a large share of production is for home consumption (subsistence farming, household services).
- This underestimates actual welfare and per capita income.

5. Long-term comparisons

- Over short periods (2–3 years), comparisons are valid.
- Over longer periods, they become misleading because new products appear and old ones disappear, making comparisons across decades inaccurate.

6. Inflation and price changes

- Rising prices may show an increase in national income even if production has fallen.
- Falling prices may show a decrease in national income even if production has increased.
- This makes it difficult to distinguish between real growth and inflationary effects.

7. Unequal distribution of income

- National income figures show aggregates but hide inequalities.
- A rise in per capita income may benefit only a small elite, while the majority remain poor.
- This makes national income a weak indicator of welfare.

8. Environmental and social costs

- National income statistics ignore environmental degradation, resource depletion, and pollution.
- For example, cutting down forests may increase GDP in the short term but reduce long-term welfare.
- Social costs such as poor working conditions or child labor are also excluded.

9. Transfer payments and informal activities

- Benefits, gifts, remittances, and aid are difficult to classify.
- Informal activities (street vending, small-scale farming, smuggling) are often excluded, leading to underestimation.

10. Regional disparities

- National income figures give averages but hide regional differences.
- Some regions may be highly developed while others remain backward, yet the statistics show only the national total.



Key Takeaway

National income figures are useful but imperfect. They provide a broad picture of economic activity but fail to capture **quality of life, inequality, environmental sustainability, informal sector contributions, and regional disparities**. Policymakers must therefore use them alongside other indicators (like Human Development Index, poverty rates, and employment statistics) to get a complete view of welfare and development.

Application activity 2.5



1. How do the factors of production (land, labour, capital, and entrepreneurship) contribute to the determination and growth of Rwanda's national income?
2. Explain the role played by each factor in increasing production and income generation.
3. How does investment in infrastructure such as modern roads, electricity supply, transport networks, and communication systems influence productivity and contribute to national income growth in Rwanda?
4. In what ways have technological developments such as internet access, mobile money services, digital banking, and innovation improved efficiency, productivity, and income generation in Rwanda?
5. How do agricultural development programmes such as crop intensification, irrigation schemes, improved seeds, fertilizers, and modern farming methods increase farmers' incomes and raise Rwanda's national income?
6. Why are exports such as coffee, tea, minerals, and tourism services important in determining Rwanda's national income, and how do imports affect the circular flow of income in an open economy?
7. How do government policies related to taxation, public expenditure, education, vocational training, and investment promotion influence national income and economic development in Rwanda?
8. National income figures are commonly used to measure economic performance. What are the limitations or shortcomings of national income statistics in measuring economic welfare in Rwanda, and what measures can be taken to improve their accuracy and reliability?

2.6 Equilibrium level of national income

Learning Activity 2.6



Scenario:

Imagine a country like Rwanda. In Rwanda, the government and businesses are trying to understand how much the country produces and spends in a year. They learn about the concepts of aggregate demand (the total amount people want to buy) and aggregate supply (the total goods and services produced).

One day, the government notices that when people and businesses spend more money, the total income of Rwanda increases. When they spend less, the total income decreases.

Task:

Help Rwanda understand how the economy reaches a balance point where everything produced is bought. Draw a simple graph showing the aggregate demand and aggregate supply curves. Find the point where they meet—that's the equilibrium level of national income.

Questions to Think About:

- What might cause people in Rwanda to start spending more or less?
- What happens if the government introduces a new policy that encourages more spending?
- Why is finding this balance important for Rwanda's economy?

Meaning of equilibrium

Equilibrium can be defined as a state of stability in economic conditions, irrespective of the forces influencing different economic agents. That is, there is no tendency for consumers and firms to change their economic behaviour. This means that the variables in an economy are equal and have no tendency to change. As already pointed out, during equilibrium, national product and national income must be equal and also national income is equal to national expenditure since it is derived from this income.

Approaches to national income equilibrium

2.6.1 National income equilibrium: Inflationary gap and deflationary gap approach

According to the **classical economists**, equilibrium income is determined where aggregate demand is equal to aggregate supply. At such a point, savings are equal to investment, and all the resources are fully employed.

Keynes, on the other hand, disagrees with this analysis. According to him, full employment level of income (Y_f) may not equal to the equilibrium income (Y_e). It is possible to have (Y_f) being greater than (Y_e) and hence the **deflationary gap**. It is also possible to have (Y_e) being greater than (Y_f) and hence the **inflationary gap**.

A **deflationary gap** is a situation where aggregate supply exceeds aggregate demand at full employment. All that is produced is not demanded. Realised investment is greater than the actual demand. A deflationary gap is caused by deficiency in demand.

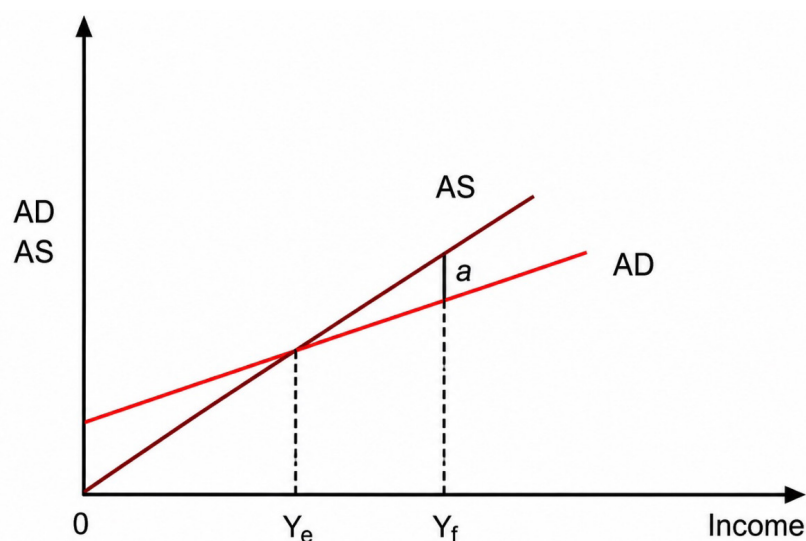


Figure 2.4.: Deflationary gap

From the figure above, gap a-b is a deflationary gap

a) How to close a deflationary gap?

It can be closed by increasing aggregate demand from aggregate demand (1) to aggregate demand (2) and this can be done through using the policies that increase demand for goods and services.

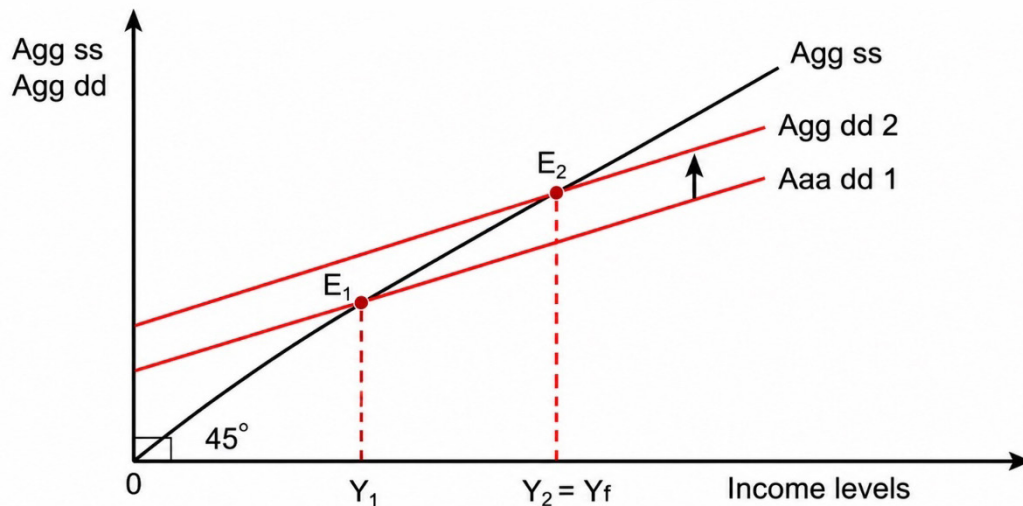


Figure 2.5.: Closing a deflationary gap

From the figure above, Agg ss- aggregate supply, Agg dd- aggregate demand, E_1 - equilibrium 1, E_2 equilibrium 2, Y_1 – income 1, $Y_2 = Y_f$ – shows full employment levels of resources.

b) Policies that can be used to increase aggregate demand

- **Increasing the volume of exports** to get rid of the surplus; Discouraging imports so as to avoid increasing the volume of goods in the country;
- **Adopting an expansionary monetary policy** – This will help to increase money in the hands of the people, hence increasing their demand.
- **Increasing government expenditure:** This will also increase money in the hands of the people, hence increasing their demand.
- **Increasing wages** such that the people can have access to income and increase demand;
- **Reducing taxes** – This will increase the disposable income in the hands of the people, hence increasing their demand.
- **Adopting price control**, especially a **maximum** price that is low
- and will encourage demand.
- **An inflationary gap** is a situation where aggregate demand exceeds aggregate supply at full employment level. Realised investment is less than actual demand. Supply is not enough.

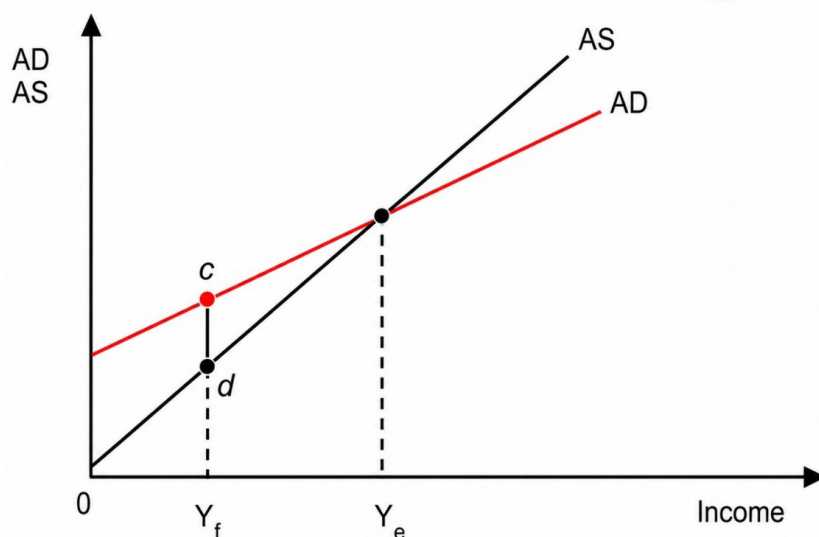


Figure 2.6.: An inflationary gap

From Figure 2.6. above, gap c-d is an inflationary gap.

At point (Y_e), national income is in equilibrium when aggregate demand is equal to aggregate supply. Before (Y_e), aggregate demand exceeds aggregate supply and this is called **an inflationary gap**.

c) How to close an inflationary gap?

An inflationary gap can be closed by reducing aggregate demand from aggregate demand (1) to aggregate demand (2) as shown below and this can be done through policies which reduce the money in the hands of the people so that because of little income, demand is also reduced.

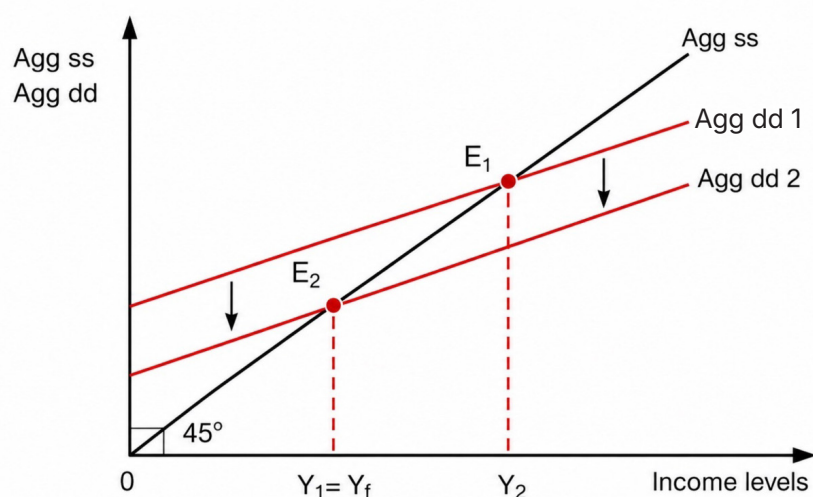


Figure 2.7.: Closing an inflationary gap

From Figure above, Agg ss- aggregate supply, Agg dd- aggregate demand, E_1 - equilibrium 1, E_2 equilibrium 2, $Y_1 = Y_f$ shows full employment levels of resources, Y_2 - income.

d) Policies that can be used to reduce aggregate demand

- **Adopting a restrictive monetary policy** – This will reduce the amount of money supply and hence lower aggregate demand.
- **Increased imports** – This can help to increase the volume of goods available when demand is greater than supply.
- **Reduced exports** – This is to reduce the amount of goods going out of the country so as to make them enough for the people.
- **Increased taxes** – This is to help reduce the disposable income of the people so that they do not have a lot to spend.
- **Reduced government spending** – This is aimed at reducing and controlling the amount of money in circulation which leads to limited liquidity and low demand.
- **Adopt price control** especially **minimum price** legislation which is a high price that can reduce the demand for goods and services.
- **Reducing wages** by adopting a tight income and wage policy.

e) National income equilibrium: leakages – injections approach

National income is in equilibrium when total leakages are equal to total injections.

Leakages refer to elements which withdraw money from the circular flow of income. They include savings -S, Consumption-C, taxation -T, Imports-M and capital outflow-Ko. These elements remove money from the circular flow of income, i.e. $C + S + T + M + K_o$.

Injections refer to elements that add to the circular flow of income. They include Investment-I, Consumption-C, government Expenditure-G, Exports-X and capital Inflow-Ki. These elements add money to the economic activities in an economy, i.e. $C + I + G + X + K_i$. Thus, national income is in equilibrium when $S + T + M + K_o = I + G + X + K_i$. This can be seen in the following ways:

1. National income equilibrium in a closed economy

A closed economy is one where transactions take place within the country without any foreign trade. The major factors, which determine the level of income (**Y**), are consumption (**C**), saving (**S**) and investment (**I**).

The country's income can be used for the consumption and saving. These are known as leakages or withdrawals. The identity is $Y = C + S$.

If the level of income is to be maintained, saving must be put back into the economy in form of investment so as to create more income.

Consumption must also take place. These are known as injections. The identity is $C + I = Y$. Therefore, we have $Y = C + S$ and $Y = C + I$;

equating the two expressions yields $C + S = C + I = Y$.

In a closed economy without government, equilibrium income is a situation where savings are equal to investments. This can be illustrated as below:

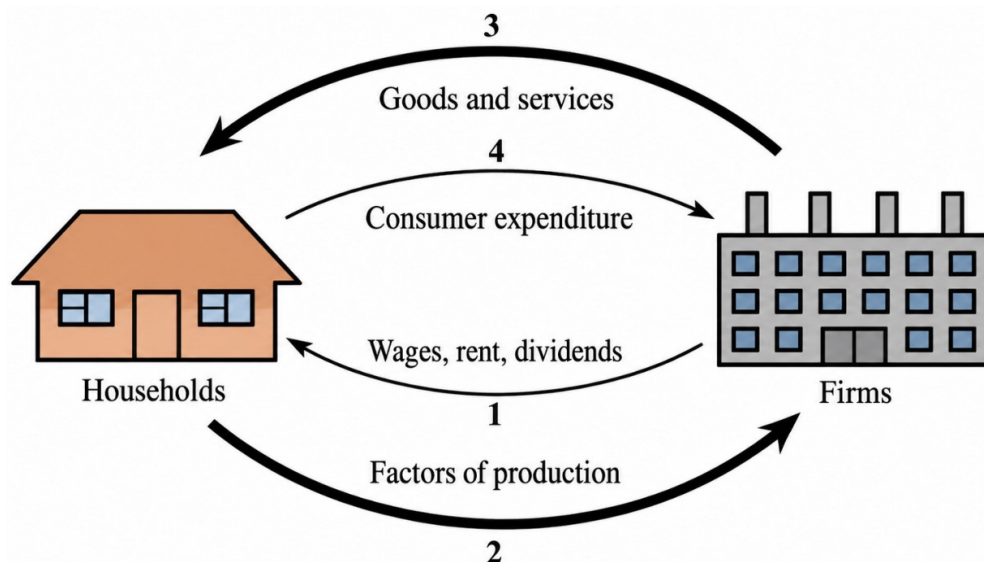


Figure 2.8.: National income equilibrium in a closed economy

Firms purchase factors of production from households (2) and provide payments for these inputs (1). These firms then combine the acquired factors to produce goods and services, which are subsequently sold to households (3). In return, households make payments for these goods and services (4).

- Arrows (2) and (3) represent real flows, meaning the movement of productive resources and finished goods/services respectively.
- Arrows (1) and (4) represent monetary or financial flows, indicating the movement of income and expenditure.
- The value of total output (output approach) is equal to total household expenditure on goods and services (expenditure approach), hence $O = E$.

Equilibrium income determining: Saving-investment approaches

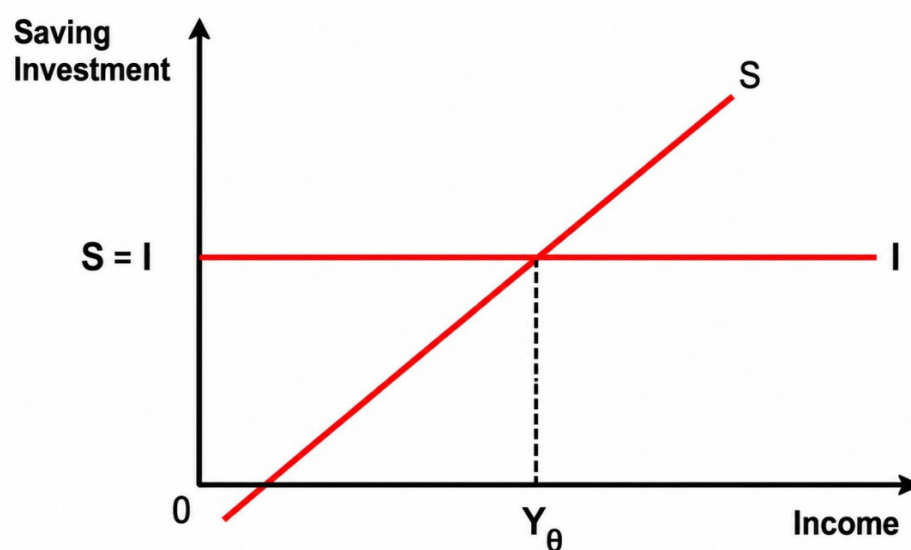


Figure 2.9.: Saving-investment approaches

From Figure 2.9. above, it is seen that equilibrium income can also be determined through the saving- investment approach. Planned saving is equal to planned investment. The intersection of the saving and investment schedules determines equilibrium income and output at Y_0 . The savings are positively related to the income while investment is autonomous.

The economy will reach equilibrium at full employment only if the amount that consumers wish to save out of their income is precisely equal to the amount that investors want to invest.

At any income below Y_0 , planned investment exceeds planned saving. Aggregate demand exceeds aggregate supply. This forces

investors to increase the rate of production until Y_0 . At any level of income beyond Y_0 , savings exceed investment. Aggregate supply exceeds aggregate demand.

Consequently, there are unsold inventories and investors will be forced to reduce the rate of production until $S = I$.

2. National income equilibrium in an open economy

An Open economy is one where there is foreign trade. So far, we have dealt with a very simple economy consisting of only households and businesses. All the income created in the process of production was passed on to households as disposable income. Thus, GNP and disposable income were identical.

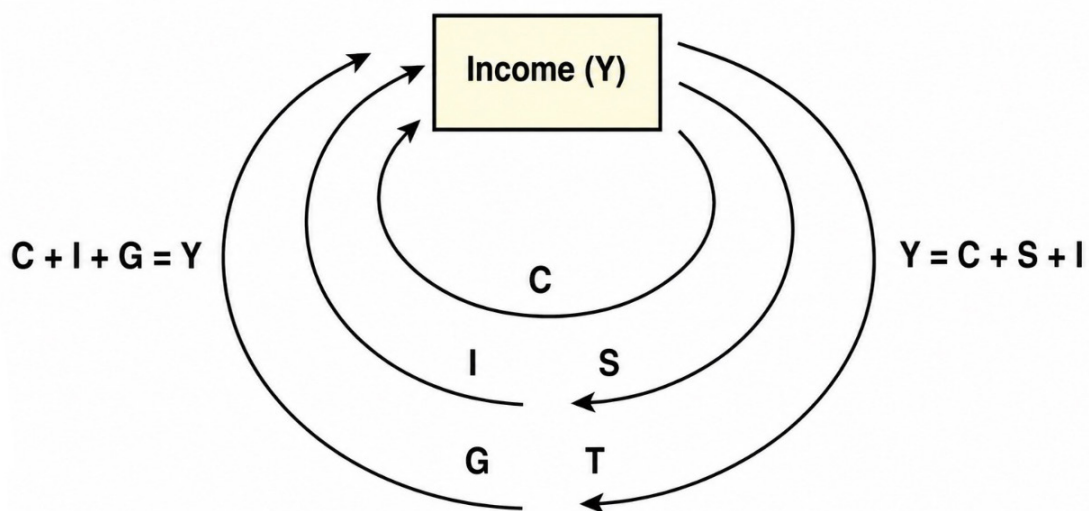


Figure 2.10.: National income equilibrium in an open economy

From Figure above, in our simple economy, when the government sector is added, it levies taxes and makes expenditures on the purchase of goods and services. The government also makes transfer payments. Aggregate demand now consists of consumption (**C**) investment (**I**) and government demand for goods and services (**G**). Therefore, $C + I + G = Y$. Not all the income from the production of output is disposable income to the households. A portion is now absorbed by the government as net tax receipts.

Net tax receipts are total tax receipts less that portion which is returned to the private sector in the form of government transfer payments. While government tax receipts reduce disposable income, government transfer payments such as unemployment allowances increase disposable income.

The income is used for consumption, saving and tax.

$Y = C + S + T$. Therefore, equilibrium income is where $S + T = I + G$, i.e. leakages = injections.

When government expenditure is added on, the equilibrium income increases from Y_1 to Y_0 where $I + G = S + T$.

2.6.2 Equilibrium income determining: The leakages-injections approach

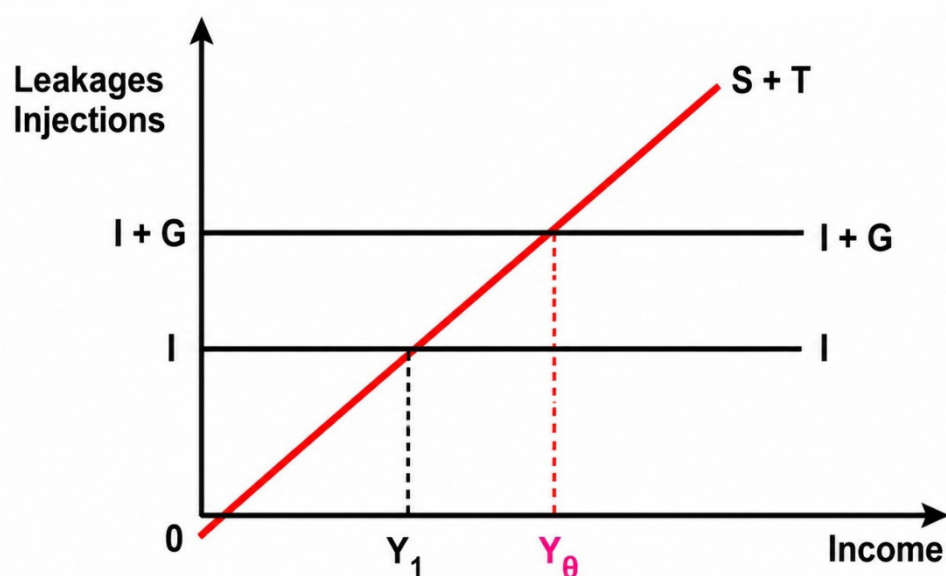


Figure 2.11.: The leakages-injections approach

When the foreign sector is introduced, income is created through consumption, investment, government expenditure and exports.

Thus, $C + I + G + X = Y$. The income is used for consumption, saving, tax and imports.

This is indicated by the following equation $Y = C + S + T + M$.

Equilibrium income is, therefore, equal to $S + T + M = I + G + X$.

Leakages = injections. The equilibrium income is indicated in Figure below;

i. Equilibrium income with the foreign sector

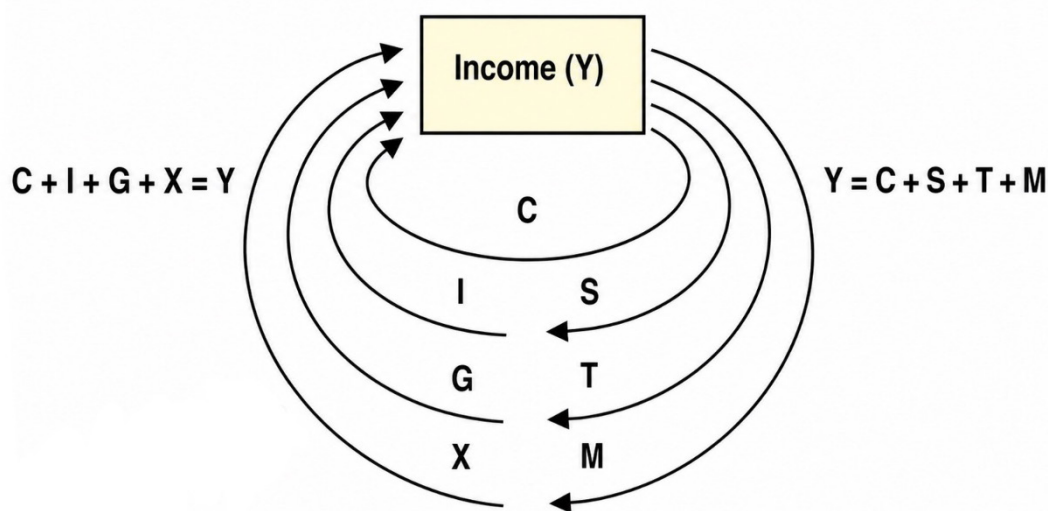


Figure 2.12.: Equilibrium income with the foreign sector

From Figure 2.3., it is seen that when the foreign sector is added on, injections become $C + I + G + X$ and leakages become $C + S + T + M$. Therefore, the equilibrium income increases to Y_e where, $I + G + X = S + T + M$.

Equilibrium income determining: Leakages-injections approach

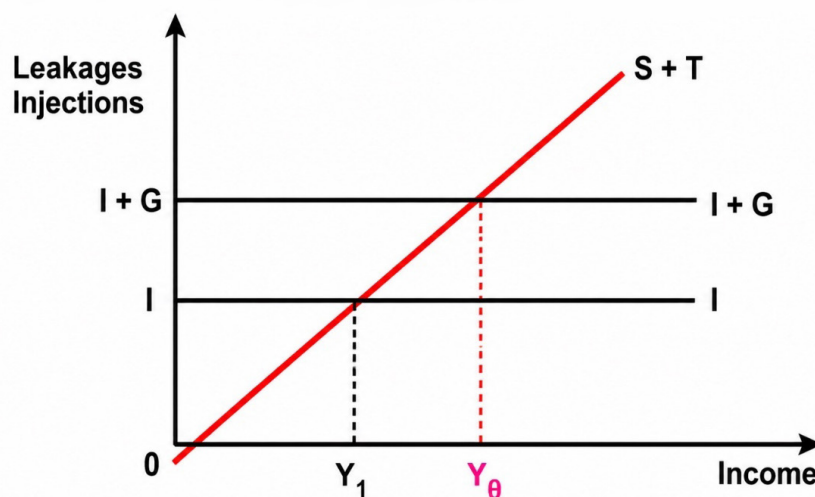


Figure 2.13.: Leakages-injections approach

From Figure above, investment, government spending and exports are known as the injections into the flow of income. They increase the circular flow of income. Savings, taxes and imports are referred to as withdrawals or leakages from the flow. They reduce the country's income — the circular flow of income.

The condition for the equilibrium income is that injections = leakages. When injections exceed leakage, the level of income increases whereas if leakages exceed injections, the income level reduces.

Experimental version

Application activity 2.6



1. Explain the concept of equilibrium level of national income. Using the aggregate expenditure and aggregate output approach, describe how equilibrium national income is achieved. What happens when aggregate expenditure is greater or less than aggregate output?
2. Distinguish between an inflationary gap and a deflationary gap. Explain their effects on output, employment, prices, and economic stability in an economy such as Rwanda.
3. Explain how the leakages–injections approach is used to determine equilibrium national income. What happens when leakages are equal to, greater than, or less than injections?
4. Rwanda experiences a situation where savings, taxes, and imports increase rapidly while investment, exports, and government expenditure remain unchanged. Using the leakages–injections approach, explain the likely impact on equilibrium national income and suggest possible solutions.

2.7 Fluctuations in national income (Business cycle)

Learning Activity 2.7



Activity Scenario

In Gatsibo District, maize farming increases during the season of September to December. During this period, employment, production, income, demand, savings, and living standards rise because many people are involved in farming and trading activities.

After the season ends, farming activities reduce, leading to lower income, employment, and living standards. When the next farming season begins, economic activities rise again, creating a cycle of ups and downs.

Questions

1. Identify the phases shown in the maize farming cycle.
2. Explain the characteristics of each phase.
3. Draw a simple graph showing the ups and downs of the cycle.
4. Explain how each phase affects people's living standards.

In any given economy, economic conditions and circumstances do not remain constant over a long period. There are always upward and downward swings in business activities or trade.

According to Professor Mitchell, it is a type of fluctuations found in aggregate economic activities of nations that organise their work mainly in business enterprises.

The course of trend of business activities passes through phases of prosperity and adversity which are referred to as trade/business cycles. They occur in every economy periodically and they follow a particular pattern. Each cycle takes a different period from the other and a time taken in each cycle is different in different economies.

A typical business cycle is generally divided into 4 phases although some scholars argue that they are five as seen below.

- Boom phase (upswing)
- Recession phase (upper turning point)
- Depression or slump or trough phase (downswing)
- Recovery phase (lower turning point)
- Prosperity

There are basically four stages of business growth although some scholars argue that they are five as seen below.

Expansion (Boom)

This is a stage where business activities are at high levels and they tend to acquire profits. The business is normally in the upswing mode as shown by the high levels of economic activities. It is characterised by the following:

- There is increase in the demand for both capital and consumer goods.
- Companies invest in more production facilities with a view of making profits from the increase in sales.
- Banks lend capital for expansion at low rates because they have confidence in the investors paying back.
- There are high rates of employment brought about by high aggregate demand and investments.
- Business is at its peak and makes supernormal profits.

Recession

This stage is the upper turning point and shows that the economy is in a decline as shown by the characteristics below:

- Level of sales and production orders start declining.
- Production facilities become underutilised.
- Companies reduce the work rate. Workers hired on casual basis are laid off.
- There is reduction in the level of output.
- Banks raise the interest rates to counter the rise in risk of default on loans.
- Most of the companies reduce the price of their goods so that they
- can increase demand.

Depression

This is the bottom of a cycle where economic activity remains at a low level. It has the following characteristics:

Demand for products and services reduce, forcing some companies to shut down some production facilities.

There are rampant cases of unemployment brought by closing of industries.

- Demand for products and services reduce, forcing some companies to shut down some production facilities.
- There are rampant cases of unemployment brought by closing of industries.
- There are high rates of poverty.
- The purchasing power in an economy becomes very low.
- The gross domestic product declines and so does the standard of living of the people. The fall in price of capital goods is more than that of consumer goods.
- Demand for loans declines because the investors become irritated by the economic situation.

Recovery

This is the stage where business begins regaining its strength. Business may sell output at very low prices to retain the operating costs; carry out some repairs; or gets some credit, among others. This helps it to begin moving from the trough. The stage is characterized by the following:

- Business become confident in the market and they begin buying goods so the business begins making profits.
- The bank rates become low so the companies can afford to borrow and finance projects.
- There is an increase in production because of increase in aggregate demand.
- Companies begin employing people and so there is a reduction in unemployment levels.
- Standards of living of the people improve since they can afford to buy goods and services.
- Profit margins of business start rising and the gross domestic product
- also begins to rise.

Prosperity

This is where business regains its strength and activities are at their peak. The following are its characteristics:

- There are high employment levels in an economy.
- There are high incomes due to the employment levels.
- High levels of production is common at this stage
- There is high aggregate demand and cost which leads to a rise in investment and prices for goods and services.
- The existing capacity of plants is underutilized.

This can be illustrated in the figure below.

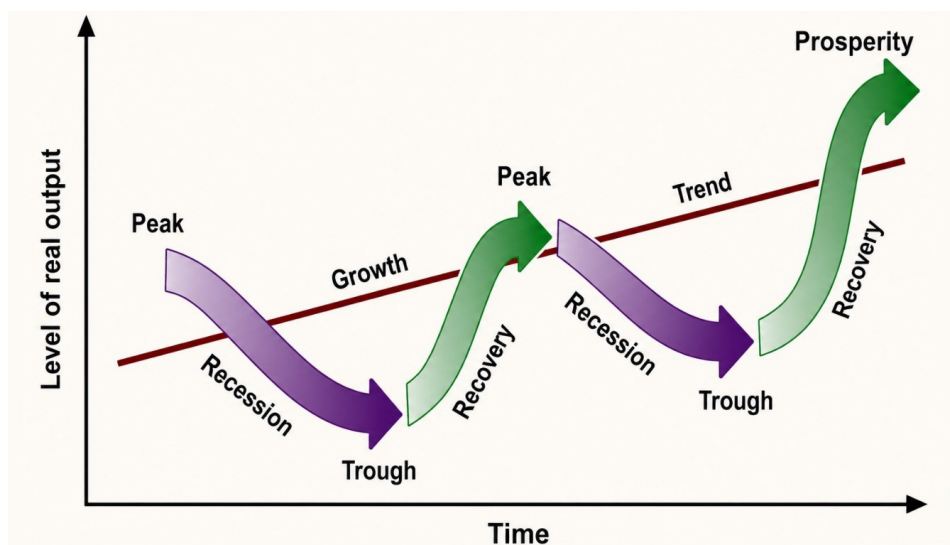


Figure 2.14.: Business cycle

Application activity 2.7



1. During the boom phase, why does the unemployment rate tend to decrease? Explain the relationship between economic growth and employment.
2. In the recession phase, why might interest rates decrease? Discuss how central banks use interest rates during economic downturns.
3. What factors could lead to a decrease in the inflation rate during the recovery phase of the business cycle? Explain the connection between economic growth and inflation.
4. Why do investments tend to decrease during a depression? Describe how economic uncertainty affects business investments.
5. How do fluctuations in the business cycle impact the overall well-being of a country's citizens? Provide examples of positive and negative effects.
6. What kinds of policies can a government implement to stabilize the economy during different phases of the business cycle? Discuss the potential benefits and challenges of these policies.

2.8 National income and standard of living

Learning Activity 2.8



In Rwanda, people earn different incomes depending on their jobs and activities. For example, a farmer earns 80,000 FRW per month while a teacher earns 220,000 FRW per month. The average income in the district is estimated at 150,000 FRW per person per month.

Although income levels differ, some households access better education, health care, housing, and clean water, while others still face poverty and unemployment. This shows that income influences living standards, but does not fully determine quality of life.

Using the scenario, answer the following questions:

1. What is the term used to describe the average income per person in a country?
2. How does income affect people's quality of life?
3. Apart from income, what other factors influence living standards in Rwanda?
4. Why can people with different incomes sometimes have similar living conditions?
5. Compare living standards between urban and rural areas in Rwanda.
6. Suggest ways to improve living standards in Rwanda.

National income and standard of living are important concepts used to assess a country's economic health and the well-being of its people. While national income measures the total income earned by a country's residents, the standard of living reflects the quality of life and material well-being of individuals. Understanding the relationship between these two helps in evaluating how economic growth translates into improved living conditions for the population.

The standard of living refers to the level of material welfare and comfort enjoyed by people in a country. It includes access to: Food and nutrition, Education, Healthcare, Housing, Clean water and sanitation, Employment opportunities, Transport and communication and Security and leisure

Generally, higher national income leads to a better standard of living, holding other factors constant.

National income of a given country reflects the way of life of its people during a specified period. The way of life people lives or hope to live is what we call **standard of living**. The higher the level of national income, the higher the standard of living and vice-versa, holding other factors constant. The standard of living will depend on the average income of the people in the country at a particular time.

i. Per capita income

Per capita income is income received per person in a period.

Per capita income = National income ÷ Total population

Example 1:

If the total national income is 200 billion francs and the population is 50 million people, then; Per-capita income = $200b \div 50m = 4000$ francs.

Example 2: If the total national income is 500 billion francs and the population is 100 million people, then:

Per-capita income = Total national income ÷ Population

$500 \text{ billion} \div 100 \text{ million} = 5,000$ francs

Study the table below showing population and GNP of countries A and B and answer the questions that follow.

Country	GNP (millions of francs)	Population (million)
X	2,400	40
Y	1,200	30

Calculate the per-capita income of countries X and Y.

Country X: Per-capita income = $GNP \div Population = 2,400 \text{ million} \div 40 \text{ million} = 60,000$ francs

Country Y: Per-capita income = $GNP \div Population = 1,200 \text{ million} \div 30 \text{ million} = 40,000$ francs

Per capital income is used as an indicator of the standard of living in a country; and to compare the standard of living between countries because it is available for all countries and it is the most relevant among other available measures. Using national income figures to compare standard of living comes with many shortcomings as seen below:

Limitations of using per capita income figures as a measure of good standard of living in a country for period of time

- Per capita income does not consider the working conditions of the people. It may be high but when the working conditions of the people are poor, it may not reflect a good standard of living.
- Per capita income does not take into account the amount of leisure.
- It may be high but when the people do not have leisure yet we know that leisure is one of the attributes of good welfare, meaning the standard of living will become low.
- Per capita income does not consider the level of subsistence sector.
- It looks at the monetary terms of the output but the subsistence sector where food is grown for home consumption is not considered yet it is a basis for good welfare in the rural areas.

- Per capita income does not consider the level of prices. It may be high because the prices of goods and services are very high implying that the people cannot afford them, hence leading to poor standards of living.
- Per capita income does not put into consideration the political climate. It may be high but when there is political instability in the country, meaning that the people are always on the run. This may not show good standards of living.
- Per capita income does not consider the quality of goods produced. It may be high but when the quality of goods produced is low. This means that the quality of life of the people will also be poor.
- Inaccurate statistical data for example, population figures. It may be high but when the population figures given are inaccurate. This means that the per capita income figures will not reflect what is actually on the ground.
- Per capita income can also be used to compare the standard of living between two different countries. The figures got from the different countries are compared and the country with a high per capita income figure is assumed to be having a high standard of living. However, this is not true in reality as seen further.

Limitations of using the figures of per capita income to compare the standard of living between countries

- It does not consider the types of goods produced. Per capital income may be high in the country, which produces many capital goods, which do not improve the welfare directly in the short-run.
- It does not take into account leisure which contributes to welfare.
- Per capita income may be high in a country where people work hard and forego leisure, which may be on top of their scale of preferences.
- Transport differences: Two countries may produce the same quantity and quality of product but may have different figures of per capita income because of the difference in transport costs.
- Per capita income figures do not reflect some factors, which influence the welfare of the people. For example, it may be high in a country where there are wars, accidents, diseases, pollution, etc.
- In Developing Countries, it is likely that population figures are inaccurate because of inadequate facilities and, therefore, the figures for per capita income are unreliable.
- Per capita income may be underestimated in a country where there are omissions in measuring GNP for example due to a large fraction of subsistence sector, high non-monetary output, etc.
- There are people who live on illegal activities such as gambling and smuggling, which are not included in national income figures.
- Countries use different concepts and definitions of national income. For example, some countries value subsistence output while others do not. Others use GNP at factor cost while others use GNP at market prices, etc.

- Income distribution: A country may have high-income per capita figures when income is in the hands of few people while the majority of the population are suffering.
- Price structures: figures of national income may be high because of inflation and this does not mean that people are better off. Also a commodity may be cheap in one country and expensive in another but generate the same welfare.
- Per capita income may be high in a country where there is unemployment which affects welfare of affected people/ households.

Application activity 2.8



1. Define "national income" and explain how it is related to the standard of living of a country.
2. List three factors other than income that can influence the standard of living of people in a country.
3. If a country's national income increases, what are the possible effects on the population's quality of life?
4. Think about your country or a country you know. How does the level of national income affect access to healthcare and education?

2.9 Income distribution and inequality

Learning Activity 2.9



1. What does income inequality mean as shown in the scenario?
2. Identify the main causes of income differences in the district.
3. What social effects are caused by the gap between the rich and the poor in the community?
4. Suggest two measures that district leaders can take to reduce income inequality.

Income distribution refers to the way income is spread among various social groups in an economy. It may be between different people in the same region or in another region or it may be how resources are distributed in different regions.

Inequality means an instance of being unequal which may imply difference in size, degree or circumstances, among others. In economics though, inequality is always talked of in terms of two major forms or types:

Types of inequality

1. **Personal inequality:** refers to the economic difference between the very poor and the rich people in society.

- 2. Regional inequality:** refers to a situation where there is a difference in terms of resource endowments, developments through infrastructure such as roads and industries, among others. These make one region appear different and more developed than others.
- 3. National inequality:** This is where some countries are richer than others in terms of resources and development.
- 4. Sector inequality:** This where some sectors are richer and developed compared to other sectors e.g Industry being more developed than agriculture.
- 5. Occupational inequality:** This where some occupations are more advanced in terms of technology used and payments etc. than other occupations.

Causes of income inequality

Income inequalities may be among different social groups or even among people in the same social group. Income inequality is caused by historical, geographical, social, economic and political factors.

Some of these factors are:

- i. Differences in the distribution of resources:** People who have access to productive resources such as fertile land, capital, modern equipment, and natural resources are likely to earn higher incomes than those living in less productive areas such as dry or marginal lands. For example, commercial farmers often earn more income than pastoralists in drought-prone regions.
- ii. Differences in social and economic infrastructure:** Areas with better infrastructure such as roads, electricity, clean water, schools, hospitals, and communication networks attract more businesses and investment opportunities. This improves production, trade, and employment opportunities, leading to higher incomes compared to poorly developed areas.
- iii. Government policies:** Certain government policies may increase income inequalities. For example, regressive taxes place a heavier burden on low-income earners than on the rich. Unequal access to public services, subsidies, and development programs can also widen income differences among people and regions.
- iv. Historical factors:** Wealth and property inherited from previous generations contribute to income differences. Individuals from wealthy families may inherit land, businesses, livestock, or houses, giving them greater economic advantages than those from poor backgrounds. In many rural areas, land inheritance is a major source of wealth.
- v. Differences in natural abilities and health conditions:** People with strong physical abilities, talents, creativity, or good health are often more productive and may earn higher incomes. On the other hand, people with disabilities or chronic illnesses may face limited employment opportunities and lower earnings.
- vi. Differences in employment opportunities:** Some occupations and professions pay higher wages and salaries than others because of differences in skills, qualifications, demand, and responsibilities. For example, doctors, engineers, and managers generally earn more income than casual labourers.

- vii. Differences in education and skills levels:** Educated and skilled people have better chances of securing well-paying jobs, starting businesses, and increasing productivity. In contrast, people with low levels of education or limited skills often remain in low-income employment or unemployment.
- viii. Uneven distribution of investment opportunities:** Most investments and industries are concentrated in urban centres where infrastructure and markets are better developed. Rural and remote areas often receive fewer investments, leading to regional income inequalities and limited employment opportunities.
- ix. Gender differences:** Women are often economically disadvantaged compared to men because they may have limited access to land ownership, credit facilities, education, employment opportunities, and decision-making positions. Cultural and social barriers also contribute to lower income levels among females in many societies.
- x. Differences in ownership of property and wealth:** Individuals who own valuable assets such as land, buildings, factories, shares, or businesses earn income through rent, interest, dividends, and profits. Those without property depend mainly on wages and salaries, which are often lower and less stable.
- xi. Population growth and family size:** Large families may experience lower income per person because available resources are shared among many dependents. High population growth can also increase unemployment and reduce average living standards.
- xii. Corruption and discrimination:** Corruption, tribalism, nepotism, and other forms of discrimination may prevent equal access to jobs, education, and economic opportunities, leading to unfair income distribution in society.

i. Lorenz curve

The Lorenz curve is a graph used to show the distribution of income or wealth in an economy. It was developed in 1905 by Otto Max Lorenz to measure income inequality.

The horizontal axis shows the cumulative percentage of the population from the poorest to the richest, while the vertical axis shows the cumulative percentage of income or wealth earned. A straight 45-degree line called the **line of perfect equality** represents equal income distribution, where everyone earns the same share of income. The Lorenz curve usually lies below this line.

The farther the Lorenz curve is from the line of perfect equality, the greater the income inequality between the rich and the poor. If the curve is closer to the line, income distribution is more equal. The Lorenz curve is also used to calculate the Gini Coefficient, which measures the level of inequality numerically.

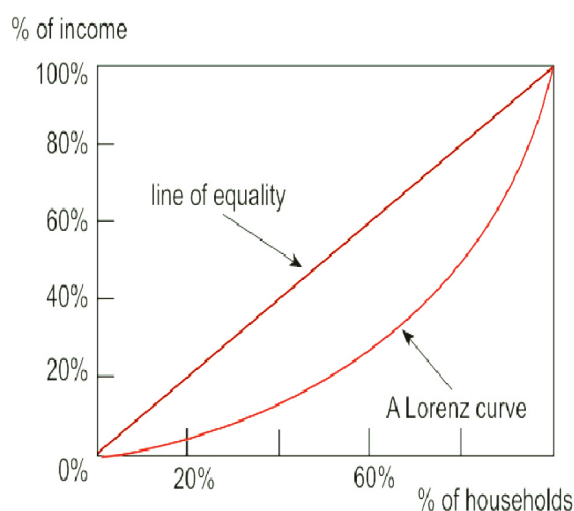


Figure 2.15.: Lorenz Curve

From Figure above, the further away the Lorenz curve from the line of perfect equality, the more the income inequality. And the nearer the Lorenz curve from the line of perfect equality, the less the income inequality.

ii. Gini coefficient

The Gini Coefficient, also called the **Gini index** or **Gini ratio**, is a statistical measure used to show the degree of income inequality among a country's population.

The Gini coefficient ranges from 0 to 1:

- A value of **0** represents **perfect equality**, where everyone earns the same income.
- A value of **1** represents **perfect inequality**, where one person receives all the income while the rest of the population earns nothing.

The higher the Gini coefficient, the greater the level of income inequality in a country.

i. Advantages of income inequality

- **Encourages investment and job creation:** Wealthy people often invest in industries, businesses, factories, and commercial agriculture. These investments create employment opportunities for poor people and improve their standards of living.
- **Increases government tax revenue:** High-income earners pay more taxes through progressive taxation. The government can use this revenue to provide public services such as education, health care, roads, and water supply, which contribute to national development.
- **Promotes hard work and innovation:** Income differences motivate people to work harder, acquire more education and skills, and become more productive in order to earn higher incomes and improve their living conditions.
- **Provides labour for different occupations:** Poor people are willing to take up low-income or less attractive jobs such as cleaners, guards, waste collectors,

and mortuary attendants, which are necessary for the functioning of society.

- **Encourages savings and capital formation:** Rich people are able to save a larger proportion of their income. These savings can be invested in productive activities, increasing capital formation and economic growth.
- **Promotes foreign exchange earnings:** Wealthy investors and export firms produce goods for export, earning foreign exchange for the country and improving the balance of payments.
- **Supports economic growth:** Investments made by high-income earners increase the production of goods and services, leading to higher national output and economic growth.
- **Improves technological development:** Rich investors may introduce modern technology and machinery in production, improving efficiency and productivity in the economy.

ii. Disadvantages of Income Inequality

- **Leads to poverty and low living standards:** Poor people may fail to meet basic needs such as food, shelter, clothing, education, and health care because of low incomes.
- **Reduces aggregate demand:** The poor have low purchasing power and cannot buy many goods and services. Since the rich spend a smaller proportion of their income on consumption, total demand in the economy may fall, discouraging investment and production.
- **Causes misallocation of resources:** The rich spend more on luxury goods and services while basic needs for the majority of poor people remain unsatisfied, leading to inefficient use of resources.
- **Promotes capital outflow:** Rich people, especially foreigners, may transfer their profits and savings to other countries. This reduces domestic investment and slows economic development.
- **Reduces government revenue:** When many people earn low incomes or remain unemployed, the government collects less tax revenue, limiting its ability to provide public services and development projects.
- **Leads to social disharmony and political instability:** Large income differences create feelings of neglect, exploitation, and unfairness among poor people, which may result in crime, strikes, riots, and political instability.
- **Causes unequal access to social services:** Poor people may fail to access quality education, health care, housing, and financial services, widening the gap between the rich and the poor.
- **Leads to unemployment and underemployment:** Poor people often lack education, skills, and capital, limiting their employment opportunities and increasing dependency levels.
- **Limits participation in development programmes:** Many government development programmes require financial contributions or resources that poor people cannot afford, leading to low participation.

- **Encourages rural-urban migration:** People from poor rural areas may move to urban centres in search of better incomes, leading to congestion, unemployment, and growth of slums in towns and cities.

iii. Measures Taken to Reduce Income Inequality in Rwanda

- **Education reforms:** The government has expanded access to education through programs such as free basic education, technical and vocational education and training (TVET), and adult literacy programs. Education equips people with knowledge and skills needed to secure employment and improve their incomes and standards of living.
- **Poverty reduction programmes:** Government programmes such as *Girinka* ("One Cow per Poor Family") have helped poor households acquire cows for milk production, manure, and income generation, improving nutrition and household earnings.
- **Land tenure reforms:** Land redistribution and land registration policies have enabled many people to access and legally own land. This encourages agricultural production, investment, and improved incomes, especially in rural areas.
- **Improvement of infrastructure:** Construction of roads, electricity networks, water supply systems, and communication facilities has improved transportation and market access. This enables producers and traders to sell goods easily and increase their incomes.
- **Progressive taxation:** The government uses progressive taxation where high-income earners pay higher taxes than low-income earners. Revenue collected is used to provide social services and support development programmes that benefit poor people.
- **Liberalisation of the economy:** Economic liberalisation has encouraged private investment, trade, and entrepreneurship. People are now able to participate more freely in economic activities, leading to increased employment opportunities and incomes.
- **Population control measures:** Family planning and awareness programmes have helped to reduce rapid population growth and dependency burdens, enabling families to better manage available resources and improve their living standards.
- **Modernisation of agriculture:** The government has promoted the use of improved seeds, fertilizers, irrigation, mechanisation, and modern farming methods. This has increased agricultural productivity, food security, and incomes for rural households.
- **Improvement of the investment climate:** Policies such as tax holidays, free economic zones like the Kigali Special Economic Zone, and simplified business registration procedures have attracted both local and foreign investors, creating jobs and reducing unemployment.
- **Political stability and good governance:** Peace, security, and political stability have created a favourable environment for production, trade, and investment. People are able to carry out economic activities without fear or disruption.
- **Promotion of small-scale enterprises:** The government supports small and medium enterprises (Small and Medium-sized Enterprises (SMEs)) through training, loans, and entrepreneurship programmes. These businesses create

employment opportunities and increase household incomes.

- **Formation of cooperatives:** Cooperatives such as Umurenge SACCOs, teachers' SACCOs, and producer cooperatives encourage savings, provide credit facilities, and improve bargaining power for members. This helps people increase investment, production, and income.
- **Expansion of financial services:** Increased access to banks, SACCOs, and microfinance institutions has enabled people to obtain loans and start income-generating activities, especially in rural areas.
- **Social protection programmes:** Programmes such as Vision 2020 Umurenge Programme (VUP) provide financial support, public works employment, and direct assistance to vulnerable groups, helping to reduce poverty and inequality.
- **Promotion of gender equality:** Policies promoting equal access to education, employment, land ownership, and financial services for women have helped reduce gender-based income inequalities in society.

Application activity 2.9



1. What is meant by income distribution in an economy?
2. What do you understand by income inequality?
3. What are the main causes of income inequality in society?
4. What are the effects of unequal income distribution on people's living standards?

2.10 Causes of low per capita income in Rwanda

Learning Activity 2.10



1. What is meant by per capita income, and why is it important in measuring living standards?
2. What are the main causes of low per capita income in Rwanda?
3. How does low productivity affect per capita income and the overall well-being of citizens?

Low per capita income in Rwanda remains a significant challenge that impacts the country's overall development. This issue is influenced by various factors that hinder economic growth and individual prosperity. Understanding the key causes behind low per capita income is essential for designing effective strategies to improve living standards and foster sustainable development in Rwanda. This discussion explores the main reasons contributing to this economic situation.

Causes of low per capita income in Rwanda

- Low levels of education: This makes the people get casual low- paying jobs leading to low levels of income and in the end the per capita income will be low.
- Under developed infrastructure especially in the rural areas: This complicates movement of people and their goods from one place to another. This further limits the level of earnings of the people and some of the output may remain unsold.
- High levels of unemployment: Lack of jobs means that the output in the country will be low leading to a low national income and per capita income figures.
- Low levels of income: Some institutions give people low incomes despite their high contributions to the economy. Some may end up getting subsistence wage that will reflect a low income per person.
- Dependence of agriculture which depends on climate: Agriculture in Rwanda depends on climate and in situations where there is a mismatch between the seasons, the farmers suffer with no output and income leading to low income per person.
- Large subsistence sector: This yields little income since only the incidental surplus is sold. Most of the foods grown here are for home consumption implying that the there are no incomes expected during the production process.
- Unbalanced development: Some areas are highly developed with many economic activities while others are lagging behind in terms of development. The least developed areas yield low productivity and the general national income will be affected, leading to a low per capita income.
- Lack of capital to invest in businesses: There is still lack of enough capital to invest in production activities. This is evidenced by the low level of manufacturing industries. This leads to low productivity, low national income and finally low per capita income.
- Low prices of agriculture products: This accompanied by high rates of price fluctuation and little earnings from the agricultural sector, leading to low national income and hence low income per person.

Application activity 2.10



1. What is meant by per capita income, and why is it important in measuring living standards?
2. What are the main causes of low per capita income in Rwanda, and how does low productivity contribute to this problem?
3. Suggest ways Rwanda can improve per capita income and raise living standards for its citizens

2.11. End of unit assessment



A. Knowledge and Understanding

1. Define the following concepts: national income, Gross Domestic Product (GDP), Gross National Product (GNP), and per capita income. Explain why Rwanda compiles annual GDP statistics.
2. Describe the methods of measuring national income. Explain their advantages, disadvantages, and the challenges faced when applying them in developing countries.
3. Illustrate and explain the circular flow of income in both closed and open economies, showing the differences between them.
4. Identify the determinants of national income and explain the concepts of equilibrium level of national income, inflationary gap, and deflationary gap.
5. Explain the meaning of leakages and injections in the economy. Describe the phases of the business cycle and explain their effects on economic activities.

B. Skills and Application

6. Draw and label a circular flow of income diagram for an open economy and explain the role of each component.
7. Using a hypothetical example, demonstrate how leakages and injections affect the level of national income.
8. Draw a business cycle graph, identify its phases, and describe what happens at each stage.

C. Attitudes and Values

9. Explain why the accurate measurement of national income is important for Rwanda's economic planning and development.
10. Discuss the importance of reducing income inequality for sustainable economic growth and development in Rwanda.
11. Assess the role of individuals, communities, and government in promoting sustainable economic growth and improving living standards.
12. Explain the limitations of using national income figures alone in measuring people's welfare and quality of life.

UNIT 3

CONSUMPTION, SAVING, INVESTMENT & MULTIPLIER



Key Unit Competence:

Analyze the relationships among consumption, saving, and investment, and their implications for the national economy via the multiplier principle.

Learning objectives

Knowledge and understanding

1

- Define and distinguish consumption, saving, investment, and the multiplier; explain their roles in national income.
- Explain propensities (APC, APS, MPC, MPS) and how they determine the size of the multiplier.
- Describe the accelerator principle and how investment interacts with income to influence growth. - Explain government policies that promote investment and sustainable growth.
- Understand the theoretical bases for investment behavior (Classical, Keynesian) and the concept of leakages in the multiplier process.
- Describe factors that limit consumption, saving, and investment in developing economies, with Rwanda-specific considerations.

Skills

2

- Analyze data to identify consumption, saving, and investment patterns.
- Calculate APC, APS, MPC, MPS, and the spending/multiplier effects from numerical problems.
- Construct and interpret the consumption function $C = a + bY$ and related graphs (consumption and saving functions).
- Use spreadsheets (Excel/Google Sheets) for calculations and graphing.
- Draw and interpret AD/AS implications for equilibrium national income in response to policy shocks.
- Evaluate investment constraints in developing economies and propose policy solutions.
- Conduct small-group research and present findings on Rwanda's investment, saving, and consumption dynamics.

Attitudes and values

3

- Appreciate the role of saving and investment in development and fiscal sustainability.
- Develop responsible consumption and saving habits aligned with long-term growth.
- Value evidence-based policy analysis and the use of data from Rwanda (NISR, BNR) for development planning.
- Foster a collaborative approach to debates on spending vs saving and investment incentives.



Introductory activity 3

In a certain country, people engage in various economic activities every day—farmers produce crops, factories manufacture goods, and service providers such as teachers, health workers, and transport operators contribute to production and income generation. The government invests in infrastructure, such as building a new road, which improves access to goods and services and stimulates further economic activity. As households spend part of their income on consumption, save for future needs, and invest in productive ventures, these actions collectively influence national income through the multiplier effect. Increased consumption raises demand for goods and services, saving provides funds for investment, and investment expands production and employment, creating a cycle of growth and development.

Questions:

1. How does consumption contribute to economic growth?
2. Why is saving important for individuals and the economy?
3. In what ways does investment promote development?
4. How does the multiplier effect link consumption, saving, and investment?

In any economy, the levels of consumption, saving, and investment play vital roles in determining overall economic performance and growth. Consumption refers to the expenditure by households on goods and services, which directly influences the demand for products in the market. Saving is the portion of income that is not spent on immediate consumption, serving as a source of funds for future investment. Investment involves the expenditure on capital goods such as machinery, buildings, and infrastructure, which contribute to increasing productive capacity and economic development.

These three components are interconnected, with savings providing the funds necessary for investment, and consumption driving demand that stimulates production. Furthermore, the concept of the multiplier explains how an initial change in investment or spending can lead to a larger overall impact on national income and

Experimental version

economic growth. Understanding how consumption, saving, investment, and the multiplier interact is essential for analysing economic policies and fostering sustainable development.

3.1 Consumption theory

Learning Activity 3.1



Mr. Mugisha of Nyarubuye Village in Nyamagabe District regularly purchases 3 kilograms of sugar, 7 kilograms of meat, 20 kilograms of rice, 35 units of electricity, and 4 sacks of charcoal each month. Meanwhile, Ms. Uwase from the same village spends 5 kilograms of sugar, 3 kilograms of meat, 25 kilograms of rice, and 45 units of electricity monthly.

Using the case study above, respond to the following questions:

1. What economic term can you use to describe the act of Mr. Mugisha and Ms. Uwase using these commodities?
2. What factors do you think influence their consumption levels?

3.1.1 Meaning of consumption, Consumption function and consumption curve

i. Meaning of consumption

Consumption in economics refer to the use of goods and services to satisfy wants, which reduces the stock of economic resources available for future use. From an individual perspective, consumption is the expenditure on goods and services for final use by households. Consumption decisions determine current living standards and, together with saving and investment, drive the flow of income in an economy.

ii. Consumption Function

Consumption function describes how current consumption depends on the factors that influence consumer spending. In general form it can be written as:

$C = F(yd., w, cr, r, dg, ex, ygf)$ where;

C: Consumer spending	yd: personal disposable income	w: Wealth
Cr: availability of credit	r: Interest rates	dg: stock of durable consumer goods
ex: expectations	ygf: income distribution.	

A commonly used linear specification is: $C = C_0 + bY_d$ where **(C₀)** is autonomous consumption (consumption when income is zero), **(b)** is the marginal propensity to consume (MPC), and **(Y_d)** is disposable income.

Autonomous and Induced Consumption

Autonomous consumption (C₀) is the minimum level of consumption that occurs even when income is zero; households may finance this through savings, borrowing, or transfers. **Induced consumption** is the portion that varies with disposable income

Experimental version

and is captured by the term (bYd). The induced part rises as income increases and falls when income declines.

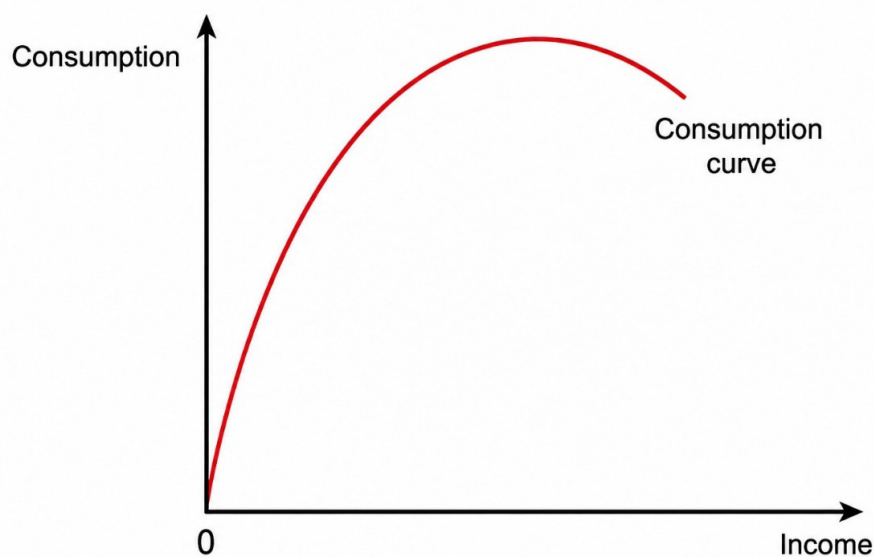


Figure 3.1.: Consumption Curve

From Figure above, as income rises, Marginal Propensity to Consume (MPC) declines (MPC is the proportion of additional income consumed). Rich individuals have a low MPC. This means that as their income increases, they consume a small proportion of the additional income. A straight-line consumption curve implies that as income rises, MPC is constant. It means that as income increases, one consumes a constant amount of the additional income. Nevertheless, this is not realistic.

3.1.2 Average & marginal propensity to consume (APC, MPC)

1. Average propensity to consume (APC)

Average Propensity to Consume (APC): This refers to the ratio of total consumption to the level of disposable income. $APC = C/Y$.

The average propensity to consume declines as income increases. This is because as income increases, more is saved than consumed.

Marginal propensity to consume (MPC): This refers to the fraction of an individual's additional income that is spent on consumption. $MPC = \Delta C / \Delta Y$

As the income rises, MPC falls. Rich people tend to consume a smaller proportion than the poor people. Normally, MPC is less than 1. It can be equal to one only when all the additional income is consumed.

Exemple:

Scenario

Kilisa is a house helper living in **Kicukiro District, Kigali City**. Initially he earned **50,000 RWF per month**, of which he spent **15,000 RWF** on consumption and saved the rest. After one year his monthly earnings rose to **75,000 RWF**, and he increased his monthly consumption to **30,000 RWF**.

Questions

Using the case study above, work out and explain the following in your exercise book:

1. The fraction of income that is put to use (consumption) before and after the income increase.
2. The fraction of the change in income that is put to use (the proportion of the income increase that went to higher consumption).
3. Suggest measures that can increase Kalisa's fraction of income put to use.

Answer

1. Fraction of income put to use before and after the income increase

The fraction of income spent on consumption is called the Average Propensity to Consume (APC).

Formula:

$$\text{APC} = C/Y$$

Before the income increase:

$$\text{APC} = 15,000/50,000 = 0.3$$

Therefore, **APC = 30%**.

So, Kalisa consumed 30% of his income and saved 70%.

After the income increase:

$$\text{APC} = 30,000/75,000 = 0.4$$

Therefore, **APC = 40%**.

So, Kalisa consumed 40% of his income and saved 60%.

2. Fraction of the change in income put to use

This is called the Marginal Propensity to Consume (MPC).

Formula:

$$\text{MPC} = \text{Change in consumption} / \text{change in income}$$

$$\text{Change in consumption} = 30,000\text{RWF} - 15,000\text{RWF} = 15,000\text{RWF}$$

$$\text{Change in income} = 75,000 - 50,000 = 25,000\text{RWF}$$

Therefore:

$$\text{MPC} = 15,000/25,000 = 0.6$$

$$\text{MPC} = 60\%$$

Kalisa spent 60% of his additional income on additional consumption and saved the remaining 40%.

3. Measures that can increase Kalisa's fraction of income put to use

The following measures can increase his consumption:

- Increase his income by acquiring skills that can help him obtain better-paid employment.
- Obtain affordable credit to enable him to purchase necessary goods and services.
- Reduce excessive saving and use some savings for essential consumption.
- Improve financial literacy so that he can plan his spending effectively.
- Start a small business or income-generating activity to increase his disposable income.
- Reduce unnecessary expenses, leaving more money available for essential consumption.

3.1.3 Measures to raise the APC and MPC

Factors That Increase Consumption

- Advertising and promotion : Well designed advertising and promotional campaigns raise consumer awareness of products and their uses, stimulate desire, and encourage higher spending.
- Improved infrastructure: Better roads, transport links, and market facilities lower distribution costs, expand market reach from production centres to consumers, and reduce final prices, which encourages purchases.
- Urbanisation and demonstration effects: Migration to towns changes tastes and living standards, exposing people to new goods and lifestyles; visible consumption by peers can prompt others to buy similar items.
- Higher wages and employment: Rising wages and more stable jobs increase disposable income and purchasing power, directly boosting household consumption.
- Government transfers and public spending : Social payments (unemployment benefits, pensions, cash transfers) and public investment reduce income uncertainty and raise effective demand.
- Accessible and affordable credit: Cheap, easy credit and microfinance enable households to smooth consumption and buy durables they otherwise could not afford.
- Income redistribution: Progressive taxation and targeted subsidies shift income toward lower income households, who typically have a higher marginal propensity to consume, raising aggregate consumption.
- Peace, security, and stable institutions: A safe, predictable environment supports steady income generation, investment, and consumer confidence, all of which sustain consumption.
- Price stability and low inflation: Stable prices preserve real purchasing power and reduce precautionary saving, encouraging households to spend rather than

Experimental version

hoard cash.

- **Financial inclusion and savings products:** Access to bank accounts, mobile money, and tailored savings or credit products helps households manage cash flow and maintain consumption during shocks.
- **Remittances and external income:** Money sent home by migrants raises household budgets and often finances consumption of food, education, and housing.
- **Technological access and ecommerce:** Mobile phones, digital payments, and online marketplaces expand access to goods and services, making it easier and cheaper to consume.
- **Demographics and household composition:** A growing workingage population or larger household sizes can raise aggregate consumption needs and demand for goods and services.
- **Consumer confidence and expectations:** Optimism about future income, employment, or prices encourages households to spend more today; pessimism has the opposite effect.
- **Access to markets and product variety:** Greater availability of affordable, quality goods—locally produced or imported broadens choices and stimulates spending.
- **Behavioral and cultural factors:** Social norms, aspirations, and advertising-driven trends influence preferences and can raise consumption of particular goods or services.

3.1.4 Factors influencing the level of consumption

Consumption in an economy is affected by several economic and social factors that determine how much households spend on goods and services:

- **General price level:** When prices are high, consumers buy less and consumption decreases. When prices are low, people buy more and consumption increases.
- **Disposable income:** Higher disposable income (income after tax) increases purchasing power, leading to higher consumption. Low disposable income reduces consumption.
- **Population size:** A large population leads to higher total consumption due to increased demand for goods and services, while a small population results in lower consumption.
- **Availability of goods and services:** When goods are readily available in markets, consumption increases. Scarcity or shortages reduce consumption levels.
- **Income distribution:** When income is evenly distributed among people, more individuals can afford goods and services, increasing overall consumption. Unequal distribution reduces consumption because only a few people have high purchasing power.
- **Liquidity preference:** When people prefer holding cash instead of investing or saving in banks, short-term consumption may increase. However, excessive hoarding of money can reduce investment and long-term consumption growth.
- **Interest rates and credit facilities:** Easy access to credit, loans, and hire

purchase schemes increases consumption. High interest rates or limited credit reduce consumption.

- **Government taxation policy:** High taxes reduce disposable income and lower consumption, while lower taxes increase disposable income and consumption.
- **Expectations about future prices (speculation):** If people expect prices to rise in the future, they tend to increase current consumption. If they expect prices to fall, they reduce current consumption and wait.
- **Marginal Propensity to Consume (MPC):** This refers to the proportion of additional income that is spent on consumption. A higher MPC leads to higher consumption, while a lower MPC results in lower consumption.
- **Consumer tastes and preferences:** Changes in fashion, culture, advertising, and personal preferences influence what and how much people consume.
- **Level of savings:** When people save more, consumption tends to decrease, and when savings are low, consumption increases.
- **Government policies and subsidies:** Government support such as subsidies on essential goods increases affordability and raises consumption levels.

Application activity 3.1



Activity Scenario

In Rwanda, households use their incomes in different ways. Some families spend most of their income on food, clothing, transport, education, and health care, while others save part of their income for future use.

For example, a family in Kigali earns 400,000 FRW per month. They spend 320,000 FRW on consumption and save the remaining amount. Another family with lower income spends almost all of its income on basic needs.

Economists study how people consume and save income using concepts such as the consumption function, consumption curve, Average Propensity to Consume (APC), and Marginal Propensity to Consume (MPC).

1. Define the term consumption and explain its role in household income utilization.
2. State the meaning of Marginal Propensity to Consume (MPC) and illustrate it with an example from Rwandan households.
3. Draw and interpret a simple consumption curve, showing the relationship between income and consumption.
4. Calculate the Average Propensity to Consume (APC) for a household that consumes 320,000 FRW out of an income of 400,000 FRW.
5. Analyze why low-income families usually have a higher MPC than high-income families, and discuss the economic implications.
6. Suggest measures that can promote responsible consumption and saving among households in Rwanda.

3.2 Saving

Learning Activity 3.2



Activity Scenario

Sam is a S5 student living in Rwanda who receives monthly pocket money from his parents. At first, he spends all his money on food, transport, and entertainment. Later, after attending a school discussion on financial management, he decides to keep part of his income for future use.

As Sam's income increases, the amount he keeps for future needs also increases. He notices that some of his friends keep a larger share of their income than others depending on their income levels, spending habits, and family background. However, many people in developing countries find it difficult to keep part of their income because of low earnings, unemployment, and rising prices of goods and services.

Using Sam's experience, learners explore how people use income for present consumption and future needs, and how income levels influence financial decisions in society.

Learning Activity:

Aline, a student at G.S. Kicukiro, received 200,000 RWF as a prize for excellent performance in the S.3 national exams. She used 160,000 RWF to buy items and kept 40,000 RWF as savings. One week later her uncle, Mr. Niyonzima, rewarded her for the achievement and gave her additional money so that her monthly total rose to 300,000 RWF. Aline then bought other items and increased her savings to 60,000 RWF.

Individually, using the case study above, work out and explain the following in your exercise book:

1. The fraction of Aline's income that is saved (before and after the gift).
2. The fraction of her additional income that is saved.
3. Practical advice you would give Aline on managing her money.
4. How Aline's Average Propensity to Consume (APC) relates to her Marginal Propensity to Consume (MPC).

Savings is an essential concept in economics that plays a crucial role in individual and national economic development. In simple terms, savings refer to the act of setting aside a portion of one's income or earnings for future use, rather than spending it all immediately.

3.2.1 Meaning of saving, saving function and saving curve

i. Meaning of saving:

Savings refers to the portion of disposable income that is not spent on current consumption but kept for future use. An individual's total income is divided into consumption and savings. After spending on goods and services, the remaining income is saved for future needs, emergencies, or investment purposes.

Savings may take different forms, including personal savings kept in banks or at home, cooperative savings through SACCOs and cooperatives, and compulsory savings such as contributions to Rwanda Social Security Board (RSSB) and community health insurance schemes like *Mutuelle de Santé*.

ii. Saving function and saving curve

Savings function: It shows the relationship between the level of saving and disposable income. $S=F(Y)$. Savings depend on the level of income whereby if the income increases, also the savings increase, other factors remaining constant. This is more so with the rich individuals as compared to the poor individuals.

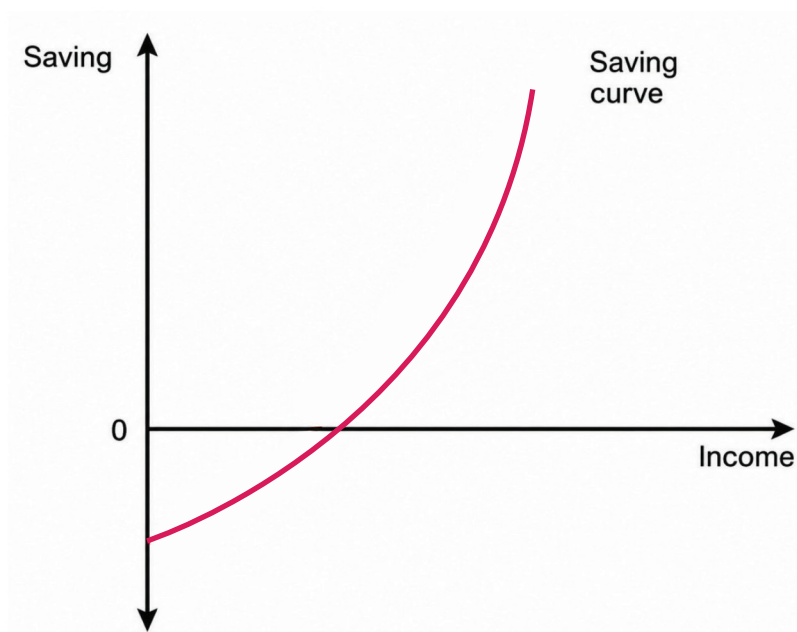


Figure 3.2.: The saving function

Figure above shows the savings curve. As income rises, marginal propensity to save (MPS) increases. Rich individuals have a high MPS. This means that as their incomes increase, they save a bigger proportion of their additional income.

3.2.2 Average propensity to save (APS) and marginal propensity to save

i. Average propensity to save (APS)

Average propensity to save (APS) refers to the ratio of saving to the level of disposable income.

The APS increases as the income of the individual increases.

Average propensity to save = **APS** = S/Y

ii. Marginal propensity to save (MPS)

This refers to the fraction of an individual's additional income that is spent on savings.

Marginal propensity to save = **MPS** = $\Delta S / \Delta Y$

Relationship between MPC and MPS

$$MPC + MPS = 1$$

$$Y = C + S$$

$$\Delta Y = \Delta C + \Delta S$$

$$\Delta Y / \Delta Y = \Delta C / \Delta Y + \Delta S / \Delta Y$$

$$1 = MPC + MPS$$

3.2.3 Factors and limitations of saving in developing countries

Key Points

Saving in developing countries is driven by income, financial access, and expectations, but is often constrained by low and volatile incomes, weak financial systems, and limited social protection — in Rwanda (Murindi, Kigali) policies that raise incomes, expand financial inclusion, and stabilize prices are the most effective ways to raise national saving.

A. Main factors that increase the level of saving

- **Higher and more stable incomes**
Regular, rising wages and steady earnings increase households' ability to set money aside.
- **Greater financial inclusion**
Wider access to banks, SACCOs, and mobilemoney services makes saving safer, more convenient, and more attractive.
- **Competitive real interest rates**
Positive returns on deposits encourage people to postpone consumption and keep funds in formal savings instruments.
- **Social protection and pensions**
Reliable transfers and pension schemes reduce the need for precautionary saving and free households to save for other goals.
- **Remittances and external inflows**
Regular money sent from relatives abroad raises household budgets and creates more scope for saving.
- **Demographic and lifecycle effects**
Workingage adults typically save more than children or the elderly; population structure therefore affects aggregate saving.
- **Cultural norms and institutional trust**
Societies that value thrift and that trust financial institutions tend to have higher

formal saving rates.

- **Overall income level**

Higher average incomes raise the capacity to save; low incomes constrain saving.

- **General price level**

High inflation erodes real balances and discourages saving; stable prices preserve the value of deposits and encourage saving.

- **Interest rate environment**

Higher nominal and real deposit rates increase the incentive to save; low rates reduce that incentive.

- **Development of financial institutions**

A welldeveloped banking and microfinance sector lowers transaction costs and increases opportunities to save.

- **Political stability**

Stable political conditions support production, investment, and confidence in saving; instability tends to reduce saving.

- **Existing capital stock and investment opportunities**

A larger capital base and profitable investment outlets raise expected returns and can boost saving for investment.

- **Marginal propensity to save (MPS)**

The higher the MPS in a population, the larger the share of additional income that is saved rather than spent.

- **Spending habits and consumption culture**

Strong consumption preferences or social pressures to spend reduce saving; frugal habits increase it.

B. Key limitations to saving in developing countries

- **Low average incomes:** many households live near subsistence and cannot set aside funds.
- **Income volatility (agriculture and informal work):** irregular earnings force households to consume or use informal buffers rather than save formally.
- **Limited access to formal financial services:** rural and poor households often lack bank accounts or face high transaction costs.
- **High inflation and currency risk:** erodes the real value of savings and discourages longterm deposits.
- **Weak social safety nets:** absence of reliable unemployment or health coverage increases precautionary consumption.
- **High dependency ratios and social obligations:** large household sizes and cultural obligations (e.g., ceremonies) reduce disposable income available to save.
- **Limited profitable investment opportunities:** if returns on local investment are low or risky, households prefer consumption or informal saving.

Application activity 3.2



1. Define the term saving, explain the meaning of the saving function and saving curve, and state the meanings of Average Propensity to Save (APS) and Marginal Propensity to Save (MPS).
2. Draw and interpret a simple saving curve, then calculate APS and MPS using given data.

Income (Y)	Saving (S)
100,000 Frw	10,000 Frw
150,000 Frw	20,000 Frw

3. Analyze how income levels, factors, and limitations affect saving behavior in developing countries such as Rwanda.
4. Discuss the importance of saving for future needs, investment, financial discipline, proper management of income and resources, and economic development, and suggest measures to improve the culture of saving in Rwanda.

3.3 Investment theory

Learning Activity 3.3



Scenario Rubavu District

In Rubavu District, a group of young entrepreneurs start a community bakery. They pool their savings and secure a microloan, giving them an initial capital of 1,200,000 RWF. This money is used to buy ovens, baking tools, and set up a stall. In the first month, the bakery earns 600,000 RWF. Showing good financial management, the owners reinvest 150,000 RWF into the business by purchasing more flour and hiring an extra worker. By the second month, sales rise to 900,000 RWF. This example demonstrates the meaning of investment—spending money on tools, equipment, or facilities to increase future production. It also shows the factors influencing investment decisions, such as available savings, access to credit, and rising demand. However, like many businesses in developing economies, the bakery faces limitations such as high costs, limited finance, and supply challenges. Identifying ways to overcome these challenges is essential for sustaining growth.

Questions

1. What does investment mean in the bakery scenario?
2. Which factors influenced the bakery's decision to reinvest part of its income?
3. What limitations might reduce the bakery's ability to expand, and how could they be improved?
4. Why is reinvestment important for the bakery's future growth?

Investment refers to the use of money to buy capital goods such as machines, buildings, roads, and equipment that help to increase future production and income. In economics, investment plays an important role in promoting economic growth, creating employment opportunities, and improving the standards of living of people. Both individuals and governments invest in different sectors such as agriculture, industry, education, and infrastructure to increase productivity and development.

3.3.1 Meaning and factors influencing the level of investment in an economy.

A. Meaning

Investment can be defined as the expenditure on capital goods with an aim of increasing production. There are mainly two classes of investments:

1. **Fixed income investment:** This involves investments in assets such as bonds, fixed deposits and preference shares, among others.
2. **Variable income investments:** This involves investments such as business ownership or property ownership. It should also be noted that expenditure on education and health is recognised as an investment in human capital.

B. Factors influencing the level of investment in an economy.

- Disposable income: The higher the disposable income, the higher the level of investment; and the lower the income, the lower the investment.
- General Price level: The higher the price, the higher the level of investments by the investors so as to enjoy the profits; and the lower the price, the lower the investment.
- Liquidity preference: The higher the liquidity preference than investing money, the lower the investment; and the lower the liquidity preference, the higher the investment.
- Level of demand: The higher the demand, the higher the level of investment; and the lower the demand for the goods and services, the lower the investment.
- Degree of speculation: When investors expect a boom, there will be high rates of investment than when they expect a slump or depression.
- Government policy of taxation and subsidization: If the government overtaxes the investors, investments will be low but if the government subsidizes people, investments will be high.
- Availability of credit facilities: Investment will be high if there are credit facilities accessed by people in form of loans but it will be low if these facilities are scarce.
- Political situation: Peace and stability encourage investment while insecurity discourages investment.
- Presence of capital: The presence of adequate capital resources increases the rate of investment and if capital is not enough, the rate of investment will be low.

- Level of entrepreneurs: Once the level of entrepreneurs is high,
- investments will be high; and once it is low, investment will be low.

3.3.2 Limitations and ways of improving investment in an economy.

i. Limitations of improving investment in an economy.

- **Limited markets** due to low incomes of the people and this limits the investors from putting up mega structures in terms of investments.
- **Underdeveloped infrastructure** in some areas especially in rural areas limits the movement of potential investors, goods and services from areas of production to the markets.
- **Unfavourable investment climate** in form of high taxes charged by the government to investors discourages them from investing. However, the trend changes where investors are given tax holidays depending on their capacities.
- **Limited capital:** There is still a problem of limited capital whereby the local people do not have the necessary capital to invest. This leads to domination of foreign investors who carry and capital,
- **Political insecurity** in some areas especially those surrounding Burundi and Congo: The instabilities in those countries cause panic in the Rwandan areas adjacent to those areas. Hence the investors fear to invest in such areas.
- **High population growth rate** leads to increase in the dependence ratio among the families, leaving little disposable income for savings and investment.
- **High level of liquidity preference:** Many people prefer to hold their money in cash or near cash form because of different motives such as precautionary and speculation, among others. This limits their capacity to invest.
- **Limited entrepreneurial skills** needed for inventing and innovating. Still the levels are low, meaning that there are no new things that may come on the market. The manpower needed to carry out the inventions is still low, hence leaving the levels of investments low. Competition from outside countries that bring in their goods on the local market at a low price compared to the one at which home made goods are sold: This diverts people's attention to buy cheap imported goods, leaving the local goods without market.

ii. Ways of improving investments in Rwanda

- **Availing capital:** This can be through the commercial banks giving loans and credit at low rates of interests to attract the potential borrowers; furthermore, requesting for lower collateral security when seeking loans. This will attract many people to get soft loans that can be used for setting up investments.
- **Expanding the market through integration and signing treaties with other countries:** This will help to solve the problem of lack of market for the home-made products and furthermore, it will bring in more foreign exchange.
- **Availing a conducive investment climate:** This is through reducing the rates of taxes that the beginner firms can pay. This has further been helped by the

government setting up investment zones in the areas of Kabuga, Rwamagana, Bugesera etc.

- **Strengthening security in areas near countries with instabilities:** This can be through working hand in hand with the respective governments or more deployments of security personnel in those areas. It will create confidence in the local people and the investors, hence increase capital and investments in those respective areas.
- **Developing more feeder roads in addition to the ones that already exist:** These will help to link the rural inaccessible areas to the main roads and will ease the movement of goods and services, hence attracting more investors to those respective areas.
- **Controlling the rate of population growth through a maximum number of children per family:** This will reduce the rate of dependence in the families and it will leave the people with enough incomes that can be used for savings and investments in the long-run. Still, it will reduce the government expenditure on social services, leaving it with enough to invest in government-owned parastatals.
- **Encouraging people to reduce the rate of liquidity and carry outsavings:** This could be through increase in the interest rates offered on savings. In the long-run it will lead to capital accumulation which can further lead to investments.
- **Improving entrepreneur skills needed for inventing and innovating.** This can be through on-job training and through seminars and workshops. Furthermore, the government has embarked on competence based form of education which helps learners to carry out research and critically think. This form will help students develop their own ideas, leading to constant innovation and inventions.
- **Competition can be reduced through economic integration and specialisation** where countries within the same region produce different goods and sell to others. Furthermore, it can be reduced through producing good quality output through research. The government can also carry out protection to shield domestic industries from competition.
- **Subsidisation policy** by the government can help to reduce the problem of inadequate capital. At the same time, the goods produced will be sold at a lower price since the cost of production is also low; meaning that they may be sold at a price lower than the foreign goods, hence increasing their demand.

3.3.3 Investment Theories

Investment theories explain why firms and individuals decide to invest in businesses, machines, buildings, and other productive activities.

Here's a **developed narrative version** of the Classical Theory of Investment, extended with context and flow for learners:

1. Classical Theory of Investment

The Classical Theory of Investment emphasizes the role of the **interest rate** in determining how much firms and individuals invest. According to this view, investment decisions are closely tied to the cost of borrowing money and the expected return on that investment.

Experimental version

Narrative Explanation

In the classical framework, savings made by households and individuals provide the pool of funds available for investment. Banks and financial institutions channel these savings into loans for businesses. The **interest rate** acts as the balancing mechanism between savings and investment:

- When interest rates are **low**, borrowing becomes cheaper. Firms can access funds at a lower cost, making it easier to invest in new projects, expand production, or purchase machinery. This stimulates economic activity and growth.
- When interest rates are **high**, borrowing is expensive. Firms hesitate to take loans because the cost of repayment reduces profitability. As a result, investment slows down, and economic expansion is limited.

Summary

1. **Savings provide funds** for investment through financial institutions.
2. **Interest rate balances** savings and investment.
3. **Profit motive** drives investor decisions.
4. **Low interest rates** encourage investment, while high rates discourage it.

Example

If banks in Rwanda reduce loan interest rates, more businesses may borrow money to buy machines, build factories, or expand production.

2. Keynesian theory of investment

The Keynesian Theory of Investment was developed by **John Maynard Keynes** during the Great Depression of the 1930s. Unlike the Classical view, which focused mainly on interest rates, Keynes argued that investment depends on several broader factors, especially expectations and confidence.

Narrative Explanation

Keynes believed that investors do not simply look at the cost of borrowing; instead, they consider whether their investment will generate **future profits**. This expectation is captured in the concept of the **Marginal Efficiency of Capital (MEC)** — the expected rate of return from investing in new capital goods.

- If the **MEC is high**, firms are confident that new machines, buildings, or projects will bring strong profits, so they invest more.
- If the **MEC is low**, firms hesitate to invest because they fear losses or weak

returns.

Another key factor is **business confidence**. When entrepreneurs and firms feel optimistic about the future, they are more willing to take risks and invest. But when confidence is low — perhaps due to political instability, falling demand, or global uncertainty — investment slows down.

The **level of demand in the economy** also plays a central role. If households and businesses are buying more goods and services, firms respond by investing to expand production. Conversely, weak demand discourages investment because firms see no need to increase output.

Finally, Keynes acknowledged that the **interest rate** still matters, but he argued it is not the only determinant. Even with low interest rates, firms may avoid investing if they lack confidence or expect poor returns.

Main Ideas Summarized

1. **Expected profits** drive investment decisions.
2. **Business confidence** shapes whether firms take risks.
3. **Demand levels** influence how much firms invest.
4. **Marginal efficiency of capital** measures expected returns from investment.
5. **Interest rate** is important, but not the only factor.

Example

A company in Rwanda may invest in producing more goods when people's incomes and demand increase, even if interest rates remain high.

Comparison between classical and keynesian theories of investment

Classical Theory

- Investment mainly depends on interest rate.
- Lower interest rates increase investment.
- Savings determine investment funds.
- Gives more importance to interest rates.
- Assumes markets work automatically.

Keynesian Theory

- Investment depends on expected profits and demand.
- High business confidence increases investment.
- Income and demand influence investment decisions.
- Gives more importance to expectations and profits.
- Believes government intervention may be needed.

Application activity 3.3



Activity Scenario

Rwanda is promoting investment in sectors such as agriculture, industry, tourism, and technology to increase production, employment, and national income. However, some investors still face challenges such as limited capital, high interest rates, inadequate infrastructure, and small markets. The government and private sector are working together to improve the investment climate through better infrastructure, access to loans, and regional trade integration.

1. Define the term investment and explain its importance in promoting employment, entrepreneurship, and economic growth in Rwanda.
2. Analyze the factors that influence the level of investment in an economy, including the role of interest rates in investment decisions.
3. Compare the Classical theory of investment and the Keynesian theory of investment, showing how each explains investment behaviour.
4. Explain two major limitations to investment in developing countries such as Rwanda and suggest possible measures to improve investment.
5. Discuss how responsible use of resources, innovation, and entrepreneurship can contribute to increased investment and economic development.
6. Evaluate the contribution of investment to economic growth and development and explain how individuals can participate in activities that promote economic progress.

3.4 Multiplier

Learning Activity 3.4



Activity Scenario

In Rwanda, the government started a road construction project in a rural district. Many local people were employed to work on the project and received wages. After getting paid, workers used part of their income to buy food, clothes, and other goods from nearby shops. Shop owners also earned more income and increased their purchases from suppliers. As a result, more people in the community got jobs and incomes increased.

Questions

1. Identify the main economic activity taking place in the scenario.
2. Explain how the road project created income for different people in the community.
3. Describe how spending by workers and shop owners increased economic activities in the area.
4. What do you understand by the multiplier effect from the scenario?
5. Suggest other projects in Rwanda that can create a multiplier effect in the economy.

The multiplier in economics refers to the process where an initial increase in investment, government spending, or income leads to a larger increase in national income and employment. When people receive additional income, they spend part of it on goods and services, which becomes income for others. As this spending continues from one person to another, the total increase in income becomes greater than the original amount invested. The multiplier therefore helps to explain how investment and spending can promote economic growth and improve living standards in an economy.

3.4.1 Meaning and types of multipliers

i. Meaning

Assuming that the economy is not in conditions of full employment, any increase in investment will not only have an initial effect upon output and income of the same amount, but will also have additional subsequent effects upon output and income. The final increase in income will be larger than the initial increase in investment expenditure. This is the process of the *multiplier*.

Multiplier, therefore, refers to the number of times initial change in expenditure multiplies itself to give a final change in income. The multiplier can be shown as:
Multiplier (M) = $\Delta Y / \Delta E$.

The size of the multiplier is determined by the proportion of extra income going on extra consumption, i.e. MPC. The higher the MPC, the higher the multiplier. The multiplier process reflects the fact that extra spending becomes extra incomes, which in turn becomes partly extra spending. The MPC determines the increase in national income. The formula for determining the actual size of the multiplier is:

$$M = \frac{1}{1-MPC} \text{ or } M = \frac{1}{MPS}$$

Example 1: Given that MPC = 0.75, calculate the multiplier magnitude.

Solution;

$$M = \frac{1}{1-MPC} \text{ or } M = \frac{1}{MPS}$$

$$MPS = 1 - 0.75 = 0.25$$

Therefore, Multiplier = $1 \div 0.25$ and Multiplier = 4

Therefore, the multiplier magnitude is **4**.

If the initial increase in capital investment was **30 million FRW** and this resulted in a final increase in total expenditure of **120 million FRW**, the multiplier would be:

$$\text{Multiplier (M)} = \Delta Y / \Delta E.$$

$$\text{Multiplier} = 120 \text{ million FRW} \div 30 \text{ million FRW}$$

$$\text{Multiplier} = 4.$$

Example 2: If the initial increase in capital investment was 20 million FRW and this brought a final increase in total expenditure of 100 million FRW, the multiplier would be:

Answer

$$\text{Multiplier (M)} = \Delta Y / \Delta E$$

$$100 \text{ million} / 20 \text{ million} = 5$$

Investment and national income at full employment

An increase in investment or spending increases national income only when the economy has unemployed resources and excess capacity. Firms can then increase production by employing more workers and using idle resources. In this case, both real output and income increase.

However, when the economy is at full employment, all resources are fully utilized and production cannot increase further. Additional spending only causes prices to rise, leading to inflation. National income may increase in money terms, but real output remains unchanged.

Key Points

- » Excess capacity allows investment to increase real national income.
- » At full employment, extra spending causes inflation instead of more output.
- » Real income refers to actual goods and services produced.
- » Governments should consider resource availability before increasing spending.

ii. Types of multipliers

Multipliers can be classified according to the form of sectors that spend. These sectors are: government, firms, and foreign sector. The types of multipliers are seen below.

- **Government multiplier:** It refers to the number of times initial government expenditure multiplies itself to give a final change in income, i.e. $M = \Delta Y / \Delta E$
- **Investment multiplier:** It refers to the number of times initial investment expenditure multiplies itself to give a final change in income. $M = \Delta Y / \Delta I$ or $1/\text{MPS}$ or $1/1-\text{MPC}$
- **Consumption multiplier:** It refers to the number of times initial change in consumption expenditure multiplies itself to give a final change in income. $M = \Delta Y / \Delta C$
- **Export multiplier:** It refers to the number of times a given change in export earnings multiplies itself to give a final change in income. $M = \Delta Y / \Delta X$
- **Employment multiplier:** It refers to the number of times a given change in employment multiplies itself to give a final change in income. $M = \Delta Y / \Delta \text{Employment}$
- **Tax multiplier:** It refers to the number of times a given change in tax expenditure multiplies itself to give a final level of income. $M = \Delta Y / \Delta \text{Tax expenditure}$
- **Income multiplier:** this is the number of times a change in total expenditure multiplies itself to give a final change in income. This explains the process by which a change in total expenditure ($E = C + I + G + X - M$) leads to a change in income.

3.4.2 Calculations of multiplier

Multiplier Calculation Activities

Activity 1: Calculating MPC and the Multiplier

Work out and interpret the following:

1. Given that the marginal propensity to save (MPS) is 0.2, calculate:
 - The marginal propensity to consume (MPC)
 - The multiplier
2. Explain what the multiplier value means for the economy.

Activity 2: Finding the Multiplier from MPC

Work out and interpret the following:

1. Given that the marginal propensity to consume (MPC) is 0.75, calculate:
 - The marginal propensity to save (MPS)
 - The multiplier
2. State the meaning of your answer.

Activity 3: Calculating the Final Level of GDP

Work out the following calculation:

The current level of GDP is 300 million FRW. Investment expenditure increases by 50 million FRW and the marginal propensity to save (MPS) is 0.2.

Calculate:

1. The multiplier
2. The total increase in national income
3. The final level of GDP

Activity 4

Given that $MPS = 0.1$:

1. Calculate MPC.
2. Calculate the multiplier.
3. If investment increases by 20 million FRW, calculate the increase in national income.

Activity 5

Given that $MPC = 0.6$

1. Calculate MPS.
2. Determine the multiplier.
3. If government expenditure increases by 40 million FRW, calculate the total increase in GDP.

Activity 6

An economy has a multiplier of 8.

1. Calculate the MPS.
2. Calculate the MPC.
3. If investment increases by 15 million FRW, determine the final increase in national income.

Example 1: Given that as a result of increase in investment expenditure from 20 million FRW to 60 million FRW, national income increased from 3000 million FRW to 7000 million FRW. What is the investment multiplier?

Solution: Investment multiplier = $\Delta Y / \Delta I$

$\Delta Y = 7000\text{million FRW} - 3000\text{million FRW} = 4000\text{million FRW}$

$\Delta I = 60\text{ million FRW} - 20\text{ million FRW} = 40\text{ million FRW}$ Thus; I.M (investment multiplier) = $4000\text{million} / 40\text{ million} = 100$

This means that the initial investment expenditure has multiplied itself 100 times to give a final change in national income.



Note: Multiplier has no units; it is simply a number of times.

Example 2: Given that the final change in investment from 3 million FRW to 8 million FRW led to an increase in income from 200 million FRW to 400 million FRW, calculate the investment multiplier.

Investment multiplier = $\Delta Y / \Delta I$

$\Delta Y = 400\text{million} - 200\text{ million} = 200\text{ million FRW}$

$\Delta I = 8\text{ million} - 3\text{ million} = 5\text{million FRW}$ Thus,

I.M = $200\text{ million} / 5\text{million FRW} = 4$ Therefore, the investment multiplier = 4.

This means that the initial investment expenditure of 3million FRW has multiplied itself 3 times to bring about a final change in income of 400 million FRW.

Application activity



A coffee processing factory in Kigali increases its investment spending to buy new machines and employ more workers. As workers receive more income, they spend more on food, transport, and clothing. Shop owners and transport operators also earn more income and continue spending in the economy.

Questions:

1. Explain how the multiplier process works in this scenario.
2. Identify the initial injection into the economy.
3. State two factors that can reduce the multiplier effect.
4. Explain why the multiplier is important in economic development.

3.4.3 Factors limiting the operation of the investment multiplier

Among all the multipliers, the **investment multiplier** is considered to have a strong effect on the level of national income because an increase in investment leads to a multiplied increase in output, employment, and income. However, in Rwanda, the operation of the investment multiplier is limited by several factors, which reduce the extent to which increased investment can generate further economic growth. These factors include the following:

1. Limited market due to low incomes

Many people in Rwanda still have low levels of income and purchasing power. As a result, demand for goods and services remains limited, discouraging investors from undertaking large-scale investments. When consumers cannot afford many products, businesses make lower sales and the multiplier effect becomes weak.

2. Underdeveloped infrastructure

In some rural and remote areas, infrastructure such as roads, electricity, water supply, and communication networks is still inadequate. Poor infrastructure increases transport and production costs and limits the movement of goods and services from production areas to markets. This discourages investors and reduces the effectiveness of investment spending.

3. Unfavourable investment climate

High taxes, complicated administrative procedures, and high costs of doing business can discourage investment. When investors expect low profits because of heavy taxation or unfavourable policies, they reduce their level of investment. However, the government sometimes encourages investment through incentives such as tax holidays and subsidies.

4. Limited capital and savings

Many local investors lack enough capital to start or expand businesses. Low levels of savings in the economy reduce the availability of funds for investment. As a result, Rwanda depends heavily on foreign investors and foreign loans. In some cases, foreign investors repatriate profits back to their home countries, causing capital outflow and reducing the multiplier effect within the country.

5. Political insecurity in neighbouring regions

Although Rwanda is politically stable, insecurity and conflicts in some neighbouring countries such as parts of the Democratic Republic of Congo may create fear among investors. Such regional instability reduces investor confidence, especially in border areas, and limits investment activities.

6. High liquidity preference

Many people prefer to keep their wealth in cash or near-cash form for precautionary or speculative reasons instead of investing it in productive activities. This reduces the amount of money available for investment and weakens the multiplier process.

7. High level of imports

A large proportion of consumer and capital goods used in Rwanda are imported. When income increases, people may spend more on imported goods instead of locally produced goods. This causes money to leak out of the economy, reducing the multiplier effect.

8. Unemployment and low skills among labour force

Some workers lack the necessary skills and training required in modern industries. This lowers productivity and limits the ability of investment projects to expand production efficiently. Consequently, the expected increase in income and employment may not fully occur.

9. Low level of industrialization

Rwanda still has a relatively small industrial sector. Limited industries reduce the economy's capacity to respond quickly to increased demand created by new investments. This weakens the chain reaction of production, income, and employment.

10. Corruption and mismanagement of resources

Misuse of public funds and corruption in some sectors can divert resources away from productive investment projects. This reduces the effectiveness of government and private investment, thereby limiting the multiplier effect on national income.

11.A high population growth rate increases the dependency ratio within families, leaving households with little disposable income for saving and investment.

Application activity 3.4



Question 1: Define the multiplier and explain its importance in national income analysis. In your explanation, identify and briefly describe the main types of multipliers: spending multiplier, tax multiplier, balanced budget multiplier, investment multiplier, and government expenditure multiplier.

Question 2: Using the multiplier formula, calculate the multiplier when $MPC = 0.8$. If the multiplier value is 5, calculate the Marginal Propensity to Consume (MPC) and explain how MPC influences the size of the multiplier effect.

Question 3: Explain the major leakages that reduce the multiplier effect (savings, taxes, and imports). Analyse how these leakages affect national income and suggest practical ways of minimizing their impact in an economy.

Question 4: Suppose government expenditure increases by FRW 100 million and the multiplier is 4. Calculate the expected change in GDP (national income). Discuss how fiscal policy helps stabilize the economy and explain the limitations and uncertainties of the multiplier in real-life situations, including ethical and distributional issues.

3.5 Meaning and calculations of accelerator principle

Learning Activity 3.5



Activity Scenario

A rise in people's income usually increases their demand for goods and services. When consumers buy more products, producers respond by increasing production capacity through purchasing more machines, building factories, employing more workers, or expanding businesses. This relationship between changes in consumption and changes in investment is explained by the accelerator principle.

In Rwanda, for example, an increase in demand for milk, cement, construction materials, or agricultural products often encourages firms to invest more in machinery, transport, storage facilities, and processing plants in order to meet the growing demand.

Questions

1. Explain the relationship between consumption and changes in national income.
2. Describe the meaning of the accelerator principle.
3. State the formula used to calculate the accelerator coefficient.
4. Work out and interpret the following calculations:
5. (i) Milk Consumption and Investment

As a result of an increase in milk consumption in the country from **80,000 litres to 100,000 litres per week**, investment in the dairy sector increased by **200 million FRW**. Given that the price of milk is **300 FRW per litre**, calculate the magnitude of the accelerator.

(ii) Consumption Expenditure and Investment

Consumption expenditure increased from 30 million FRW to 70 million FRW, leading to an increase in investment from 500 million FRW to 800 million FRW.

Calculate the accelerator coefficient and interpret your answer.

3.5.1 Meaning of the accelerator principle

The **accelerator principle** is an economic concept which states that a small increase in consumer demand or consumption can lead to a larger increase in investment.

It explains the relationship between changes in consumption and changes in investment. When the demand for goods and services rises, producers increase investment in machines, factories, transport, and other capital goods in order to

increase production. Likewise, when demand falls, investment also decreases.

Therefore, the accelerator principle shows that investment depends on the rate of change in consumption or national income.

3.5.2 calculations of accelerator principle

The accelerator principle is the number of times the original change in consumption multiplies itself to give a final change in investment.

$$\text{Accelerator principle} = \frac{\text{change in investment}}{\text{change in consumption}}$$

$$\text{Accelerator} = \Delta I / \Delta C$$

Example 1: For instance, if the consumption of beans increases from 15 kg to 22 kg and increase in investment moves from 100 FRW to 150 FRW, the accelerator principle would be: $(150-100)/(22-15) = 50 \div 7$

$$\text{Accelerator} = 7.1$$

The accelerator is 7.1, meaning investment increases three times more than the change in consumption.

Example 2: If consumption increases from 50 million FRW to 70 million FRW and investment increases from 100 million FRW to 160 million FRW:

- $\Delta C = 20$ million
- $\Delta I = 60$ million

$$\text{Accelerator} = \Delta I / \Delta C. \quad 60\text{million} / 20 \text{ million} = 3$$

Interpretation: The accelerator is 3, meaning investment increases three times more than the change in consumption.

Interpretation of the Accelerator

- If $v > 1$, investment changes more than consumption (strong accelerator effect).
- If $v = 1$, investment changes proportionally with consumption.
- If $v < 1$, investment changes less than consumption (weak accelerator effect).

Application activity 3.5



Question 1: What is the meaning of the Accelerator Principle in economics?

Question 2: How is the Accelerator Principle calculated?

Example Calculation:

Suppose the change in consumption (ΔC) is 60 million FRW and the change in investment (ΔI) is 450 million FRW. Calculate the accelerator coefficient and interpret the result.

3.6. End of unit assessment



A. Knowledge and Understanding

1. Define the term **consumption** and explain the consumption function.
2. Describe the meaning of **saving** and outline the saving function.
3. Explain the meaning of **investment** and state any four factors influencing investment in an economy.
4. Define **multiplier** and distinguish between types of multipliers.

B. Skills and Application

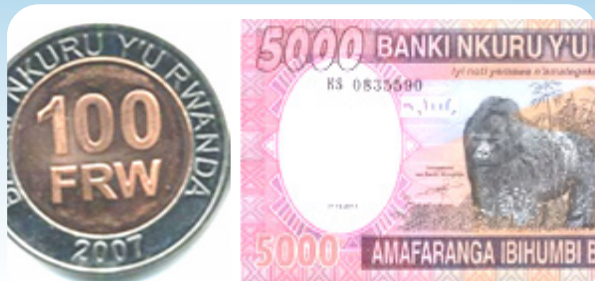
1. Given income of 200,000 FRW and consumption of 150,000 FRW, calculate **Average Propensity to Consume (APC)**.
2. If consumption increases from 100,000 FRW to 130,000 FRW when income increases from 150,000 FRW to 200,000 FRW, calculate **Marginal Propensity to Consume (MPC)**.
3. An investment of 50 million FRW leads to a national income increase of 250 million FRW. Calculate the **multiplier** and interpret your answer.
4. Consumption increases from 40 million FRW to 60 million FRW, while investment rises from 100 million FRW to 180 million FRW. Calculate the **accelerator coefficient**.

B. Skills and Application

1. Explain why households should develop a habit of **saving regularly** for future investment and stability.
2. Discuss the importance of being **financially responsible in consumption decisions** to avoid wasteful spending.
3. Suggest why both government and private investors should promote **ethical investment practices** for sustainable development.
4. Explain how positive attitudes towards **work, entrepreneurship, and investment** can contribute to economic growth in Rwanda.

UNIT 4

MONEY



Key Unit Competence:

Evaluate the functions, characteristics, and value of money, and its role in facilitating exchange and maintaining economic stability.

Learning objectives

Knowledge and understanding

1

- Explain the evolution of money from barter to electronic forms; distinguish different money types (commodity, metallic, paper, electronic, deposit, near money, token, fiat).
- Describe the qualities and functions of good money; assess reasons for and against barter economies.
- Explain Keynes's motives for holding money (transaction, precautionary, speculative).
- State and interpret Fisher's equation of exchange; discuss its applicability and criticisms in developing countries.
- Distinguish monetary aggregates (M1, M2, M3) and identify determinants of money supply in Rwanda.
- Explain the meaning of value of money and determinants of its value.
- Understand money supply, its determinants, and the effects of an increased money supply.
- Identify types of interest rates and factors that influence them; analyze the relationship between money supply, inflation, and interest rates for macroeconomic stability.

Skills

2

- Analyze the evolution, types, and functions of money using data and case studies.
- Assess barter trade vs monetary economies through evidence-based arguments.
- Assess barter trade vs monetary economies through evidence-based arguments.

- Apply Keynesian and Fisherman frameworks to real-world money demand and money supply issues.
- Compute and interpret interest rates; explain determinants of lending and saving behavior.
- Use graphs and tables to illustrate money supply-demand relationships, inflation, and interest rate effects.
- Conduct policy analysis on monetary stability and financial inclusion in Rwanda.
- Use digital tools (spreadsheets, charts) for calculations and visualization.

Attitudes and values

3

- Advocate for the regulation of money demand and supply to maintain purchasing power and economic stability.
- Appreciate the importance of stable interest rates and choose the most appropriate type of interest for different economic situations.
- Value transparency and accountability in financial institutions and government monetary policies.
- Promote responsible saving and investment habits to support personal and national development.
- Develop a positive attitude toward using money efficiently as a tool for growth and welfare.
- Respect the role of the central bank and financial institutions in safeguarding monetary stability.
- Encourage fairness in income distribution and recognize the impact of money management on social equity.
- Choose the best type of interest



Introductory activity

Activity Scenario

In the past, many communities in Rwanda exchanged goods directly through barter trade. For example, a farmer could exchange beans for milk or bananas for clothes. However, this system created difficulties because people had to find someone who wanted exactly what they were offering.

Today, people can easily buy goods and services, pay transport fares, purchase airtime, and store wealth for future use through banks and mobile payment systems. Trade and business activities have become faster, easier, and more efficient than before.

Study the situation above and answer the following questions.

Questions

1. What is barter trade?
2. Identify any two problems faced under barter trade.
3. Explain how trade has become easier in modern economies.
4. State any four functions performed by the item commonly used in buying and selling goods and services.
5. Explain how mobile payment systems facilitate transactions in Rwanda today.
6. Why do people keep part of their income in banks instead of spending it immediately?
7. Explain how the modern system of exchange promotes trade and economic development.

Money is an important tool in the economy because it facilitates the exchange of goods and services and overcomes the problems of barter trade. It plays a central role in trade, saving, investment, and economic decision-making.

Money performs four major functions:

- It acts as a **medium of exchange** for buying and selling goods and services.
- It serves as a **store of value** for future use.
- It is a **unit of account** used to measure and compare values.
- It acts as a **standard of deferred payment** in borrowing and lending.

Money is important because it helps allocate resources efficiently, supports investment through financial institutions, and enables governments and central banks to manage the economy through monetary policy. Its effectiveness depends on trust, stable institutions, and sound economic policies.

4.1 Meaning and evolution of money

Learning Activity 4.1



Scenario

Imagine a village in Rwanda many years ago. People exchanged goods directly: one farmer gave **beans** to another in return for **milk**. Sometimes this worked, but other times it was difficult what if the person with milk did not need beans?

Later, people began using **salt and cattle** as a common medium of exchange because everyone valued them. With time, **metals like silver and gold** replaced these commodities, and eventually, coins and paper notes were introduced. Today, we use **Rwandan Francs (FRW)** in the form of coins, banknotes, and even **mobile money** on our phones.

Answer the following questions:

1. What makes modern money more convenient than exchange of goods and services?
2. Why do people trust paper notes or mobile money even though they have no intrinsic value?

Money is a tool for exchange, a store of value, and a unit of account. It helps people trade goods and services efficiently, without bartering.

A. Meaning of money

Money is anything that people generally accept and use to buy goods and services and to pay debts. It can take different forms depending on the time and place. In the past, people used items such as beads, salt, stones, gold, silver, cattle, and paper as money. Today, money mainly exists as coins, notes, and electronic balances in banks.

All sorts of articles or objects have served as money throughout the ages. These include, among others, beads, salt, stones, gold, silver, paper and cattle.

B. Evolution of money

Money did not develop suddenly. It evolved gradually over many years as people searched for easier and more convenient ways of carrying out trade and exchange. The development of money passed through several stages, beginning with barter trade and later advancing to modern forms such as paper money and bank money. The major stages in the evolution of money are explained below:

- **Barter trade:** This was the earliest system of exchange where goods and services were exchanged directly for other goods and services. For example, a farmer could exchange cassava for a sheep. However, barter trade faced many problems such as the lack of double coincidence of wants, difficulty in measuring value, and problems of storing wealth.

- **Commodity money:** To overcome the problems of barter trade, people began using certain valuable commodities as money. Items such as salt, tobacco, cattle, skins, and grains were widely accepted because they had value and were demanded by many people.
- **Durable commodity money:** Later, societies started using more durable items such as iron, copper, beads, and cowrie shells as money. These items lasted longer and were easier to transport than earlier commodities. However, many of them became too common and lost their effectiveness as reliable money.
- **Metal money and coins:** Precious metals such as gold and silver later replaced ordinary commodities because they possessed the qualities of good money. They were durable, portable, divisible, scarce, and widely accepted. These metals were eventually shaped into coins of different sizes and values.

At first, the actual value of the metal was equal to the value written on the coin. Such coins were called **full-bodied coins**.

Later, governments introduced **token coins**, whose face value was greater than the actual value of the metal used to make them.

- **Paper money:** The use of paper money began with goldsmiths who kept people's gold safely. In return, they issued receipts showing the amount of gold deposited. These receipts started circulating as money because people trusted that they could exchange them for gold at any time.
Over time, goldsmiths issued more receipts than the gold they actually possessed. This led to the development of **fiduciary money**, which is paper money accepted on the basis of trust and government authority rather than being fully backed by gold.
- **Banknotes and bank money:** Governments and central banks later took over the responsibility of issuing paper currency in the form of banknotes of different denominations such as 10, 20, 50, and 100. These notes became widely accepted for buying goods and services.
Modern economies also use **bank money**, which includes deposits in commercial banks that can be transferred through cheques, debit cards, mobile banking, and electronic transfers.
- **Electronic and digital money:** Today, money has further evolved into electronic forms. People can now make payments using mobile phones, internet banking, debit and credit cards, and other digital payment systems without physically carrying cash. Electronic money has made transactions faster, safer, and more convenient.

Barter is a system of exchange where goods or services are traded directly for other goods or services without using money. It depended on the **double coincidence of wants**, meaning that each trader had to want what the other person was offering.

For example, a person with bananas could exchange them for cattle only if the cattle owner also wanted bananas. This system was difficult because it was hard to find matching needs, determine the value of goods, and transport products over long distances.

Due to these challenges, people began using commonly accepted commodities such as sea shells, cattle, salt, metals, and bananas as mediums of exchange, leading to the development of money.

Conditions necessary for barter trade

For barter trade to take place, the following conditions must exist:

- There must be people who possess goods or services that others need.
- There must be at least two parties willing to exchange their goods or services.
- The commodities to be exchanged must be available in the right quantities and quality.
- There must be a **double coincidence of wants**, where each party wants what the other is offering.
- The parties involved must agree on the value or rate of exchange of the commodities.
- There is no generally accepted medium of exchange such as money.

Advantages of barter system

- Barter system is very simple, without any complications and suitable in international trade when countries exchange commodities for commodities.
- In this system the shortage of foreign exchange and imbalance in trade does not occur.
- In a barter system, there is no wastage which occurs in monetary economy because goods are not overproduced or under produced. Only the required quantity is produced.

The economic power is not concentrated at one particular place in the barter system because people do not gather wealth.

Disadvantages of barter system

- Problem of portability: It was difficult to transfer certain items from one place to another. For instance, it would be difficult moving around with sheep looking for market.
- Limited production: Large scale production was limited since
- demand for goods was never certain.
- Needs differentiate from man to man. If either party gets what they need, then it is double coincidence. If not, it is a trouble as they do not get what they need.
- In the barter system, it was difficult to divide and sub-divide goods. Possibility of exchanging one's good for only one good not for two or more.
- The value of the goods could not be distributed equally as there was no measure during that period.
- The exchange of goods worth a service of a person was impossible as it was inconvenient to measure it. The barter system cannot exchange goods for services.

- Perishable goods could not serve as wealth for their future expenses. The economy could not contribute anything towards capital formation of the country, too.
- There was a lot of cheating during the transaction. There was a high possibility of cheating in transactions under the barter system because there was no standard unit of measurement or agreed method of valuing goods and services.

Application activity 4.1



In a rural community, people originally exchanged goods through barter trade. Farmers exchanged beans for cattle, while fishermen exchanged fish for bananas. Over time, traders faced difficulties because not everyone wanted the goods being offered. Later, commodities such as salt, cowrie shells, and metals began to be used as mediums of exchange. Today, people use coins, paper notes, bank accounts, mobile money, and electronic payments to carry out transactions easily.

Questions

1. Define money in simple terms.
2. Explain what is meant by barter trade and State any three problems faced under barter trade.
3. Explain the meaning of double coincidence of wants.
4. Identify and describe any four stages in the evolution of money.
5. Why were precious metals like gold and silver preferred to ordinary commodities as money?
6. Distinguish between barter trade and money economy.
7. Explain any three advantages of using money in modern economies.
8. Explain how the introduction of money promoted trade and specialisation

4.2 Types of money

Learning Activity 4.2



Scenario

In a busy Rwandan market, Aline wants to buy bananas. She notices that people are paying in different ways:

- One customer hands over coins.
- Another gives a colorful note.
- A third writes something on paper and gives it to the seller.
- Later, someone simply uses their phone to complete the payment.

Aline is surprised and asks herself: *"How can all these different things be accepted as payment for the same bananas?"*

Tasks

1. Identify the different types of forms of payment Aline observed.
2. Explain why each of these forms is accepted as money.
3. Share examples from your own community where people use different ways to pay.

Money comes in several forms, from tangible items with value to digital entries and symbolic notes. Broadly, types include commodity money, representative money, fiat money, and near money each differs in what backs its value and how it's used in exchanges.

1. Commodity money

Commodity money was the earliest form of money, valued for its **intrinsic worth**. People accepted items that were naturally useful or scarce such as **sea shells, cattle, salt, tobacco, and grains** as mediums of exchange. Over time, **precious metals** like gold and silver became preferred because they were durable, divisible, and rare. These metals were later **minted into coins**, marking the transition from barter to metallic currency.

2. Convertible paper money

The invention of **convertible paper money** arose from the practices of **goldsmiths**. Individuals deposited their gold with trusted goldsmiths and received **receipts** promising redemption on demand. These receipts gradually became accepted as payment since people trusted the goldsmith's promise. Thus, paper money began as a **claim to real gold**, representing a major innovation in trade convenience and security.

3. Fractionallybacked paper money

Banks later issued notes exceeding their gold reserves, meaning not every note could be redeemed for gold simultaneously. This system allowed banks to expand credit and stimulate economic activity, though it required careful regulation to maintain public confidence.

4. Fiat money

Fiat money has no intrinsic or commodity backing. Its value comes from government authority and public trust. Modern currencies, including the Rwandan Franc (FRW), are fiat money — they circulate because the government declares them legal tender and people accept them for transactions.

5. Deposit money

Deposit money refers to funds held in commercial banks that can be withdrawn on demand. These include current and savings accounts, accessible through cheques, debit cards, or mobile banking. Although cheques do not circulate freely like banknotes, they represent a vital part of modern monetary systems.

6. Near or quasi money

Near money includes financial assets easily convertible into cash, such as treasury bills, bonds, and fixed deposits. They serve as temporary stores of value, not direct mediums of exchange, but can quickly be turned into spendable money.

7. Token money

Token money has a face value greater than its material value. Modern coins and notes are examples their worth lies in the denomination printed, not in the metal or paper itself.

8. Fullbodied money

Fullbodied money is the opposite of token money: its real value equals its face value. Historically, gold or silver coins were fullbodied because the metal content matched their declared worth.

9. Fiduciary money

Fiduciary money is accepted because of trust in the issuing authority, not because it is backed by precious metals. It relies on confidence in the central bank or government to maintain stability and value.

10. Electronic or digital money

In today's world, money increasingly exists in electronic form — stored and transferred through computers, mobile phones, and online banking systems. Examples include mobile money services, internet banking, and digital payment cards, which make transactions faster and more secure.

11. Legal tender money

Legal tender is money that must be accepted by law for payment of debts and transactions within a country. In Rwanda, banknotes and coins issued by the National Bank of Rwanda (BNR) serve as legal tender.

Summary Table: Types of money

Type of Money	Main Feature	Examples
Commodity Money	Has intrinsic value	Salt, cattle, gold
Metal Money	Made of durable metals	Copper, silver coins
Paper Money	Issued by authority	Banknotes
Token Money	Face value > real value	Modern coins
Deposit Money	Held in bank accounts	Savings, current accounts
Near Money	Easily converted to cash	Treasury bills, bonds
Fiat Money	Value from government decree	Rwandan Franc
Fiduciary Money	Based on trust	Central bank notes
Electronic Money	Stored digitally	Mobile money, ebanking
Legal Tender	Must be accepted by law	BNR notes and coins

Advantages of using money

Money plays an important role in the economy because it simplifies exchange and improves economic activities. The following are some of the major advantages of using money:

- **Facilitates exchange:** Money acts as a medium of exchange, making it easier to buy and sell goods and services. Unlike barter trade, money removes the problem of double coincidence of wants.
- **Increases consumer satisfaction:** When people receive income in the form of money, they are free to spend it on the goods and services that give them the greatest satisfaction according to their needs and preferences.
- **Facilitates distribution of goods and services:** In modern economies, goods and services are distributed through the price system. People can obtain products easily by paying money for them.
- **Promotes division of labour and specialisation:** Money encourages workers and producers to specialise in specific tasks because they can exchange their earnings for other goods and services they need. This increases efficiency and productivity.
- **Facilitates Lending and Borrowing:** Money makes it possible for individuals, businesses, and governments to borrow and lend funds. Loans and credit systems support investment and economic growth.
- **Enables Deferred Payments:** Money allows payments to be made in the future, making contracts, credit purchases, and instalment payments possible. This promotes trade and investment.
- **Acts as a Store of Value:** Money enables people to save wealth for future use. Instead of storing perishable goods, individuals can keep money and use it later when needed.
- **Provides a Common Measure of Value:** Money serves as a unit for measuring the value of goods and services. It makes it easier to compare prices and determine the worth of different products.
- **Facilitates Transfer of Property and Wealth:** Money makes it easier to buy, sell, and transfer ownership of assets such as land, houses, and vehicles from one person to another.
- **Encourages Savings and Investment:** By saving money in banks and other financial institutions, people can accumulate funds for future expenditure and investment projects.
- **Promotes Economic Growth:** Through easier exchange, trade, investment, and specialisation, money contributes to increased production, employment, and overall economic development.

Application activity 4.2



Activity Scenario

People in modern societies use different forms of money in their daily transactions. Some use coins and paper notes to buy goods, while others use bank accounts, debit cards, mobile money, and electronic transfers. The use of money has simplified trade, promoted savings, and encouraged specialisation and investment.

Questions

1. Identify and explain any four types of money used in modern economies.
2. Distinguish between commodity money and paper money.
3. Explain any four advantages of using money in economic activities.
4. Describe how electronic money has improved trade and transactions in modern economies.

4.3 Qualities of good money and its functions.

Learning Activity 4.3



Introductory Scenario

A group of Senior Five students visited a local market in Kigali to observe how trade takes place. They noticed that people were using paper notes, coins, mobile money, and bank cards to buy goods and services. During the discussion, one trader explained that money is accepted because it is durable, portable, divisible, and trusted by everyone. The students also observed that money helps people buy goods, measure value, save wealth, and make future payments.

Using the scenario above and your knowledge of economics, answer the following questions.

Questions

1. What is meant by good money?
2. State any four qualities of good money.
3. Explain why durability is an important quality of money.
4. Why should money be portable and easily divisible?
5. Identify any four functions of money in an economy.
6. Explain how money acts as:
 - a) a medium of exchange
 - b) a store of value

7. Describe how mobile money and bank cards perform the functions of money in Rwanda today.

8. Explain the relationship between the qualities of good money and the effective performance of its functions.

Money should be reliable, easy to use, and widely accepted. Good money helps people trade smoothly, store value, and measure worth over time.

i. Qualities of good money.

- **Acceptability:** Good money should be acceptable by everyone as a medium of exchange. This is the prime requirement for money. The use of money is based on confidence. One is prepared to accept money provided one is confident that others will also accept it.
- **Durability:** The goods used as money should be durable. It would not be acceptable if it could deteriorate. For instance, if a metal is used, it should not wear away. It should be noted that the money's durability lies in its value.
- **Scarcity:** Good money should be scarce because if it is common, it would lose value due to increase in demand. Its supply must be less than the demand but it must be available.
- **Homogeneity:** Good money should be similar. The features on the same denominations should be the same as on another denomination. Varying degrees of quality will lead to confusion and uncertainty in the public and eventually there will be loss of confidence.
- **Divisibility:** Good money should easily be divisible in small units. Whatever physical commodity is being used as money, it must be capable of being divided into smaller amounts to make possible smaller transactions.
- **Portability:** One should be able to carry good money from one place to another. It must not be heavy in relation to its value. It must be transportable
- in terms of bulk and weight. Modern money consists of coins, bank notes, cheques and bank drafts and these can easily be carried without attracting attention.
- **Difficult to forge:** Good money should be hard to forge. It should be made of features and quality that cannot be easily forged. Otherwise, forged money will increase money in an economy which leads to inflation and in turn money will lose value. Money which is forged is called **counterfeit money**.

ii. Functions of Money

Money plays a central role in modern economies by facilitating exchange, improving efficiency, and enabling economic planning. Its main functions include the following:

Medium of exchange: All transactions are made through the use of money. Previously in the old ages, there was use of commodity and barter system of exchange. The introduction of money eased the system of exchange because of its good qualities such as portability, divisibility, among others. Therefore, it is used as a medium of exchange.

Measure of value: The value of goods and services and factors of production are expressed in terms of money. Determining the value of a commodity is based on how much an individual is prepared to pay for it. The higher the amount paid for a commodity, the more valuable it is; and the lower the amount paid, the less the value, other factors remaining constant.

Store of value or wealth: Wealth or goods can be stored for future use in form of money than assets. It is easy for Musoni in Musanze District to sell his Irish potatoes and store one million FRW for a period of one year than storing one hundred sacks of Irish potatoes. For example, it is easier to store one million FRW than storing one cow.

Standard of deferred payments: Many transactions are conducted on the basis of credit where goods and services are sometimes given out on credit. When paying for the goods, it is more convenient to express these future obligations in terms of money.

Distribution: In a modern society, money is a mechanism through which most goods and services are distributed through the pricing mechanism. Goods can be moved from one area to another mainly from areas of low price to areas of high price through a process called *arbitrage*. All this can only be done through price mechanism which uses money.

Transfer of immovable property: Money can facilitate the physical transfer of property. For example, it may be hard to move a building from Nyamaseke to Gisenyi but the owner can sell the building and easily moves with his money to the new area of location.

Unit of account: All business transactions and accounting are made possible by use of money. When computing business transactions, statistics of national income, to mention but a few, money is the most suitable medium. It may be easy to record the amount got after selling a commodity than recording the actual commodity.

Application activity 4.3



Activity Scenario

In Rwanda today, people use different forms of money such as coins, paper notes, bank cards, and mobile money to buy goods and services. Traders in markets, shops, and transport services easily accept money because it possesses qualities that make it suitable for exchange. Money also performs several important functions that support trade and other economic activities. Using the knowledge acquired in this lesson, answer the following questions.

Questions

1. Define the term good money.
2. State and explain any four qualities of good money.
3. Explain any three functions of money in an economy.

4. Why is portability and durability important for money?
5. Distinguish between money as a medium of exchange and money as a store of value.
6. Explain how mobile money and electronic payments perform the functions of money in Rwanda today.

4.4 Demand for Money

Learning Activity 4.4



Case Study

Mbabazi, an entrepreneur working in Gasabo District, needed money to carry out various business activities and to support her personal welfare. As her income increased, she began demanding more goods and services. Over time, this rising demand contributed to an increase in the prices of goods, which made her feel disappointed with the outcomes of her business activities.

Using the case study above, answer the following questions:

The need for money by Mbabazi is known as _____.

1. Explain the reasons why Mbabazi demanded more money.
2. Describe what happened in the market when Mbabazi increased her demand for goods and services.
3. Examine the relationship between money supply and the prices of commodities in the economy.

Demand for money refers to the amount of wealth individuals wish to hold in the form of cash or easily convertible assets at a particular time. This preference is influenced by several motives, including the need to carry out daily transactions, the desire to safeguard against unexpected expenses, and expectations about future income opportunities or changes in interest rates.

Money is demanded because it serves as a widely accepted asset that can be used to settle payments and easily exchanged for goods and services. For this reason, individuals do not only hold money for immediate spending but also as a convenient form of wealth storage.

In economics, the demand for money is also referred to as **liquidity preference**. This is the tendency of people to prefer holding their wealth in cash or near-cash form rather than investing it in less liquid assets.

i. Keynesian theory of demand for money

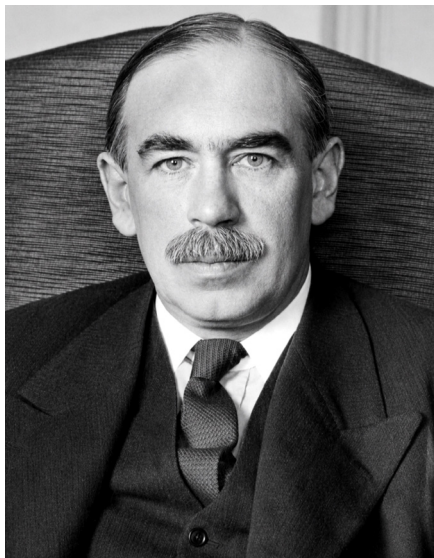


Image 4.1: John Maynard Keynes

According to Keynes, the demand for money arises from three main motives: the transactions motive, the precautionary motive, and the speculative motive.

Transactions motive:

People hold money in order to carry out daily economic activities such as buying food, clothing, paying transport costs, rent, utility bills, and other basic needs. Firms and households also keep money in bank deposits to facilitate regular payments. Keynes explained that the amount of money held for transactions mainly depends on the level of income—higher income leads to higher transaction needs—and on institutional arrangements such as payment systems and banking facilities.

Precautionary motive:

Individuals demand money to meet unexpected or emergency situations such as illness, accidents, sudden travel, or urgent repairs. For example, a traveler may budget for transport, accommodation, and food, but still keep extra cash for unforeseen events like medical emergencies or fare increases. This motive is influenced by income level, uncertainty in the economy, and individual risk attitudes. The higher the uncertainty or insecurity, the more money people prefer to hold for precautionary purposes.

Speculative motive:

People also hold money to take advantage of future changes in interest rates, prices, or investment opportunities. When they expect bond prices to fall or interest rates to rise, they prefer holding money rather than investing in bonds, so that they can invest later at more favorable conditions. This allows them to earn capital gains or avoid losses. Keynes assumed that wealth held for speculative purposes can be kept either in cash balances beyond transaction and precautionary needs or invested in bonds.

In summary, Keynes argued that money is not only a medium of exchange but also a store of value held strategically based on income, uncertainty, and expectations about the future.

Interest rate

Meaning of interest and interest rate

Interest is the **payment made for the use of borrowed capital**. It represents the **cost of borrowing money** or the **reward for saving**. When individuals or businesses borrow funds from financial institutions, they pay interest as compensation to the lender. Similarly, banks pay interest to depositors for keeping their money in savings accounts.

An **interest rate** is the **percentage charged or paid on the principal amount** over a specific period. It varies depending on the type of loan, the financial institution, and prevailing economic conditions.

In Rwanda, commercial banks currently offer **loan interest rates ranging between 14% and 18% per annum**, while **deposit interest rates** often range from **6% to 9%**, depending on the account type and duration.

Types of interest

Interest can be classified into different forms depending on how it is calculated and applied. The main types include **simple interest** and **compound interest**.

Simple interest

Simple interest is the interest calculated only on the original amount of money borrowed or invested, known as the principal. It does not take into account any interest that has already been earned.

It is calculated using the formula:

$$\text{Simple Interest (SI)} = (P \times R \times T) / 100$$

Where:

- P = Principal amount
- R = Rate of interest
- T = Time period

Compound interest

Compound interest is the interest calculated on both the initial principal and the accumulated interest from previous periods. This means that interest is added to the principal, and future interest is earned on the new total. This process is known as *compounding*.

For example, if a loan of 30,000 FRW earns 30% interest per month, after the first month the interest is 9,000 FRW, making a total of 39,000 FRW. In the second month, interest will be calculated on 39,000 FRW instead of the original principal.

Compound interest is calculated as:

$$\text{Compound Interest} = A - P$$

Where:

- P = Principal amount
- A = Total accumulated amount

Experimental version

The total amount (A) is obtained using the formula:

$$A = P(1 + r)^t$$

Where:

- **P** = Principal amount
- **r** = Interest rate per period (in decimal form)
- **t** = Time period

In summary, simple interest is based only on the original amount, while compound interest grows over time because interest is added to the principal.

Nominal interest

Nominal interest refers to the rate of interest expressed in monetary terms without adjusting for inflation. It shows the actual percentage return earned on savings or paid on borrowed money over a given period.

For example, if a person deposits 100,000 FRW in a bank for one year and earns 10,000 FRW as interest at the end of the year, the nominal interest rate is:

$$(10,000 \div 100,000) \times 100 = 10\% \text{ per annum}$$

This means the nominal interest rate is **10% per year**, before considering the effect of inflation.

Real interest

Real interest is the interest rate adjusted for inflation. It reflects the actual increase in purchasing power of money after accounting for the rise in the general price level. It shows the true cost of borrowing or the real return on savings.

For example, if the nominal interest rate is **15% per annum** but the inflation rate during the same period is **9%**, then the real interest rate is: $[15\% - 9\% = 6\%]$.

Although the saver earns 15%, the purchasing power of money has decreased by 9% due to inflation, meaning the real gain is only **6%**.

In summary, nominal interest shows money terms, while real interest reflects the actual economic benefit after considering inflation.

Summary Table: Interest concepts

Concept	Definition	Example (FRW)
Simple Interest	Based only on principal	50,000 at 10% for 2 years → 10,000
Compound Interest	Interest earns interest	30,000 at 12% for 2 years → 7,632
Nominal Interest	Before inflation adjustment	100,000 deposit → 10%
Real Interest	After inflation adjustment	10% nominal – 6% inflation → 4%

Reasons why interest is paid

Interest is the payment made for the use of borrowed money or as a reward for saving money in financial institutions. It is usually expressed as a percentage of the amount borrowed or saved over a given period of time. Interest is paid for several reasons, which include the following:

Reward for saving and postponing consumption

People who save money in banks and other financial institutions postpone their current consumption in order to consume more in the future. Instead of spending their income immediately on goods and services, they allow financial institutions to use their money for lending and investment purposes. Interest is therefore paid as a reward for sacrificing present consumption.

For example, if a person saves **500,000 FRW** in a bank instead of spending it immediately, the bank may pay interest as compensation for waiting.

Payment for parting with liquidity

When people lend or save money, they give up immediate access to cash that could have been used for daily activities or emergencies. Since money is highly liquid and easily usable, lenders expect compensation for temporarily surrendering it. Interest is therefore paid for parting with liquidity.

Price for the use of credit

Interest acts as the price paid by borrowers for using loanable funds belonging to other people or financial institutions. Just as rent is paid for the use of a house, interest is paid for the use of money over a period of time.

For example, a businessperson who borrows money to expand a shop must pay interest to the lender for using the borrowed capital.

Reward for risk bearing

Lending money involves the risk that the borrower may fail to repay the loan or may delay repayment. Financial institutions and lenders charge interest partly to compensate for such risks. The greater the risk of default, the higher the interest rate that may be charged.

Compensation for administrative and management costs

Lending money involves administrative expenses such as processing applications, keeping records, printing documents, communication costs, monitoring borrowers, and managing accounts. Interest helps financial institutions recover these operational and management costs.

Reward for inconvenience

When a lender gives out money, he or she may experience inconvenience because the funds cannot be used for personal needs during the lending period. Interest is therefore paid as compensation for the inconvenience caused by giving up the use of money temporarily.

To cater for the time value of money

Money loses value over time because of inflation and changes in purchasing power. A certain amount of money today can buy more goods and services than the same amount in the future. Therefore, interest is paid to compensate the lender for the fall in the value of money over time.

For example, **10,000 FRW** today may buy more goods and services than **10,000 FRW** one year later if prices rise. This means that money today is more valuable than money in the future.

To encourage savings and investment

Interest encourages people to save money in financial institutions because they expect to earn returns on their savings. Increased savings provide funds that can be lent to investors and businesses, promoting economic growth and development.

To control the use of borrowed funds

Interest discourages unnecessary borrowing because borrowers must pay an additional cost on top of the principal amount borrowed. This helps ensure that borrowed funds are used carefully and productively.

In summary, interest is paid because money has value, lending involves risk and sacrifice, and savings are necessary for investment and economic development.

Determinants of the rate of interest

The rate of interest refers to the cost of borrowing money or the reward for saving money, usually expressed as a percentage of the principal amount over a given period of time. Interest rates are influenced by several economic and financial factors, including the following:

Amount of money in circulation

The supply of money in the economy greatly affects interest rates. When there is a large amount of money in circulation, banks and financial institutions have more funds available for lending, which lowers the rate of interest. On the other hand, when money supply is limited, loanable funds become scarce, leading to higher interest rates.

Duration of the loan

The period taken to repay a loan also determines the interest charged. Long-term loans usually attract higher interest rates because they involve greater uncertainty, risk, and the possibility of inflation over time. Short-term loans generally have lower interest rates because the lender recovers the money within a shorter period.

Size of the loan

The amount of money borrowed can influence the interest rate. Large loans may attract relatively lower interest rates because they generate more business for financial institutions and reduce administrative costs per unit of money lent. Small loans may have relatively higher interest rates due to higher administrative and processing costs.

Competition among financial institutions

When many banks and financial institutions compete to attract customers, they tend to reduce interest rates on loans and sometimes increase interest on savings to attract depositors. In contrast, limited competition may result in higher lending rates and lower savings rates.

Risks involved in lending

The greater the risk that a borrower may fail to repay a loan, the higher the interest rate charged. Borrowers with unstable incomes, poor credit history, or insufficient collateral are often charged higher interest rates to compensate lenders for the risk involved.

Productivity of capital

If borrowed money is expected to generate high profits in productive activities such as business or investment, lenders may charge higher interest rates because borrowers can afford to pay more. When the productivity of capital is low, interest rates tend to be lower.

Rate of inflation

Inflation affects the purchasing power of money over time. When inflation is high, lenders charge higher interest rates to protect themselves against the loss in the value of money. Low inflation usually leads to relatively lower interest rates.

Government monetary policy

Central banks, such as the National Bank of Rwanda, influence interest rates through monetary policy measures. For example, increasing the central bank rate may lead commercial banks to raise lending rates, while lowering the policy rate may encourage lower interest rates and increased borrowing.

Demand and supply of loanable funds

Interest rates are also determined by the interaction between the demand for loans and the supply of savings. High demand for borrowing with limited savings leads to higher interest rates, while low demand for loans and abundant savings result in lower interest rates.

Level of economic activity

During periods of rapid economic growth, businesses and individuals demand more credit for investment and consumption, which may increase interest rates. During economic slowdowns, demand for loans may decrease, causing interest rates to fall.

In summary, interest rates are determined by economic conditions, financial market forces, government policies, inflation, and the risks associated with lending and borrowing.

Application activity 4.4



Case Study: Money demand and interest in Rwanda

In 2026, the National Bank of Rwanda (BNR) reported that commercial banks were experiencing changes in money demand. Many people preferred keeping cash for daily transactions and emergencies, while others invested in savings accounts and treasury bills. At the same time, the average lending interest rate was 16%, and the deposit rate was 7%. Inflation was rising, and the government encouraged saving and investment to stabilize the economy. Keynes explained that people hold money for different motives and that the interest rate influences how much money people choose to hold or invest.

Activity instructions

Work in small groups and read the case study carefully. Discuss and answer the following questions:

1. Identify the three motives for holding money according to Keynes and give examples from Rwanda's daily life for each.
2. Explain how interest rate affects the speculative motive for holding money.
3. Discuss the meaning of interest and interest rate using examples from Rwandan banks.
4. State reasons why interest is paid by borrowers and received by savers.
5. Identify determinants of interest rate in Rwanda's financial system (e.g., inflation, demand for loans, central bank policy).
6. Explain the relationship between money demand and interest rate in the context of Rwanda's economy.

4.5 Quantity theory of money

Learning Activity 4.5



Case Study

In Nyagatare District, farmers and traders noticed that during one year, the amount of money circulating in the economy increased rapidly due to increased bank lending and higher government spending. However, the quantity of goods and services produced in the district remained almost unchanged. As people had more money to spend, the prices of commodities such as maize flour, beans, transport, and clothes started rising steadily.

Teachers at a nearby secondary school explained that when the amount of money in circulation increases faster than the production of goods and services, the general price level in the economy is likely to rise.

Questions

1. From the case study, identify the major cause of the increase in prices in Nyagatare District.
2. Explain the relationship between money supply and the general price level in an economy.
3. What is meant by the velocity of circulation of money?
4. Explain what may happen when the amount of money in circulation increases while the supply of goods and services remains constant.

4.5.1 Quantity theory of money (Fisher's equation of exchange) and its assumptions

i. Quantity theory of money (Fisher's Equation of Exchange)

The Quantity Theory of Money was first developed by the Classical economists and later improved by the American economist Irving Fisher. In his famous book, **The Purchasing Power of Money published in 1911**, Fisher explained how the amount of money in circulation affects prices and the value of money in an economy.

According to the theory, *the value of money depends mainly on the quantity of money available in the economy. When the amount of money in circulation increases while the supply of goods and services remains the same, prices rise. As prices rise, the purchasing power of money falls because more money is needed to buy goods and services.*



Image 4.2 : Irving Fisher

On the other hand, *when the quantity of money decreases while the supply of goods and services remains unchanged, prices fall. As prices fall, the purchasing power or value of money increases because the same amount of money can buy more goods and services.*

The theory is expressed using Fisher's Equation of Exchange:

$$MV = PT$$

Where:

- **M** = Quantity of money in circulation
- **V** = Velocity of circulation of money
- **P** = General price level
- **T** = Volume of transactions or output of goods and services

The equation means that the total amount of money spent in the economy (**MV**) is equal to the total value of goods and services sold (**PT**).

The theory also explains the following important factors:

- **Quantity of money in circulation (M):**
When money supply increases, prices tend to rise. When money supply decreases, prices tend to fall.
- **Velocity of circulation of money (V):**
This is the number of times money changes hands in a given period. If money circulates faster, prices may rise.
- **Volume of transactions or output (T):**
This refers to the amount of goods and services produced and exchanged in the economy. Higher production can help reduce price increases.
- **General price level (P):**
This is the average level of prices of goods and services in the economy.

The theory assumes that:

- The velocity of money remains constant.
- The level of output remains constant in the short run.
- Money is mainly used as a medium of exchange.
- The economy operates under full employment.

For example, if the amount of money in circulation doubles while the quantity of goods and services remains unchanged, prices are expected to double. This means the value of money falls because the same amount of money buys fewer goods and services. Similarly, if the quantity of money is reduced by half, prices may fall and the value of money increases.

The equation can also be written as: **$P = MV/T$**

This shows that the general price level depends on the quantity of money and the speed at which money circulates compared to the amount of goods and services produced.

The Quantity Theory of Money is important because it helps economists and governments understand the relationship between money supply, inflation, prices, and purchasing power in the economy.

Example 1: Effect of changes in money supply on price level

Assume that the velocity of circulation of money (**V**) and the volume of transactions (**T**) remain constant.

Given:

- Quantity of money (**M**) = 100
- Velocity of money (**V**) = 10
- Volume of transactions (**T**) = 50

Using Fisher's equation: **$P = MV/T$**

Experimental version

Substituting the values:

$$P = (100 \times 10) \div 50 = 20$$

Therefore, the general price level is **20**.

When the quantity of money increases

If the quantity of money increases from **100 to 200** while **V** and **T** remain constant:

$$P = (200 \times 10) \div 50 = 40$$

The price level rises from **20 to 40**. This shows that an increase in money supply leads to a rise in prices and a fall in the value of money.

When the quantity of money decreases

If the quantity of money decreases from **100 to 50** while **V** and **T** remain constant:

$$P = (50 \times 10) \div 50 = 10$$

The price level falls from **20 to 10**. This means that a decrease in money supply reduces prices and increases the value of money.

Example 2: Increase in velocity of money

Assume:

- Quantity of money (**M**) = 150
- Velocity of money (**V**) = 8
- Volume of transactions (**T**) = 60

Initial price level:

$$P = (150 \times 8) \div 60 = 20; \text{ So, the initial price level is } \mathbf{20}.$$

If the velocity of money increases from **8 to 12** while **M** and **T** remain constant:

$$P = (150 \times 12) \div 60 = 30.$$

The price level increases from **20 to 30**. This shows that when money circulates faster in the economy, prices tend to rise.

Example 3: Increase in Output of Goods and Services

Assume:

- Quantity of money (**M**) = 240
- Velocity of money (**V**) = 5
- Volume of transactions (**T**) = 60

Initial price level:

$$P = (240 \times 5) \div 60 = 20. \text{ So, the initial price level is } \mathbf{20}.$$

If the volume of transactions or output increases from **60 to 120** while **M** and **V** remain constant:

$$P = (240 \times 5) \div 120 = 10$$

The price level falls from **20 to 10**. This shows that an increase in the production of goods and services can reduce prices when the money supply remains unchanged.

ii. Assumptions of Fisher's quantity theory of money

According to Irving Fisher, the Quantity Theory of Money is based on several assumptions:

- The general price level depends mainly on the quantity of money, the velocity of money, and the volume of transactions in the economy.
- The velocity of circulation of money (**V**) is assumed to remain constant in the short run.
- The volume of transactions or output (**T**) is also assumed to remain constant in the short run.
- Money is mainly held for transaction purposes, that is, to buy and sell goods and services.
- The supply of money is determined by the government or central bank and is independent of changes in prices or output.
- The theory mainly applies in the long run, where prices are flexible and can fully adjust to changes in money supply.
- The economy is assumed to operate at full employment, meaning all available resources are fully utilized.
- Prices and wages are flexible and adjust easily to restore equilibrium in the economy.
- Any increase in money supply is spent on goods and services rather than being hoarded or saved.
- Real output is assumed to be fixed in the short run, so changes in money supply mainly affect prices rather than production.
- Interest rates, financial markets, and expectations are not given much importance in the basic equation:

$$MV = PT$$

- The theory assumes there are no major financial problems such as liquidity traps, credit shortages, or currency substitution that could affect the demand for money.

iii. Modern view of the quantity theory of money

Modern economists have modified Fisher's theory because some of its assumptions are not realistic in today's economies.

- i. The **velocity of money** is not always constant because it can change due to **interest rates, technology, financial innovation, and expectations**.
- ii. The **demand for money** depends not only on transactions, but also on **income levels, interest rates, and future expectations**.

- iii. **Commercial banks** can create money through **lending**, making money supply partly **endogenous**.
- iv. **Prices** and **wages** may be rigid in the short run, so changes in money supply may not immediately affect prices.
- v. **Government policies, fiscal policy, inflation expectations, and central bank credibility** influence economic outcomes.
- vi. If an economy has **unemployed resources**, an increase in money supply may increase **output** and **employment** before causing **inflation**.
- vii. In the long run, increases in **money supply** do not always lead to proportionate increases in **prices** because **velocity** and **money demand** may change.
- viii. Modern economies are more complex due to developed **financial markets, credit systems, and international transactions**.

4.5.2 Criticisms of the quantity theory of money

Although the Quantity Theory of Money developed by Irving Fisher is important in explaining the relationship between money supply and prices, the theory has been criticised by many economists, especially John Maynard Keynes. The major criticisms are explained below:

1. The Theory is a truism

Keynes argued that the Quantity Theory of Money is simply a truism because Fisher's equation only states that the total money spent on goods and services (**MV**) is equal to the total value of goods and services sold (**PT**).

$$MV = PT$$

The equation is mathematically true, but it does not clearly explain how or why changes in money supply directly cause proportional changes in prices in real life.

2. Other factors do not remain constant

The theory assumes that other factors such as the velocity of money (**V**) and the volume of transactions (**T**) remain constant. In reality, these factors frequently change.

For example:

- An increase in money supply may change the velocity of circulation of money.
- Output and production may also increase or decrease.
- Therefore, changes in prices may not be directly proportional to changes in money supply.

3. Constants relate to different time periods

Economist Emil Halm criticised Fisher for combining variables that relate to different time concepts.

- **M** (money supply) refers to a point in time.
- **V** (velocity of money) refers to a period of time.

According to Halm, multiplying these two different concepts together is technically inconsistent.

4. Failure to measure the true value of money

The theory mainly measures the amount of money used in transactions rather than the actual purchasing power of money.

It focuses on all business transactions, including financial transactions, instead of concentrating only on goods and services consumed by people. As a result, the theory does not accurately measure the real value of money.

5. The theory is weak during trade cycles

Economist Geoffrey Crowther argued that the theory gives too much importance to money supply as the main cause of price changes.

According to him:

- During a depression, prices may remain low even if money supply increases.
- During a boom, prices may remain high even if money supply decreases.

This means that other economic factors such as demand, investment, and production also influence prices.

6. Neglect of the rate of interest

The theory ignores the role of the interest rate in influencing prices and economic activity.

In reality:

- Interest rates affect borrowing, saving, investment, and spending.
- Changes in interest rates can influence the demand for money and the price level.

Therefore, ignoring interest rates makes the theory incomplete.

7. Unrealistic assumptions

Keynes criticised the theory for relying on unrealistic assumptions such as:

- Full employment
- Constant velocity of money
- Constant output
- Flexible prices and wages

In modern economies, unemployment, price rigidity, and economic instability are common. Therefore, the assumptions do not always hold in practice.

8. Velocity of money is not constant

The theory assumes that the velocity of circulation of money remains constant. Keynes argued that velocity changes frequently depending on:

Experimental version

- Income levels
- Business conditions
- Interest rates
- People's expectations

Thus, it is unrealistic to assume that velocity is always stable.

9. Neglect of the store of value function of money

The theory mainly considers money as a medium of exchange and ignores its store of value function.

In reality, people may hold money for:

- Future use
- Precautionary motives
- Speculation

Therefore, the demand for money is not always constant as Fisher assumed.

10. Neglect of the role of Government

The theory ignores government intervention in the economy.

For example:

- Governments may control prices through regulations.
- Central banks may use monetary policy to influence inflation and interest rates.
- Fiscal policies such as taxation and government spending can also affect prices.

Thus, an increase in money supply may not always lead to higher prices if the government controls prices or production.

11. Ignores the role of banks and credit creation

Modern economists argue that commercial banks create money through lending and credit creation. Fisher's theory mainly focuses on the quantity of physical money and ignores the important role of the banking system in determining money supply.

12. Ignores international influences

The theory mainly explains a closed economy and ignores international trade and financial flows.

In modern economies:

- Exchange rates
- Imports and exports
- Foreign investment can influence prices and the value of money.

Summary

Despite its weaknesses, the Quantity Theory of Money remains important because it helps explain the relationship between money supply and inflation. However, modern economists believe that prices are influenced not only by money supply, but also by factors such as interest rates, production, expectations, government policies, and financial markets.

Application activity 4.5



- Q1: What is the Quantity Theory of Money according to Fisher's equation of exchange?
- Q2: What are the main assumptions of the Quantity Theory of Money?
- Q3: What are two criticisms of the Quantity Theory of Money?
- Q4: What happens if money supply increases faster than output according to the theory?

4.6 Value of money

Learning Activity 4.6



1. If you had 10,000 RWF today, what kinds of things could you buy, and how might that be different from what your parents could buy with the same amount years ago?
2. When the prices of goods and services increase, how does that affect what people can afford with the same amount of money?
3. What do you think happens when more money is printed and put into circulation in the country?
4. Can you give an example from Rwanda where rising prices made it harder for families to buy the same items they used to?
5. How does people's confidence in the Rwandan franc influence their willingness to use it in trade and savings?

Value for money means getting the best possible outcomes for the price paid, balancing quality, usefulness, and durability. It also considers long-term costs and opportunity costs, not just the upfront price.

4.6.1 Meaning and determinants of value of money

i. Meaning of value of money

The **value of money** refers to the amount of goods and services that a given amount of money can buy. It is also known as the **purchasing power of money**.

Money can be understood in two ways:

a) Nominal money: This is the face value of money expressed in currency units such as **Rwandan francs (FRW), dollars, or shillings**. For example, 1,000 FRW is nominal money.

b) Real money: This refers to what that money can actually buy in terms of goods and services. It shows the **true purchasing power** of money.

The value of money depends on **time**, meaning whether it is used today or in the future. Money has a concept known as **time value of money**, which explains that the value of money changes over time.

For example, **1,000 FRW today is more valuable than 1,000 FRW in the future**, because:

- You can use it immediately to buy goods and services.
- You can invest it and earn extra income over time.
- Prices may rise in the future due to inflation, so the same amount will buy fewer goods later.

Example

If 2,000 FRW today can buy **1 kilogram of sugar**. But after one year, due to an increase in prices (inflation), the same 2,000 FRW may buy only **half a kilogram (0.5 kg) of sugar**.

This shows that the **value of money has fallen over time**.

Key Idea

Money available today is more valuable than the same amount in the future because it can be used immediately to consume, invest, or earn returns. This is known as the **time value of money principle**.

ii. Determinants of the time value of money

The **time value of money (TVM)** means that a sum of money today is worth more than the same sum in the future, because of its potential earning capacity. Several key determinants explain why:

1. Consumption preference

- People generally prefer to consume now rather than later if satisfaction is equal.
- However, many are willing to postpone current consumption if they expect greater benefits in the future.
- A higher **rate of return** encourages saving and investment instead of immediate spending.

Experimental version

- Some individuals, fearing uncertainty, still choose present consumption even if future gains could be larger.

2. Uncertainty of the future

- The future is unpredictable: health, income, and economic conditions may change.
- Because of this, people often prefer **certain cash flows today** over uncertain ones tomorrow.
- Risk and uncertainty increase the preference for immediate consumption or guaranteed returns.

3. Inflation

- Inflation reduces the **purchasing power** of money over time.
- In economies with high inflation, investors demand higher returns to compensate for the loss in value.
- Consumers also prefer spending now rather than later, since goods and services will cost more in the future.

4. Investment opportunities

- Money received today can be reinvested to generate additional returns.
- For example, periodic cash inflows can be reinvested to earn compound interest.
- This reinvestment potential makes present cash flows more valuable than future ones.

5. Interest rates

- Interest rates directly affect the time value of money.
- Higher interest rates increase the opportunity cost of holding money idle, encouraging investment.
- Lower interest rates reduce the incentive to save, making current consumption more attractive.

6. Liquidity preference

- People value money that is readily available for emergencies or opportunities.
- The more liquid the asset, the higher its perceived value today compared to uncertain future returns.

Application activity 4.6



Scenario:

In Rwanda, the value of money changes over time due to several economic factors. Traders and consumers are often concerned about what determines the strength or purchasing power of money and how it is affected.

Question 1: Define the term “value of money.”

Question 2: What is meant by the real value of money?

Question 3: State and explain any four determinants of the value of money.

Question 4: How does inflation affect the value of money?

4.7 Money supply

Learning Activity 4.7



Scenario

Imagine the National Bank of Rwanda decides to increase the amount of money circulating in the economy. Shops, banks, and households begin noticing changes in prices, savings, and borrowing. Students are asked to explore what this means for the economy.

Questions

1. When economists talk about “money supply,” what do they mean, and why do they classify it into levels such as M1, M2, and M3? How do these levels differ in terms of what they include, and why might it be important to distinguish between them?
2. How does the National Bank of Rwanda influence the amount of money circulating in the economy, and why is its role critical in maintaining stability?
3. In what ways do commercial banks, through lending and deposits, and government policies such as spending and taxation, affect the overall money supply in Rwanda?
4. If more money becomes available in the economy, what changes might occur in the prices of goods and services, and how could this influence people’s savings and borrowing behavior?

4.7.1 Meaning and levels of money supply

a) Meaning of money supply

Money supply refers to the total amount of money available in an economy at a given period of time. It includes money circulating among the public in the form of notes and coins, together with money kept in commercial banks such as deposits and other near-money assets (quasi money). In simple terms, money supply is the total stock of money that people, businesses, and financial institutions can use for spending, saving, and investment.

Money supply mainly consists of:

- Currency in circulation such as notes and coins held by the public,
- Demand deposits in commercial banks that can be withdrawn at any time,
- Time and savings deposits,
- Other liquid financial assets that can easily be converted into cash.

The central bank controls and regulates the money supply in the economy through monetary policy instruments such as bank rate policy, open market operations, reserve requirements, and selective credit controls.

b) Levels of money supply

In modern economies, money supply is commonly classified into different levels or measures according to liquidity (the ease with which assets can be used for payments).

1. M1 (Narrow money)

M1 is the most liquid form of money because it can easily be used for day-to-day transactions. It mainly performs the function of a medium of exchange.

M1 includes:

- Currency notes and coins held by the public,
- Demand deposits or current account deposits in commercial banks.

Therefore: **$M1 = \text{Currency (notes and coins)} + \text{Demand deposits}$**

M1 is called narrow money because it only includes money that can immediately be used for buying goods and services.

2. M2 (Broad money)

M2 is a wider measure of money supply. It includes all components of M1 plus savings and time deposits that may require notice before withdrawal.

M2 includes:

- Currency notes and coins,
- Demand deposits,
- Savings deposits,
- Time deposits.

Therefore: **$M2 = M1 + \text{Time Deposits} + \text{Savings Deposits}$**

M2 is referred to as broad money because it includes both highly liquid money and less liquid bank deposits.

3. M3 (Broader Money)

M3 is the broadest commonly used measure of money supply. It includes all components of M2 together with other large liquid financial assets and money market instruments.

M3 includes:

- Currency notes and coins,
- Demand deposits,
- Savings and time deposits,
- Fixed deposits,
- Money market instruments such as certificates of deposit and treasury-related instruments.

Therefore: **$M3 = M2 + \text{Fixed Deposits} + \text{Money Market Instruments}$**

M3 gives a comprehensive measure of the total money available in the economy and is useful in analyzing inflation, credit creation, and overall economic activity.

4.7.2 Importance of measuring money supply and types of money supply

a) Importance of measuring money supply

Measuring money supply is important because it:

- Helps the government and central bank control inflation,
- Influences interest rates and credit creation,
- Affects investment, production, and employment,
- Helps maintain economic stability,
- Guides the formulation of monetary policies.

An excessive increase in money supply may lead to inflation, while a very low money supply may reduce investment and slow down economic growth.

b) Types of money supply

1. Exogenous (Discretionary) money supply

This is money supply that is determined and controlled by the central bank or monetary authority. It is fixed independently of the level of economic activities in the economy. The central bank regulates it through monetary policy instruments such as printing money, reserve requirements, and open market operations.

Example: The central bank may increase currency in circulation to influence economic conditions.

2. Endogenous (Automatic) money supply

This is money supply that changes according to the level of economic activities in the economy. When production, trade, and investment increase, demand for loans rises and commercial banks create more credit, increasing the money supply. When economic activities decline, money supply also decreases.

Example: During economic expansion, businesses borrow more from banks, leading to an increase in money supply.

4.7.3 Determinants of money supply

The **money supply** in an economy depends on several interrelated factors. These determinants explain why the amount of money circulating can rise or fall:

1. Level of economic activity

- When production, trade, and investment are high, more money circulates to support transactions.
- During recessions or low activity, money supply contracts because fewer transactions occur.

2. Central Bank operations

- The buying and selling of securities (like bonds) by the central bank directly influences liquidity.
- Buying securities injects money into the economy, while selling them withdraws money.
- This is part of **open market operations**.

3. Balance of payments

- A surplus (exports > imports) increases foreign currency inflows, boosting money supply.
- A deficit reduces available money as more funds flow out to pay for imports.

4. Printing of money

- When governments print more currency (financial accommodation), circulation rises.
- Excessive printing without matching production leads to inflation.

5. Capital inflows and outflows

- Tourism, foreign investment, and remittances increase money supply.
- Capital flight or large outflows reduce available money.

6. Monetisation of the economy

- Economies that rely heavily on money as a medium of exchange have higher money supply.
- In contrast, reliance on barter reduces monetary circulation.

7. Credit creation by commercial banks

- Banks expand money supply by lending more (creating deposits).
- Restrictive lending policies reduce money supply.

8. Government borrowing

- When governments borrow heavily from banks, money supply increases.
- Low borrowing reduces the injection of new money.

9. Interest rates

- Low interest rates encourage borrowing and spending, increasing money supply.
- High interest rates discourage borrowing, reducing circulation.

10. Public Confidence and expectations

- If people trust the currency and banking system, they deposit and invest, sustaining money supply.
- Lack of confidence leads to hoarding or shifting to foreign currencies, reducing domestic supply.

4.7.4 Effects of increased money supply

Positive effects

- **Encourages investment:** When extra money is channelled through banks as loans, businesses expand and new projects are financed.
- **Boosts demand:** Greater purchasing power allows households to buy more goods and services, stimulating markets.
- **Creates employment:** Increased demand encourages firms to expand production, leading to more job opportunities.
- **Stabilises recessions:** By raising aggregate demand, it helps economies recover from depression or slowdown.
- **Raises tax revenue:** Higher incomes and spending generate more taxes, provided inflation is controlled.
- **Asset accumulation:** People may invest in fixed assets (land, houses, machinery) instead of holding idle cash.
- **Encourages innovation:** With more funds available, firms can invest in new technologies and processes.

Negative Effects

- **Inflationary pressure:** Too much money chasing limited goods raises prices.
- **Interest rate adjustments:** Banks may increase lending rates to offset inflation risks.
- **Currency depreciation:** Excess supply reduces the external value of the currency, making imports more expensive.
- **Loss of confidence:** If inflation is severe, people may hoard goods or revert to barter trade.
- **Reduced bank lending:** Fear of repayment in devalued currency discourages banks from issuing loans.
- **Shift in production:** Producers may focus on exports (with stable global prices) rather than local markets.

- **Tax erosion:** Inflation reduces the real value of tax revenue, weakening government budgets.
- **Higher production costs:** Rising input prices increase costs, discouraging local production.
- **Income inequality:** Inflation often benefits asset owners but hurts wage earners, widening inequality.
- **Speculative behaviour:** Excess liquidity may lead to speculation in land, housing, or foreign currency instead of productive investment.

Application activity 4.7



Question 1: Define money supply and explain its significance in the economy. What are the different levels of money supply?

Question 2: Why is it important for governments and policymakers to measure the money supply? Mention at least two types of money supply.

Question 3: List and explain two determinants of the money supply.

Question 4: What are the potential effects of an increase in the money supply on the economy?

4.8. End of unit assessment



Knowledge and Understanding

1. Meaning and Evolution: Explain how money evolved from barter trade to modern forms such as fiat and digital money.
2. Types of Money: Distinguish between commodity money, fiat money, deposit money, and near money, giving examples of each.
3. Quantity Theory: State Fisher's equation of exchange and explain its assumptions.
4. Value of Money: Identify and explain the main determinants that influence the purchasing power of money.

Skills

5. Using a case study, analyze how the qualities of good money (durability, divisibility, stability) support its functions in Rwanda's economy.
6. Apply Keynesian theory to explain why households and firms in Rwanda hold money for transaction, precautionary, and speculative motives.
7. Given that the money supply (M) is 50 million RWF, velocity of circulation (V) is 4, and total transactions (T) are 200 million, calculate the average price level (P) using Fisher's equation of exchange.

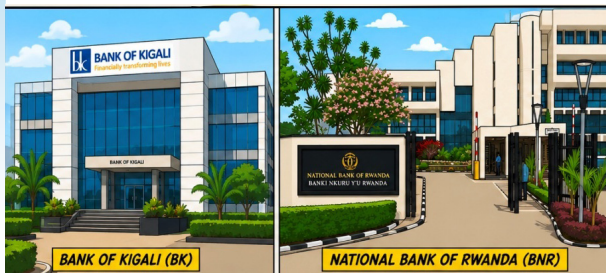
8. If the marginal propensity to consume (MPC) is 0.8, calculate the multiplier. Then, explain how an increase in investment of 10 million RWF would affect national income.

Attitudes and Values

9. Discuss why maintaining trust in the Rwandan franc is important for economic stability and national pride.
10. Evaluate the importance of responsible government borrowing in sustaining long-term growth and avoiding inflation.
11. Reflect on how promoting savings through SACCOs and banks can strengthen household resilience and national development.
12. Debate why citizens should avoid speculative hoarding or misuse of money, and instead channel funds into productive investments for Rwanda's future.

UNIT 5

FINANCIAL INSTITUTIONS



Key Unit Competence:

Analyse the role of financial institutions in mobilizing savings, creating credit, promoting financial inclusion, and supporting economic development.

Learning objectives

Knowledge and understanding

- 1
 - Analyze and differentiate types of financial institutions (banking and non-banking).
 - Evaluate the role of financial institutions in economic growth and stability.
 - Examine operations of central and commercial banks, including assets, liabilities, and credit creation.
 - Assess financial intermediation and its impact on investment and development.
 - Apply financial literacy concepts to real-life decision-making.
 - Evaluate monetary policy tools and their effectiveness in developing economies.
 - Compare money and capital markets and their economic functions.
 - Analyze Rwanda's financial sector structure and inclusion initiatives.
 - Assess opportunities, risks, and ethical issues of AI in financial services.
 - Evaluate challenges facing financial institutions in developing countries.

Skills

- 2
 - Analyze financial data (assets, liabilities, performance).
 - Interpret monetary policy effects on the economy.
 - Compare financial institutions and markets.
 - Apply financial literacy in real-life contexts.

- Evaluate financial inclusion and development outcomes.
- Conduct research, present findings, and debate economic issues.
- Assess risks and ethical implications of financial technologies.

Attitudes and values

3

- Appreciate the role of financial institutions in development
- Recognize their contribution to economic growth, stability, and poverty reduction.
- Develop a culture of saving and responsible financial behaviour
- Value planning, discipline, and long-term financial security.
- Support financial inclusion and equal access to services
- Promote fairness and participation of all groups in the financial system.
- Promote responsible borrowing and lending
- Understand the importance of avoiding over-indebtedness and ensuring ethical practices.
- Value well-regulated financial markets
- Appreciate the need for rules to ensure transparency, trust, and stability.
- Encourage entrepreneurship and investment
- Recognize the role of finance in business growth and job creation.
- Demonstrate ethical use of technology (AI) and data privacy
- Respect confidentiality, integrity, and responsible innovation.
- Promote accountability and trust in financial systems
- Build confidence in banks and financial institutions.



Introductory activity

Scenario

In a certain district in Rwanda, different people and businesses use various organisations to meet their financial needs. Jean, a coffee farmer, keeps his savings in a commercial bank and receives a loan to buy fertilizers and improve production. Claudine, a teacher, contributes money every month to a savings and credit cooperative so that she can borrow money to build a house in the future. Amina insures her shop against fire and theft through an insurance company. Eric sends and receives money using a mobile money service. Meanwhile, a large construction company obtains long-term funds from an investment organisation to expand its activities.

These organisations help people to save money, borrow funds, make payments, insure property, and support investment activities that promote economic development in Rwanda.

Questions

1. What do the institutions mentioned in the scenario mainly deal in?
2. Identify the different organisations used by the people in the scenario.
3. Why do people keep their money in such organisations instead of keeping it at home?
4. How do these organisations support businesses and economic activities in Rwanda?
5. Apart from giving loans, what other services are provided by these organisations?

Financial institutions are organizations that deal with money and other financial services in an economy. They help people, businesses, and the government to save money, borrow funds, make payments, and invest. Financial institutions play an important role in promoting economic growth and development by mobilizing savings and providing credit for production and trade activities.

In Rwanda, financial institutions include commercial banks, the central bank, microfinance institutions, savings and credit cooperatives (SACCOs), insurance companies, and development banks. These institutions help to improve people's standards of living by providing financial services such as loans, savings accounts, money transfers, and insurance services.

This unit introduces learners to the meaning, types, functions, and importance of financial institutions in Rwanda. Learners will also understand how financial institutions contribute to economic development, employment creation, poverty reduction, and financial inclusion in the country.

5.1 Meaning & types of financial institutions

Learning Activity 5.1



Case Study

In Rwanda, different organisations help people and businesses manage money and other financial services. Farmers save money and obtain loans from commercial banks and SACCOs to improve production. Traders insure their businesses against risks through insurance companies. Some workers contribute money to pension funds for future retirement benefits. Investors buy shares and securities through the capital market to raise funds for businesses. Mobile money companies also help people to send and receive money quickly.

These organisations play an important role in mobilising savings, providing credit, facilitating payments, and promoting investment and economic growth.

Questions

1. What is meant by a financial institution?
2. Identify the financial institutions mentioned in the case study.
3. Classify the institutions identified above into:
 - a) Banking financial institutions
 - b) Non-banking financial institutions
4. Explain any four roles played by financial institutions in Rwanda's economy.



Image 5.1.: Some financial institutions in Rwanda

5.1.1 Meaning of Financial institutions

Financial institutions are organisations that facilitate the flow of money in an economy by bringing together savers and borrowers. They act as financial intermediaries between individuals or organisations with surplus funds and those who need funds for investment, production, or consumption purposes.

These institutions collect savings and deposits from the public, especially through savings and fixed deposit accounts, and then lend these funds to businesses, households, and governments in the form of loans and advances. In this way, they help to mobilise savings and channel them into productive economic activities.

Financial institutions also provide a safe place for keeping money, facilitate payments and money transfers, create credit, and promote investment and economic growth. They deal in financial assets and securities such as shares, bonds, treasury bills, and debentures, which may be classified as primary and secondary securities.

Examples of financial institutions in Rwanda include commercial banks, development banks, SACCOs, insurance companies, pension funds, microfinance institutions, and mobile money service providers.

A) Types of financial institutions

Financial intermediaries are generally divided into two main categories:

1. Banking financial intermediaries

These are institutions that accept deposits from the public and provide loans and other banking services. Examples include commercial banks, the central bank, development banks, and SACCOs.

2. Non-banking financial intermediaries

These are institutions that provide financial services but do not operate as traditional banks. Examples include insurance companies, pension funds, microfinance institutions, investment companies, and mobile money service providers.

B) Functions of Financial Intermediaries

Financial intermediaries perform several important functions in a modern economy, including the following:

- Linking savers with borrowers by transferring funds from surplus units to deficit units.
- Mobilising savings and providing loans for investment and consumption.
- Purchasing and selling government securities such as treasury bills and bonds.
- Increasing liquidity by enabling people to easily convert assets into cash.
- Spreading and reducing financial risks through diversification and insurance services.
- Promoting efficient use of scarce resources by financing businesses and development projects.
- Maintaining stability in financial and capital markets.
- Encouraging economic growth through credit creation and investment financing.

Application activity 5.1



Question 1:

Explain the meaning of a financial institution and give two types with examples.

Question 2:

Classify the financial institutions operating in Rwanda and explain one role performed by each category.

Question 3:

Discuss four roles played by financial institutions in Rwanda's economy, using examples from the case study.

Section 1: Banking financial institutions

Banking financial institutions are formal entities that accept deposits, provide loans, and enable payments to help individuals and businesses manage money. They play a central role in storing value, allocating credit, and supporting economic activity, with various types serving different needs.

5.2 Central Bank

Learning Activity 5.2



Case Study

In recent years, Rwanda experienced changes in prices of goods and services, fluctuations in exchange rates, and an increase in the use of digital payments. To maintain economic stability, the National Bank of Rwanda introduced policies such as controlling the amount of money in circulation, supervising commercial banks, and encouraging the use of electronic banking systems.

For example, when inflation increased, the Central Bank raised the bank rate to reduce excessive borrowing and spending in the economy. It also ensured that commercial banks had enough reserves to protect customers' deposits. In addition, the Bank promoted financial inclusion by supporting mobile money services and digital financial transactions across the country.

Through these measures, the Central Bank of Rwanda plays an important role in maintaining price stability, regulating financial institutions, managing the country's currency, and promoting economic development.

Questions

1. Identify the institution described in the case study and explain its main role in Rwanda's economy.
2. Describe any two functions performed by the Central Bank of Rwanda mentioned in the case study.
3. Explain why the Central Bank may increase the bank rate during periods of high inflation.
4. Assess the importance of the Central Bank of Rwanda in promoting financial stability and economic development in Rwanda.

5.2.1 Meaning, functions and roles of the central bank

i. Meaning of central bank

A central bank is the highest monetary authority in a country responsible for managing and controlling the monetary and financial system. It regulates the supply of money and credit, supervises commercial banks and other financial institutions, and helps to maintain economic and financial stability.

The central bank is established by the government and has the sole authority to issue the national currency. In Rwanda, the central bank is the National Bank of Rwanda, commonly known as NBR.



Image 5.2.: National Bank of Rwanda (BNR)

ii. Functions of the Central Bank

1. Banker, fiscal agent and advisor to Government

The central bank keeps government accounts, receives and makes payments on behalf of the government, manages public debt, and advises the government on financial and economic matters. It also handles government transactions involving foreign exchange.

2. Banker to commercial banks

The central bank acts as a bank for commercial banks by keeping their deposits and facilitating transfers between them. It also helps commercial banks settle payments and maintain financial stability.

3. Issuer of currency

The central bank has the monopoly of issuing notes and coins in the country. This ensures a uniform and controlled currency system and helps regulate money supply in the economy.

4. Lender of last resort

The central bank provides emergency loans to commercial banks and other financial institutions facing temporary financial difficulties. This helps to prevent banking failures and maintain confidence in the financial system.

5. Controller of credit and money supply

The central bank controls the amount of money and credit in the economy using monetary policy tools such as:

- Open market operations
- Bank rate policy
- Legal reserve requirements
- Moral suasion

These measures help to control inflation and promote economic stability.

6. Custodian of foreign exchange reserves

The central bank keeps and manages the country's foreign exchange reserves. This helps to maintain a stable exchange rate and support international trade and payments.

7. Clearing and settlement house

The central bank facilitates the clearing, settlement, and transfer of funds between commercial banks. This simplifies financial transactions and improves efficiency in the banking system.

8. Supervisor of financial institutions

The central bank supervises and regulates banks and other financial institutions to ensure they operate safely, fairly, and according to the law.

iii. Role of central banks

Apart from performing traditional banking functions, central banks also play an important role in promoting economic growth and development. Their major roles include the following:

- Assisting the government in **economic planning by providing financial and economic data** needed for policy formulation and national development plans.
- Developing and strengthening the financial sector by encouraging the growth of commercial banks, microfinance institutions, SACCOs, and other financial institutions that mobilise savings and provide credit.
- Directing credit to priority sectors of the economy such as agriculture, industry, education, housing, and infrastructure through monetary policy measures like selective credit control.
- Regulating the supply of money and credit in the economy to achieve important economic objectives such as price stability, low inflation, full employment, and sustainable economic growth.
- Promoting savings and investment through stable interest rate and exchange rate policies, which encourage both local and foreign investment.
- Maintaining stability in the financial system by supervising and regulating financial institutions to ensure public confidence in the banking sector.
- Managing foreign exchange reserves and promoting stable exchange rates to support international trade and balance of payments stability.
- Educating and training bankers and financial experts to improve professionalism, efficiency, and innovation in the financial sector.
- Encouraging financial inclusion by supporting policies and systems that enable more people to access banking and financial services, especially in rural areas.

5.2.2 Monetary policy

i. Meaning and objectives of monetary policy

a) Meaning of monetary policy

Monetary policy refers to the measures and actions taken by a central bank to regulate and control the supply of money, availability of credit, and rate of interest in the economy in order to influence the level of economic activities.

It is one of the major tools used by governments and central banks to achieve economic stability and sustainable development.

Monetary policy is implemented by the central bank through various instruments such as:

- Open market operations
- Bank rate policy
- Legal reserve requirements
- Selective credit control
- Moral suasion

In Rwanda, monetary policy is implemented by the National Bank of Rwanda to maintain financial and economic stability.

b) Objectives of monetary policy

The major objectives of monetary policy include:

- Maintaining price stability by controlling inflation and stabilising the value of money.
- Promoting full employment and reducing unemployment in the economy.
- Achieving a favourable balance of payments position.
- Maintaining stability of the foreign exchange rate.
- Influencing the level and direction of investment and economic activities.
- Encouraging the growth and development of the financial sector.
- Promoting economic growth and development through increased production and investment.
- Encouraging savings and efficient use of financial resources.
- Ensuring availability of credit to productive sectors of the economy at affordable interest rates.
- Maintaining stable and reasonable interest rates in the financial system.
- Supporting government borrowing and creating a stable market for government securities such as treasury bills and bonds.
- Promoting financial stability and public confidence in the banking system.

ii. Tools/ Instruments of monetary policy

The central bank uses different monetary policy instruments to regulate the supply of money and credit in the economy. These tools help control inflation, stabilise prices, encourage investment, and promote economic growth.

1. Bank rate (Discount rate)

This is the interest rate at which the central bank lends money to commercial banks.

- When the central bank increases the bank rate, borrowing becomes expensive for commercial banks, which reduces lending and decreases money in circulation.
- When the bank rate is reduced, commercial banks borrow more easily and extend more loans to the public, increasing money supply in the economy.

2. Open Market Operations (OMO)

This involves the buying and selling of government securities such as treasury bills and bonds by the central bank.

- To increase money supply, the central bank buys securities from the public, injecting money into the economy.
- To reduce money supply, it sells securities to the public, withdrawing money from circulation.

3. Legal Reserve Requirement (LRR)

This is the proportion of deposits that commercial banks are legally required to keep with the central bank.

- Increasing the reserve requirement reduces the amount of money available for lending, thereby reducing money supply.
- Reducing the reserve requirement increases the funds available for loans and increases money circulation.

4. Special deposits

The central bank may require commercial banks to keep additional deposits apart from the legal reserve requirement.

This reduces the amount of money commercial banks can lend to the public and is mainly used during inflationary periods.

5. Moral suasion

This refers to informal appeals, advice, or directives given by the central bank to commercial banks to influence their lending behaviour.

For example, the central bank may request banks to reduce lending during inflation or increase lending during economic recession.

6. Selective credit control

The central bank may direct commercial banks to give priority to certain sectors while restricting credit to others.

This helps encourage productive sectors such as agriculture, industry, and exports while limiting loans for less essential activities.

7. Marginal reserve requirement (Margin requirement)

This is the minimum collateral or security required before banks issue loans.

- Lower margin requirements encourage borrowing and increase money supply.
- Higher margin requirements discourage borrowing and reduce money circulation.

8. Currency reform

Currency reform involves replacing old currency notes with new ones, especially when inflation becomes severe or the currency loses value.

People exchange old notes for new currency within a given period. This helps control excess money in circulation, reduce counterfeit money, and restore confidence in the currency system.

iii. Applicability and limitations of monetary policy tools in developing countries.

Applicability of monetary policy tools in developing countries.

The National Bank of Rwanda uses different monetary policy tools to regulate money supply, maintain price stability, and promote economic growth. Some of the major tools applied in Rwanda include the following:

1. Open Market Operations (OMO)

Open market operations involve the intervention of the National Bank of Rwanda in the money market to either reduce excess liquidity or inject money into the banking system.

This is done to maintain the desired level of reserve money and ensure stability in the financial system. The main instruments used include:

- Treasury bills and government bonds
- Repurchase agreements (repos) and reverse repos
- Standing lending facilities
- Standing deposit facilities
- Refinancing facilities for commercial banks

When the central bank buys securities, money supply increases, while selling securities reduces money circulation.

2. Reserve requirements

Commercial banks and other deposit-taking institutions are required to keep a certain percentage of their deposits as reserves with the central bank.

Reserve requirements are imposed for:

- Monetary control
- Liquidity management
- Prudential and financial stability purposes

Changes in reserve requirements affect the ability of banks to create credit:

- Higher reserve requirements reduce lending and money supply.
- Lower reserve requirements increase lending and money supply.

In Rwanda, the reserve requirement ratio is determined by the central bank according to economic conditions.

3. Foreign exchange intervention

The National Bank of Rwanda intervenes in the foreign exchange market to maintain exchange rate stability and manage international reserves.

The central bank may buy or sell foreign currencies to influence the value of the Rwandan franc and control liquidity in the economy. This helps to:

- Stabilise the exchange rate
- Promote international trade
- Maintain confidence in the currency
- Improve the balance of payments position

Foreign exchange intervention directly affects reserve money and overall money supply in the economy.

Challenges facing the effectiveness of monetary policy in developing countries

The implementation of monetary policy in developing countries like Rwanda faces several challenges that reduce its effectiveness in controlling money supply and achieving economic stability. These challenges include the following:

- Large subsistence sector: A big proportion of the population still operates in the subsistence economy where barter trade and non-monetary transactions are common. This limits the influence of monetary policy because many economic activities do not pass through the banking system.
- Foreign ownership of some commercial banks: Some banks operating in Developing countries are branches or subsidiaries of foreign banks. Their major financial decisions and liquidity support may depend on their parent companies abroad, making local monetary control more difficult.
- Corruption and lack of professionalism: Corruption and weak commitment among some financial institutions can make monetary policy tools such as selective credit control ineffective, as loans may be diverted to non-priority sectors.
- High liquidity preference: Many people prefer keeping money in cash rather than depositing it in banks due to limited financial literacy and low confidence in financial institutions. This reduces savings mobilisation and weakens monetary policy effectiveness.
- Lack of collateral security: Many people lack assets that can be used as collateral for loans. As a result, even when banks are encouraged to lend more, many individuals and businesses fail to qualify for credit.
- Limited investment opportunities: High interest rates, heavy taxation, and an unfavourable business environment discourage borrowing and investment, reducing the effectiveness of expansionary monetary policies.
- Poorly developed money markets: Underdeveloped financial and capital markets limit the effectiveness of tools such as open market operations because there are few securities available for buying and selling.
- Preference for real assets: Many people prefer investing in land, livestock, houses, and other physical assets instead of saving money in banks due to low interest rates on deposits. This limits the amount of funds available in the banking system.
- Conflict between political and monetary objectives: Government policies such as wage increases or price support measures may conflict with monetary policy objectives like controlling inflation.
- Weak supervisory capacity: Limited monitoring and supervision of commercial banks by the central bank may lead to excessive lending and liquidity creation beyond desired levels.
- Economic insecurity and uncertainty: Insecurity and economic instability reduce public confidence, discourage savings, and lower investment activities, thereby affecting monetary policy implementation.

- Poor infrastructure: Inadequate transport, communication, and banking infrastructure in some rural areas limit access to banking services and reduce savings mobilisation.
- Foreign influence and external conditions: International financial institutions such as the International Monetary Fund and the World Bank may impose conditions that influence monetary policy decisions and limit policy independence.

Application activity 5.2



1. Define financial institutions and explain the main types of financial institutions operating in Rwanda.
2. Explain the meaning and functions of the central bank and describe its role in Rwanda's financial system.
3. Identify and explain the tools of monetary policy used by the National Bank of Rwanda (BNR) and how they help control inflation and lending.
4. Evaluate the importance of monetary policy and financial institutions in promoting economic growth and financial stability in Rwanda.

5.3 Commercial banks

Learning Activity 5.3



Scenario

Imagine a group of Senior 5 students in Rwanda discussing how their families and communities use banks. Some students mention that their parents keep money in SACCOs, others say they use mobile money, while a few talk about visiting a bank branch to deposit or withdraw money. The teacher wants to explore what learners already know about commercial banks and their role in the economy.

Questions

1. Why do people choose to keep their money in a commercial bank instead of at home?
2. What services do you think a commercial bank provides to individuals and businesses?
3. If you deposit 100,000 RWF in a bank, what do you expect the bank to do with that money?
4. How do you think a commercial bank differs from the central bank?

Commercial banks in Rwanda today are modern financial institutions offering retail banking services such as deposit accounts, loans, and payment systems. As of 2026, Rwanda has 28 supervised banks, including major players like Bank of Kigali Plc, BPR Bank Rwanda Plc (part of KCB Group), Equity Bank Rwanda Plc, Ecobank Rwanda Plc, Access Bank Rwanda Plc, Guaranty Trust Bank Plc, I&M Bank Rwanda Plc, and NCBA Rwanda Plc.

5.3.1 Meaning and functions of commercial banks

i. Meaning of commercial banks

A commercial bank is a financial institution that accepts deposits from the public, safeguards money, and provides loans and other financial services to individuals, businesses, and government institutions with the aim of making profit. Commercial banks act as financial intermediaries by mobilising savings from surplus units and lending them to deficit units for investment and consumption purposes.

Commercial banks play an important role in the economy by promoting trade, investment, savings, capital formation, employment creation, and economic development. Examples in Rwanda include Bank of Kigali, Equity Bank Rwanda, and I&M Bank Rwanda.

ii. Functions of commercial banks

1. Credit creation

Commercial banks create credit by accepting deposits from customers and lending part of these deposits to borrowers in form of loans and advances. Through this process, banks increase the money supply in the economy. Banks earn profit by charging interest on loans while paying lower interest on deposits.

Credit creation promotes investment, production, trade, and economic growth.

2. Provision of credit facilities

Commercial banks provide different forms of credit to individuals, businesses, and organisations. These include: Loans, Advances, Bank overdrafts, Cash credits, Mortgages and Trade finance

These credit facilities help customers to start businesses, expand production, buy assets, and meet short-term financial needs.

3. Transmission and transfer of money

Commercial banks provide safe and convenient methods of transferring money from one place to another within the country and internationally. Money transfer services are carried out through:

- Cheques
- Electronic funds transfer (EFT)
- Standing orders
- Direct debits
- Mobile and internet banking

- Telegraphic transfers
- SWIFT international transfers

Banks also facilitate international trade by making payments abroad on behalf of customers and exchanging foreign currencies.

4. Safe custody of valuables

Commercial banks provide secure facilities for safeguarding customers' valuables and important documents such as: **land titles, academic certificates, legal documents, jewellery and wills and contracts**

Some banks provide safety deposit boxes and strong rooms for maximum security.

5. Opening and operating different accounts

Commercial banks open and manage various types of accounts for customers according to their financial needs.

a) Current accounts

These are accounts mainly used by business people and organisations for frequent transactions. Features include:

- Use of cheques
- No restriction on withdrawals
- Direct salary deposits
- Standing orders and direct debits
- Usually little or no interest paid

b) Savings accounts

These accounts encourage customers to save money regularly. Features include:

- Small minimum deposits
- Interest earned on savings
- Limited withdrawals
- Suitable for individuals and small savers.

c) Fixed Deposit accounts

Under these accounts, customers deposit money for a fixed period without withdrawing it before maturity. Features include: **Higher interest rates, fixed repayment period, encourages long-term savings and Withdrawal before maturity may attract penalties.**

6. Advisory services

Commercial banks provide financial and business advice to their customers. These advisory services include: **Investment guidance, foreign exchange advice, financial planning, business management consultancy and trusteeship services.**

Banks may charge fees for some professional advisory services.

7. Provision of other financial products

Commercial banks offer additional financial products and services to meet customer needs. These include:

- Insurance services, Mortgage financing,
- Debit and credit cards,
- Mobile banking, Internet banking,
- Foreign exchange services and Agency banking.
- These services improve financial inclusion and convenience.

8. Encouraging savings

Commercial banks mobilise savings from the public by providing safe places for keeping money and paying interest on deposits. **Savings mobilisation increases capital formation and promotes investment in the economy.**

9. Facilitating trade and commerce

Commercial banks support local and international trade through: **issuing letters of credit, foreign exchange transactions, trade financing, money transfers and discounting bills of exchange.**

These services simplify business transactions and promote economic activities.

10. Creation of employment opportunities

Commercial banks employ many people such as accountants, managers, cashiers, ICT specialists, auditors, and security personnel. They therefore help to reduce unemployment levels in the economy.

11. Collection of payments

Commercial banks collect payments on behalf of customers such as: dividends, salaries, rent, utility bills, school fees and taxes

This simplifies financial transactions and saves time for customers.

12. Promotion of economic development

By mobilising savings, extending credit, encouraging investment, and facilitating trade, commercial banks contribute greatly to economic growth and development in the country.

5.3.2 Assets and liabilities of commercial banks

Commercial banks prepare balance sheets showing their financial position. A balance sheet contains two main parts namely assets and liabilities.

Meaning of assets

Assets are all the possessions, properties, and claims owned by a commercial bank or owed to the bank by other institutions and customers. Assets are important because they generate income and determine the financial strength of the bank.

Examples of assets of commercial banks

1. Cash in hand and reserves with the central bank

Commercial banks keep part of their money as cash in their vaults and as reserves with the National Bank of Rwanda. These reserves help banks meet customers' daily withdrawals and satisfy legal reserve requirements.

2. Deposits with other banks and financial institutions

Commercial banks keep some funds with other commercial banks and financial institutions for transaction and security purposes.

3. Loans and advances to customers

These are funds lent to individuals, businesses, and organisations in form of: Loans, Overdrafts, Mortgages and Advances

Loans are major income-earning assets because banks charge interest on them.

4. Investments

Commercial banks invest in government securities such as treasury bills and bonds, shares, and other long-term investments to earn profits.

5. Fixed assets

These are physical properties owned by banks such as: office buildings, furniture, computers, motor vehicles and Automated Teller Machines (ATMs)

6. Foreign exchange assets

Commercial banks hold foreign currencies and foreign financial assets used in international trade and foreign exchange transactions.

Meaning of liabilities

Liabilities are financial obligations or claims by outsiders against the commercial bank. They represent money owed by the bank to customers, government, or other financial institutions.

Examples of liabilities of commercial banks

1. Customer deposits

These are the largest liabilities of commercial banks and include: **current account deposits and savings account deposits.**

2. Fixed deposit accounts

The bank must repay these deposits whenever customers demand them according to account conditions.

3. Deposits from other banks

Commercial banks may borrow or receive deposits from other banks and financial institutions.

4. Government deposits

Government ministries, departments, and agencies keep money in commercial banks, which becomes a liability to the bank.

5. Borrowings from the central bank

Commercial banks may borrow money from the National Bank of Rwanda to solve temporary liquidity shortages.

6. Shareholders' capital

This refers to money invested by shareholders to start and operate the bank.

7. Retained profits

These are profits kept by the bank for future expansion instead of distributing them to shareholders.

5.3.3 Liquidity and profitability in commercial banks

Commercial banks aim at achieving two important objectives namely liquidity and profitability. These objectives help banks operate efficiently and remain financially stable.

i. Meaning of liquidity

Liquidity is the ability of a commercial bank to meet customers' withdrawals and other short-term obligations quickly without loss.

Liquid assets are assets that can easily be converted into cash. Examples include:

- Cash in hand
- Reserves with the central bank
- Treasury bills
- Deposits with other banks

Liquidity is important because it helps banks: Meet withdrawal demands and Gain public confidence

- Avoid financial problems

However, highly liquid assets earn low profits.

ii. Meaning of profitability

Profitability is the ability of a commercial bank to earn profits from its operations.

Banks earn profits through:

- Interest on loans
- Investments
- Service charges
- Foreign exchange transactions

Profitability helps banks to: Pay workers and shareholders, Expand business operations and Improve banking services

However, highly profitable investments are usually less liquid.

iii. Liquidity profitability dilemma

Liquidity and profitability conflict with each other.

- Keeping too much cash improves liquidity but reduces profits.
- Investing most funds in long-term loans increases profits but reduces liquidity.

Commercial banks must therefore balance the two objectives carefully.

How commercial banks' balance liquidity and profitability

Commercial banks face a constant challenge in balancing **liquidity** the ability to meet short-term obligations and **profitability**, which ensures sustainable growth and returns for shareholders. Because these objectives often conflict, banks must act strategically to achieve an optimal balance. To manage this trade-off, commercial banks employ several measures:

Strategies used to balance liquidity and profitability

- **Maintaining cash reserves**
Banks keep a specific percentage of deposits in cash form (known as the **cash reserve ratio**) to meet customer withdrawal demands. This ensures liquidity, while the remaining funds are lent out to generate interest income, supporting profitability.
- **Holding reserves at the central bank**
Commercial banks maintain mandatory reserves with the central bank. In times of liquidity shortages, they can draw on these reserves to meet depositor needs, thus safeguarding stability.
- **Investing in liquid assets**
Banks invest in short-term securities such as treasury bills, government bonds, and interbank placements. These assets earn returns while remaining easily convertible to cash, ensuring both liquidity and profitability.
- **Diversifying loan portfolios**
By offering short-term, medium-term, and long-term loans, banks create a steady inflow of repayments. This diversification helps maintain liquidity while earning interest income over time.
- **Setting minimum deposit balances**
Customers are required to maintain a minimum balance in their accounts. This accumulated amount strengthens the bank's liquidity position and ensures funds are available to meet withdrawal demands.
- **Borrowing from the central bank**
When facing temporary liquidity constraints, banks can borrow from the central bank, which acts as the **lender of last resort**. This facility prevents disruptions in customer service and maintains confidence in the banking system.
- **Engaging in non-banking investments**
To enhance profitability, banks may invest in sectors such as real estate, transport, and manufacturing. These ventures generate additional income streams that supplement core banking revenues.

➤ **Charging service fees**

Banks impose fees for services like ATM withdrawals, cheque clearing, account statements, and safe deposit boxes. These charges increase non-interest income, improving overall profitability.

➤ **Discounting bills of exchange**

Banks purchase bills of exchange at a discount and earn profits upon maturity. This short-term investment supports liquidity while generating income.

➤ **Interest rate management**

By paying lower interest rates on deposits and charging higher rates on loans, banks maintain a favorable interest margin that boosts profitability. They also demand collateral to minimize credit risk and protect liquidity.

Summary

Balancing liquidity and profitability is a delicate act that requires prudent financial management, regulatory compliance, and strategic investment decisions. A well-managed bank ensures it can meet customer needs promptly while generating sustainable profits for long-term growth.

5.3.4 Credit creation by commercial banks

Probing Questions on Credit Creation

- » What is meant by credit creation?
- » Why do commercial banks create credit?
- » What factors determine the ability of commercial banks to create credit?
- » What difficulties are faced in the process of credit creation by commercial banks in Rwanda?



i. Meaning of credit creation

Credit creation is the process by which commercial banks create additional money supply by lending out deposits to borrowers. It occurs when banks issue loans and advances, and the borrowed money is redeposited into the banking system, leading to multiple expansion of deposits.

In simple terms, credit creation means that banks create money through lending activities.

Commercial banks are required to keep a certain percentage of deposits as reserves to meet customers' daily withdrawals. This reserve is called the cash reserve ratio or liquidity ratio.

ii. Assumptions of credit creation

Credit creation is based on the following assumptions:

1. There is one banking system or one bank with many cooperating branches.
2. A fixed cash reserve ratio is maintained.

3. Banks are willing to lend excess reserves.
4. Borrowers are willing to borrow from banks.
5. All payments are made through banks using cheques or electronic transfers.
6. Money borrowed is redeposited into the banking system.
7. There is public confidence in the banking system.
8. There is political and economic stability.
9. There is no excessive government interference.
10. Banks operate efficiently without corruption or loan default problems.

iii. Process of credit creation

Assume there is one commercial bank with an initial deposit of **100 million FRW** and a cash reserve ratio of **20%**.

Formula for credit multiplier

The credit multiplier is calculated as:

Credit Multiplier = $1 \div \text{Cash Reserve Ratio}$, Since the reserve ratio is 20%:

$$\text{Credit Multiplier} = 1 \div 0.20 = 5$$

Therefore, the banking system can create total deposits equal to five times the initial deposit.

Illustration of credit creation

Customer	New Deposit (FRW)	Reserve 20% (FRW)	New Loan (FRW)
A	100,000,000	20,000,000	80,000,000
B	80,000,000	16,000,000	64,000,000
C	64,000,000	12,800,000	51,200,000
D	51,200,000	10,240,000	40,960,000
E	40,960,000	8,192,000	32,768,000
F	32,768,000	6,553,600	26,214,400
...	...	...	...
Total Final Deposits	500,000,000		

Explanation of the process

Customer A deposits **100 million FRW** in the bank. The bank keeps **20 million FRW** as reserves and lends out **80 million FRW** to Customer B.

Customer B deposits the **80 million FRW** into the banking system. The bank keeps **16 million FRW** as reserves and lends out **64 million FRW** to Customer C.

Customer C also deposits the borrowed money back into the bank. The process continues repeatedly until no excess reserves remain for lending.

As loans are redeposited, new deposits are created many times, leading to multiple expansion of money supply in the economy.

Experimental version

Final Deposit Calculation

The final total deposits are calculated as:

Final Deposits = Initial Deposit × Credit Multiplier

Thus: 500,000,000 = 100,000,000 × 5

Therefore, an initial deposit of **100 million FRW** can generate total deposits of **500 million FRW** under a 20% reserve ratio.

Key Notes on Credit Creation

- Commercial banks do not lend all deposits; they keep part as reserves.
- The lower the reserve ratio, the greater the ability of banks to create credit.
- Credit creation increases money supply in the economy.
- Credit creation promotes investment, trade, production, and employment.
- Excessive credit creation may lead to inflation.

Central banks control credit creation through monetary policy instruments such as:

- Bank rate policy
- Open market operations
- Legal reserve requirements
- Selective credit controls

iv. Factors determining the ability of commercial banks to create credit

The ability of commercial banks in Rwanda to create credit depends on several economic, institutional, and social factors. These factors influence how much money banks can lend out and how effectively they can support economic growth.

Key Determinants

- **Size of customer deposits**
The larger the volume of deposits mobilized from the public, the greater the capacity of banks to extend loans and create credit.
- **Cash reserve ratio**
The central bank requires commercial banks to hold a certain percentage of deposits as reserves. A higher ratio reduces lending capacity, while a lower ratio increases credit creation.
- **Public willingness to borrow**
The demand for loans depends on people's confidence in their ability to repay and their willingness to invest. Low borrowing appetite limits credit expansion.
- **Level of economic activity**
A vibrant economy with active trade, investment, and production increases demand for loans and enhances banks' ability to create credit.
- **Interest rates**
High interest rates discourage borrowing, while moderate rates encourage loan uptake. The spread between deposit and lending rates also affects profitability and credit creation.

Experimental version

- **Public confidence in banking**
Trust in banks encourages people to deposit savings and borrow funds. Lack of confidence reduces deposits and limits credit creation.
- **Availability of collateral security**
Banks require collateral such as land, property, or business assets to secure loans. Limited collateral reduces access to credit and constrains lending.
- **Government monetary policy**
Policies such as credit controls, interest rate ceilings, and lending restrictions directly affect banks' ability to create credit.
- **Level of technology and banking development**
Modern banking technologies (e.g., mobile banking, digital platforms) improve efficiency, expand outreach, and enhance credit creation capacity.
- **Political and economic stability**
Stability fosters confidence among savers and borrowers, while instability discourages deposits and borrowing, limiting credit creation.
- **Regulatory environment**
Strict regulations on lending, capital adequacy, and risk management can restrict credit creation, while supportive policies enhance it.
- **Foreign exchange availability**
Access to foreign currency reserves enables banks to finance international trade and investment, indirectly boosting credit creation.

v. Limitations of commercial banks in creating credit in developing countries (Rwanda)

While commercial banks play a crucial role in mobilizing savings and extending credit, their ability to create credit in Developing countries faces several challenges. These limitations stem from structural, economic, and social factors that constrain their operations.

Key limitations

- **Low savings mobilization**
Widespread poverty reduces disposable income, making it difficult for banks to attract deposits. Limited savings mean fewer funds available for lending.
- **High illiteracy levels**
Many people lack financial literacy and knowledge of banking operations. This discourages them from keeping money in banks, reducing the deposit base.
- **Government interference**
Excessive regulation and political influence can hinder banks' independence, limiting their flexibility in credit creation and lending decisions.
- **Dominance of subsistence economy**
A large portion of Rwanda's population engages in subsistence farming, which generates low incomes. This reduces the ability to save or access credit.

➤ **Poor creditworthiness of borrowers**

Many borrowers default on loans, leading to losses for banks. Weak enforcement of contracts and lack of reliable credit history systems exacerbate this problem.

➤ **Liquidity preference**

People often prefer holding cash rather than depositing it in banks, reducing available funds for lending.

➤ **Low demand for loans**

Many potential borrowers lack collateral such as land or property. Without security, banks are reluctant to extend credit, limiting loan uptake.

➤ **Impact of inflation**

Inflation erodes the value of money, discouraging savings. People prefer investing in tangible assets rather than depositing money in banks.

➤ **Urban-centric banking**

Most banks are concentrated in urban areas, leaving rural populations underserved. This creates competition for a limited customer base and reduces overall deposits.

➤ **Poor infrastructure**

Inadequate roads, electricity, and telecommunications hinder banking operations, especially in rural areas. This limits access to banking services and credit creation.

➤ **High operational costs**

Running branches in remote areas is costly, discouraging banks from expanding services to rural communities where savings potential is low.

➤ **Weak financial inclusion**

A significant portion of the population remains outside the formal banking system. Limited access to banking services reduces deposits and credit creation capacity.

5.3.5 Role of commercial banks in Rwanda's economic development

Commercial banks are vital institutions in Rwanda's financial system. Beyond offering traditional banking services such as deposits, withdrawals, and loans, they play a significant role in promoting economic growth and development. Their contributions extend across employment, investment, trade, and policy implementation.

Key roles of commercial banks

➤ **Employment creation**

Commercial banks generate employment opportunities for a wide range of professionals — auditors, accountants, managers, tellers, IT specialists, drivers, security guards, and cleaners. This helps reduce unemployment and improves living standards in Rwanda.

➤ **Mobilization of Savings**

By offering interest on deposits, banks encourage individuals and businesses to save. These savings are then channeled into productive investments, fueling entrepreneurship and industrial growth, which enhances national economic development.

- **Provision of Credit Facilities**
Banks provide loans to individuals, businesses, and government entities to finance projects such as agriculture, manufacturing, and infrastructure. This access to credit stimulates production, innovation, and job creation.
- **Financial Advisory Services**
Commercial banks offer technical and professional advice to investors and entrepreneurs. This guidance helps clients make informed decisions on viable projects, risk management, and efficient business operations, contributing to a more stable economy.
- **Tax Contribution**
Banks pay corporate taxes and other levies to the government from their profits. These revenues support public expenditure on education, healthcare, and infrastructure, strengthening Rwanda's fiscal capacity.
- **Facilitation of Trade**
Through foreign exchange services, letters of credit, and international money transfers, banks promote both domestic and international trade. This enhances the flow of goods, technology, and expertise, boosting Rwanda's competitiveness in global markets.
- **Implementation of Monetary Policy**
Acting as intermediaries between the central bank and the public, commercial banks help implement monetary policies such as interest rate adjustments and credit controls. This supports macroeconomic stability and helps manage inflation.
- **Efficient Payment Systems**
Banks facilitate quick and secure transactions through cheques, standing orders, mobile banking, and electronic transfers. These systems enhance convenience and reduce the need for physical cash circulation.
- **Custody of Valuables and Documents**
Commercial banks provide safe deposit boxes and secure storage for valuable items and documents such as certificates, wills, and property titles, ensuring safety and peace of mind for customers.
- **Promotion of Financial Inclusion**
By expanding branch networks and digital banking services, commercial banks reach rural and underserved populations. This inclusion empowers citizens to participate in formal financial systems, fostering equitable economic growth.
- **Support for Government Development Programs**
Banks collaborate with the government to finance national development initiatives, including infrastructure projects, small and medium enterprise (SME) support, and agricultural modernization.

Summary

Commercial banks are not only financial intermediaries but also engines of economic transformation. Through their diverse roles from mobilizing savings to financing development projects — they contribute significantly to Rwanda's vision of becoming a middle-income, knowledge-based economy.

The Role of foreign commercial banks in Rwanda

Meaning of foreign commercial banks

Foreign commercial banks are commercial banks whose headquarters are located outside the host country but operate branches or subsidiaries in other countries. In Rwanda, these banks play an important role in financial development, trade, investment, and economic growth.

Examples of foreign commercial banks operating in Rwanda in 2026 include:

- | | |
|----------------------|----------------------|
| » Equity Bank Rwanda | » Access Bank Rwanda |
| » Ecobank Rwanda | » GTBank Rwanda |
| » I&M Bank Rwanda | » NCBA Bank Rwanda |

Foreign commercial banks have both positive and negative effects on the development of Rwanda.

a) Positive role of foreign commercial banks in Rwanda

1. Promoting foreign investment

Foreign commercial banks attract foreign investors by making transfer of funds easier, faster, and safer. This increases investor confidence because their money and investments are protected through international banking systems.

2. Increasing foreign exchange

Foreign banks help to increase the availability of foreign exchange needed for international trade. Importers and exporters can easily obtain foreign currencies such as dollars, euros, and pounds for business transactions.

This promotes international trade and economic growth.

3. Promoting modern banking technology

Foreign commercial banks introduce modern banking technologies and innovations such as:

- » Automated Teller Machines (ATMs)
- » Mobile banking
- » Internet banking

- » Agency banking
- » Electronic funds transfer
- » Visa and debit cards

These technologies improve efficiency and quality of banking services in Rwanda.

4. Increasing competition and efficiency

Foreign commercial banks increase competition in the banking sector. Competition forces local banks to improve their services, customer care, and financial products.

As a result, customers benefit from:

- » Better banking services
- » Faster transactions

- » Lower transaction costs
- » Improved financial products

5. Creating employment opportunities

Foreign commercial banks provide jobs to Rwandans such as: managers, accountants, auditors, loan officers, cashiers, ICT specialists and security personnel

Employees gain modern banking skills and experience which improve the quality of labour in the country.

6. Mobilising savings

Foreign banks encourage people to save money through different savings accounts and digital banking services.

Increased savings help to:

- » Promote capital formation
- » Increase investment
- » Expand economic activities

7. Facilitating International Trade

Foreign commercial banks simplify international trade through services such as:

- » Letters of credit
- » Foreign exchange transactions
- » Bank drafts

- » International money transfers
- » Trade financing

These services support importers and exporters in Rwanda.

8. Providing loans and credit facilities

Foreign commercial banks provide loans to businesses and individuals for: agriculture, trade, manufacturing, construction, transport and small and medium enterprises (Small and Medium-sized Enterprises (SMEs)). This promotes production, employment, and economic development.

9. Increasing government revenue

Foreign banks contribute to government revenue through payment of: Corporate taxes, license fees, employee taxes and Value Added Tax (VAT) on services

The government uses this revenue to provide public services.

10. Promoting financial inclusion

Foreign commercial banks extend banking services to more people through:

- | | |
|-------------------|---------------------------------|
| » Digital banking | » International money transfers |
| » Mobile banking | » Trade financing |

This helps more people access financial services.

11. Supporting community development

Many foreign commercial banks participate in corporate social responsibility programmes by:

- | | |
|--------------------------------------|--|
| » Sponsoring education programmes | » Supporting sports activities |
| » Supporting health services | » Assisting vulnerable groups during disasters |
| » Promoting environmental protection | |

12. Improving infrastructure development

Foreign commercial banks finance infrastructure projects such as: Roads, Housing, Energy projects and Communication networks

This supports national development.

b) Negative role of foreign commercial banks in Rwanda

1. Repatriation of profits

Foreign commercial banks often transfer part of their profits back to their home countries. This reduces the amount of capital remaining in Rwanda for domestic investment.

2. Outcompeting local banks

Foreign banks usually have: more capital, better technology, skilled manpower and strong international connections

This may make local banks fail to compete effectively, leading to loss of customers and reduced growth of local financial institutions.

3. Urban concentration

Most foreign commercial banks are concentrated in urban areas such as Kigali and other towns.

This creates regional imbalances because many rural areas remain underserved by banking services.

4. Promoting rural urban migration

People may move from rural areas to towns in search of better banking services and employment opportunities offered by foreign banks.

This increases rural urban migration.

5. Employment discrimination

Some foreign commercial banks may prefer expatriates for top management positions while local workers occupy lower-level jobs.

This limits promotion opportunities for local skilled workers.

6. Conflict with government policies

Foreign commercial banks may follow policies and decisions from their headquarters abroad which may sometimes conflict with the monetary and financial policies of the government of Rwanda.

7. High interest rates and strict conditions

Some foreign banks charge high interest rates and demand strict collateral requirements before giving loans.

This limits access to credit by small businesses and low-income earners.

8. Encouraging dependence on foreign technology

Heavy reliance on foreign banking systems and technologies may slow the development of local banking innovations and indigenous technology.

9. Unequal service provision

Some foreign banks may favour multinational companies and large businesses more than small local enterprises in providing loans and financial services.

10. Risk of external financial shocks

Foreign commercial banks may be affected by economic crises in their home countries. Such problems can negatively affect banking operations and financial stability in Rwanda.

Summary

Foreign commercial banks play an important role in the economic development of Rwanda by promoting investment, trade, savings, employment, and modern banking services. However, they may also create challenges such as profit repatriation, stiff competition to local banks, and regional imbalances.

The government and the National Bank of Rwanda should therefore regulate and supervise foreign commercial banks properly to ensure that their activities contribute positively to national development.

5.3.6 Limitations of commercial banks in developing countries

Although commercial banks contribute significantly to economic growth in developing countries, they face numerous challenges that limit their efficiency and ability to expand credit creation. These limitations include:

Key limitations

- **High Liquidity Preference in the citizens**
Many people prefer to hold cash rather than deposit it in banks. This reduces the volume of deposits available for lending, limiting banks' profitability and credit creation capacity.
- **Unfavourable Taxation Policies**
Heavy taxes imposed on commercial banks reduce their profit margins, discouraging expansion and innovation in banking services.
- **Shortage of Skilled Personnel**
Banks often struggle to recruit well-trained, competent, and trustworthy employees. Cases of corruption and mismanagement further weaken efficiency and erode public trust.
- **Stiff Competition**
Intense competition forces banks to lower lending rates, raise deposit rates, and introduce multiple services. This increases operational costs and reduces profitability, sometimes leading to closures.
- **Low Savings Levels**
Widespread poverty limits the ability of people to save. Insufficient deposits weaken banks' ability to sustain operations and extend credit.
- **Subsistence Economy**
A large portion of the population engages in subsistence farming, generating little surplus income. This reduces both savings and demand for banking services.
- **Fear of Borrowing**
Many people are reluctant to take loans due to limited investment opportunities, fear of indebtedness, and lack of collateral. This reduces loan demand and banks' profitability.

- **Poor Infrastructure**

Weak road networks, unreliable electricity, and poor internet connectivity hinder banking operations, especially in rural areas, limiting outreach and efficiency.

- **Bureaucracy and Inefficiency**

Arrogance and inefficiency among some bank officials discourage potential customers, reducing trust and limiting banking penetration.

- **Discouraging Banking Conditions**

High minimum deposit requirements and complex account-opening procedures discourage low-income earners from accessing banking services.

- **High Marginal Propensity to Consume (MPC)**

With most income spent on consumption, savings remain low. This limits banks' ability to mobilize funds for lending.

- **Government Interference**

Policies such as fixing high interest rates to control inflation often conflict with banks' profit motives, discouraging borrowing and reducing credit creation.

- **Political and Economic Insecurity**

Insecurity and instability discourage people from depositing money in banks, reducing liquidity and weakening operations.

- **Limited Financial Inclusion**

Many rural populations remain outside the formal banking system due to lack of awareness, infrastructure, and tailored financial products.

5.3.7 Difference between central bank and commercial banks

Differences between a central bank and commercial banks

A central bank and commercial banks are both important financial institutions in the economy, but they differ in their objectives, ownership, functions, and operations.

In Rwanda, the central bank is the National Bank of Rwanda while examples of commercial banks include Bank of Kigali, Equity Bank Rwanda, and KCB Bank Rwanda.

The following are the major differences between a central bank and commercial banks.

Central Bank	Commercial Banks
A central bank is established mainly to serve public interest and ensure economic stability.	Commercial banks are established mainly to earn profits.
A central bank is owned and controlled by the government.	Commercial banks may be owned by the government, private individuals, or foreign investors.
A central bank is responsible to the government and parliament.	Commercial banks are responsible to shareholders and owners.
A central bank controls, supervises, and regulates other banks.	Commercial banks are supervised and regulated by the central bank.

A central bank is the only institution legally allowed to issue currency notes and coins.	Commercial banks cannot issue currency.
A central bank formulates and implements monetary policy.	Commercial banks only follow the monetary policies set by the central bank.
A central bank acts as the banker to the government by keeping government accounts and managing public debt.	Commercial banks mainly keep accounts for individuals, firms, and organisations.
A central bank acts as a banker's bank by keeping reserves of commercial banks.	Commercial banks keep deposits for the public.
A central bank acts as the lender of last resort to commercial banks facing liquidity problems.	Commercial banks borrow from the central bank during financial difficulties.
A central bank generally does not deal directly with the public.	Commercial banks deal directly with the public on a daily basis.
A central bank aims at maintaining price stability and controlling inflation.	Commercial banks aim at maximising profits through loans and other services.
A central bank manages the country's foreign exchange reserves and exchange rate policies.	Commercial banks mainly provide foreign exchange services to customers under central bank regulations.
A central bank does not compete with commercial banks for customers.	Commercial banks compete among themselves for customers and profits.
A central bank keeps the national reserves of gold and foreign currencies.	Commercial banks keep customer deposits and business funds.
A central bank influences credit creation through reserve requirements and interest rate policies.	Commercial banks create credit by lending out customer deposits.
A central bank promotes financial stability in the entire economy.	Commercial banks mainly provide banking and financial services to customers.
A central bank operates at the national level.	Commercial banks operate through branches in towns and rural areas.
A central bank may conduct open market operations by buying and selling government securities.	Commercial banks may buy government securities mainly for investment purposes.

Application activity 5.3



1. Define commercial banks and explain their main functions in the economy.
2. Distinguish between the assets and liabilities of commercial banks and give examples of each.
3. Explain the liquidity and profitability dilemma faced by commercial banks and describe how they balance the two.
4. Evaluate the role of commercial banks in promoting economic growth, financial inclusion, and national development.

Section 2: Non-banking financial institutions

5.4 Specialized Banks, microfinance institutions, insurance companies and savings and credit cooperatives (SACCOs).

Learning Activity 5.4



1. Are there non-banking financial institutions operating in Rwanda? If yes, mention some that you know.
2. Discuss how specialized banks, microfinance institutions, insurance companies, and SACCOs contribute to economic stability and support households and businesses during financial difficulties.
3. Evaluate the importance of savings, credit access, and risk protection provided by SACCOs, microfinance institutions, insurance companies, and specialized banks in promoting economic development.

Various financial institutions play an important role in promoting economic growth and improving the welfare of individuals and businesses.

- Specialized banks provide long-term finance to sectors such as agriculture, housing, education, and industry.
- Microfinance institutions provide small loans and financial services to low-income earners and small entrepreneurs.
- Insurance companies protect individuals and businesses against financial risks and uncertainties.
- Savings and Credit Cooperative Organisations (SACCOs) mobilise savings and provide affordable loans to members.
- Together, these institutions promote:
- Financial inclusion

- Savings mobilisation
- Investment
- Employment creation
- Poverty reduction
- Sustainable economic development

5.4.1 Meaning and characteristics of non-banking financial institutions (NBFIs)

i. Meaning of Non-Banking Financial Institutions (NBFIs)

Non-banking financial institutions are financial organisations that mobilise funds or savings from the public but do not create credit like commercial banks. Instead, they provide specialised financial services such as savings mobilisation, investment financing, insurance, and risk management.

ii. Key characteristics of NBFIs

Non-banking financial institutions:

Non-banking financial institutions (NBFIs) have several important characteristics that make them different from commercial banks. These institutions mainly provide specialised financial services to individuals, businesses, and communities.

- **Do not create money through lending:**
Unlike commercial banks, NBFIs do not create money by issuing demand deposits. They mainly use funds collected from savings, premiums, shares, or investments to provide financial services.
- **Provide long-term financing and investment services:**
NBFIs offer long-term loans and investment opportunities for activities such as housing, agriculture, business expansion, and infrastructure development.
- **Offer insurance and savings facilities:**
Many NBFIs help people to save money safely and provide insurance services that protect individuals and businesses against risks such as accidents, illness, fire, and loss of property.
- **Operate in different sectors of the economy:**
NBFIs serve various sectors including agriculture, trade, transport, construction, education, and small-scale industries, thereby supporting economic activities.
- **Complement commercial banks by serving specialised financial needs:**
They work alongside commercial banks by providing financial services that banks may not easily offer, especially to small businesses, farmers, cooperatives, and low-income earners.
- **Promote economic growth and national development:**
By mobilising savings and providing credit and investment services, NBFIs encourage production, employment creation, income generation, and capital formation in the economy.

- **Improve financial inclusion, especially among low-income earners and rural populations:**

NBFIs extend financial services to people who may not easily access commercial banks, particularly those in rural areas and informal sectors, helping to reduce poverty and improve living standards.

5.4.2 Categories and types of non-Banking financial institutions in Rwanda

Categories of non-banking financial institutions in Rwanda

i. Specialized banks

Meaning of specialized banks

Specialized banks are financial institutions that provide long-term finance to specific sectors such as agriculture, industry, housing, education, and infrastructure development.

Case Study: Development Bank of Rwanda (BRD)

The Development Bank of Rwanda (BRD) was established in 1967 as a long-term financial institution aimed at financing national development projects.

By April 2011:

- Total assets were approximately 72 billion FRW.
- Shareholders' equity was about 25 billion FRW.

Following restructuring, recapitalisation, and government support after the 1994 Genocide against the Tutsi, BRD expanded significantly.

As of **31/12/2025**, BRD's total assets stood at approximately **1.015 trillion FRW**, while shareholders' equity was about **164.37 billion FRW** as at 31 December 2025. (https://rse.rw/uploads/reports/1774982734_333aad6034fc3d05b611.pdf?utm_source=chatgpt.com)

Major sectors financed by BRD

The Development Bank of Rwanda (BRD) finances key sectors of the economy to support sustainable development and improve the welfare of citizens.

- **Agriculture and agribusiness:**
BRD provides loans and financial support to farmers, cooperatives, and agribusiness enterprises to increase agricultural production, improve food security, and promote exports.
- **Renewable energy:**
The bank finances projects such as solar energy, hydro power, and biogas to increase access to clean and affordable energy in both urban and rural areas.
- **Manufacturing and industry:**
BRD supports industries and manufacturing firms to increase production, add value to local products, promote exports, and reduce dependence on imports.
- **Affordable housing:**
The bank provides financing for housing projects to help citizens access decent and affordable homes while supporting urban development.

- **Education and student loans:**

BRD offers education financing and student loans to support access to higher education and skills development for national human resource growth.

- **ICT and innovation:**

The bank finances information and communication technology projects and innovative businesses to promote digital transformation and technological advancement.

- **Small and medium-sized enterprises (SMEs) and entrepreneurship:**

BRD supports small and medium enterprises by providing capital and business financing to encourage entrepreneurship, innovation, and business expansion.

- **Infrastructure and rural development:**

The bank finances roads, water projects, transport, and rural development activities that improve public services and economic activities in different regions.

Importance of specialized banks

Specialized banks play an important role in the economic development of a country by providing financial services to priority sectors.

- **Promote economic growth:** They provide financial resources needed for production, trade, and investment, which increases national income and economic activities.
- **Support industrialisation:** Specialized banks finance industries and manufacturing projects that help transform raw materials into finished goods and promote industrial development.
- **Finance long-term development projects:** They provide long-term loans for projects such as infrastructure, housing, agriculture, and energy which require large amounts of capital.
- **Encourage investment:** By providing credit facilities and investment support, specialized banks motivate individuals and businesses to invest in productive activities.
- **Create employment opportunities:** The projects financed by specialized banks lead to the establishment and expansion of businesses, which generate jobs for the population.
- **Promote financial inclusion:** They extend financial services to groups and sectors that may not easily access commercial bank financing, especially rural communities and small businesses.

ii. Insurance companies

Meaning of insurance companies

Insurance companies are financial institutions that protect individuals and businesses against financial losses and risks in exchange for premium payments.

Compensation roles of insurance companies (NBFIs)

- **Accidents:** Provide financial support to cover medical bills, repairs, or loss of income after accidents.
- **Illness:** Pay for hospital treatment, medication, and other health-related expenses.
- **Death:** Offer life insurance benefits to dependents, ensuring family security after the loss of a breadwinner.
- **Fire:** Compensate for destruction of homes, businesses, or assets caused by fire outbreaks.
- **Theft:** Cover losses from burglary, robbery, or stolen property.
- **Natural disasters:** Provide relief for damages caused by floods, earthquakes, or other calamities.
- **Property damage:** Pay for repairs or replacement of damaged assets, helping households and businesses recover quickly.

Insurance companies operate on the principle of risk pooling, where many people contribute premiums into a common fund used to compensate affected members.

Types of insurance services

Insurance companies provide:

- | | |
|---------------------------|----------------------------------|
| » Life insurance | » Agricultural insurance |
| » Health insurance | » Fire and accident insurance |
| » Motor vehicle insurance | » Marine and travel insurance |
| » Property insurance | » Pension and retirement schemes |

Examples of insurance companies in Rwanda

Examples include:

- | | |
|-----------------------------|------------------------|
| » SONARWA General Insurance | » AAR Health Services |
| » SORAS/Sanlam | » MMI Rwanda |
| » Prime Insurance Company | » RAMA |
| » Phoenix Assurance Rwanda | » Prime Life Insurance |

Importance of insurance companies

Insurance companies:

Point	Short notes
Compensate people against risks and losses	Insurance companies provide financial compensation to individuals and businesses that suffer losses due to accidents, fire, theft, illness, death, or natural disasters. This helps people recover from unexpected events.
Promote savings and investment	Some insurance policies, such as life insurance, encourage people to save money regularly. Insurance companies also invest collected premiums in productive sectors of the economy.
Encourage entrepreneurship	Insurance gives confidence to investors and business owners because their businesses and properties are protected against possible risks and losses.
Reduce business uncertainty	Businesses operate with less fear of unexpected losses because insurance helps cover risks such as fire, accidents, or theft, leading to stable business operations.
Promote social welfare and financial security	Insurance improves the living standards of people by providing financial support during difficult situations such as sickness, accidents, disability, or death.
Support long-term investment in the economy	Insurance companies mobilize large amounts of funds from premiums and invest them in long-term projects such as infrastructure, housing, industries, and government securities.

iii. Savings and Credit Cooperative Organisations (SACCOs)

Meaning of SACCOs

Savings and Credit Cooperative Organisations (SACCOs) are member-owned financial cooperatives formed to mobilise savings and provide loans to members at affordable interest rates.

Members save regularly and part of these savings is lent to members.

Services provided by SACCOs

Savings and Credit Cooperatives (SACCOs) provide important financial services to their members, especially people in rural and low-income communities.

- **Savings facilities:**
SACCOs encourage members to save money regularly by providing safe and convenient savings accounts. These savings help members develop a saving culture and accumulate funds for future use.
- **Small and medium loans:**
SACCOs provide affordable loans to members for activities such as farming, trade, school fees, business expansion, and household needs. The loans are usually offered at lower interest rates compared to informal lenders.

- **Money transfer services:**

Many SACCOs offer money transfer and payment services that help members send, receive, and manage money more easily and securely.

- **Financial advice and support:**

SACCOs guide members on proper financial management, savings, borrowing, and investment decisions in order to improve their economic well-being.

Membership of SACCOs

Membership in SACCOs is usually based on common interests or shared relationships among members.

- **Common geographical location:**

Some SACCOs are formed by people living in the same village, sector, district, or community who join together to improve their financial welfare.

- **Profession or occupation:**

Membership may include people engaged in the same economic activity or profession such as farmers, teachers, traders, or transport operators.

- **Workplace or institution:**

Certain SACCOs are established for employees of a particular organisation, company, school, or institution to promote saving and access to credit among workers.

Examples of SACCOs in Rwanda

Examples include:

- Umurenge SACCO
- Umwalimu SACCO

Importance of SACCOs

- **Encourage a Saving Culture:** SACCOs encourage members to develop the habit of saving money regularly. Through small but consistent savings, members are able to plan for future needs, emergencies, education, and investments. This helps improve financial discipline among individuals and communities.
- **Provide Affordable Credit:** SACCOs provide loans to members at relatively low interest rates compared to many other financial institutions. These loans help members finance business activities, pay school fees, invest in agriculture, build houses, and solve emergency financial problems.
- **Reduce Poverty and Unemployment:** By providing access to loans and financial services, SACCOs help people start small income-generating activities and expand existing businesses. This creates employment opportunities, increases household income, and improves living standards, thereby reducing poverty.
- **Promote Rural Development:** SACCOs play an important role in rural areas where commercial banking services may be limited. They provide financial services to rural populations, support local economic activities, and encourage development projects such as farming, trade, and small enterprises.

- **Support Small Businesses and Agriculture:** SACCOs provide financial assistance to small-scale entrepreneurs and farmers who may not easily access loans from commercial banks. The credit obtained helps them buy inputs, increase production, expand businesses, and improve productivity.
- **Improve Financial Inclusion:** SACCOs bring financial services closer to low-income earners and people in remote areas. They enable more people to access savings accounts, loans, money transfers, and financial advice, thereby increasing participation in the formal financial system.

iv. Microfinance Institutions (MFIs)

Meaning of Microfinance Institutions (MFIs)

Microfinance institutions are financial institutions that provide small-scale financial services to low-income earners, small entrepreneurs, and people who cannot easily access loans from commercial banks.

Microfinance or microcredit refers to the provision of small loans and related financial services to individuals whose main security is their regular income or small business activity.

Target Groups of MFIs

MFIs mainly serve:

- **Small-scale traders:** They provide working capital to help them stock goods and expand their businesses.
- **Farmers:** They offer credit for farm inputs such as seeds, fertilizers, and tools to improve productivity.
- **Women and youth entrepreneurs:** They support income-generating projects and promote self-employment among these groups.
- **Low-income households:** They help families meet basic needs such as education, health, and housing through small loans.
- **Rural communities:** They bring financial services closer to people in remote areas where banks are limited.

Services Provided by MFIs

MFIs provide:

- **Small loans and credit facilities:** These help clients start or expand small businesses and activities.
- **Savings accounts:** They encourage members to save small amounts regularly for future use.
- **Financial literacy training:** They educate clients on budgeting, saving, and proper use of loans.
- **Money transfer services:** They help clients send and receive money safely and conveniently.

- Business support services: They guide clients in planning, managing, and growing their businesses.

Characteristics of MFIs

MFIs are characterised by:

- Small loan sizes: They offer limited amounts suitable for low-income borrowers.
- Flexible repayment conditions: Clients are allowed to repay in small, manageable installments.
- Limited collateral requirements: Loans are often given without strict security conditions.
- Community-based operations: They operate close to local communities to serve clients effectively.
- Focus on poverty reduction and self-employment: They aim to improve living standards and create jobs.
- Use of group lending systems: Borrowers are grouped and guarantee each other's loans.
- Use of mobile banking and digital financial services: They use technology to improve access and efficiency.

Reasons for high interest rates in MFIs

Interest rates may be higher because:

- Loans are small in size: Managing many small loans increases operational costs.
- Administrative costs are high: Processing and monitoring many clients is expensive.
- Borrowers often lack collateral: This increases the risk faced by MFIs.
- Risk of default is relatively high: Some borrowers may fail to repay loans on time or at all.

Examples of MFIs in Rwanda

Examples include:

- Urwego Finance
- VisionFund Rwanda
- Village savings and loan associations
- Rotating savings and credit associations

Importance of MFIs

MFIs:

- **Promote self-employment and entrepreneurship**
MFIs provide small loans that enable individuals to start or expand businesses, encouraging people to create their own jobs instead of relying on formal employment.
- **Reduce poverty**
By giving access to credit and financial services, MFIs help low-income people improve their living conditions and break the cycle of poverty.
- **Improve household incomes**
The financial support offered allows families to engage in income-generating activities, which increases their overall earnings.
- **Increase access to finance**
MFIs reach people who are excluded from commercial banks, especially those without collateral or stable income.
- **Encourage savings**
They promote a saving culture by offering simple and accessible savings facilities for low-income earners.
- **Promote rural and community development**
MFIs support rural populations by financing agriculture and small businesses, leading to overall community growth and development.
- **Enhance financial inclusion**
They integrate more people into the formal financial system, ensuring that even marginalized groups can access financial services.

Summary Table of categories of NBFIs in Rwanda

Category of NBFi	Meaning / Main function	Examples in Rwanda
Specialized Banks	Provide long-term finance for development projects	BRD, Housing Bank of Rwanda
Insurance Companies	Provide protection against risks and losses	SONARWA, SORAS/ SANLAM, Prime Insurance
SACCOs	Mobilise savings and provide affordable loans	Umurenge SACCO, Umwalimu SACCO
Microfinance Institutions	Provide small loans and financial services to low-income earners	Urwego Finance, VisionFund Rwanda
Other NBFIs	Provide specialised financial services	Post office savings banks, discount houses

5.4.3 Intervention sectors of NBFIs in Rwanda

Non-banking financial institutions operate in many sectors of the economy.

i. Agriculture and Livestock

They finance:

- » Crop production
- » Livestock farming
- » Irrigation
- » Agricultural machinery
- » Crop and livestock insurance

ii. Manufacturing and Industry

They support:

- » Factories
- » Agro-processing industries
- » Small and Medium-sized Enterprises (SMEs)
- » Industrial expansion

iii. (Education and Health

They finance:

- » Schools and universities
- » Hospitals and clinics
- » Education loans
- » Health insurance services

iv. ICT and innovation

They finance:

- » Technology startups
- » Innovation hubs
- » Fintech services
- » Digital businesses

v. Transport and Infrastructure

They finance:

- » Roads and bridges
- » Public transport
- » Construction equipment
- » Energy projects

vi. Housing and Real Estate

They provide:

- » Mortgage loans
- » Housing finance
- » Real estate development finance

vii. Trade and Commerce

They support:

- » Traders and retailers
- » Import and export businesses
- » Working capital financing

viii. Tourism and Hospitality

They finance:

- » Hotels
- » Restaurants
- » Tourism facilities
- » Eco-tourism projects

ix. Energy and Environmental Protection

They support:

- » Solar energy projects
- » Biogas projects
- » Green investment
- » Environmental conservation

x. Community Development and Microfinance

They:

- » Support rural households
- » Promote self-employment
- » Reduce poverty
- » Improve living standards

5.4.4 Functions, roles, and limitations of nonbanking financial institutions (NBFIs)

i. Functions of Non-Banking Financial Institutions (NBFIs)

Non-banking financial institutions (NBFIs) perform important financial and economic functions in Rwanda and other economies. They complement the work of commercial banks by mobilising savings, providing credit, supporting investment, and promoting economic development.

Acting as Brokers of Loanable Funds

NBFIs act as intermediaries between savers and investors. They collect savings from individuals and institutions and channel these funds to businesses, farmers, entrepreneurs, and government projects that need capital for investment.

They reduce financial risk by spreading investments across different sectors and borrowers. Losses from some investments can be balanced by gains from others, thereby protecting savers and improving confidence in the financial system.

Mobilisation of Savings

NBFIs encourage people to save by providing safe, convenient, and accessible financial services. They offer various saving schemes, insurance policies, pension plans, and investment opportunities that attract funds from the public.

By mobilising savings, NBFIs increase the amount of money available for investment and economic activities. In Rwanda, SACCOs and microfinance institutions have greatly improved the saving culture, especially in rural areas.

Channeling Funds into Productive Investments

Funds collected from savers are directed into productive sectors such as:

- Agriculture and livestock
- Manufacturing industries
- Housing and construction
- Education and health
- ICT and innovation
- Trade and transport

This promotes capital formation, employment creation, income generation, and economic growth.

Stabilisation of capital markets

NBFIs contribute to the stability of financial and capital markets by investing in different financial assets and operating under government regulations and financial laws.

Their activities help to balance the demand and supply of financial resources, increase investor confidence, and protect the interests of savers and borrowers.

Provision of liquidity

NBFIs provide liquidity by making funds available to individuals and businesses when needed. They offer short-term and long-term loans, insurance compensation, and other financial services that help customers meet urgent financial obligations.

This improves the smooth flow of economic activities and supports business operations.

Promoting financial inclusion

NBFIs extend financial services to people who may not easily access commercial bank services, especially:

- Rural populations
- Low-income earners
- Women and youth
- Small-scale traders and farmers

Institutions such as SACCOs and MFIs help reduce financial exclusion and improve living standards.

Supporting Entrepreneurship and Small and Medium-sized Enterprises (SMEs)

NBFIs provide financial support to small and medium enterprises (Small and Medium-sized Enterprises (SMEs)) and startups through loans, microcredit, leasing, and business financing. This encourages innovation, self-employment, and private sector development.

Providing Insurance and Risk Protection

Insurance companies protect individuals and businesses against risks such as accidents, illness, fire, theft, and crop failure. This reduces uncertainty and encourages investment and production.

Supporting Government Development Programmes

Some NBFIs, especially development banks such as Development Bank of Rwanda, support national development goals by financing strategic sectors and public development projects.

Creating Employment Opportunities

NBFIs create jobs directly by employing workers and indirectly by financing businesses and development projects that generate employment opportunities.

Promoting Rural Development and Poverty Reduction

Through microfinance services, agricultural loans, and SACCO operations, NBFIs help improve rural incomes, reduce poverty, and promote balanced regional development.

Encouraging Investment and Economic Growth

By mobilising savings and financing productive activities, NBFIs stimulate investment, increase production, improve incomes, and contribute to overall economic growth and national development.

Roles of NonBanking Financial Institutions (NBFIs)

NonBanking Financial Institutions (NBFIs) play a crucial role in complementing commercial banks and promoting inclusive financial development.

- **Employment creation:** NBFIs such as insurance firms, pension funds, and microfinance institutions provide jobs for accountants, managers, loan officers, and other professionals.
- **Risk management:** Insurance companies protect property and lives, reducing vulnerability to accidents, illness, and disasters.
- **Financial intermediation:** They mobilize savings from households and channel them into productive investments, creating assets and liabilities.
- **Support for underserved sectors:** NBFIs extend credit to smallscale industries, agriculture, cooperatives, and rural enterprises often neglected by commercial banks.
- **Development financing:** They serve as conduits for donor funds, government programs, and NGO initiatives aimed at poverty reduction and community development.

- **Rural transformation:** By mobilizing rural savings and offering loans, NBFIs help shift communities from subsistence to a monetized economy.
- **Financial inclusion:** Credit unions, SACCOs, and microfinance institutions spread across Rwanda enhance access to financial services for marginalized groups.
- **Indigenisation of the economy:** They empower local entrepreneurs, women, and youth by providing affordable loans, fostering selfreliance and inclusive growth.
- **Revenue generation:** NBFIs contribute to government revenue through taxes, which finance public investment and infrastructure.
- **Capital market development:** Pension funds and investment companies stimulate longterm savings and investment, supporting Rwanda's capital market growth.
- **Social protection:** Pension schemes and insurance services provide security for old age, disability, and unforeseen risks.

ii. Challenges and limitations of NBFIs in Rwanda

Despite their importance, NBFIs face several constraints that hinder their effectiveness:

- **Governance issues:** Corruption, mismanagement, and weak accountability reduce efficiency and trust.
- **Loan defaults:** Many clients lack creditworthiness, leading to high nonperforming loans.
- **Low financial literacy:** Limited awareness and high illiteracy prevent many citizens from saving or using NBFi services effectively.
- **Government interference:** High taxes, interest rate controls, and regulatory burdens sometimes discourage growth.
- **Low savings capacity:** Widespread poverty and high consumption levels (high MPC, low MPS) limit savings mobilization.
- **Insecurity and uncertainty:** Political or economic instability reduces confidence in financial institutions.
- **Poor infrastructure:** Inadequate transport and ICT systems in rural areas hinder service delivery and savings mobilization.
- **Subsistence economy:** Heavy reliance on barter trade reduces the scope for financial transactions.
- **Intense competition:** NBFIs face rivalry from commercial banks and among themselves, limiting profitability.
- **External influence:** Conditionalities from international institutions may restrict flexibility in policy and operations.
- **Cultural barriers:** Fear of indebtedness and limited investment opportunities discourage borrowing, reducing NBFIs' lending potential.

- **Lack of well qualified:** There is a shortage of well-qualified, competent, and trustworthy employees in non-banking financial institutions. When qualified personnel are available, they are sometimes prone to corruption and may misuse the institutions' resources.



Insight

NBFIs are indispensable for Rwanda's **Vision 2050**, as they promote financial inclusion, rural transformation, and longterm investment. However, their sustainability depends on tackling governance issues, improving financial literacy, strengthening infrastructure, and fostering a savings culture.

Application activity 5.4



1. What are Non-Banking Financial Institutions (NBFIs)?
2. What are the different types of NBFIs found in Rwanda?
3. How do NBFIs help people access financial services in Rwanda?
4. How do NBFIs support sustainable economic development in Rwanda?

5.5 Financial markets

Learning Activity 5.5



Imagine you are a young investor interested in growing your savings through Rwanda's financial investment opportunities. You have heard about the buying and selling of shares, bonds, and other securities, and want to understand how these investment options work within Rwanda's financial system. You are curious about how these investments can help support economic growth and what risks and benefits they involve.

Questions:

1. What are the main types of investments you can make in Rwanda's financial system?
2. How do buying and selling securities contribute to economic development?
3. Why is investing in securities important for individuals and the country's economy?
4. What are some potential risks and benefits associated with investing in these financial instruments?

Financial markets are places or systems where people and organizations can buy and sell financial assets such as money, investments, or other valuable items. These markets help connect those who need money with those who have money to invest. They play a vital role in the economy by facilitating the flow of funds, supporting businesses, and helping individuals save and grow their wealth.

In Rwanda, as in many other countries, financial markets are important for economic development because they provide opportunities for investment, savings, and financial growth.

5.5.1 Meaning, types, and terms used in financial markets

i. Meaning of financial markets

Financial markets are institutions or systems where people and organisations with surplus funds provide money to those who need funds for investment and production. They include the money market for short-term finance and the capital market for long-term finance. Financial markets help in mobilising savings, allocating resources, and promoting economic development.

Financial markets are often associated with raising finance for businesses and governments. They provide platforms for trading **stocks, bonds, foreign exchange**, and other instruments. Examples include **capital markets, money markets, derivative markets**, and **currency markets**.

ii. Types of financial markets

1. Money Markets

- Deal with **shortterm debt instruments** (less than one year).
- Instruments include **treasury bills, certificates of deposit, commercial paper, and trade credit**.
- Mainly accessed by large corporations, governments, and financial institutions to meet shortterm liquidity needs.

2. Capital markets

- Handle **longterm financial securities** (more than two years).
- Provide funding for expansion and development through **stocks and bonds**.
- Subtypes:
 - **Primary markets:** Deal in newly issued securities.
 - **Secondary markets:** Provide continuous trading of existing securities, ensuring liquidity.

iii. Terms used in money markets

- **Bank overdraft:** Credit extension allowing withdrawals beyond account balance.
- **Shortterm loan:** Loan repayable within one year, often used for working capital.

- **Promissory note:** Written promise to pay a specified sum at a future date.
- **Bill of exchange:** Unconditional order to pay a sum, functioning like a negotiable draft.
- **Trade credit:** Agreement allowing delayed payment for goods purchased.
- **Certificate of deposit:** Savings certificate with fixed interest and maturity date.
- **Treasury bill:** Shortterm government security sold at a discount.
- **Banker:** Individual engaged in banking; an **investment banker** specializes in raising capital for firms and governments.

iv. Terms Used in Capital Markets

- **Equity:** Ownership interest in a company, representing net worth or shareholders' funds.
- **Stocks:** Shares issued by corporations, traded on stock exchanges.
- **Bonds:** Longterm debt instruments where issuers commit to repay investors with interest.
- **Debenture:** Unsecured corporate bond backed by reputation rather than collateral.
- **Balanced schemes:** Investment plans combining shares and fixed income securities for both growth and income.
- **Collective investment schemes:** Pooled funds managed by banks or trust companies, offering diversification and professional management.



Key Insight

Financial markets are the backbone of modern economies. **Money markets** ensure shortterm liquidity, while **capital markets** provide longterm financing for development. Understanding their instruments and terms is essential for analyzing how savings are mobilized and investments are financed in Rwanda and globally.

Comparative Table: Money market vs capital market

Market Type	Instruments	Examples in Rwanda
Money Market	- Treasury bills (shortterm government securities) - Certificates of deposit - Commercial paper - Trade credit - Bank overdrafts - Promissory notes - Bills of exchange	- Treasury bills issued by the National Bank of Rwanda (BNR) to manage liquidity. - Shortterm loans offered by SACCOs and microfinance institutions. - Trade credit arrangements between local wholesalers and retailers. - Certificates of deposit issued by commercial banks like Bank of Kigali .

Capital Market	- Equity (shares) - Stocks - Bonds - Debentures - Balanced schemes - Collective investment schemes	- Shares traded on the Rwanda Stock Exchange (RSE) (e.g., Bank of Kigali, Bralirwa). - Government bonds issued through the BNR for infrastructure financing. - Corporate bonds by institutions like I&M Bank Rwanda. - Collective investment schemes managed by Rwanda Capital Market Authority (CMA). - Balanced mutual funds offered by licensed investment companies.
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5.5.2 Functions and limitations of financial markets in developing countries

i. Functions of financial markets

1. Financing trade

Financial markets play an important role in financing both domestic and international trade. Traders obtain short-term funds through instruments such as bills of exchange, commercial papers, and bank credit. Financial institutions such as commercial banks and discount houses assist traders in carrying out business transactions smoothly.

2. Financing industrial development

Financial markets support the growth and expansion of industries in several ways

i. Provision of short-term finance

The money market provides industries with short-term loans to meet working capital requirements such as payment of wages, purchase of raw materials, and transportation costs.

ii. Provision of long-term finance

The capital market provides long-term funds needed for investment in machinery, factory construction, and expansion projects through shares, bonds, and debentures.

3. Providing profitable investment opportunities

Financial markets enable commercial banks and other financial institutions to invest their excess funds in profitable and liquid assets. Investments in treasury bills and short-term securities help institutions earn income while maintaining liquidity to meet customer withdrawals.

4. Promoting self-sufficiency of commercial banks

A developed money market enables commercial banks to obtain short-term funds easily from other financial institutions instead of depending entirely on the central bank. This improves efficiency and stability in the banking system.

5. Assisting the central bank in monetary policy

Financial markets support the work of the central bank in implementing monetary policy.

i. Providing economic indicators

Interest rates in the money market help the central bank to assess the monetary and banking conditions of the economy.

ii. Enhancing implementation

Well-developed financial markets help the central bank to spread its monetary policy measures quickly and effectively throughout the economy.

6. Mobilising savings

Financial markets encourage individuals and institutions to save money. These savings are then channelled into productive investments such as industries, agriculture, and infrastructure development, leading to capital formation.

7. Promoting economic growth and development

Financial markets ensure efficient allocation of resources by directing funds to productive sectors of the economy. This increases production, employment opportunities, income levels, and national output.

8. Linking savers and investors

Financial markets bring together surplus spending units (savers and lenders) and deficit spending units (borrowers and investors). This interaction facilitates borrowing, lending, and investment activities in the economy.

9. Promoting liquidity of financial assets

Financial markets provide a mechanism through which financial assets such as shares and bonds can easily be bought and sold. This increases investor confidence because assets can quickly be converted into cash.

10. Encouraging investment and entrepreneurship

By making finance available, financial markets encourage entrepreneurs to establish new businesses and expand existing enterprises, thereby promoting innovation and economic progress.

ii. Limitations of financial markets in developing countries

1. Lack of coordination among financial institutions

Many financial institutions operate independently and follow different policies. This creates duplication of services, delays in lending, and inefficiency in serving the industrial and commercial sectors.

2. Political interference and bureaucracy

In some developing countries, financial institutions are influenced by political leaders and bureaucrats rather than financial experts. This weakens decision-making and reduces efficiency in financial management.

3. Shortage of skilled and professional personnel

Qualified professionals often leave financial institutions for more profitable sectors, creating a shortage of skilled manpower. This affects service delivery and proper management of financial institutions.

4. Limited financial services in rural areas

Many financial institutions are concentrated in urban centres, leaving rural populations with limited access to banking and financial services. This reduces savings mobilisation and investment in rural areas.

5. Poor loan assessment and risk management

Some financial institutions provide loans without carefully evaluating the borrower's repayment ability. This increases loan defaults and financial losses.

6. High levels of loan default

Failure by borrowers in both the public and private sectors to repay loans reduces the ability of financial institutions to provide further credit and weakens financial stability.

7. Low levels of income and savings

In many developing countries, low household incomes limit people's ability to save and invest. This reduces the amount of funds available in financial markets.

8. Underdeveloped financial infrastructure

Poor technology, limited financial innovation, and weak communication systems reduce the efficiency and effectiveness of financial markets.

9. Limited public awareness about financial services

Many people lack financial literacy and knowledge about financial products and services, which limits participation in financial markets.

10. Economic instability

Problems such as inflation, unemployment, currency instability, and economic shocks discourage investment and reduce confidence in financial markets.

11. Weak legal and regulatory framework

Weak laws and poor enforcement mechanisms may encourage corruption, fraud, and mismanagement within financial institutions, reducing public trust in the financial system.

Application activity 5.5



1. What is meant by the term financial market?
2. What are the different types of financial markets and some common terms used in them?
3. Explain the functions of financial markets in an economy.
4. Discuss the limitations of financial markets in developing countries like Rwanda.

5.6 Financial sector in Rwanda

Learning Activity 5.6



1. How has the expansion of SACCOs, microfinance institutions, and mobile money services improved access to financial services for rural communities in Rwanda?
2. Why do most commercial banks in Rwanda focus on trade finance rather than long-term investment financing, and what are the implications for economic development?
3. How does the Rwanda Stock Exchange and Capital Markets Authority contribute to mobilizing long-term funds for businesses and government projects?
4. What opportunities exist in Rwanda's financial sector to support SMEs and farmers?

Historical background and overview of Rwanda's financial sector

Historical background

Rwanda's modern financial sector began developing in the 1960s with the establishment of early commercial banks such as Bank of Kigali (BK) and other banking institutions. The National Bank of Rwanda (BNR) was established in 1964 to regulate and supervise the country's financial system, maintain monetary stability, and issue national currency.

After the 1994 Genocide against the Tutsi, Rwanda rebuilt and reformed its financial sector through strong government policies and economic reforms. Today, the sector is aligned with Vision 2050 and NST2 (2024–2029), which aim at promoting inclusive growth, financial stability, investment, and poverty reduction.

Overview

Rwanda's financial sector has expanded and diversified significantly. It now includes commercial banks, microfinance institutions, SACCOs, insurance companies, pension funds, forex bureaus, capital markets, and digital financial service providers. By June 2025, total financial sector assets had risen to about FRW 14.3 trillion, while financial inclusion reached about 96% of the adult population.

The sector plays an important role in mobilising savings, financing businesses and development projects, promoting entrepreneurship, expanding access to credit, and supporting national economic development. Digital banking, mobile money, and fintech innovations have further improved access to financial services, especially in rural and underserved communities.

Regulation and policy direction

The National Bank of Rwanda is the main regulator of the financial sector, while the Capital Markets Authority (CMA) regulates the capital market.

The financial sector is aligned with Rwanda's long-term development vision, including Vision 2050, NST2, and the new National Financial Inclusion Roadmap 2026–2030, which was jointly led by BNR and MINECOFIN. (bnr.rw)

Composition of the financial sector in Rwanda

Rwanda's financial sector in 2026 is modern, diversified, and rapidly expanding. It plays a major role in promoting economic growth, investment, financial inclusion, savings mobilisation, and poverty reduction. The sector is regulated mainly by the National Bank of Rwanda (BNR), which supervises banks, microfinance institutions, insurance companies, pension schemes, forex bureaus, and payment service providers to ensure financial stability and public confidence.

The financial sector in Rwanda consists of banking institutions, non-banking financial institutions, capital markets, insurance companies, pension funds, and digital financial service providers.

Commercial banks

Rwanda has about 11 licensed banks, including commercial banks, a cooperative bank, and a development bank. Major commercial banks include Bank of Kigali (BK), BPR Bank Rwanda, I&M Bank Rwanda, Equity Bank Rwanda, KCB Bank Rwanda, and Access Bank Rwanda. These banks provide services such as savings accounts, loans, money transfers, foreign exchange services, and digital banking.

Microfinance institutions and SACCOs

The microfinance sector remains important in promoting financial inclusion, especially in rural areas. Rwanda has over 500 microfinance institutions, including cooperatives, limited companies, and Umurenge SACCOs. There are about 416 active Umurenge SACCOs operating across the country. These institutions provide savings facilities, small loans, and financial services to low-income earners, farmers, and small businesses.

Development bank

The Development Bank of Rwanda (BRD) is Rwanda's main development bank. It finances long-term projects in sectors such as agriculture, housing, energy, education, infrastructure, manufacturing, and entrepreneurship.

Insurance companies

Rwanda's insurance sector continues to grow steadily. The country has about 15 insurance companies, including life and non-life insurers as well as public insurance institutions. Insurance companies provide protection against risks such as accidents, illness, fire, crop losses, and property damage while promoting social and economic security.

Pension funds

The pension sector is led by the Rwanda Social Security Board (RSSB) together with more than 40 private pension schemes. These institutions mobilise long-term savings and provide retirement benefits and social protection to workers.

Forex bureaus

Foreign exchange services are provided by approximately 78 licensed forex bureaus operating under BNR supervision. These institutions facilitate the buying and selling of foreign currencies and support international trade and tourism.

Capital market

The Rwanda Stock Exchange (RSE), supervised by the Capital Market Authority (CMA), supports the buying and selling of shares and bonds. The capital market helps businesses and government institutions raise long-term capital for investment and development projects.

Digital financial services

Digital finance has expanded rapidly in Rwanda through mobile banking, internet banking, and electronic payment systems. Mobile money services and fintech innovations have improved access to financial services, especially among rural populations and small businesses.

Current performance and key data

Financial inclusion

Rwanda has made major progress in financial inclusion. More than 70% of adults now use formal financial services compared to about 42% in 2012. This growth has been supported by SACCOs, mobile money services, digital banking, and financial literacy programmes.

Banking sector performance

The banking sector remains profitable and stable. Average Return on Equity (ROE) in the banking sector was estimated at about 12–14% in 2025, showing improved efficiency and strong financial performance.

Economic growth

Rwanda recorded strong economic growth of about 9.4% in 2025. Growth was mainly driven by services, industry, agriculture, construction, tourism, and ICT activities. Economic growth for 2026 is projected at around 7.5% due to continued investment in infrastructure, industrialisation, and digital transformation.

Capital market development

Rwanda's capital market is growing steadily under CMA supervision. In 2025, CMA reported broad-based market performance, including FRW 46 billion raised through corporate bonds and FRW 246 billion mobilised through 14 Treasury bond auctions and reopenings. This shows increasing confidence in long-term financing instruments. (cma.rw)

Digital finance and inclusion trends

Digital finance continues to strengthen access to financial services. A BNR open finance paper noted high mobile phone penetration, strong mobile money use, and widespread digital payments, while the National Financial Inclusion Roadmap highlights remaining gaps in savings, credit, and financial health for some groups, especially women and youth.

Opportunities in the sector

The main opportunities in Rwanda's financial sector include mortgage finance, agricultural credit, SME and start-up financing, green finance, fintech innovation, and open finance. These areas can deepen financial inclusion, support job creation, and channel more savings into productive investment. (bnr.rw)

Application activity 5.6



1. Explain the structure of the financial sector in Rwanda and identify its main components.
2. Analyse the role of the financial sector in promoting economic growth, financial inclusion, and stability in Rwanda.

5.7 Artificial intelligence (AI) in finance and economics

Learning Activity 5.7



Scenario

Imagine a Rwandan bank decides to implement Artificial Intelligence (AI) systems to offer personalized financial advice to its customers, automate loan approvals, and detect fraudulent transactions. While this technology promises to make banking more efficient and accessible, there are also concerns about potential risks such as data privacy breaches, biases in decision-making, and the loss of jobs for some employees.

Questions:

1. How can AI improve banking services and economic activities in Rwanda?
2. What are some potential risks associated with using AI in the financial sector?
3. How might biases in AI algorithms affect customers and the economy?
4. What measures can banks and policymakers take to ensure AI is used responsibly?
5. Why is it important for financial institutions in Rwanda to balance innovation with ethical considerations?

Artificial Intelligence (AI) is transforming the fields of finance and economics by enabling faster decision-making, improving accuracy, and creating new opportunities for innovation. From automated trading algorithms to personalized financial advice, AI offers many benefits. However, along with these advantages come risks such as biases, security concerns, and ethical dilemmas. It is essential to understand how to use AI responsibly to maximize its positive impact while minimizing potential harms. This presentation will explore the role of AI in finance and economics, discuss the associated risks, and highlight the importance of responsible usage.

5.7.1 Meaning of artificial intelligence (AI) in finance

Artificial Intelligence (AI) in finance refers to the use of computer systems and advanced technologies such as machine learning, big data analytics, and automation to perform financial tasks that normally require human intelligence.

AI is widely used in financial institutions such as banks, insurance companies, and fintech firms. It helps in activities like fraud detection, credit scoring, customer support through chatbots, risk assessment, investment forecasting, and mobile banking services.

5.7.2 Applications of AI in financial institutions

AI is transforming how financial institutions operate and serve customers.

i. Fraud detection and security

AI systems analyse transaction patterns to detect unusual activities and prevent fraud in real time.

ii. Credit scoring and loan approval

Banks use AI to assess customers' creditworthiness based on income, spending habits, and financial history.

iii. Customer service (Chatbots)

AI-powered chatbots provide instant responses to customer questions, improving service delivery and reducing waiting time.

iv. Investment and financial advice

AI tools help investors make decisions through automated financial analysis and predictions (robo-advisors).

v. Mobile Banking and digital payments

AI supports secure mobile money transactions and enhances digital financial services used by SACCOs and banks.

5.7.3 Benefits of AI in finance and economics

AI provides several advantages to financial institutions and the economy:

Improves efficiency: It automates routine tasks such as loan processing, record keeping, and customer service.

- **Enhances accuracy:** AI analyses large volumes of data quickly, leading to better financial decisions.
- **Promotes financial inclusion:** Digital platforms powered by AI make financial services accessible to rural and underserved communities.
- **Encourages innovation:** It supports new financial products such as mobile wallets, robo-advisors, and digital insurance services.
- **Reduces operational costs:** Automation reduces the need for manual work and improves service delivery.

5.7.4 Risks of AI in Finance

Despite its benefits, AI also presents several risks in the financial sector:

- **Data privacy risks:** Personal financial information may be misused or leaked.
- **Cybersecurity threats:** Financial systems may be exposed to hacking, fraud, and cyber-attacks.
- **Bias and discrimination:** AI systems may make unfair decisions if trained on biased data.
- **Job displacement:** Automation may reduce employment opportunities in traditional banking roles.
- **Over-reliance on technology:** Excessive dependence on AI may reduce human judgment in financial decision-making.

5.7.5 Responsible use of AI in finance

To ensure safe and fair use of AI, financial institutions and users must adopt responsible practices:

- **Ethical use:** AI systems should be fair, transparent, and accountable.
- **Strong regulation:** Institutions such as the National Bank of Rwanda (BNR) should enforce rules guiding AI use in finance.
- **Human supervision:** Important financial decisions should be reviewed by qualified professionals.
- **Data protection:** Strong systems should be used to protect customer information.
- **Consumer education:** Users should be trained in digital and financial literacy to safely use AI-based services.

Summary

AI is transforming finance and economics by making services faster, smarter, and more accessible. However, it must be used responsibly to avoid risks such as fraud, inequality, and job loss. Proper regulation, education, and ethical practices are essential for sustainable development in an AI-driven financial system.

Application activity 5.7



Case Study: AI in Rwanda's Financial Sector

In Rwanda, financial institutions are increasingly adopting AI technologies to improve service delivery and efficiency. The National Bank of Rwanda (BNR) together with banks such as Bank of Kigali (BK) and other commercial banks use AI for fraud detection, credit scoring, mobile banking, and customer support.

These innovations have improved financial inclusion and efficiency; however, they also raise concerns about data privacy, fairness in loan approvals, and cybersecurity risks, especially for rural SACCO users and low-income clients.

Discussion Questions:

- How does AI improve access to financial services in Rwanda?
- What risks may SACCO members face when using AI-based mobile money services?
- How can regulators ensure the safe use of AI in the financial sector?
- What skills do students need to work in an AI-driven financial system?

5.8. End of unit assessment



Financial Institutions

Knowledge and Understanding

1. a) Define a central bank.
b) Explain any two functions performed by the National Bank of Rwanda.
2. a) What is a commercial bank?
b) Describe any three functions performed by commercial banks in Rwanda.
3. a) Define Non-Banking Financial Institutions (NBFIs).
b) Identify and explain any two types of NBFIs operating in Rwanda.
4. a) What are financial markets?
b) Explain any three roles of financial markets in Rwanda's economy.

Skills

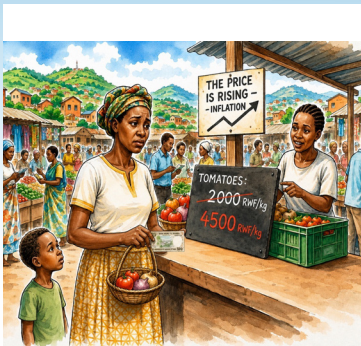
5. Explain how commercial banks create credit and show how this process promotes economic growth in Rwanda.
6. Compare the roles of the central bank and commercial banks in maintaining financial stability in Rwanda.
7. A farmer borrowed FRW 600,000 from a SACCO at a simple interest rate of 20% per annum for 2 years.
 - a) Calculate the simple interest and compound interest, then compare the results.
 - b) Determine the total amount to be repaid.

Attitudes and Values

8. Explain why responsible and ethical use of AI is important in Rwanda's financial sector.

UNIT 6

INFLATION



Key Unit Competence:

Evaluate how inflation affects key economic variables, agents, and policies, and judge effective measures for its control.

Learning objectives

Knowledge and understanding

1

- Give the meaning of inflation and related concepts.
- Identify key terms: price index, Consumer Price Index (CPI) and inflation rate.
- Describe methods used to measure inflation (CPI, GDP deflator).
- Explain types (creeping, galloping, hyperinflation) and causes.
- Describe fiscal, monetary, and supply-side policies.
- Explain importance of measuring inflation accurately in an economy
- Explain effects of inflation on consumers, producers, and government
- Describe historical trends and major inflationary events
- Explain the inverse relationship between inflation and unemployment.

Skills

2

- Calculate inflation rate using data
- Examine the causes and effects of inflation on the Rwandan economy.
- Evaluate who gains or losses from inflation
- Evaluate effectiveness of policies and propose improvements

Attitudes and values

3

- Value accuracy and transparency in economic data.
- Appreciate the importance of price stability

- Show empathy toward affected populations
- Value disciplined policy and responsible governance
- Appreciate data-driven policy-making and economic growth.
- Value balanced economic growth and informed policy trade-offs.



Introductory activity

John visits the same shop his parents regularly buy from over several years. He notices a clear pattern: last year a loaf of bread cost 1,000 RWF, but this year it has risen to 1,300 RWF, and the year before it was 900 RWF. His father explains that this is not only happening to bread but also to other goods like rice, soap, and cooking oil, which keep increasing in price every year while his salary has remained the same. John becomes curious and wonders why prices keep rising continuously while incomes do not change situation where prices keep on increasing time after time?

Based on John's experience and the changes he observes in prices over time, reflect on the situation by answering the following questions:

1. What pattern of price changes does John observe over the different years?
2. Which other goods, apart from bread, are mentioned as continuously increasing in price?
3. How does the continuous rise in prices affect the purchasing power of John's family?
4. What economic term is used to describe a general and persistent increase in prices over time?

In Rwanda households have increasingly experienced a steady rise in the prices of essential goods and services such as food, transport, and education materials, while their incomes have remained relatively stable. This persistent imbalance has gradually reduced the purchasing power of money, meaning that families are now able to buy fewer goods with the same amount of income compared to earlier periods such as 2021–2025. As a result, many people find it difficult to maintain their usual standard of living, which raises important economic questions about the changing value of money over time. This situation reflects a broader macroeconomic trend in which the general level of prices continues to increase in the economy. In this unit, we explore this phenomenon, known as inflation, and examine its causes, effects, and implications for economic stability.

6.1 Meaning of inflation and related terminologies

Learning Activity 6.1



In a trading centre in Rwanda, prices of basic goods such as rice, sugar, transport fares, and school materials have been increasing over time. A student, Eric, notices that the same items he bought last year now cost more, while his pocket money has remained the same. Different people describe the situation in different ways: a parent says “life is becoming expensive,” a trader says “prices keep rising,” and a worker says “money has lost value.” An economics teacher refers to the situation as inflation. Eric becomes curious about what these different expressions mean.

Based on Eric’s observation and the different ways people describe the situation, answer the following questions:

1. What change has Eric observed in prices over time?
2. What does it mean when people say money has lost value?
3. What do you think inflation means based on this case?

In economics, a clear understanding of key concepts is essential for analyzing changes in the general price level within an economy. These concepts provide a foundation for interpreting how inflation affects economic agents and overall economic performance. The following section outlines the meaning of inflation and related terminologies.

Meaning of inflation

Inflation is the sustained increase in the general price level of goods and services in an economy over a period of time, which leads to a fall in the purchasing power of money.

Terms used in inflation

- **Stagflation:** A situation where an economy experiences both inflation and unemployment at the same time, often leading to rising prices alongside slow economic growth.
- **Deflation:** A continuous fall in the general price level of goods and services in an economy.
- **Reflation:** A deliberate government policy to increase prices and stimulate economic activity after a period of economic decline, usually through measures like tax cuts, increased government spending, or expansionary monetary policy.
- **Suppressed inflation:** A condition where demand is higher than supply, but prices are controlled by the government, preventing them from rising freely.
- **Open inflation:** Inflation where prices are allowed to rise freely without government control or restrictions.

- **Underlying inflation:** This is inflation calculated by not considering the prices of foodstuffs.
- **Headline inflation:** This is inflation calculated by considering the prices of foodstuffs.
- **Inflationary spiral:** A situation where rising prices lead to higher wage demands, which increase production costs, causing further price increases in a continuous cycle.

Application activity 6.1



1. Explain the meaning of the following concepts:

- a) Deflation
- b) Stagflation

2. Multiple Choice Questions

a) Inflation is best defined as:

- i. A fall in general prices of goods and services
- ii. A sustained increase in the general price level
- iii. A decrease in population growth
- iv. A rise in government expenditure

3. Fill in the blanks

- a) Inflation calculated by considering foodstuffs is called _____.
- b) Income adjusted for inflation is called _____ income.

6.2 Measurement of inflation

Learning Activity 6.1



In a local market, a basket of basic goods such as rice, sugar, and soap cost 10,000 RWF in 2022. The same basket costs 13,000 RWF in 2025, even though the items have not changed. Shoppers notice that they now spend more money to buy the same goods they used to buy a few years ago.

Questions:

- 1. What change is observed in the price of the basket between 2022 and 2025?
- 2. What does this change suggest about the prices of goods in general?
- 3. Are prices increasing, decreasing, or remaining constant over time?

Inflation cannot be measured directly because there is no single price for all goods and services in an economy. Instead, economists first measure changes in the general price level using a **price index**, which tracks the average movement of prices over time. By comparing price index values across different periods, it becomes possible to determine the rate of inflation. Therefore, price indices provide the key tool for accurately measuring and understanding inflation in an economy.

6.2.1 Meaning and types of price indices

Price index can be defined as the measure of the variation in the prices of a given class of goods and services over a period in a given region. It shows average changes in price over a period, i.e. from a base year to a current year.

Types of price indices

Price changes do not affect all goods and services equally, so different price indices are used to measure them from various perspectives. Each index focuses on a specific area of the economy, helping us better understand how price changes impact consumers, producers, and overall economic activity. The following are types of price indices:

1. **Consumer price index (CPI):** This measures the changes in the price level of a market basket of consumer goods and services bought by households. The National Institute of Statistics of Rwanda (NISR) collects the CPI price information and calculates the CPI statistics for Rwanda on a monthly basis.
2. **Producer price index (PPI):** This measures the average changes in the prices received by domestic producers for their output.
3. **Gross Domestic Product deflator (GDP deflator):** It is sometimes referred to as implicit price deflator. It is a measure of the level of prices of all new, domestically produced final goods and services in an economy it can be measured by:
4. **GDP Deflator = $(\text{Nominal GDP} \div \text{Real GDP}) \times 100$**
5. **Retail price index (RPI):** This is the measure of the variation of consumer prices or consumer retail goods and services over a period. RPI is computed on a monthly basis but an annual rate is also published which serves as a yardstick for adjusting salaries and wages, tax allowances and pensions, among others.

6.2.2 The procedures for computing cost of living index

1. **Selecting the base year:** This is the year when prices were relatively stable and it is equated to 100% so as to show the change in the prices.
2. **Selecting the basket of commodities:** This is a basket which represents commodities that were wholly consumed by the people in a country during a period of time.
3. **Collecting information about the basket of commodities.**

4. Calculating the simple price index/price relative.

This is calculated by:

$$\text{SPI} = \frac{P_1}{P_0} \times 100 \quad \text{or} \quad \text{SPI} = \left(\frac{P_1 - P_0}{P_0} \times 100 \right) + 100$$

Where P1 is current year price, P0 is base year price.

5. Attaching weights: Weights are attached to show the relative value of the goods. The commodity with the biggest weight has more value than the one with a small weight. Weighted index = SPI x Weights.

6. Calculate Average Simple Index(A.S.I).

$$\text{A.S.I} = \frac{\text{Sum of Simple Price Index}}{\text{Total number of items}}$$

7. Calculating the cost of living index/average weighted index.

$$\text{COL (AWI)} = \frac{\text{Sums of SPI x weight}}{\text{Sum of weight}}$$

8. Interpreting the findings. It helps to show whether consumer prices have risen or fallen over time. In conclusion, price indices convert raw price data into meaningful measures of price level changes, helping to identify inflation, deflation, or price stability in the economy.

6.2.3 Use of the price indices

- They measure changes in the value of money by showing changes in purchasing power.
- They are used to determine inflation or deflation in an economy.
- They guide adjustments of wages, salaries, and living standards during price changes.
- They help governments set regional taxes, wages, and allowances based on cost of living.
- They allow comparison of living standards across different countries and time periods.
- They assist governments, firms, and central banks in economic planning, pricing, and policy making.

6.2.4 Problems of computing index numbers

- Different methods and formulas give different results, making comparison difficult.
- Choosing a stable base year is difficult due to continuous inflation.

- It is impossible to include all commodities, so results may not fully represent cost of living.
- Weights change over time because of changing tastes, income, and consumption patterns.
- Changes in product quality may be wrongly interpreted as inflation.
- One index cannot fully reflect cost of living differences for all income groups.

Application activity 6.2



1. A consumer basket in a given region consists of five commodities: maize flour, rice, sugar, cooking oil, and beans. The prices and weights for the base year and current year are given below:

Commodity	BaseYear Price	Current Year Price	Weight
Maize flour	1000	1400	5
Rice	2000	2600	3
Sugar	1500	1800	2
Cooking oil	3000	3600	4
Beans	1200	1500	3

Tasks:

- a) Compute the Simple Price Index (SPI) for each commodity.
 - b) Calculate the Weighted Index for each commodity.
 - c) Compute the Average Simple Index (ASI).
 - d) Calculate the Cost of Living Index (COLI) using weights.
 - e) Interpret your results by stating whether:
 - i. There is inflation or deflation
 - ii. The cost of living has increased or decreased
2. Explain the problems with compiling price indices in an economy.
 3. Assess the importance of computing price indices in an economy.

The retail price index records changes in the prices of a variety of goods and services that enter into the average household's budget weighted according to their relative significance in the household expenditure.

The inflation rate is the percentage rise in the price level. That is:

$$\text{Inflation rate} = \frac{P_1 - P_0}{P_0} \times 100$$

Where P_1 is the current year's price level and P_0 is last year's price level.

If for example, a kilogramme of meat did cost 2,000 FRW in 2015 and it costs 2,500 FRW in 2016, the rise in prices is $2500 - 2000 = 500 / 2000 = (5 : 200)100 = 250\%$

6.3 Classification of inflation

Learning Activity 6.3



In a certain economy, people are experiencing rising prices, but not in the same way. In some areas, prices are increasing slowly over a long period, while in other cases, prices are rising very fast within a short time, making it difficult for households to afford basic needs. At the same time, some traders report that prices are rising because many people want to buy goods, while others say prices are increasing because the cost of production has gone up.

Task: Why do you think prices in the same economy can increase in different ways and at different speeds, and what does this suggest about how economists may classify inflation?

Inflation can be classified in different ways depending on the criteria used to analyse its nature and behaviour in the economy. These classifications help to clearly distinguish the various forms inflation may take and make it easier to understand how it operates under different economic conditions. By grouping inflation into specific categories, economists are able to examine its causes, intensity, and effects more systematically, which supports better interpretation and policy response. The different classifications of inflation are discussed below.

Classifications of inflation:

6.3.1 According to the rate or degree of intensity

Under this, we have the following types of inflation:

i. Creeping or mild or gradual inflation

This refers to the slow increase in the price of goods and services at a rate which is less than 3%. Creeping inflation is the mildest form and is conducive for economic progress and growth. It is a good state of inflation because a small rise in price acts as an incentive to producers. Savings, investment, output and employment opportunities all increase.

ii. Walking or trotting or moderate inflation

This refers to a moderate increase in the general price level of goods and services, usually at a rate of less than 10% per year. It indicates that prices are rising steadily but not rapidly, and it serves as a warning sign that inflation may become more serious if not controlled.

iii. Running inflation

This refers to a relatively high and accelerating increase in the general price level of goods and services, usually ranging between 10% and 20% per year. At this stage, prices rise quickly, reducing the purchasing power of money and creating economic instability.

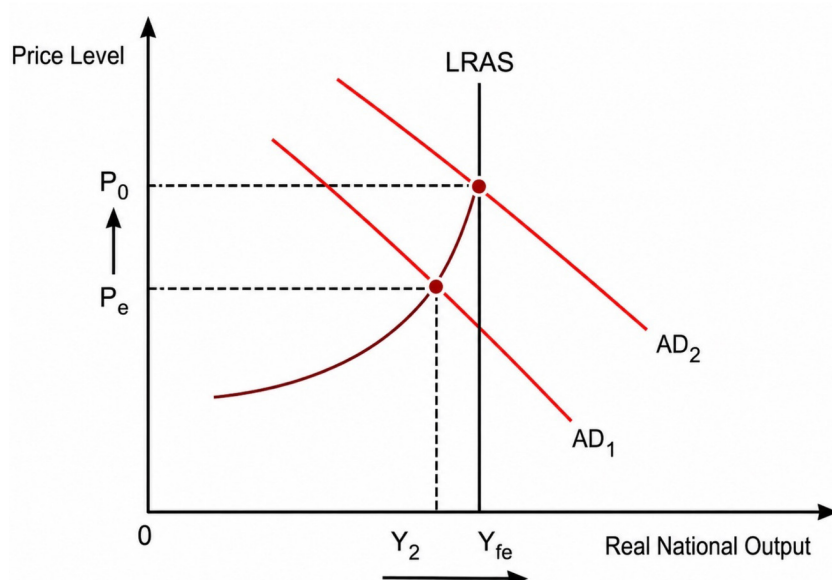
iv. Hyperinflation or galloping or runaway inflation

Hyperinflation is an extremely rapid increase in prices, often ranging from about 20% to over 100%, where prices rise continuously and uncontrollably. As a result, money quickly loses its value, sometimes leading to the introduction of a new currency. It is easily noticed by the public, discourages production and consumption, and leads to a decline in savings, investment, output, and employment opportunities.

6.3.2 According to the causes

1. Demand-pull or excess demand inflation

This is a continuous rise in the general price level caused by aggregate demand exceeding the available supply of goods and services, especially when the economy is operating at full employment. In this situation, production cannot increase further, so when demand keeps rising, it pushes prices upward.



- P represents price.
- Y represents output.
- AD represents aggregate demand.
- LRAS represents long-run aggregate supply.

Figure 6.1: Demand pull or excess demand inflation

From Figure the above, it can be seen that when aggregate demand increases from AD_1 to AD_2 , the price level increases from P_e to P_0 .

Causes of demand-pull inflation

Aggregate demand for goods and services can exceed aggregate supply at full employment due to various factors in the economy. **Some of those factors are the following:**

- **Trade union activities:** Strong trade unions demand higher wages and better conditions, increasing workers' income and raising demand for goods and services.
- **Deficit financing:** When government increases spending by borrowing or printing money, more money enters circulation, increasing demand in the economy.

- **Increase in population:** A growing population raises demand for goods and services, which can exceed supply and push prices up.
- **Credit expansion:** When banks increase lending, people and firms borrow more, increasing spending and raising demand in the economy.
- **Increase in public expenditure:** Higher government spending increases money in circulation, leading to higher demand for goods and services.
- **Expansion of exports:** When more goods are exported, fewer remain for domestic use, creating excess demand at home and increasing prices.
- **Reduction in taxation:** Lower taxes increase disposable income, allowing people to spend more, which raises demand and can increase prices.

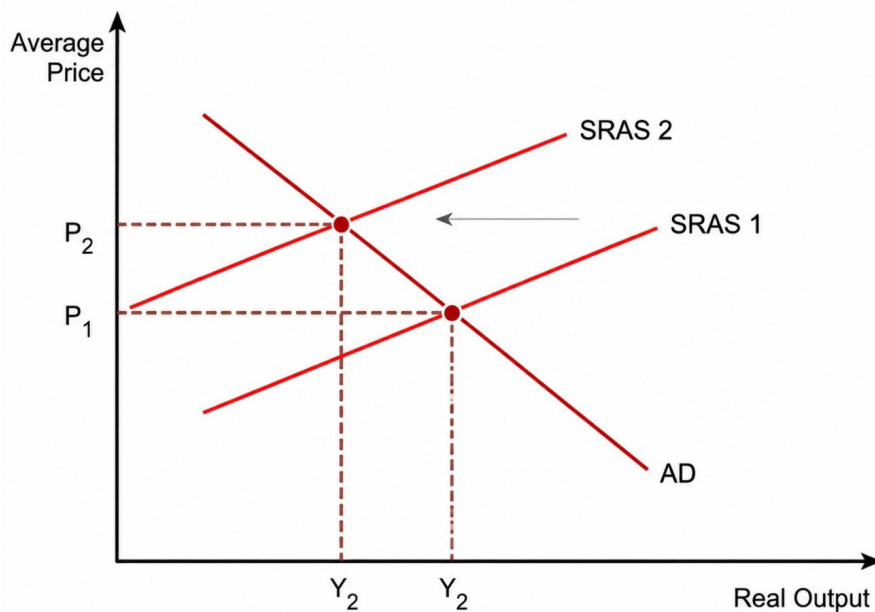
Policies for demand-pull inflation

To control demand-pull inflation, governments use various policies aimed at reducing excessive demand in the economy. These measures are designed to limit unnecessary spending by households, firms, and the government so that aggregate demand is brought in line with available supply, helping to stabilize prices. The following are some of the key policies used to achieve this:

- **Restrictive monetary policy:** The central bank reduces money supply by using tools like open market operations, higher reserve requirements, and increased bank rates.
- **Credit control measures:** Commercial banks are guided to limit lending through selective credit control, margin requirements, and moral persuasion.
- **Reduction in government expenditure:** Cutting public spending reduces income in the economy and lowers overall demand.
- **Increase in taxation:** Higher taxes reduce disposable income, leading to lower consumption and investment.
- **Trade and import policies:** Reducing tariffs or increasing import quotas increases availability of goods and helps reduce pressure on domestic prices.
- **Price and income policies:** Government may regulate prices and wages to control excessive increases, although this method is less effective due to challenges like black markets.

2. Cost push inflation

Cost-push Inflation is also called supply inflation or sellers' inflation. It is the persistent increase in the prices of goods and services brought about by increase in the cost of production.



- P represents price.
- Y represents output.
- AD represents aggregate demand.
- SRAS represents short-run aggregate supply.

Figure 6.2.: Cost-push inflation

From Figure it can be seen that when the costs of production increase, aggregate supply reduces from SRAS1 to SRAS2, P_1/P_2 . Output reduces from Y_1 to Y_2 , leading to price increase from P_1 to P_2 .

Cost push inflation arises because of the following reasons:

- **Wage increases:** When wages rise, firms spend more on paying workers. To cover these higher labour costs, they increase the prices of goods and services, leading to cost-push inflation.
- **Government taxes:** When taxes on production (such as fuel or excise duties) increase, firms face higher production costs. They pass these costs to consumers by raising prices.
- **Changes in interest rates:** When interest rates increase, borrowing money becomes expensive for firms. This raises their production costs, causing them to increase prices of goods and services.
- **Exchange rate changes:** When the local currency depreciates, imported raw materials become more expensive. Firms then increase prices to cover the higher import costs.
- **Increase in operating costs:** When costs such as raw materials, rent, electricity, transport, or advertising rise, firms face higher total production costs. They respond by increasing the prices of their products.

Forms of cost push inflation:

- Price- wage inflation:** This occurs when the increase in the price of goods and services forces the workers to demand for higher wages to meet the increasing cost of living and the producers will react by increasing the prices of the goods and the process will continue.

b) Wage-price inflation: This occurs when workers demand for high wages through their trade unions and this increases the cost of production hence producers react by increasing the prices of the goods.

c) Wage-wage inflation: This occurs when workers of a particular firm demand for high wages as those earned by a different similar firm and this increases the cost of production, hence price increase.

Measures to reduce cost-push inflation:

- **Wage control policies:** Limiting excessive wage increases helps reduce production costs and prevents firms from raising prices.
- **Tax reduction on production inputs:** Lowering taxes on raw materials and fuel reduces firms' production costs, helping to stabilize prices.
- **Stable exchange rate policies:** Maintaining a stable currency reduces the cost of imported raw materials and prevents price increases.
- **Control of interest rates:** Reducing interest rates lowers the cost of borrowing for firms, making production cheaper.
- **Improving production efficiency:** Encouraging use of better technology and skills increases productivity and reduces unit costs of production.
- **Diversification and stable supply of inputs:** Developing local sources of raw materials reduces dependence on imports and protects the economy from supply shocks.

3. Profit push or price markup inflation

This is a type of inflation that occurs when firms deliberately increase the prices of goods and services in order to earn higher profits, even when production costs have not significantly increased. Profit push is commonly in monopoly or oligopoly where firms have the power to influence prices.

4. Bottleneck /scarcity / demand shift or structural inflation

Structural inflation is a type of inflation that arises from supply constraints or rigidities which limit the production of goods and services. It is a form of cost-related inflation and is commonly experienced in developing countries, where structural weaknesses in the economy restrict output and create upward pressure on prices.

Causes of bottleneck inflation

- **Poor infrastructure:** Weak transport and storage systems make it difficult to produce and distribute goods efficiently, leading to shortages and higher prices.
- **Low agricultural productivity:** Limited use of modern farming methods reduces food supply, causing food prices to rise when demand increases.
- **Shortage of skilled labour:** Lack of skilled workers lowers production efficiency, reducing output and increasing production costs.
- **Limited industrial development:** Overdependence on a few sectors (like agriculture) limits supply of goods, leading to imbalances and rising prices.

- **Market inefficiencies:** Poor distribution systems and lack of market information cause delays and shortages, which push prices upward.
- **Weak financial systems:** Limited access to finance makes it difficult for firms to invest in production, reducing output and causing shortages that increase prices.
- **Rapid population growth:** When population grows faster than production, demand for goods and services increases beyond supply, leading to continuous price increases.

Policies for controlling bottleneck inflation

Policies for controlling bottleneck inflation focus on reducing supply shortages that cause prices to rise in the economy. These policies aim at improving production efficiency, easing supply chain problems, and increasing the availability of goods and services. They are important for stabilizing prices and supporting economic growth.

- **Improvement of infrastructure:** Developing better roads, transport, and storage systems helps increase production and efficient distribution of goods, reducing shortages and prices.
- **Modernization of agriculture:** Using improved farming methods and technology increases food production, helping to stabilize food prices.
- **Industrial development:** Promoting industries increases the supply of goods and reduces overdependence on limited sectors.
- **Education and skill development:** Training workers improves productivity and efficiency, leading to higher output and lower production costs.
- **Diversification of the economy:** Expanding into different sectors reduces reliance on a few industries and ensures a more stable supply of goods and services.
- **Improving market systems:** Strengthening distribution channels and market information systems helps reduce delays and shortages, ensuring goods reach consumers efficiently and at stable prices.
- **Encouraging local production:** Supporting local industries reduces dependence on imports and increases the supply of goods, helping to control rising prices.

5. Imported inflation

According to this theory, imported inflation occurs when the prices of goods and services rise due to an increase in the cost of imported goods or raw materials. This often happens when a country depends heavily on imports for production and consumption needs. As import prices increase, firms face higher production costs and respond by increasing the prices of final goods and services in the domestic market.

Policies for controlling imported inflation

Imported inflation is hard to fight because it is exogenous to the economy. It originates from outside the economy. **However, there are some solutions:**

- **Import restrictions:** Certainly, this will make the country go without certain commodities although prices are reduced.
- **Import-substitution strategy:** Setting up industries to produce goods formerly imported. However, some commodities cannot be import-substituted, for instance, petroleum products.
- **Subsidisation of importers by the government:** If importers are subsidised, they will sell their produce at a lower price and hence, a fall in the inflation rate.
- **Diversification of import sources:** This involves importing goods from different countries instead of relying on one source to reduce the impact of price increases from any single country.

6. Monetary inflation

According to the monetary economists, it is the persistent increase in the prices of goods and services brought by increase in money supply without corresponding increase in goods and services.

Policies to Control Monetary Inflation

Monetary inflation can disrupt price stability, so governments and central banks use various policies to control it and maintain economic balance. Some of those policies are described below:

- a) **Increase in interest rates:** The central bank raises interest rates to make borrowing expensive. This discourages borrowing and spending, reducing the money supply in circulation and helping to control inflation.
- b) **Open market operations:** The central bank sells government securities to the public. When people buy them, money is withdrawn from circulation, reducing the money supply and lowering inflationary pressure.
- c) **Increase in reserve requirements:** Commercial banks are required to keep a higher proportion of deposits with the central bank. This limits the amount they can lend, reducing money in circulation and controlling inflation.
- d) **Credit control measures:** The central bank restricts excessive lending by commercial banks. By controlling the availability of credit, it reduces spending in the economy and helps stabilize prices.

7. Expectation inflation

This is a type of inflation that arises when producers and consumers expect prices to rise in the future. Based on this expectation, producers may reduce the supply of goods or create artificial shortages (speculation) in order to sell at higher prices later, which in turn contributes to an actual increase in the general price level.

Policies to Control Expectation Inflation

Inflation may result not only from current economic conditions but also from people's expectations about future price increases. When consumers, workers, and producers expect prices to keep rising, their actions can further increase inflation. Therefore, governments and monetary authorities introduce measures to reduce such expectations and maintain price stability. **The following are some policies used to control expectation inflation.**

- **Price control measures:** The government can regulate prices of essential goods to prevent producers from increasing prices unnecessarily due to future expectations of inflation.
- **Strict laws against hoarding and speculation:** The government can punish traders and producers who create artificial shortages by hiding goods in order to sell them later at higher prices.
- **Public awareness and confidence building:** The central bank and government can assure the public about stable economic policies to reduce fear and panic buying, which often fuels inflation expectations.
- **Increase in production and supply of goods:** Encouraging higher production helps ensure enough goods are available in the market, reducing shortages and limiting expectations of future price increases.
- **Stable monetary and fiscal policies:** Maintaining controlled money supply and responsible government spending helps build confidence that inflation will remain low, reducing inflationary expectations among producers and consumers.

Application activity 6.3



1. Define the terms:

- a) Suppressed inflation b) Underlying inflation

2. Fill in the Gaps

- a) Inflation that increases slowly at a rate of less than 3% is called _____ inflation.
- b) Inflation caused by an increase in aggregate demand when the economy is at full employment is called _____ inflation.
- c) When prices rise due to an increase in the cost of imported goods, it is known as _____ inflation.

3. Analyze how an increase in aggregate demand can lead to demand-pull inflation in an economy operating at full employment, and evaluate the effectiveness of at least four government policies used to control it.
4. Explain how people's expectation that prices will rise in the future can actually make prices increase now.
5. Explain how an increase in money supply can lead to higher prices in an economy. Under what conditions may this not happen?

6.4 Causes, consequences and measures to solve inflation

Learning Activity 6.4



1. What do you think might cause prices of goods and services to keep increasing in an economy over time?
2. How do rising prices (inflation) affect the daily life of households, businesses, and the government?
3. If prices continue to rise rapidly in a country, what actions do you think the government or central bank could take to control the situation?

6.4.1 General causes of inflation in an economy

Inflation in an economy does not occur randomly; it results from several interacting factors that push the general level of prices upward over time. These causes may arise from the demand or supply side. To understand this phenomenon more clearly, the following are general causes of inflation in an economy:

- **Increase in aggregate demand:** When total spending in the economy rises faster than the supply of goods and services.
- **Rising production costs:** Increases in wages, raw materials, transport, or taxes that make production more expensive.
- **Increase in money supply:** When more money is circulated in the economy without a matching increase in goods and services.
- **Rising prices of imported goods:** When import costs increase, local prices also rise.
- **Population growth:** A growing population increases demand for goods and services.
- **Low production capacity:** Limited output or productivity leads to shortages and higher prices.
- **Government budget deficits:** High government spending financed by borrowing or money creation increases circulation of money.
- When people expect prices to rise, they may increase prices or demand higher wages in advance.
- **Weak infrastructure:** Poor transport, storage, and distribution systems increase costs and reduce efficiency in production and supply.
- **Market inefficiencies:** Poor competition or monopoly power allows firms to raise prices above normal levels.
- **External shocks:** Events such as global crises, wars, or natural disasters disrupt supply chains and increase prices.
- **Exchange rate depreciation:** When the local currency loses value, imported

Experimental version

goods become more expensive, increasing overall prices.

6.4.2 Consequences of inflation

Inflation is regarded as beneficial only when it results in higher income levels, increased output, and improved employment. However, if it fails to achieve these objectives, it becomes harmful to the economy. As a result, inflation has wide-ranging effects on the economic, social, and political life of a country. In general, inflation is undesirable because it leads to serious social and economic challenges. It affects individuals, businesses, and the overall performance of the economy. The main consequences of inflation include:

- **Low purchasing power:** Inflation makes prices of goods increase, so the same amount of money cannot buy as many goods and services as before.
- **Loss in real value of savings:** Money saved loses its **real value** over time because rising prices reduce its purchasing power.
- **Higher cost of living:** Inflation increases the prices of essential goods and services, leading to a higher **cost of living** for consumers.
- **Uncertainty for investors:** Frequent price changes create **economic uncertainty**, making firms delay or reduce investment decisions.
- **Unequal distribution of income:** Inflation reduces the real income of people with fixed earnings, while **debtors benefit** as they repay loans with less valuable money.
- **Demand for higher wages:** Workers demand higher wages to maintain their **real income** as prices of goods and services increase.
- **Lower exports:** Higher domestic prices make locally produced goods less competitive in the international market, reducing **export demand**.
- **Poor economic planning:** Price instability makes it difficult for government and firms to carry out **effective economic planning and budgeting**.

6.4.3 Measures to solve inflation

Governments use different measures to control inflation and maintain price stability in the economy. These measures are designed to regulate money supply, influence the level of demand, and encourage increased production of goods and services. The following are some of the measures used to solve inflation.

- **Monetary policy:** The central bank reduces the supply of money in circulation by increasing interest rates, selling government securities, or increasing bank reserve requirements to reduce excessive demand.
- **Fiscal policy:** The government reduces inflation by decreasing public expenditure and increasing taxes in order to reduce the amount of money spent in the economy.
- **Increasing production:** The government encourages producers to increase the supply of goods and services so that demand can match supply and

prices stabilize.

- **Importation of goods:** The government imports scarce essential goods to increase supply in the market and reduce price increases.
- **Price control policy:** The government sets maximum prices on essential commodities to prevent excessive price increases.
- **Wage control policy:** The government controls unnecessary wage increases to avoid increasing production costs and consumer demand.
- **Encouraging saving:** People are encouraged to save money in financial institutions so that less money remains in circulation, helping to reduce inflationary pressure.

Application activity 6.4



Read the following situation carefully:

In recent years, the prices of basic commodities such as food, transport, fuel, and school materials have continuously increased in many countries, including Rwanda. Some traders supply fewer goods to the market while consumers continue to demand more products. At the same time, the government and the central bank have introduced different measures to control inflation and stabilize prices.

Task:

1. What economic problem is described in the passage above?
2. Identify any five possible causes of the problem mentioned in (1).
3. What are the possible consequences of inflation on:
 - a) Consumers?
 - b) Producers?
 - c) Savers?
 - d) The government?
4. Which measures can the government and central bank use to reduce inflation?

6.5 Circumstances under which an increase in money supply may not lead to inflation

Learning Activity 6.5



Question 1: What are some economic conditions or policies that can prevent inflation despite an increase in the money supply? Explain how these conditions help control inflation.

Question 2: Describe how factors such as increased production, improved productivity, or monetary policies can mitigate the inflationary effects of an expanded money supply. Provide examples to support your explanations.

An increase in money supply does not always cause inflation, depending on how the extra money is used in the economy. If it leads to higher production, investment, and use of idle resources, prices may remain stable. Therefore, certain economic conditions can prevent inflation despite an increase in money supply. These conditions include the following:

- **Effective taxation system:** When the government has a strong tax system, any increase in money supply is partly absorbed through taxation. This reduces the amount of money left in circulation, limiting excess demand and preventing prices from rising, thus controlling inflation.
- **High marginal propensity to save (MPS):** When individuals tend to save a large proportion of their income, any additional money received is mostly saved rather than spent. This reduces consumption demand in the economy and helps keep prices stable.
- **Increase in production of goods and services:** If the rise in money supply is matched by a rise in output, more goods enter the market. Since supply increases alongside demand, there is no shortage, and inflation is avoided.
- **Use of money for external debt repayment:** When additional money is used to pay external debts, it flows out of the country. This reduces the money circulating within the domestic economy, limiting demand and preventing inflation.
- **Government price control policies:** When the government sets maximum prices for essential goods, producers cannot raise prices beyond the set limit even if demand increases. This helps to stabilize prices and control inflation.
- **Improved productivity and efficiency (added point):** When increased money supply is used to improve technology, infrastructure, and efficiency in production, output rises faster than demand. This ensures that goods remain abundant in the market, preventing inflationary pressure.

Application activity 6.5



In 2025, the government of Country X increased the money supply by injecting a large amount of money into the economy through public spending on infrastructure projects such as roads, schools, and hospitals. Many people expected that this would lead to a rise in general price levels (inflation). However, after several months, the prices of goods and services remained stable.

Task

1. Explain the possible reasons why the increase in money supply did not lead to inflation in Country X.

6.6 Rwanda inflation rates (2020 – 2025)

Learning Activity 6.6



Question:

Study the trend of inflation rates in Rwanda from 2020 to 2025. Describe how inflation changed during this period and explain the possible economic factors that caused these changes. Use relevant economic concepts to support your explanation.

Year	Annual Inflation Rate (%)	Primary economic drivers and context	Source Basis
2020	7.7%	COVID-19 pandemic logistical disruptions and supply chain bottlenecks.	NISR (CPI annual average) https://statistics.gov.rw/statistical-publications/price-indices-cpi-ppi/consumer-price-index-cpi-may-2025
2021	0.8%	Economic slowdown and baseline corrections following pandemic lockdown measures.	NISR (very low inflation year)
2022	13.9%	Global surge in energy and food prices following geopolitical conflicts.	NISR / BNR reports

2023	12.3%	Severe domestic climate shocks affecting agricultural harvests and driving food inflation.	NISR / BNR (slight easing from 2022 peak) https://www.bnr.rw/documents/Monetary_Policy_Report_Feb2025.pdf
2024	6.3%	Strict central bank monetary policy tightening and improved local crop yields.	BNR monetary policy updates (moderation trend) https://www.bnr.rw/documents/Monetary_Policy_Report_Feb2024.pdf
2025	7.04%	Moderate resurgences in hospitality and core fresh food costs.	https://www.bnr.rw/documents/Monetary_Policy_Report_May2025.pdf

Source: Nation Bank of Rwanda /NISR(Annual average headline inflation)

Based on the macroeconomic drivers outlined in this data, Rwanda predominantly experiences **cost-push inflation** (specifically **structural inflation** and **imported inflation**) rather than demand-pull inflation. According to historical reports from the National Bank of Rwanda (BNR), the country's general price level is highly sensitive to **supply-side shocks**. These include domestic climate shocks, such as droughts or excessive rainfall that lower agricultural yields and cause fresh food prices to increase as well as external structural disruptions. Because Rwanda relies significantly on imported fuel, fertilizers, and consumer items, international logistical bottlenecks and **geopolitical conflicts** instantly trigger imported cost-push inflation, raising local production and transport costs. Consequently, aggregate demand plays a secondary role; instead, volatile supply disruptions in food and energy serve as the primary structural roots of high inflation in the Rwandan economy.

Application activity 6.6



Instructions: Use the information on Rwanda's inflation rates from 2020 to 2025 to answer the following questions:

Question 1: State and explain two possible reasons for the increase or decrease in Rwanda's inflation rate during this period.

Question 2: Explain how the government and the central bank can control inflation using fiscal policy and monetary policy. Give examples of measures that can be used.

6.7. End of unit assessment



Knowledge and Understanding

1. Define the terms **inflation** and **spiral inflation**.
2. Classify inflation according to its rate and briefly explain each type.
3. Differentiate between **demand-pull inflation** and **cost-push inflation**.
4. Distinguish between **imported inflation**, **monetary inflation**, **expectation inflation**, and **headline inflation**.

Skills

5. A government increases money supply to promote economic growth, but prices continue to rise. Analyse how this policy may cause inflation.
6. In a country where prices of essential goods rise because demand is high and supply is limited, suggest measures that can help restore price stability.

Attitudes and Values

7. As a responsible citizen, explain how people in the community can help reduce the negative effects of inflation.
8. State the positive attitudes and values that government officials should demonstrate to help control inflation in the economy.

UNIT 7

UNEMPLOYMENT



Key Unit Competence:

Assess the impact of unemployment on economic development.

Learning objectives

Knowledge and understanding

1

- Distinguish between voluntary and involuntary unemployment.
- Explain the types, causes, effects and solutions of unemployment.
- Distinguish between under and full employment.
- Explain the measures to attain full employment in Rwanda.
- Explain the Keynesian theory of unemployment.
- State the causes and effects of rural urban migration.
- Explain the concept of the unemployment multiplier
- Explain Okun's Law with the aid of an illustration.
- Explain employment multiplier
- Describe the economic implications of the unemployment multiplier on national income and growth.

Skills

2

- Analyze possible solutions or policies to reduce involuntary unemployment.
- Assess the causes and effects of unemployment in the Rwandan economy.
- Analyze the factors that may lead to full employment in an economy.
- Analyze the reasons why it's difficult to attain full employment in Rwanda.
- Calculate the employment multiplier.
- Use illustrations to analyze the applicability of the Keynesian theory of unemployment.
- Examine the causes, effects and measures to control rural urban migration in Rwanda
- Calculate the employment multiplier.

Attitudes and values

3

- Be aware of the effects of unemployment in an economy and advocate for measures to control unemployment.
- Take part in job creation.
- Appreciate the balance needed between full and underemployment in sustaining economic growth.
- Develop awareness that both over- and under-utilization of labor have social and economic costs.
- Show commitment to supporting initiatives that promote efficient use of human resources.
- Value teamwork and cooperation in addressing employment challenges.
- Cultivate a realistic and proactive outlook toward labor market issues.



Introductory activity

In different districts of Rwanda, many young people complete their studies in secondary schools, TVET centres, and universities and return home expecting to find jobs. However, some of them actively seek employment opportunities for a long time without securing. In both urban and rural areas, some adults are also seen without regular work or income-generating activities. At the same time, there are various economic activities such as construction works, shops, transport services, and agricultural activities taking place in these districts, yet some employers report that they are not able to get enough people to take up the available jobs. Despite the presence of these activities, many people still do not have stable employment.

Questions based on the scenario:

1. What do you observe about young people and employment in different districts of Rwanda?
2. Why do some people remain unemployed despite the existence of various economic activities in the economy?
3. What challenges do individuals face as a result of remaining unemployed for a long period?
4. What measures can be taken to improve employment opportunities for people in such situations?

Introduction

Work plays an important role in improving people's standards of living and promoting economic development. In every economy, individuals engage in different productive activities in order to earn income and satisfy their needs. However, some people who are willing and able to work fail to obtain employment opportunities.

This situation results in unemployment, which has become a major economic and social challenge in many countries due to its effects on income levels, production, living standards, and social welfare. Understanding unemployment requires examining its forms, causes, effects, and the measures that can be taken to reduce it.

7.1 Meaning, forms, measurement of unemployment and employment multiplier

Learning Activity 7.1



After completing their studies, a group of youths in Rwanda started looking for jobs in different institutions and businesses. Although they were willing and able to work, many of them stayed for several months without getting employment opportunities. Some depended on their families for support while continuing to search for jobs.

Questions

1. What were the youths in the scenario looking for after completing their studies?
2. What situation were the youths facing despite being willing and able to work?
3. What term can be used to describe a situation where people who are willing and able to work fail to get jobs?

i. Meaning and forms of unemployment

Unemployment is a state in which some members of the labour force are idle. Or it can be defined as a situation where a factor of production is able and willing to work at the ongoing wage rate but fails to find a job.

According to Lord Keynes, unemployment takes two forms;

Forms	Causes
» Voluntary Unemployment: This refers to the situation where jobs are available but people are not willing to work at the ongoing wage rate.	» Low wages
» Jobs exist, but people choose not to work at current wages	» Desire to live on personal or family wealth
	» Target workers
	» Poor working conditions
	» Preference for leisure
	» Remote job locations
	» Risky jobs
	» Early retirement

- | | |
|--|--|
| <ul style="list-style-type: none"> » Involuntary Unemployment: This is a situation where members of the labour force are idle but are willing to work in the existing situations at the current wage. Individuals want to work but there is no employment. » People are willing to work at current wages, but jobs are unavailable | <ul style="list-style-type: none"> » Low demand for goods and services. » Economic recession. » Use of machines and technology. » Rapid population growth. » Seasonal changes in economic activities. |
|--|--|

ii. Measurement of unemployment and the employment multiplier

Unemployment is not only observed in everyday life when people are seen without jobs, but it is also carefully measured to understand its actual level in an economy. Measuring unemployment helps policymakers and economists determine how many people who are willing and able to work are actively searching for jobs but cannot find them. This measurement provides useful information for assessing the performance of the economy, identifying labour market problems, and designing appropriate policies to create employment opportunities and improve living standards.

Governments and economists need accurate information to plan and make sound decisions. Reliable data helps them understand economic conditions and design effective policies.

Key points on measuring unemployment

Labor Force: refers to all people within the working-age group, typically aged **15 to 64 years**, who are either employed or actively seeking employment. It includes both individuals who are currently working and those who are willing and able to work but are unemployed.

The labour force comprises of all unemployed (**U**) and employed (**E**) persons.

Labour force = U + E

People who are **not members of the labour force** are those within the working-age group (usually 15–64 years) who are **neither employed nor actively seeking employment**. This includes students, prisoners, full-time homemakers who are not looking for jobs, retired persons, and individuals who are unable or unwilling to work.

Employed people: are individuals within the working-age group (usually 15–64 years) who are engaged in paid work or self-employment. They are individuals who are actively working in jobs or economic activities from which they earn income.

Unemployment Rate Formula:
$$\text{Unemployment rate} = \frac{\text{NPUE}}{\text{NPLF}} \times 100$$

Where:

NPUE: is the number of people unemployed

NPLF: is the number of people in the labour force.

Exercises 7.1

In **Kicukiro District**, a labour market survey was conducted in 2024. The results showed that:

- Total labour force = 50,000 people
- Employed persons = 42,000 people
- Unemployed persons = 8,000 people

Required:

Calculate the unemployment rate in **Kicukiro District**.

Employment multiplier

When there is a change in development because of increased government expenditure, it generates additional change in employment in related production activities and sectors and as a result, the initial change in employment results in greater in total employment. For example, if the government spends on road construction, it directly employs the people. This is called **primary employment**. The road works will lead to increased demand for cement, bitumen, stones etc. this will create additional jobs in the cement, bitumen production and stone quarries. This is called **secondary employment**.

The employment multiplier therefore is the number of times initial employment will multiply itself to give a final change in total employment. If the initial jobs created are 7 and as a result the total employment created is 35, then the multiplier is: $\frac{35}{7} = 5$

$$\text{Employment multiplier} = \frac{\text{Total employment}}{\text{Jobs created}}$$

Application activity 7.1



1. Explain the meaning of unemployment.
2. Identify and explain forms of unemployment according to Keynes.
3. Explain how unemployment is measured in an economy.

if:

Labour force = 40,000

Employed = 34,000

Calculate:

- i) Number of unemployed persons
- ii) Unemployment rate

4. Explain how the establishment of Gisakura Tea factory located in Nyamasheke district created additional employment in other sectors.

7.2 Underemployment and Full Employment

7.2.1 Difference between underemployment and Full Employment

Aspect	Underemployment	Full unemployment
Definition	– This is a situation where the marginal productivity of a worker is less than what it should be. Or a situation where the person's capacity to work is underutilized.	– According to Lord Keynes, full employment is a situation where a person who wants a job gets it i.e. there is no involuntary unemployment. – F.W Parish says that full employment is a situation where 3% or less of the people is unemployed.
Examples	– A Senior Six MEG leaver working in a kiosk rather than applying his Economics, Mathematics, and Geography knowledge. – A university graduate working only a few hours a week as a street vendor because they cannot find a full-time job that matches their qualifications and skills.	– A situation where almost everyone willing and able to work has a job, except a few people changing jobs or voluntarily unemployed. – Jobs available equal job seekers

7.2.2 How to achieve full employment

Achieving full employment is one of the major goals of any economy because it ensures that available human resources are effectively utilized and living standards are improved. It requires deliberate efforts from both the government and the private sector to create enough job opportunities for all people who are willing and able to work. Various strategies can be applied to reduce unemployment and move the economy closer to full employment. Some of those strategies are mentioned below:

- **Promoting industrialization:** Establishing more industries creates many job opportunities for both skilled and unskilled labour.
- **Improving education and skills training:** Providing relevant skills through technical and vocational education helps workers match available jobs.
- **Encouraging entrepreneurship:** Supporting small and medium enterprises enables people to create their own jobs instead of waiting for formal employment.
- **Government investment in infrastructure:** Building roads, energy, and communication systems creates direct jobs and supports business growth.

- **Attracting foreign and local investment:** Investment increases production activities, which leads to higher demand for labour.
- **Effective labour market policies:** Policies that support job creation, reduce unemployment barriers, and improve job matching help reduce joblessness and move towards full employment.
- **Encouraging rural development:** Developing agriculture and rural industries creates jobs outside cities, reducing unemployment pressure in urban areas.
- **Improving access to finance:** Providing loans and credit facilities helps individuals start businesses, increasing self-employment opportunities.
- **Strengthening labour market information systems:** Providing accurate job information helps workers find suitable jobs faster, reducing structural unemployment and improving employment levels.

7.2.3 Reasons why it is difficult to attain full employment

Full employment in an economy is often difficult to attain due to a range of economic, social, and structural factors that limit job creation and restrict access to available employment opportunities. These factors affect both the demand and supply of labour, making it hard for all willing and able individuals to be fully employed. The main reasons are explained below:

- **Rapid population growth:** The labour force increases faster than job creation, making it difficult for the economy to provide employment for everyone.
- **Skills mismatch:** Many job seekers lack the skills required by employers, leading to unemployment even when jobs are available.
- **Technological advancement:** Use of machines and automation replaces human labour, reducing the number of available jobs.
- **Economic fluctuations (business cycles):** During recessions or economic downturns, businesses reduce production and lay off workers, increasing unemployment.
- **Limited investment:** Low levels of local and foreign investment result in fewer industries and fewer job opportunities.
- **Structural changes in the economy:** Shifts from agriculture to industry or services can make some skills obsolete, leaving workers unemployed.
- **Low government expenditure:** Limited job creation despite programs like One Cow per Family.
- **Contractionary monetary policy:** High interest rates discourage borrowing and investment.
- **Tight fiscal policy:** Heavy taxation forces small businesses to close or deters new investors.
- **Regional insecurity:** Instability in neighboring countries (DRC, Burundi) affects border areas.

- **Residual unemployment:** this is unemployment among people who face specific barrier to available jobs, skills, location, health or accessibility constraints.

Application activity 7.2



1. Despite government efforts to create jobs, full employment is still difficult to achieve. What do you think are the main obstacles preventing full employment, and how do they affect different groups of people in the economy?
2. If you were an economic advisor to the Rwandan government, what practical measures would you propose to move the economy closer to full employment? Explain how your proposals would solve unemployment problems.

7.3 Types and causes of unemployment

Learning Activity 7.3



1. Why do some people remain unemployed even when they are willing and able to work?
2. Do all unemployed people become jobless for the same reasons? Explain.
3. What factors in an economy can increase the level of unemployment?

7.3.1 Types of unemployment

- Unemployment in an economy does not occur in a single form, but rather appears in different categories depending on the conditions and causes behind it. These variations reflect the complex nature of the labour market and the diverse situations faced by job seekers. For this reason, economists classify unemployment into distinct types in order to better understand its nature and behaviour within an economy. **The main types of unemployment are outlined below.**

- **Open Urban Unemployment:** This occurs when people in urban areas are willing and able to work but cannot find jobs, either voluntarily or involuntarily, often due to rural–urban migration.

Example: A person who moves from **Nyamagabe District** to **Kigali City** in search of employment but remains jobless because available jobs in the city are limited and highly competitive

- **Seasonal / Occasional unemployment:** Seasonal unemployment refers to a situation where **workers are unemployed during certain times of the year** because their jobs depend on seasonal activities. So, at that time, they are **not working and not earning income**, which is why it is called unemployment.

Example: In Rwanda, many farm workers are employed during the planting and harvesting seasons for crops such as maize and beans. However, after

the harvesting period, these workers become unemployed because farming activities slow down until the next planting season begins.

- **Disguised unemployment:** This is a situation where more people are employed in a job than is actually needed. The extra workers appear to be working, but they do not add any extra output. If they are removed, total production does not change, and those workers may lose their jobs and become unemployed since their work was not truly required.

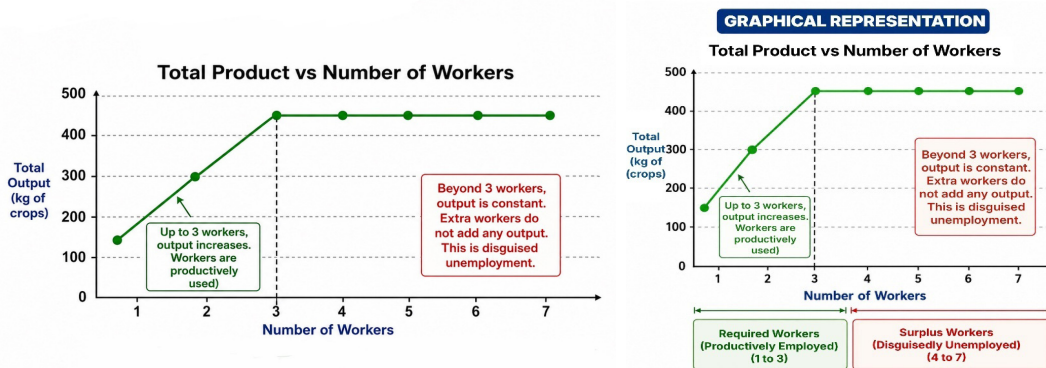


Image 7.1: Disguised unemployment

- **Frictional unemployment :** Frictional unemployment is essentially **short-term unemployment** that occurs when people are switching from one job to another. This may occur when people constantly change their jobs, some quit or others are fired. These constant changes result in frictional unemployment. There are always some firms with unfilled vacancies and some people looking for work. The period of unemployment between losing one job and finding another is included under frictional unemployment.
- **For example:** A teacher in **Kigali City** may decide to resign from her current school at the end of the term in search of better pay, improved working conditions, or a school closer to home. After resigning, she does not immediately get another job and spends some time searching, attending interviews, and waiting for job responses from different schools. During this short period, she is unemployed, even though she is qualified and willing to work. This temporary gap between leaving one job and starting another is what is referred to as

frictional unemployment.

There are a variety of reasons to explain the existence of frictional unemployment but these can be explained in the two categories below;

- **Normal labour market turnover.** This arises from two sources. First, people are constantly changing their economic activities. Young people are leaving school and joining the labour force and secondly, old people are retiring and leaving the labour force temporarily for some other reason, and then joining it.
- **The fortunes of businesses are constantly changing:** some are closing down and laying off their workers and new firms are starting up and are hiring.

1. Secular/Structural unemployment

This is unemployment that occurs when changes in the economy (such as changes in demand for goods and services, technology, or production methods) reduce the need for certain skills or types of work, creating a mismatch between the skills workers have and the skills required in available jobs. **For example**, as an economy develops, there is a move away from employment in the primary sector towards employment in secondary and tertiary sectors and as a result of such economic change many workers in the primary sector become unemployed.

Technological unemployment



Image 7.2.: Technological unemployment

This is a type of unemployment that happens when workers lose their jobs because machines, computers, or new technologies replace human labour.

As firms adopt new technologies (**like automation, artificial intelligence, or modern machines**), they can produce more efficiently with fewer workers. As a result, some employees are laid off or cannot find jobs in the same sector because their skills are no longer needed

In Rwanda, technological unemployment caused by AI or other technologies is seen in different sectors of economy. For instance, banks use AI systems to handle customer service and transactions through mobile apps and call centers also some farmers use machine to increase production which create unemployment to large groups of people.

Possible solutions to technological unemployment:

- **Skill development and retraining:** Workers should be trained to acquire new skills that match modern technologies, such as digital literacy, **AI** management, and technical skills.
 - **Education system reform:** Schools and training institutions should update curricula to focus more on science, technology, engineering, and problem-solving skills.
 - **Encouraging entrepreneurship:** Governments should support people to start their own businesses, creating self-employment instead of depending only on formal jobs.
 - **Government job creation policies:** The government can invest in public projects and industries that create new employment opportunities for displaced workers.
 - **Balanced use of technology:** Firms should adopt technology gradually and combine it with human labour where possible, so that machines complement rather than fully replace workers.
- **Casual/ Erratic unemployment:** This is a form of unemployment that results from expiry of contracts and target workers. It commonly occurs to people like private doctors, lawyers, and car washers, who work if they have clients. The private doctor works when he has a patient to attend to. If he has no patients, he becomes unemployed and hence casual unemployment. This type of unemployment is hard to cure because it comes at irregular intervals. It cannot be predicted unlike the seasonal unemployment.
 - **Residual unemployment:** This is a type of unemployment that occurs to a factor of production especially labour because it is mentally or physically handicapped like the lame, mad people among others. Labour may become unemployed when the work that is available needs someone who is physically and mentally able.
 - **Transitional (Normal) Unemployment:** This is a type of unemployment that occurs when production or business activities stop temporarily for a short period. This happens because firms may close for maintenance such as repairing breakdowns, renovating buildings, painting, or other short-term adjustments. During this time, workers are not actively working, but they are expected to return once production resumes.
 - **Cyclical / Keynesian unemployment:** This is caused by a decline in demand for goods and services in the economy, which forces businesses to reduce production and lay off workers.
 - **Cost-push unemployment** This is unemployment that occurs when the cost of production increases, forcing firms to reduce production and lay off workers. It may happen because of rising wages, high prices of raw materials, increased taxes, or expensive fuel and electricity. As production becomes more costly, firms employ fewer workers, leading to unemployment.

7.3.2 General causes of unemployment

Experimental version

Unemployment does not occur without reasons. In every economy, there are factors and conditions that may prevent people from getting jobs even when they are willing and able to work. These factors explain why unemployment exists and why its level may increase or decrease over time. Therefore, it is important to study the causes of unemployment in order to better understand the unemployment problem in an economy. **These causes are explained as below:**

- **High population growth rate:** The high population growth rate which does not correspond with increase in job creation has left many unemployed.
- **Use of capital-intensive technology:** The desire to produce high quality out and more in the shortest time possible has led to the use of machines than people hence people have been left out of jobs.
- **Slow growth of economies:** These especially in developing countries due to low incomes, low savings and low investments which makes it difficult to generate employment opportunities.
- **Existence of a large subsistence sector:** Agriculture in developing countries is still backward and still uses primitive methods which leads to low production, underemployment and unemployment.
- **Government policy towards employment:** In developing countries there is lack of national policies regarding employment and because of this, jobs have been advertised to people who already have them leading to unemployment of the newly qualified.
- **Defective education system:** This creates more job seekers than creators and it imparts little skills to them thus looking for jobs instead of creating them.
- **Social discrimination:** This is done by the employers where they offer jobs basing on race, tribe and religious grounds leading to discrimination and unemployment.
- **Uneven distribution of development:** The social and economic infrastructure is lacking in rural areas hence forcing the people to move to urban centers causing open urban unemployment.
- **Seasonal changes:** This occurs in the agricultural sector where crops are grown in particular seasons hence during unproductive seasons, the farmers become unemployed.
- **Political instability:** This causes uncertainty which discourages production, limits investment and job creation.

Application activity 7.3



1. Explain the different types of unemployment found in an economy.
2. Discuss the main causes of unemployment in a developing economy.

Fill in the gaps below:

3. Unemployment that occurs when workers move from one job to another is called _____ unemployment.
4. Unemployment caused by a fall in demand for goods and services in the economy is known as _____ unemployment.
5. Unemployment that results from a mismatch between workers' skills and available jobs is called _____ unemployment.

6. Read the scenario below and answer the questions that follow.

"In a certain country, many workers in the agricultural sector are employed only during planting and harvesting seasons. After the harvest period, they remain jobless until the next farming season begins. At the same time, some factory workers lose their jobs because machines have replaced human labour.

- a) Identify the types of unemployment illustrated in the scenario.
- b) Explain the causes of each type of unemployment identified above.

7.4 Effects and policies of unemployment

Learning Activity 7.4



Read the situation below and answer the questions:

In a certain district, many young people remain unemployed for a long time despite being able and willing to work. As a result, they struggle to meet basic needs, depend on family support, or engage in low-income informal activities, leading to poverty and difficult living conditions.

Questions:

1. Why do some young people remain unemployed even though they want to work?
2. Identify the economic effects of unemployment mentioned or implied in the situation.
3. Identify the social effects of unemployment mentioned or implied in the situation.
4. Which solution do you think is better for reducing unemployment? Give reasons for your answer.

7.4.1 Effects of unemployment

Unemployment is not only a challenge for individuals who lose income and livelihood, but it also has far-reaching consequences on the overall performance of the economy and society. The effects of unemployment can be grouped into economic and social effects as follows:

Economics effects:

- **Reduces government revenue:** When many people are unemployed, they do not earn income and therefore pay little or no income tax. This reduces the amount of money the government collects, limiting its ability to fund public services such as education, health, and infrastructure.
- **Decrease in national output (GDP):** Unemployment means that part of the labour force is not engaged in production. As a result, fewer goods and services are produced, leading to a decline in the country's Gross Domestic Product (GDP) and slowing economic growth.
- **Increase in government expenditure:** The government may need to spend more money on supporting unemployed people through welfare programs, job creation schemes, and other social assistance. This increases public spending and can strain the national budget.
- **Loss of skills and productivity:** When people remain unemployed for a long time, they may lose their skills or become less productive due to lack of practice. This reduces their efficiency when they eventually get jobs and lowers overall labour productivity in the economy.
- **Reduced consumer spending:** Unemployed individuals have little or no income, so they reduce their spending on goods and services. This leads to lower demand in the economy, which can negatively affect businesses and slow down economic activity.
- **Wastage of human resources:** Unemployment means that available labour is not being used in production. This results in a waste of human potential, as people who could contribute to economic development remain inactive instead of working productively.

Social effects:

- **Increase in poverty levels:** When people are unemployed, they have no regular income to meet basic needs such as food, clothing, and shelter. This leads to widespread poverty and a lower standard of living in society.
- **Rise in crime and insecurity:** Unemployment may push some individuals, especially youth, to engage in illegal activities such as theft or fraud in order to survive. This increases crime rates and makes communities less safe.
- **Family breakdown and conflict:** Lack of income can create conflict within families. This may lead to disagreements, domestic conflicts, or even separation, weakening family unity and stability.
- **Psychological problems (stress and depression):** Unemployed individuals often experience frustration, stress, low self-esteem, and depression because they cannot find work or support themselves financially.

- **Reduced access to social services:** Without income, unemployed people struggle to afford services such as healthcare, education, and proper housing, leading to poor living conditions and social exclusion.
- **Increase in dependency:** Unemployed individuals often depend on relatives, friends, or government support for survival. This increases the burden on families and society, especially when unemployment is widespread.

7.4.2 Measures to reduce unemployment

- **Expansionary fiscal policy:**
 - **Increasing government spending:** The government increases spending on projects such as roads, schools, and hospitals. These projects create jobs directly for workers and indirectly for suppliers, reducing unemployment.
 - **Tax reduction for businesses and investors:** Reducing taxes encourages both local and foreign investors to start or expand businesses. When businesses grow, they hire more workers, creating employment opportunities.
- **Population control and family planning:** The government promotes family planning and awareness programs to ensure that population growth does not exceed the number of available job opportunities, helping to reduce pressure on the labour market.
- **Rural development and infrastructure improvement:** Developing rural areas through better roads, electricity, and small industries reduces rural–urban migration and creates employment opportunities in rural communities.
- **Education reform and vocational training:** Education systems are improved to emphasize practical and vocational skills that match labour market needs, enabling graduates to become more employable and reduce skill mismatch.
- **Government investment in public works:** The government creates jobs directly by investing in projects such as road construction, schools, hospitals, and environmental programs, which also stimulate economic activity.
- **Economic diversification and industrial development:** Expanding different sectors such as agriculture, mining, tourism, and manufacturing increases job opportunities and reduces dependence on a single economic sector.
- **Promotion of private investment and tax incentives:** By offering tax holidays and investment incentives, governments attract local and foreign investors who establish businesses and create employment opportunities.
- **Support for SMEs and self-employment:** Providing loans, grants, and training to small and medium enterprises helps individuals start their own businesses and reduces dependence on formal employment.
- **Modernisation of agriculture and labour-intensive industries:** Improving agricultural methods and encouraging industries that use more labour increases productivity and creates more jobs, especially in rural areas.

- **Improvement of job information services and labour mobility**
Job centers, media advertising, and better transport and communication systems help workers access job information easily and move to areas where jobs are available.
- **ICT and AI-based job matching and skills development systems**
The use of online job platforms and mobile applications helps connect job seekers with employers more efficiently, reducing the time people spend searching for jobs. In addition, artificial intelligence can analyze labour market trends to forecast future job demands and provide personalized training recommendations. This combination improves access to job information, enhances relevant skills, and increases employability, thereby reducing unemployment.

Application activity 7.4



1. How has the unemployment problem affected the development process in Rwanda?
2. What do you think the Rwandan government should do to reduce the unemployment problem?

7.5 Theories of unemployment

Learning Activity 7.5



Unemployment is a complex economic problem that affects different people in different ways. Although it is a common issue in many economies, people often do not agree on its causes or solutions because they look at it from different perspectives and experiences. Why might economists, government officials, and ordinary citizens have different views on the causes of unemployment? Explain the reasons for these differences.

Unemployment can be explained in different ways depending on its causes and the state of the economy. Economists have developed various theories to better understand why unemployment occurs and why it may persist in society. These theories provide different perspectives on the problem and help in designing appropriate solutions to address.

In this section, we are going to focus on the following theories:

7.5.1 Keynesian unemployment theory

This theory was developed by the economist John Maynard Keynes. It explains unemployment as a result of low aggregate demand in the economy, especially during recessions. When demand for goods and services falls, firms reduce production and lay off workers, leading to cyclical unemployment. Keynes also emphasized that higher consumption and investment help create employment opportunities, while low consumption increases unemployment.

Experimental version

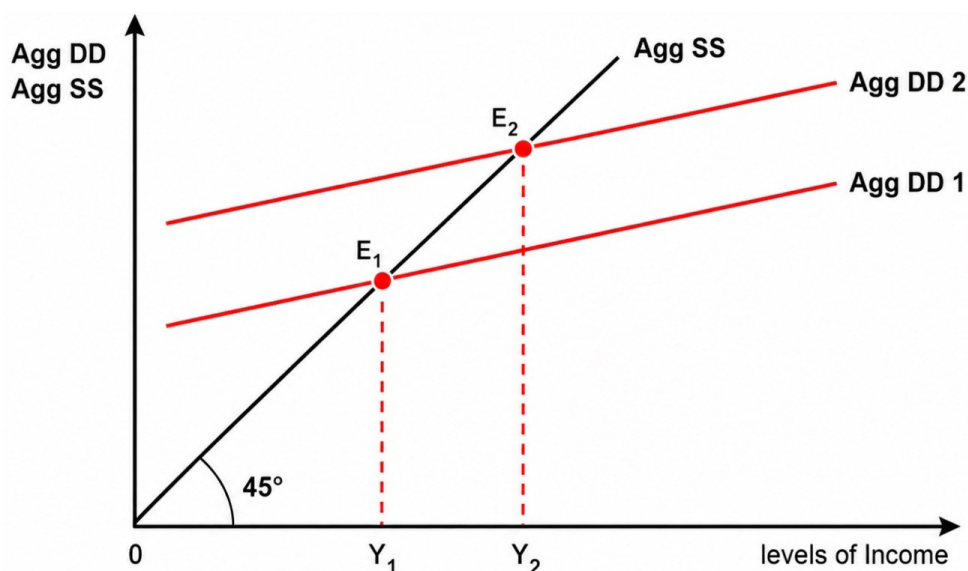


Figure 7.1.: Keynesian unemployment

From Figure above, E1 is the equilibrium point with employment level Y1; some of the workers willing to work have not been absorbed. It means that E1 (effective demand point) is an under-employment equilibrium and Y1 is under employment level of income. The unemployed workers can be absorbed if the aggregate demand increases from Agg dd1 to Agg dd2 and a new equilibrium point E2 is established with employment level Y2 which we assume is the full employment level. Hence increase in aggregate demand from Agg dd1 to Agg dd2 leads to increase in employment ($Y_2 - Y_1$)

Possible solutions to cyclical unemployment

To reduce cyclical unemployment, governments and other stakeholders can introduce measures that increase aggregate demand and create employment opportunities. Some of these policies are mentioned below:

- **Increasing government expenditure:** The government can spend more on public projects such as roads, schools, and hospitals to create jobs and increase demand.
- **Reducing taxes:** Lower taxes increase people's disposable income, encouraging consumption and investment.
- **Lowering interest rates:** Lower interest rates encourage borrowing and investment by businesses, leading to more employment opportunities.
- **Encouraging private investment:** Providing incentives to investors helps firms expand production and employ more workers.
- **Increasing aggregate demand:** Policies that stimulate demand for goods and services help firms increase production and reduce unemployment.
- **Relevance of the Keynesian theory of unemployment to less developed countries**

- As suggested by Keynes, in developing countries, demand for labour is derived demand. This implies that a fall in demand for goods and services will lead to a fall in incomes of the employers because their products have no market and in the end they will react by laying off workers.
- Through exports, new markets are got for the local goods and this implies that the local investors will earn more income. They will be able to set up more investments in order to satisfy the market and hence they will need to employ more workers.
- Reduction in tax as suggested by Keynes can lead to increase in household income and expenditure. This will in turn increase aggregate demand, the investors will react by needing to produce more to satisfy the increased demand and in the end, they will need to hire and employ more workers leading to increased employment opportunities.
- Increasing government expenditure as suggested by Keynes increases money supply in the economy. This will further increase aggregate demand provoking producers to increase production which they can only do after employing more workers.
- A conducive investment climate as suggested by Keynes encourages investment through attracting investors. These will in turn need to increase production and, in the end, they will end up employing more workers.

Limitations of Keynesian unemployment theory

- Keynes focuses on the demand side, yet in developing countries unemployment is mainly caused by inadequate capital rather than low aggregate demand.
- Increasing government expenditure may lead to inflation, raising prices and reducing purchasing power, which can still cause unemployment.
- The theory ignores institutional and structural problems such as poor infrastructure and weak land tenure systems that limit investment and job creation.
- Increased government spending may not create jobs in agriculture because people often prefer white-collar jobs, so employment may not rise.
- The theory assumes a fully monetarised economy, but Developing countries have large subsistence sectors where many people do not use money markets.
- Industrial investment may not absorb all labour because firms may adopt capital-intensive methods, reducing employment opportunities.
- Developing countries have inelastic demand for imports, so increased demand may benefit foreign producers rather than creating local jobs.
- Increasing money supply may not raise aggregate demand because people may hoard money as cash or wealth instead of spending it.
- The theory focuses only on demand-deficient unemployment and ignores other types and causes of unemployment, making it incomplete.

7.5.2 Rural urban migration theory

Rural–urban migration is the movement of people from rural areas to urban areas in search of better living conditions, employment opportunities, education, and services. The process of rural-urban migration is brought by push (from villages) or pull factors (from towns) and depends on a number of factors, which are both economic and non-economic.

Rural–urban migration theory of unemployment explains unemployment as a result of people moving from rural areas to urban areas in search of better job opportunities, but finding limited employment in towns and cities.

According to this theory, many workers leave rural areas where jobs are often seasonal, low-paid, or based on agriculture, and migrate to urban areas expecting more employment opportunities in industries and services. **However**, because the number of job seekers in urban areas increases faster than the available jobs, many migrants fail to get employment immediately or remain unemployed for a long time.

As a result, unemployment in urban areas increases due to an oversupply of labour caused by continuous rural–urban migration.

❖ Causes of rural urban migration

Push factors are the negative conditions in rural areas that force or encourage people to leave and move to urban areas	Pull factors are the positive conditions in urban areas that attract people to move from rural areas to towns and cities
» Lack of employment opportunities	» Employment opportunities
» Low incomes and poverty	» Higher wages and incomes
» Land shortage and land fragmentation	» Better education facilities
» Poor social services (education and health)	» Improved healthcare services
» Natural hazards such as drought and floods	» Better social services and living standards

Consequences of rural urban migration

- **Overpopulation in urban areas:** Cities become crowded due to high inflow of people.
- **Unemployment in towns:** Limited job opportunities lead to joblessness.
- **Growth of slums:** Poor migrants settle in informal and unsafe housing areas.
- **Rural labour shortage:** Loss of young and active workers affects rural production.
- **Increased crime rates:** Poverty and unemployment may lead to criminal activities.

- **Pressure on social services:** Schools, hospitals, water, and sanitation become overstrained.
- **Increase in urban pollution:** More people lead to higher waste production, traffic congestion, and environmental degradation.
- **Increased cost of living in urban areas:** high demand for housing, food, and services raises prices.

Measures to control rural –urban migration

- **Improving rural employment opportunities:** Governments can promote industries, agro-processing, and small enterprises in rural areas so that people get jobs near their homes.
- **Modernizing agriculture:** Providing improved seeds, irrigation, mechanization, and extension services increases farm productivity and incomes, reducing the need to move to towns.
- **Developing rural infrastructure** – Building good roads, electricity, water supply, and communication systems improves living conditions in rural areas.
- **Providing social services in rural areas** – Establishing schools, health centres, and training institutions reduces the need for people to move to urban areas for these services.
- **Promoting balanced regional development** – Government should invest equally in both rural and urban areas to reduce economic inequality that pushes migration.
- **Encouraging rural industrialization** – Setting up small and medium industries in rural areas creates job opportunities and reduces rural poverty.
- **Improving access to credit and investment in rural areas** – Providing loans and financial support enables people to start businesses and income-generating projects locally, reducing the need to migrate.
- **Improving rural housing conditions** – Providing better and affordable housing in rural areas improves living standards and reduces the desire to move to towns.
- **Promoting rural education and awareness** – Educating people about opportunities in rural areas and providing quality education reduces the belief that cities are the only place for success.
- **Encouraging rural development policies** – Implementing policies that focus on rural development (such as subsidies, rural development funds, and community projects) helps improve livelihoods and reduce migration pressure.

Application activity 7.5



Using the theories of unemployment discussed above, answer the following questions:

Explain:

1. Explain the meaning of Keynesian unemployment.
2. Describe the solutions to unemployment according to Keynesian theory.
3. Discuss the relevance and limitations of Keynesian theory in developing countries.
4. Explain the rural–urban migration theory of unemployment.

7.6 Okun's law

Learning Activity 7.6



In your own opinion, explain the relationship between unemployment and production in an economy, and show how changes in production can affect employment levels and how unemployment can also influence the level of output of goods and services.



Image 7.3.: Arthur Melvin Okun

Arthur Okun was a Yale professor and an economist born in November 1928. In his law (Okun's law), he analysed and developed a relationship between unemployment and national output (economic growth). He further said that, because of unemployment, some people are not engaged in production and so do not contribute to national income.

This implies that unemployment reduces the level of national income. If the unemployed people were to be employed, national income would increase. National planners therefore would want to determine the exact relationship between unemployment and gross national product for proper planning and plan implementation.

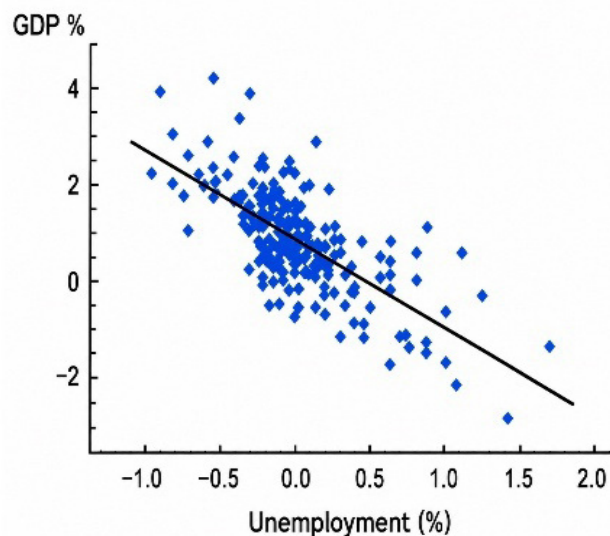


Figure 7.2.: Relationship between unemployment and GDP

Considering Figure above, it is seen that an increase in the unemployment rate to 2.0%, the percentage of GDP reduces to even negative. While a reduction in the unemployment rate to -1.0%, there is an increase in the GDP to more than 2%. One version of Okun's law has stated very simply that when unemployment falls by 1%, GNP rises by 3%.

For example, a fall in the rate of unemployment by 1% from 14% to 13%, causes gross national product to increase by 3% from 100 billion francs to 103 billion francs. Another version according to Okun's states that, a percentage increase in unemployment causes a 2% fall in GDP and a percentage fall in unemployment, causes an increase in GDP by 2%.

Application activity 7.6



Briefly explain what Okun's Law states about the relationship between unemployment and GDP growth.

7.7 Unemployment and inflation: The Phillips curve

Learning Activity 7.6



1. Why do you think prices of goods and services sometimes rise when many people are employed?
2. Imagine a country where unemployment is falling rapidly. What changes might occur in prices, wages, and demand for goods and services?



Image 7.4.: Alban W. Phillips

A.W. Phillips (1914 – 1975) was a New Zealand economist who introduced the Phillips Curve in 1958. He showed the relationship between unemployment and inflation, explaining that low unemployment is often associated with rising wages and prices. His work became important in the study of macroeconomics and government economic policies.

The coexistence of persistent inflation and widespread unemployment is a relatively new phenomenon. Before 1950, many economists believed that unemployment and inflation could not exist simultaneously. A. W. Phillips, economist was among the first to call attention to the relation between the rate of change in wages and the level of unemployment. He developed what he called the Phillips curve.

The Phillips curve expresses an inverse relationship between the rate of increase in wages and the unemployment rate. He explained that workers are less likely to demand wage increases when there is a large army of unemployed persons, hence a lower inflation rate. When unemployment is low, competition among employers for scarce labour services, forces wage up, hence a higher inflation rate.

This inverse relationship between the rate of inflation and the unemployment rate is called the Phillips curve. He then concluded that: **"The higher the rate of unemployment, the lower the rate of inflation; and the lower the rate of unemployment, the higher the rate of inflation."**

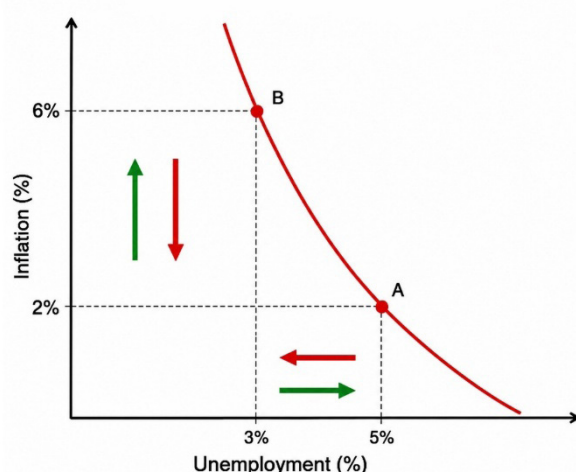


Figure 7.3.: Phillips curve

From Figure above, point A shows a lower rate of employment at 2% while the rate of unemployment is high at 5%. Point B shows a higher rate of interest of 6% while the unemployment rate is low at 3%.

Application activity 7.7



Read the scenario below and answer the questions that follow.

A country experiences rapid economic growth after the government invests heavily in industries and infrastructure. Many businesses expand their production and employ more workers. As employment increases, workers begin demanding higher wages, and the prices of goods and services also start rising.

1. Explain how the increase in employment may lead to higher wages and prices in the economy.
2. If the government wants to reduce inflation in this country, what might happen to unemployment? Explain your answer.
3. Draw and interpret a simple Phillips Curve showing the relationship between inflation and unemployment.

7.8. End of unit assessment



Knowledge and understanding

1. Distinguish between voluntary unemployment, involuntary unemployment, underemployment, and full employment.
2. Explain the major causes and effects of unemployment in developing countries such as Rwanda.
3. Describe the measures that can be used to achieve full employment and reduce unemployment in an economy.
4. Explain the Keynesian theory of unemployment and show how government policies can reduce cyclical unemployment.
5. Explain the relationship between unemployment, inflation, and national output using the Phillips Curve and Okun's Law.

Skills

1. A district has a labour force of 80,000 people, with 68,000 employed.
 - a) Calculate the number of unemployed persons.
 - b) Calculate the unemployment rate.
 - c) Interpret your results in relation to the labour market situation in the district.
2. A government project initially creates 20 jobs, but total employment later rises to 100 jobs across related sectors.
 - a) Calculate the employment multiplier.
 - b) Explain how this project has stimulated secondary employment in the economy.

3. "A manufacturing company replaces many workers with modern machines to increase efficiency."
 - a) Identify the type of unemployment involved.
 - b) Explain its likely effects on workers and the economy.
 - c) Suggest appropriate solutions to this problem.
4. Using economic reasoning, analyze how rural–urban migration and skills mismatch contribute to unemployment in developing economies, and suggest practical policy measures to address these challenges.

Values and attitudes

5. Explain why it is important for young people to develop positive attitudes toward self-employment, innovation, and job creation instead of depending only on formal employment opportunities.
6. Do you agree that unemployment is both an economic and social problem? Give reasons for your answer.

UNIT 8

PUBLIC BUDGET



Key Unit Competence:

Analyze the role of public debt management and budget in economic development.

Learning objectives

Knowledge and understanding

- Explain the meaning of public finance.
- Describe the meaning and types of public budget.
- Analyse the causes of budget deficit in developing countries.
- Explain the problems of budgeting in developing countries.
- Propose possible solutions to reduce budget deficit.
- Explain the role of government in the economy.
- Identify sources of public revenue.
- Explain methods of expanding government revenue.
- Describe the national budget and types of budgets.
- 1 • Explain the meaning of public debt.
- Classify types of public debt.
- Analyse the causes of public debt.
- Assess the consequences of public debt.
- Explain the meaning of public debt management.
- Describe the objectives of public debt management.
- Explain methods of clearing public debt.
- Analyse the burden of public debt.
- Explain the meaning and types of public expenditure.
- Analyse reasons for and purposes of public expenditure in Rwanda and other developing countries.
- Propose ways of reducing government expenditure.

Skills

2

- Analyse government budgets and interpret economic data.
- Evaluate causes and effects of budget deficits and public debt.
- Propose practical solutions to fiscal challenges.
- Apply knowledge of public finance to real-life situations.
- Work collaboratively in discussions and presentations.

Attitudes and values

3

- Appreciate the role of budgeting in an economy.
- Show the concern, transparency and Honesty to contribute to national development through responsible payment of taxes and proper use of public resources.
- Appreciate the need of government expenditures
- Develop reasoned opinions on how effective budgeting, debt management, and expenditure can promote good governance and socio-economic development.
- Advocating for responsible borrowing and spending.
- Show patriotism and willingness to contribute to national development through taxes and responsible citizenship.
- Show understanding that public finance decisions affect the welfare of all citizens.
- Expresses concern for ethical fiscal management



Introductory activity

Read the scenario and answer the questions that follow:

A government of a developing country is planning its annual budget. It wants to build new hospitals, improve roads, increase teachers' salaries, and support farmers. However, the money collected from taxes and other sources is not enough to cover all these planned expenses.

1. What do you think the government should do in this situation?
2. Why might the government's money be less than its planned spending?
3. Where else can a government get money apart from taxes?
4. What problems can arise when a government borrows a lot of money?

Introduction

Every government has the responsibility of providing essential services to its citizens, such as education, healthcare, security, roads, and clean water. To provide these services, the government must raise money and decide how it will be spent to meet the needs of the population. The study of how the government collects revenue, spends money, and manages public resources is known as **public finance**.



Image 8.1.: National Budget Brief case

Public finance therefore helps the government to plan and control the use of national resources in order to promote economic growth and improve the welfare of citizens. One of the major tools used in public finance is the **public budget**, which shows the government's estimated revenue and planned expenditure for a given financial year. Through the public budget, the government is able to allocate resources to different sectors and implement its development plans effectively.

8.1 Public budget

Learning Activity 8.1



Imagine that you are a leader of a school, family, or village with limited money but many needs to satisfy. Answer the following questions:

1. If you were given a fixed amount of money to manage the needs of your school or family for one month, how would you decide what to spend on first?
2. Why is it important to plan how money will be used before spending it?
3. What problems might arise if money is spent without a proper plan?
4. How can proper planning of income and expenditure help a government develop a country?

The public budget of Rwanda is a financial plan prepared annually by the government, through the Ministry of Finance and Economic Planning (**MINECOFIN**), and approved by Parliament before implementation. It outlines estimated government revenue and planned expenditure for a financial year. The budget is guided by national development goals such as **Vision 2050** and the National Strategy for Transformation (**NST 1&2**), aiming to promote economic growth, reduce poverty, improve social welfare, and ensure efficient use of national resources.

Government revenue mainly comes from taxes, non-tax revenues, grants, and loans, while expenditure is allocated to key sectors like education, health, agriculture, infrastructure, social protection, and technology development. For the **2024/2025** fiscal year, the budget was estimated at about Frw **5,690.1 billion**, with a significant share directed toward development projects that improve living standards and create employment opportunities.

Experimental version

8.1.1 Meaning and objectives of public budget

Meaning of public budget

A public budget is a financial plan prepared by the government that shows its expected revenue (income) and planned expenditure (spending) over a specific period, usually one fiscal year. It is a macroeconomic tool used to guide the collection of government revenue and its allocation to public expenditure in order to manage national resources and achieve key objectives such as economic growth, poverty reduction, and improved public service delivery.

Objectives of government budget

A public budget is more than just a financial statement of government income and expenditure; it is a key policy tool that reflects a country's economic priorities and development direction. It involves making difficult choices about how limited public resources should be allocated among competing needs in society. Because of this, it directly influences economic stability, efficiency in resource use, and the well-being of citizens. Understanding this role is essential before examining the specific objectives of a public budget, as it provides the foundation for why governments prepare and implement budgets in the first place. **Therefore, the main objectives of a public budget include the following:**

- To raise revenue for providing social services.
- To reduce income inequality and achieve equitable distribution of income.
- To maintain a favourable balance of payments position of the country.
- To maintain a high level of employment.
- To protect home infant industries.
- To discourage the consumption of undesirable commodities.
- To trace the sources of revenue.
- To cater for the sectors that need urgency towards the development of the country.
- To control government expenditures.
- To stimulate the rate of economic growth.
- To control inflation

8.1.2 Types of public budget

The relationship between government revenue and expenditure is not always the same in every financial year. In some cases, revenue may be equal to expenditure, in others it may be higher or lower. These differences are important because they are used to classify public budgets into different types. This classification helps economists and policymakers understand the government's financial position and make better decisions in planning and managing public resources. **Therefore, public budgets are commonly grouped into the following classifications/types:**

1. According to economic effect



Image 8.2.: A. Balanced Budget

This is the type of government budget where the estimated government revenue is equal to expected expenditure in a financial year.

Estimated Government Revenue = Estimated Government Expenditure

Example: If a government plans to collect **RWF 3 trillion** in revenue and also spends exactly **RWF 3 trillion** on public services such as education, health, and infrastructure during the same year, this is a balanced budget.

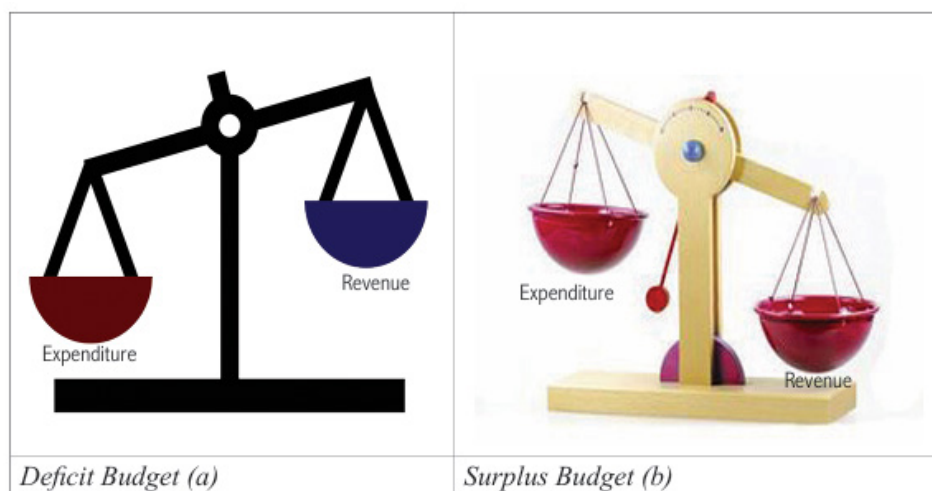


Image 8.3.: B. Unbalanced budget

This is the type of budget where the estimated revenue is either greater or less than the expected expenditure in a financial year.

It has two categories namely:

i. Surplus Budget

This is the type of budget where the estimated revenue is greater than the expected expenditure in a financial year.

Example: If a government plans to collect **RWF 6 billion** in revenue and spends **RWF 5 billion** on public services such as education, health, and infrastructure during the same financial year, this is a **surplus budget**, because revenue exceeds expenditure by **RWF 1 billion**.

Experimental version

A government may plan a surplus budget for various economic and financial reasons, including the following:

- To reduce money in circulation and reduce aggregate demand thus controlling inflationary tendencies.
- To accumulate reserves for future investment needs.
- To reduce government expenditure.
- To improve on the BOP position by reducing expenditure on imports.
- To set aside funds so as to give grants and loans to others countries.

ii. Deficit Budget

This is the type of budget where the expected expenditure is greater than the estimated revenue which is common in developing countries.

Example: If a government plans to collect **RWF 5 billion** in revenue and spends **RWF 6 billion** on public services such as education, health, and infrastructure during the same financial year, this is a deficit budget, because expenditure exceeds revenue by **RWF 1 billion**.

Deficit budget can be financed in the following ways:

- External borrowing from developed countries or international organisations.
- Internal borrowing by selling securities to the people.
- Increasing taxation by widening the base or the tax rate.
- Privatisation to generate more income.
- Borrowing from the central bank through financial accommodation.
- Use of the sinking fund and drawing from the reserves held by other countries.

Government may deliberately plan for a deficit budget because of the following reasons;

- To finance development projects such as roads, schools, and hospitals.
- To stimulate economic growth during periods of low economic activity.
- To create employment opportunities and reduce unemployment.
- To provide public services when government revenue is insufficient.
- To respond to emergencies such as wars, pandemics, or natural disasters.
- To reduce poverty and improve citizens' welfare through social programs.
- To encourage investment and increase aggregate demand in the economy.
- To correct BOP disequilibrium by increasing expenditure on and subsidisation of exports.

Causes of deficit budget in developing countries

Governments may plan for a budget deficit when planned expenditure exceeds expected revenue, often due to economic and policy conditions that limit government income or increase spending needs. This situation usually arises when public finances are under pressure and the government is required to respond to urgent economic and social demands. The following are the main causes of a budget deficit:

- **Low government revenue collection:** When tax revenues and other sources of income are insufficient due to a weak tax base or low economic activity, government expenditure may exceed income, leading to a deficit.
- **High public expenditure:** Excessive spending on services such as education, health, defense, and infrastructure can surpass available revenue, creating a budget gap.
- **Tax evasion and avoidance:** When individuals and firms fail to pay taxes or illegally reduce their tax liability, government revenue decreases, increasing the likelihood of a deficit.
- **Economic recession or slowdown:** During periods of low economic growth, incomes, production, and employment fall, reducing tax revenue while government spending (such as welfare payments) increases.
- **Large development projects:** Governments may undertake costly infrastructure and development projects that require spending beyond current revenue, resulting in borrowing and deficits.
- **High public debt servicing costs:** When a large portion of revenue is used to pay interest on existing debt, less money remains for other expenditures, often causing a deficit.
- **Unexpected emergencies:** Events such as natural disasters, pandemics, or wars force governments to increase spending urgently, often beyond planned revenue levels.
- **Corruption and mismanagement of public funds:** When government resources are misused, stolen, or poorly managed, revenue is lost and spending efficiency decreases, leading to budget deficits.
- **Dependence on external borrowing and grants:** Heavy reliance on loans and donor funds can increase expenditure commitments faster than domestic revenue growth, resulting in persistent deficits.

Possible solutions to budget deficit in developing countries:

- **Improving the value of domestic products** by processing raw materials into finished goods to increase export earnings and foreign exchange.
- **Restoring peace, security, law, and order** in all parts of the country to reduce military and security expenditure.
- **Reducing corruption and fraud** in tax assessment and administration through transparency and strong accountability systems.
- **Implementing civil service reforms** such as retrenchment to reduce expenditure on an oversized public sector.

- **Promoting population control programs** through family planning and education of women to reduce pressure on public expenditure.
- **Modernising agriculture** to increase productivity and produce higher-value products that meet market demand.
- **Increasing capital development expenditure** instead of spending on unproductive recurrent projects.
- **Reducing the subsistence sector** by monetising the economy to broaden the tax base.
- **Diversifying economic activities** to expand the tax base and increase export earnings.
- **Privatising public enterprises** to reduce government spending and generate revenue through taxation of private firms.
- **Promoting export-oriented industrialisation** to increase foreign exchange earnings and improve the balance of payments

2. According to Function / Purpose

A. Development budget

A development budget also known as a capital development budget is where government outlines estimated expenditure on long-term projects that can directly or indirectly increase productive capacity. Such expenditures include financing of industries, road and railway construction, construction of energy facilities, mining, hydroelectric dams etc.

Purpose: Promote long-term economic growth.

B. Recurrent / Operational Budget

This is where estimated government revenue and expenditures are devoted to maintaining the existing capacities or day to day programs e.g. payment of wages of public servants, day to day office expenses, cost of wear and tear capital assets etc.

Purpose: Ensures smooth functioning of public services.

3. According to Policy Approach

A. Expansive / Stimulative Budget: is a financial plan where the government spends more money than it collects in taxes to boost the economy.

Purpose: Reduce unemployment and boost production.

B. Restrictive / Contractionary Budget: is a financial plan where the government spends less money and increases taxes to slow down an overheating economy.

Purpose: Control demand and prevent overheating of the economy.

8.1.3 Problems and solutions of budgeting in developing countries

A. Problems of budgeting in developing countries

Budgeting in developing countries is a complex process influenced by various economic, institutional, and structural factors. Governments often face difficulties in ensuring effective planning and implementation of financial resources to meet national development goals. These challenges affect the efficiency and reliability of public financial management and can limit the achievement of sustainable economic growth. **The following are some of the main problems of budgeting in developing countries:**

- **Inadequate revenue collection:** Governments often fail to collect sufficient revenue due to a narrow tax base, weak tax administration, and a large informal sector. This limits available funds for public expenditure and makes it difficult to implement planned budget activities.
- **Poor forecasting and planning:** Budget preparation is often based on inaccurate data and weak statistical systems. This leads to unrealistic estimates of both revenue and expenditure, resulting in frequent budget imbalances.
- **Corruption and mismanagement of funds:** Misuse of public resources through corruption, fraud, and poor accountability reduces the amount of money available for development projects and distorts budget implementation.
- **Large budget deficits:** In many cases, government spending exceeds revenue, forcing authorities to rely on borrowing. This creates long-term financial pressure and affects economic stability.
- **Heavy dependence on foreign aid and loans:** Reliance on external funding makes budgeting uncertain because aid and loans may come with conditions or may not be consistent over time.
- **High levels of poverty and unemployment:** A large poor population reduces taxable income, while unemployment increases government spending on social support programs, putting pressure on the budget.
- **Political interference in budgeting:** Budget decisions are often influenced by political interests rather than economic priorities, leading to inefficient allocation of resources.
- **Rapid population growth:** Fast population increase raises demand for public services such as education, health, and infrastructure, putting additional strain on limited government resources.
- **Weak institutional capacity:** Many developing countries lack strong institutions, skilled personnel, and efficient systems for budgeting and financial management. This leads to poor coordination, weak implementation of budget plans, and ineffective monitoring of public funds.
- **Inflation and economic instability:** High inflation rates and unstable economic conditions make it difficult to predict government revenues and expenditures accurately. This results in frequent budget revisions and unreliable financial planning.

- **External shocks and dependence on global markets:** Developing countries are often affected by changes in global prices, foreign aid flows, and international economic crises. These external factors can disrupt planned revenues and increase uncertainty in budgeting.
- **Inadequate qualified, motivated, and incorruptible workers:** Many developing countries lack skilled, committed, and honest personnel to effectively plan, implement, and monitor government budgets, leading to inefficiency and misuse of public funds.
- **Unpredictable and unreliable sources of income:** Many developing countries depend on unstable sources of revenue such as foreign aid, loans, and primary commodity exports, making it difficult to prepare and implement accurate budgets.

B. Possible solutions to budgeting problems in developing countries

- **Strengthening institutional capacity:** Training budget officers and improving institutions helps ensure proper planning, implementation, and monitoring of budgets.
- **Improving data collection and forecasting systems:** Reliable statistical data enables governments to prepare realistic and accurate budgets.
- **Enhancing transparency and accountability:** Open budgeting systems and strict auditing reduce misuse of funds and improve public trust.
- **Reducing political interference in budgeting:** Allowing technical experts to guide budgeting ensures that resources are allocated based on economic priorities.
- **Modernizing budgeting systems:** Introducing ICT and digital financial management systems improves efficiency and reduces errors in budgeting processes.
- **Improving coordination among government institutions:** Better coordination ensures consistency in planning, budgeting, and implementation across sectors.
- **Strengthening public participation in budgeting:** Involving citizens and stakeholders improves accountability and ensures that budgets reflect real needs.

Application activity 8.1



1. What is the difference between public finance and public revenue?
2. Explain the different types of budgets.
3. Why government may plan for
 - a) Surplus budget
 - b) Deficit budget

4. Analyse the reasons as to why there are budget deficits in your country.
5. Suggest possible solutions to Rwanda's budget deficit.

8.2 Public expenditure

Learning Activity 8.2



1. Identify different activities in your area on which Rwandan government can spend her resources.
2. What factors should a government consider when deciding how to allocate limited public funds among competing needs in society?

Public expenditure plays a crucial role in the functioning and development of a country's economy. It reflects how government resources are allocated to meet public needs and achieve national development goals. Through spending on various sectors such as education, health, infrastructure, and security, governments aim to improve the welfare of citizens and promote economic growth.

8.2.1 Meaning and types of public expenditure

Public expenditure refers to the money spent by the government to provide goods and services aimed at meeting the needs of society and promoting economic development.

Types of public expenditure

1. **Development expenditure (Capital expenditure):** This refers to government spending on long-term projects that increase the productive capacity of the economy. It includes investment in infrastructure such as roads, schools, hospitals, energy projects, and industries, which promote economic growth and development.
2. **Recurrent expenditure (Current expenditure):** This is government spending on day-to-day operations and maintenance of public services. It includes payment of salaries to public servants, purchase of consumable goods, and maintenance of existing government facilities to ensure smooth service delivery.

8.2.2 Reasons for public expenditure and ways of reducing expenditure in developing countries.

i. Reasons/purpose for public expenditure

Government expenditures serve as the primary tool for a nation to manage its economy and support the well-being of its citizens. So, the following are reasons for government expenditure:

- **Provision of public goods and services:** Government spending provides essential services like health, education, and security.
- **Promotion of economic development:** Expenditure supports infrastructure and projects that stimulate growth.
- **Reduction of income inequality:** Spending on welfare and subsidies supports poorer groups.
- **Maintenance of economic stability:** Helps control inflation and unemployment during economic fluctuations.
- **Support for administration:** Finances government operations and public institutions.
- **Creation of employment opportunities:** Public projects and services generate jobs in the economy.
- **Encouragement of private sector growth:** Infrastructure spending creates a favorable environment for investment.

ii. Ways of reducing expenditures in developing countries.

It is important for governments to identify and apply appropriate measures that can help control and reduce unnecessary spending while ensuring efficient use of public resources. Therefore, the following measures can be adopted to reduce public expenditure in developing countries:

- **Reducing corruption and wastage of public funds:** Strengthening accountability and transparency ensures that public money is not misused or lost through fraud and inefficiency.
- **Prioritizing essential expenditure:** Governments should focus spending on key sectors such as health, education, and infrastructure while cutting non-essential projects.
- **Civil service reforms:** Reducing an oversized public sector through retrenchment and improving efficiency helps lower wage and administrative costs.
- **Privatisation of public enterprises:** Transferring some government-owned enterprises to the private sector reduces government spending and improves efficiency.
- **Improving public financial management:** Better budgeting, planning, and monitoring systems help avoid unnecessary or unplanned expenditures.
- **Controlling debt servicing costs:** Managing public debt effectively reduces the burden of interest payments and frees resources for development.
- Merging of some ministries so as to reduce expenditure on a single ministry.
- Demobilisation of soldiers and retrenchment of civil servants to reduce government expenditures on salaries and allowances.

Application activity 8.2



1. Discuss the different types of public expenditure, giving examples for each type.
2. Give two examples of development expenditure in your country.
3. Explain why salaries of government employees are classified as recurrent expenditure.
4. Explain how privatising government companies can help reduce public expenditure.

8.3 Public revenue

Learning Activity 8.3



Scenario:

The Governor of one province in Rwanda has observed that the demand for public services such as schools, health centers, clean water, and road maintenance has increased rapidly. However, the revenue collected within the province is not enough to finance all planned development projects. As a result, the provincial leadership is considering different ways of increasing available funds to meet these growing needs.

Questions:

1. Why do you think the revenue collected in the province is not enough to meet all public needs?
2. What problems might arise if development projects are not fully funded?
3. In your opinion, what can the provincial authorities do to increase the amount of revenue available for development?
4. Why is it important for local governments to look for new ways of raising revenue?

Public revenue is essential for government operations and development activities. It provides the funds needed to finance public services, support development projects, and maintain economic stability. Through various sources of income, governments are able to meet both current and long-term national needs.

8.3.1 Meaning and sources of public revenue

Public revenue is the income the government collects from different sources to finance public expenditure and development activities.

The following are sources of public revenue:

- **Taxes:** A tax is a compulsory contribution imposed by a public authority, irrespective of the exact amount of service rendered to the tax-payer in return.

This is the most important source.

- **Licenses:** This is a payment made to the government to secure permission to carry out any gainful activity.
- **Fees:** Payment made by an individual for personal services rendered to them by the government. Examples of fees are postal services, medical fees charged in private wards of government hospitals, weighing vehicles, surveying land for a person etc.
- **Fines:** Penalties imposed on citizens who break laws e.g. traffic offenders.
- **Loans:** Governments can borrow either internally or externally.
- **Compulsory savings:** Like insurance payments, social security funds.
- **Rent of Government Property:** Income earned by renting out government-owned land or buildings.
- **Gifts and Grants:** Voluntary contributions from individuals, societies, or foreign donors for specific projects, relief, or development; uncertain and irregular.
- **Privatisation:** Revenue from the sale of public enterprises to private individuals or companies.
- **Deficit Financing:** Raising money by issuing new currency or borrowing from the central bank.
- **Earnings from Gambling:** Revenue from government-run lotteries or similar activities.
- **Profit from Commercial Ventures:** Income from government-owned businesses or enterprises.

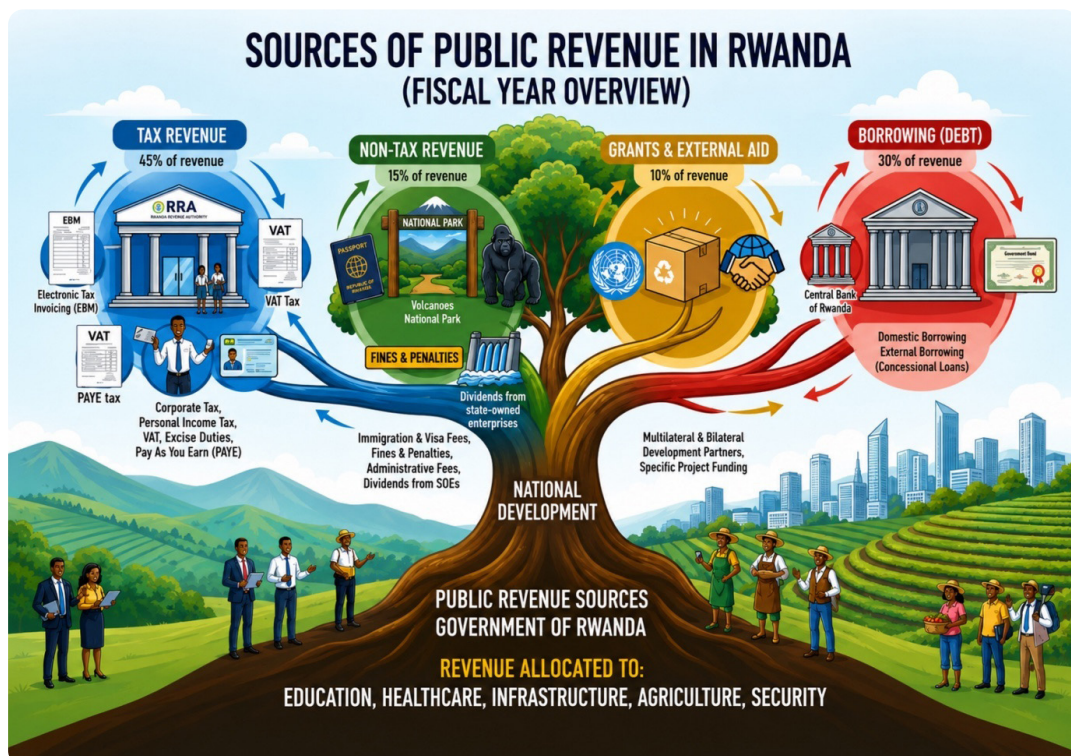


Image 8.4.: Sources of public revenue

8.3.2 Methods of expanding sources of public revenue

- Ensuring political stability and security in all parts of the country to encourage smooth production and business.
- Attracting more investors both local and foreign so as to expand the tax base.
- Monetising the economy so as to increase income earners who contribute to the revenues of the economy in different ways.
- Train and retrain manpower to properly assess taxpayers and collect tax revenue.
- Diversification of the economy so as to widen the tax base and increase tax revenue as the major source of government revenue. Introducing new forms of taxes both direct and indirect.
- Tax education to the public through radios, televisions, posters and seminars to reduce tax defaulters.

Application activity 8.3



1. Identify the different sources of government or public revenue in Rwanda.
2. The Government of Rwanda is planning to expand important services such as healthcare, education, and infrastructure. However, to achieve these goals, the government needs to increase its revenue. Suggest five possible ways the government of Rwanda can increase its revenue.

8.4 Public debt

Learning Activity 8.3



A country wants to improve public services by building more schools, hospitals, roads, and energy facilities. However, the government does not have enough money to finance all these development projects using the revenue it collects. To fill this financial gap, the government decides to borrow money from different sources.

Read the scenario above and answer the questions that follow:

1. Why do you think the government may decide to borrow money?
2. From where can a government borrow money?
3. What problems may arise if a government borrows too much money?
4. How can borrowing affect the development of a country?

Public debt is an important aspect of public finance that enables governments to obtain additional funds for development projects and public services when available revenue is insufficient. Although borrowing can support economic growth and national development, it requires proper management to avoid financial difficulties.

8.4.1 Meaning and classification of public debt

A public debt is the total amount of money borrowed by the government from internal or external sources to finance its expenditure and development activities. While Public borrowing refers to the act of the government obtaining money from internal or external sources.

Classifications of debts

1. According to the source of borrowing:

Under this we have;

a) Internal debt:

This is raised from the public within the country especially in developed countries where there are rich people and companies which can afford to lend to the government by buying its securities like bonds.

b) External debt:

This is raised by developing countries from abroad to raise money for development normally got from individual countries or organisations like **IMF, World Bank** etc.

2. According to how a debt is going to be used:

a) **Reproductive debt/ self-liquidating debt:** This is a debt which is used to finance projects that bring returns to the government. It is self-liquidating because it can be able to repay itself in the long-run.

b) **Dead weight debt/ non-reproductive debt:** is a type of public debt used to finance activities that do not directly generate income or increase production in the economy, such as wars or emergency expenditures. It is mainly borrowed for short-term purposes and does not create productive assets that can help repay the debt in the future.

3. According to the methods of debt repayment:

a) **Funded debt:** This is where the specific date of paying back the debt (redemption) is not known. The government keeps on paying annual interest on it until when it feels it is ready to pay the principal.

b) **Unfunded debt/ floating debt:** This is where the specific period and date of paying back is known. It consists of short-term debts like treasury bills.

4. According to the maturity period;

a) **Short-term debt:** This is a debt which has to be paid within one year or less than one year.

b) **Medium term debt:** This is a debt which has to be paid between 2-10 years.

c) **Long-term debts:** This is a debt which has to be paid in a period of more than 10 years.

8.4.2 Causes and consequences of public debt in developing countries

A. Causes of public debt in developing countries

Public debt in developing countries arises from various economic and financial challenges that limit government revenue while increasing expenditure needs. As a result, governments often resort to borrowing to finance development and public services. **The following are the main causes of public debt in developing countries:**

- **Fill foreign exchange gaps** by bringing in external funds.
- **Reduce tax burden** on citizens to avoid discouraging work, savings, and consumption.
- **Gain political support** by avoiding unpopular taxation.
- **Combat economic depression** by increasing money supply and boosting demand.
- **Finance balance of payments deficits** when imports exceed exports.
- **Cover saving-investment gaps** due to low domestic savings.
- **Repay existing debts** that have matured.
- **Build government reserves** to support monetary policy.
- **Control inflation** by withdrawing excess money via securities.
- **Finance budget deficits** to meet government expenditure.
- **Handle emergencies** like wars or natural disasters not covered in the budget.
- Borrowing is often used to **support development and stabilize the economy**.

B. Consequences of public debt in developing countries

The consequences of public debt in developing countries may be positive or negative depending on how the borrowed funds are used and managed. Proper use of debt can support development, while poor management can create economic challenges.

Positive consequences of public debt:

- **Improved public services:** It enables governments to provide better services like healthcare, education, and transport, improving citizens' welfare.
- **Job creation:** Borrowed funds used in development projects create employment opportunities in both public and private sectors.
- **Infrastructure development:** Public debt supports the construction of roads, hospitals, schools, and energy facilities that boost development.
- **Bridging budget gaps:** Public debt helps governments cover the gap when public expenditure exceeds available revenue, ensuring continuity of services.
- **Promotion of private sector growth:** Borrowed funds invested in infrastructure and services create a favorable environment for business and investment.

- **Support during emergencies:** It helps governments respond to crises such as wars, pandemics, or natural disasters when local revenue is insufficient.
- **Recover economy from depression** because increased money supply escalates investment and demand.

Negative consequences of public debt:

Public debt, when it becomes excessive, can create serious challenges for an economy by putting pressure on government finances and limiting overall economic performance. These challenges are discussed below.

- **High debt servicing burden:** A large share of government revenue is used to pay interest and repay loans, leaving less for development projects.
- **Reduced public investment:** Heavy debt obligations limit funds available for infrastructure, education, and health services.
- **Inflationary pressure:** If debt is financed through money creation, it can increase money supply and lead to rising prices.
- **Dependence on external lenders:** Excessive borrowing increases reliance on foreign countries and institutions, reducing economic independence.
- **Increased taxation in future:** Governments may raise taxes to repay debt, reducing citizens' disposable income.
- **Debt trap risks:** Countries may borrow new loans to repay old ones, leading to a cycle of increasing debt.
- **Slower economic growth:** High debt levels can discourage investment and strain public finances, slowing down overall economic development.

8.4.3 Public debt management

Public debt management is an important part of public finance that focuses on how governments plan, use, and repay borrowed funds. It helps ensure that debt is handled properly to maintain economic stability and support development.

i. Meaning and objectives of public debt management

Public debt management is the process of planning, organizing, and controlling government borrowing and repayment in order to ensure that borrowed funds are used efficiently and do not create economic difficulties. **Or** refers to the ways and means of how the public debts are administered.

Objectives/ aims of public debt management:

- To support economic development and growth of the country.
- To reduce the debt burden.
- To minimize debt servicing costs.
- To enhance the country's credit reputation and ability to borrow.
- To ensure timely repayment of debt.

- To ensure efficient use of borrowed funds.
- To ensure debt repayment and servicing.

ii. Methods of clearing public debts and burden of the public debt

A. Methods of clearing public debts

Managing public debt involves strategies that help governments reduce and repay their debts in a sustainable way. These methods reduce financial pressure, support economic stability, and ensure continued public service delivery while gradually lowering debt levels.

Government usually clears off their debts whether internal or external through either debt servicing or debt redemption.

- 1. Debt Servicing** refers to the regular payment of interest and part of the principal on an existing loan without fully clearing the debt. It is done periodically (e.g., annually) to keep the loan active.
- 2. Debt Redemption/ Retirement** refers to the complete repayment and cancellation of a public debt, where the government fully clears the outstanding loan and ends the debt obligation. **Debt servicing and redemption is usually through the following methods:**

Ways of clearing internal debts

- **Using a sinking fund:** This refers to that fixed amount of money set aside annually for the purposes of paying debts.
- **Taxation:** Government can widen the tax base, increase the tax rate so as to raise money to pay the debts.
- **Debt conversion:** This refers to the government act of borrowing new loans from a cheaper source so as to pay the old debts.
- **Borrowing** from the central bank through financial accommodation.
- **Using promissory notes:** These are documents issued by monetary authorities promising to pay creditors at a future date.
- **Reduction in government expenditure** to save money and clear off the debts.
- Through acquiring **gifts and grants**.
- **Privatisation** to raise money etc.

Ways of clearing an external debt

- **Debt conversion:** This involves borrowing from new low-interest rate source to pay off old debts.
- **Using sinking funds:** Here government sets aside some money in every financial year and the accumulated funds are used to service or repay the debt.
- **Dis-investment:** The government may sell its foreign investments through privatisation and uses the proceeds to raise foreign exchange to service or pay back the debt.

- **Printing more money** for payment of internal debts. However this causes inflation.
- Adoption of **export promotion** to raise money for servicing the debts.
- Adoption of import substitution to save foreign exchange.
- **Using foreign reserves** that may be held in other countries.
- **Debt cancellation/ debt relief:** Here a country negotiates for relief from the lender.
- **Debt rescheduling:** This is the act of extending the debt repayment date in case the recipient country is unable to pay immediately.
- **Repudiation:** This is the complete refusal of the debtor country to pay debts.
- **Local currency servicing strategy:** This is when the donor country allows the recipient country to service the debt in local currency and this helps to reduce the demand for foreign exchange.
- **Unrequited exports:** This involves exporting goods to the lender without expecting any payment and using the value of the goods to offset the debt.
- **International liquidity (SDRs):** A country may use its Special Drawing Rights at the IMF to pay off debts.

B. Burden of public debt

The burden of public debt refers to the cost of borrowing (effects) on the present and future generation.

Burden of an internal public debt

The burden of the internal public debt is indirect and comes about in form of the cost of taxation to finance the debt. These include;

- Taxation used to finance internal debt reduces private investment and capital formation in the economy.
- Debt repayment through taxation may discourage work effort among taxpayers due to higher tax burdens.
- Government may resort to printing money to finance debt, leading to inflation.
- Internal debt can increase income inequality, as the rich benefit more from government securities while the poor are still taxed.

Burden of an external debt

- External debt reduces a country's foreign exchange reserves due to repayment of principal and interest.
- It may force the government to reduce spending on essential social services such as health and education.
- External aid often comes with conditions that limit government flexibility in using the funds.

- Future generations may bear the burden of repayment without benefiting from the borrowed funds (dead weight debt).
- It can create dependency on foreign support and worsen balance of payments problems, increasing poverty.

Application activity 8.4



Question 1: Explain why it is important to have a well-organised public budget and propose two measures a government in a developing country can take to solve budgeting challenges.

Question 2: Define public debt and explain two strategies a government can use to manage and sustainably reduce it.

Question 3: State three sources of public revenue and describe two strategies governments can use to increase revenue generation.

Question 4: Explain two reasons for public expenditure in developing economies and suggest two ways to minimise unnecessary government spending to improve fiscal stability.

8.5. End of unit assessment



A. Knowledge and Understanding

1. Explain the meaning of a public budget and describe its types.
2. Identify the correct answers:
 - a) A balanced budget occurs when government revenue is equal to expenditure.
 - b) Which of the following is an example of development expenditure: salaries, textbooks, hospital construction, or office supplies?
 - c) Public debt is internal when it is owed to citizens within the country.
 - d) One major effect of excessive public debt is reduced funds for development due to high interest payments.
3. State three causes of budget deficits in developing countries.

B. Skills

4. A government is experiencing a budget deficit due to increased healthcare spending during a pandemic. Suggest two practical ways of raising funds without significantly increasing taxes.
5. A country borrows heavily to finance infrastructure but struggles to repay its debts. Explain the possible economic consequences and suggest two strategies to manage the situation.

6. Explain five ways governments can reduce public expenditure.
7. Identify two methods governments can use to manage or repay public debt.

C. Values and Attitudes

8. Discuss the importance of maintaining a balanced budget in promoting economic stability, giving relevant examples.
9. Evaluate the impact of corruption on government budgeting and public debt management, and suggest measures to improve accountability and responsible use of public funds

UNIT 9

TAXATION



Key Unit Competence:

Analyze the role of Tax and Fiscal policy in economic development.

Learning objectives

Knowledge and understanding

1

- Explain the common terminologies used in taxation.
- Explain key terms in taxation (tax base, tax rate, evasion, incidence, etc.)
- Describe the principles (canons) of a good tax system
- Explain different types and forms of tax used in Rwandan economy
- Identify the problems encountered in collecting taxes and policies to improve tax collection in Rwanda and in Developing countries.
- Explain the functions, tools and forms of fiscal policy.

Skills

2

- Analyze the meaning and purpose of taxation in the context of Rwanda's economic development.
- Analyze the principles of tax system used in Rwanda to evaluate the role of taxation in economic development in an economy. (on individuals, on economic activities in general)
- Examine the problems encountered when assessing and collecting taxes in Rwanda and in Developing countries.
- Analyze innovative strategies (e.g., digital tax systems, taxpayer education) to promote voluntary tax compliance.
- Analyze debt financing and taxation financing.
- Examine the impact of fiscal policy in an economy.

Attitudes and values

3

- Appreciate the role of taxation in government revenue and advocate for tax assessment and collection improvement.
- Appreciate the importance of taxation as a tool for government revenue and social welfare improvement and Advocate for Public education as the best way to improve tax collection in Rwanda”.
- Advocate for ethical tax behavior and citizens’ responsibility in contributing to Rwanda’s fiscal sustainability.
- Appreciate the importance of paying taxes truthfully and advocate for avoiding corruption or evasion.
- Demonstrate collaboration and critical thinking in group discussions on fiscal policy challenges in Rwanda.



Introductory activity

In Rwanda, people earn income in different ways. Some run businesses such as shops and transport services, while others work in offices and earn salaries. The government requires individuals and businesses to pay part of their income to support public services and national development.

To improve transparency in revenue collection, the government uses Electronic Billing Machines (EBMs) and mobile payment systems. However, some people still avoid making these payments because they believe they reduce their income.

Questions

1. Identify the payment described in the passage and state the government institution responsible for collecting it in Rwanda.
2. Using examples, explain three roles of these payments in the economic development of Rwanda.
3. Analyze the positive and negative effects of these payments on people’s standards of living and suggest measures to reduce tax evasion.

In Rwanda, the Rwanda Revenue Authority (RRA) is responsible for tax collection and management. Established in 1997 under the Ministry of Finance and Economic Planning, RRA assesses and collects taxes, customs duties, and other government revenues. It has improved efficiency and transparency through the use of Electronic Billing Machines (EBMs) and mobile payment systems in businesses and public transport. Taxation in Rwanda mainly aims to raise government revenue, support economic development, reduce income inequalities, and regulate the consumption of certain goods and services.

9.1 Meaning of taxation and related terminologies

Learning Activity 9.1



1. In your community, people use roads, hospitals, schools, and security services every day. In your opinion, where does the government get the money to provide these public services?
2. Imagine that all workers, business owners, and drivers in Rwanda stopped making payments to the government for one year. What effects do you think this would have on the country's development and public services?
3. Some businesses use Electronic Billing Machines (EBMs) to record sales and payments. Why do you think the government encourages the use of such systems, and what might happen if people hide their real income or refuse to make these payments?

Taxation is a process of collecting tax revenue from different sources to be used by the government in developing of the economy. It is not a payment for a direct service, but a legal obligation for residents who are eligible.

Who Are taxpayers?

Taxation involves the participation of different groups within an economy who are expected by law to contribute to government revenue. These contributors play an important role in supporting public expenditure and national development through the payment of various forms of taxes. The following are the main categories of taxpayers.

1. Individuals: All residents who earn income, own property, or engage in business activities are legally required to pay taxes.
2. Businesses and Companies: Corporations, partnerships, and sole proprietorships must pay taxes on profits, sales, and other applicable areas.
3. Foreign Entities Operating in the Country: Non-resident companies or individuals earning income in the country are often taxed on local earnings.
4. Consumers: Pay indirect taxes like VAT when purchasing goods or services.

Terminologies used in taxation

Concept	Definition
Tax Base	The activities, income, or items on which taxes are imposed (e.g. salaries, business activities, land, goods). It determines whether a tax is direct or indirect.
Taxable Income	The portion of income that is subject to tax after exemptions according to tax laws.
Taxable Capacity	The ability of a taxpayer to pay taxes without reducing their standard of living.

Tax Evasion	The illegal refusal to pay taxes or underreporting income. It is punishable by law.
Tax Avoidance	The legal use of tax rules to reduce tax liability (e.g. deductions).
Exemptions / Tax Holidays	Exemptions are legal exclusions from tax; tax holidays are temporary periods of reduced or no taxation to encourage investment.
Tax Compliance	The willingness to correctly declare income and pay taxes on time.
Tax Transformation	Substituting a taxed commodity with a non-taxed one to avoid tax effects.
Tax Rebate	A reduction in tax rates given to encourage investment.
Tax Burden	The loss or sacrifice felt by the person paying tax due to reduced disposable income.
Tax Shifting	Transferring tax burden from the original payer to another person.
Forward Shifting	Transferring tax burden to the next person in the production chain (e.g. producer to consumer).
Backward Shifting	Transferring tax burden to suppliers by paying lower prices for inputs.
Taxation System	The method or structure used by a government to collect taxes.
Tax Authority	The institution responsible for collecting and managing taxes (e.g. RRA).
Tax Audit	An official examination of financial records to verify correct tax payment.
Tax Relief	A reduction in the amount of tax payable.
Tax Refund	Money returned when a taxpayer has overpaid tax.
Tax Assessment	The process of determining the amount of tax to be paid.
Tax Impact	The initial burden of a tax on the person or firm on whom it is imposed. Tax incidence: the final burden after tax shifting.
Tax Incidence	The final person who bears the tax burden.
Effect of Tax	The economic and social consequences of taxation on individuals and society (e.g. reduced living standards).

Reasons for tax evasion

Weaknesses in tax administration or enforcement that make it difficult to detect and follow up tax evasion. In this context, understanding why such practices occur is important in order to strengthen tax administration and improve compliance among taxpayers.

So, the following are reasons why people evade taxes:

- Discontent with service provided
- Lack of adequate information about taxes

Experimental version

- Low income level
- Desire to retain all earnings
- Loopholes in tax system not to follow up people who evade taxes.
- Opposition by the political leaders

Effects of tax evasion

Tax evasion has significant implications for both government operations and the overall performance of the economy. When individuals or businesses fail to meet their tax obligations, it reduces the amount of revenue available for public expenditure and disrupts the smooth functioning of fiscal policy. **The consequences of this practice are outlined below.**

- 1. Loss of government revenue:** reduces funds available for public services like education, health, and infrastructure.
- 2. Poor public services:** leads to inadequate provision of social amenities due to lack of funds.
- 3. Increased tax burden on honest taxpayers:** those who pay taxes may be forced to pay more to cover the shortfall.
- 4. Unfair income distribution:** widens the gap between the rich and the poor.
- 5. Slows economic development:** limits government ability to invest in development projects.

Application activity 9.1



A. Multiple Choice Question

1. The illegal act of reporting lower income to pay less tax is called:
 - A. Tax avoidance
 - B. Tax evasion
 - C. Tax relief
 - D. Tax exemption

B. Fill in the blanks

2. The government institution responsible for collecting taxes is called _____.
3. The process of determining how much tax a business should pay is called _____.
4. A temporary period where businesses are not taxed is called _____.
5. The tax added to goods and paid by consumers is called _____.
6. The legal way of reducing tax using tax laws is called _____.

C. Open Questions

7. Explain two differences between tax evasion and tax avoidance.
8. Using the case study, explain two ways taxation affects businesses and consumers in Rwanda.
9. Explain the meaning of tax as a **non-quid pro quo payment**, and state what this means for taxpayers in relation to government services.
10. Identify 3 groups of people or entities that are required to pay tax in Rwanda.

9.2 Purposes and principles of taxation

Learning Activity 9.2



Scenario

In Kigali, the government collects taxes from employees, traders, and businesses to finance public services such as roads, schools, and hospitals. However, some taxpayers complain that they are not clearly informed about how much tax they should pay or when to pay it. Others argue that tax collection sometimes takes a large share of their income, especially for low-income earners. At the same time, tax officers from Rwanda Revenue Authority aim to ensure that tax collection is simple, fair, and does not cost too much to administer.

Questions

1. From the scenario above, identify the main problems faced by taxpayers in the tax system.
2. Suggest three qualities that a good tax system should have to solve the problems mentioned in the scenario.
3. Why do you think the government insists that tax collection should be fair, simple, and economical?

A. Purpose of taxation

Taxation is an essential function of government in any economy, as it involves the collection of compulsory payments from individuals and businesses. Before examining its specific purposes, it is important to understand that taxation is not only a source of government revenue but also a key tool used to influence economic activity, promote fairness, and support national development. So, **the following are the main purposes of taxation in an economy:**

1. **To raise government revenue:** Taxes are the main source of income used by the government to finance public services and development projects.

- 2. To protect home industries (infant industries):** Taxes on imports make foreign goods more expensive, helping local industries compete and grow.
- 3. To fight inflation:** By reducing people's disposable income, taxes reduce excessive demand and help control rising prices.
- 4. To correct balance of payments deficit:** Taxes on imports reduce import demand and help improve the country's external trade position.
- 5. To discourage importation of certain goods:** High taxes are used to reduce the consumption of unwanted or luxury imported goods.
- 6. To discourage dumping:** Taxes prevent foreign firms from selling goods cheaply in local markets, protecting domestic producers.
- 7. To discourage consumption of demerit goods:** Taxes on goods like alcohol and cigarettes reduce their consumption and protect public health.
- 8. To redistribute income:** Taxes help reduce income inequality by transferring resources from the rich to support public welfare programs.
- 9. To control inflation:** Taxes reduce excess money in circulation, helping stabilize prices in the economy.
- 10. To control monopoly power:** Taxation reduces excessive profits of monopolies and limits their market dominance.
- 11. To encourage hard work and saving:** People are motivated to work harder and save more since they retain part of their income after tax.
- 12. To control population growth:** Taxes based on number of children can discourage large family sizes in some policy systems.

B. Principles / canons maxims of taxation

The principles of taxation were proposed in **1776** by **Adam Smith in his book Wealth of nations**, an economist who suggested that a good tax system should be fair and efficient. The **canons or principle of taxation** are the basic rules or qualities that a good tax system should follow in order to make tax collection fair, effective, and acceptable to taxpayers.

Governments should follow these principles because they help to ensure fairness among taxpayers, reduce resistance to tax payment, promote economic stability, and improve efficiency in revenue collection.

The following are principle of taxation:

1. Equity (Fairness)

Taxes should be fair and based on people's ability to pay.

- **Horizontal equity:** People with the same income pay the same tax
- (e.g., two teachers earning the same salary pay the same tax).
- **Vertical equity:** Richer people pay more than poorer people
- (e.g., a millionaire pays a higher rate than a low-income worker).

2. Certainty

Taxpayers should know clearly **how much to pay, when, and how.**

Example: Rwanda Revenue Authority publishes clear income tax rates and deadlines.

3. Convenience

Taxes should be easy to pay at a convenient time and place.

Example: Farmers pay tax when selling crops; office workers pay through payroll deductions.

4. Economy

The cost of collecting tax should be low compared to revenue collected.

Example: If collecting a tax cost 1 million but brings in 50 million, it is efficient.

5. Productivity (Adequacy)

The tax must generate enough revenue to meet government needs.

6. Elasticity

Taxes should increase automatically as national income rises.

7. Simplicity

The tax system should be easy to understand and administer.

Example: Clear forms and simple filing procedures for taxpayers.

8. Comprehensiveness

Taxes should cover multiple sources of income, not just one.

9. Flexibility

The system should adjust to economic changes (e.g., increase in boom, relief in depression).

10. Neutrality

Taxes should not discourage production, investment, or consumption.

Example: Avoid high taxes on essential goods or business investments.

11. Political Acceptability

Tax policies should be acceptable to the public to avoid resistance.

12. Consistency

Taxes should align with national economic goals, like resource allocation and development planning.

Application activity 9.2



SECTION A: FILL IN THE BLANKS

1. The principle that requires taxes to be fair according to people's ability to pay is called _____.
2. The principle that states taxpayers should clearly know how much and when to pay is called _____.
3. The principle that ensures tax collection is cheap compared to revenue collected is called _____.
4. The principle that requires taxes to be easy to understand and administer is called _____.
5. The principle that ensures tax changes with changes in national income is called _____.

SECTION B: OPEN QUESTIONS

6. Explain other principles of taxation that are not included in section (a) and discuss how they help to improve the efficiency and fairness of tax systems.
7. What are the main purposes of taxation in an economy, and how do they contribute to national development?

9.3 Classifications of taxes

Learning Activity 9.3



Scenario

In a small town, the local government is trying to improve public services such as roads, schools, and hospitals. To do this, they collect different types of taxes from citizens. Some people pay tax directly from their salaries, while others pay tax when buying goods like sugar, clothes, or fuel. A group of learners is discussing why taxes are collected differently and how they are grouped.

Scenario based questions:

1. Why do you think the government uses different ways of collecting taxes from people instead of using only one method?
2. How would you distinguish between taxes paid directly from income and those paid when purchasing goods and services?
3. In what ways can grouping or classifying taxes help the government and citizens better understand the tax system?

Taxation systems are organized in different ways to make it easier to understand how taxes are imposed and collected. One important way of studying taxation is by classifying taxes according to their nature, purpose, and how they are levied on taxpayers.

This classification helps to distinguish different types of taxes and understand how each one contributes to government revenue and economic management.

9.3.1 Classification of taxes according to the rate

Taxes can be classified in different ways, including according to the rate at which they are charged. This classification shows how the tax burden changes with income or value. **The following are the classifications of taxes according to the rate.**

Type of Tax	Brief Explanation
Progressive Tax	A tax where the rate increases as income increases. Higher-income earners pay a larger percentage of their income.
Regressive Tax	A tax where the rate decreases as income increases. Lower-income earners bear a heavier burden relative to their income.
Proportional Tax	Also called a flat tax. The same tax rate is applied to all taxpayers regardless of income level.
De-regressive Tax	A tax that is progressive up to a certain income level, then becomes proportional (constant rate) beyond that level.

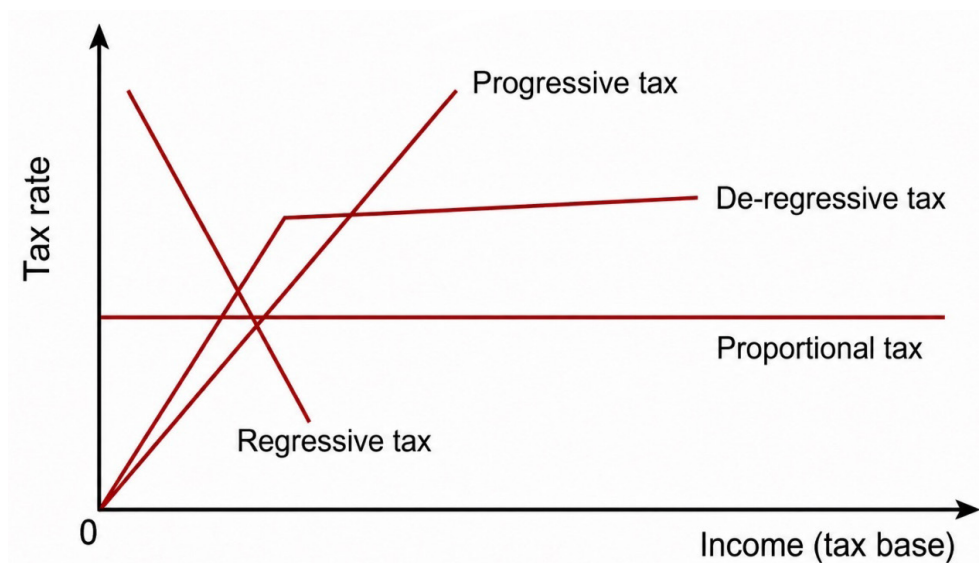


Figure 9.1.: classifications of taxes according to the rate

9.3.2 Classification of taxes according to the incidence

Taxes can also be classified according to their incidence, which refers to the final person who finally bears the burden of a tax. This classification helps to understand how taxes affect different groups in the economy. **These classifications include the following:**

ACCORDING TO INCIDENCE

TYPES EXPLANATION

Direct Tax	Is a tax levied on the individual's or company's income or wealth and the burden cannot be shifted to another person.
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Experimental version

Indirect Tax	Is a tax levied on goods and services and paid by customers when making a purchase. They are included in the price of goods and services, so the buyer pays the tax indirectly. The incidence can be shifted forward or backward.
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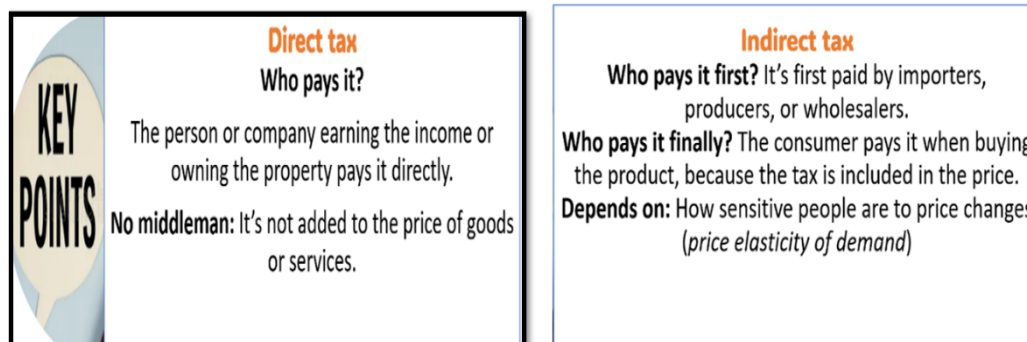


Image 9.1.: Classification of taxes according to the incidence

Examples of Direct Taxes:

- 1. Income tax:** Tax on a person's earnings above a certain level.
- 2. Corporate tax:** Tax on profits made by companies or businesses.
- 3. Estate duty (death tax):** Tax on a person's wealth after they die.
- 4. Gift tax:** Tax on property or money transferred to another person as a gift.
- 5. Capital gains tax:** Tax on profit made from selling an asset at a higher price.
- 6. Poll tax:** Fixed tax paid equally by every individual regardless of income.
- 7. Graduated tax:** Tax on income that varies based on earnings, mainly for workers.
- 8. Property tax:** Tax on ownership of assets like land and houses.
- 9. Inheritance tax:** Tax paid by those who receive property from a deceased person.

Advantages of Direct Taxes	Disadvantages of Direct Taxes
Ensures equity since higher income earners pay more (progressive system).	May discourage work and investment if tax rates are too high.
Transparent because taxpayers clearly know how much they pay.	Tax evasion is common when enforcement is weak.
Helps reduce income inequality by redistributing wealth.	Collection is costly and administratively complex for the government.
Provides a stable source of government revenue.	Can create dissatisfaction as taxpayers feel a direct financial burden.

Encourages accountability as citizens see the link between taxes and public services.

May face political resistance from taxpayers.

Examples of indirect taxes:

1. **Value Added Tax (VAT):** Tax added at each stage of production and included in the final price.

VAT can be calculated for a VAT exclusive or inclusive price as follows;

A. VAT Inclusive: This means the price of the goods or services including VAT.

Goods and services supplied by VAT registered taxpayers must always be sold at the VAT-inclusive price.

$$\text{VAT inclusive} = \frac{\text{Total invoice or amount}}{118} \times 18 \Rightarrow (\text{selling price} \times 18) / 118$$

VAT Exclusive: This means that the price of goods and services does not include VAT, as it has not been incorporated into the final cost

$$\text{VAT exclusive} = \frac{\text{Invoice with no VAT}}{100} \times 18 \Rightarrow (\text{selling price} \times 18) / 100$$

1. **Customs duty:** Tax on goods imported into or exported out of a country.
2. **Excise duty:** Tax on goods produced within a country.
3. **Sumptuary tax:** Tax on certain goods to reduce their consumption.
4. **Octroi tax:** Tax on goods entering a particular area or region.
5. **Specific tax:** Fixed tax charged per unit of a product.
6. **Ad valorem tax:** Tax based on the value of a good.
7. **Sales tax:** Tax charged on the sale of goods and collected by the seller.
8. **Consumption tax:** Tax paid by consumers when purchasing goods, included in the price.

Advantages of Indirect Taxes	Disadvantages of Indirect Taxes
Easy to collect since tax is included in the price of goods and services.	Regressive in nature, affecting low-income earners more than the rich.
Convenient for taxpayers as payment is made automatically during purchase.	Increases the cost of living by raising prices of goods and services.
Has broad coverage because it is paid by many consumers.	Less transparent as consumers may not realize the tax they are paying.
Encourages saving since tax is only paid when spending.	May reduce consumption of essential goods if prices become too high.
Difficult to evade because it is built into transactions.	Government revenue may fluctuate depending on consumption levels.

Application activity 9.3



1. Which of the following is an example of a proportional tax?

- A. Income tax where the rate increases with income
- B. A flat tax where everyone pays the same rate
- C. Tax that decreases as income increases
- D. Tax that applies only to imported goods

2. True or False Questions

- a) Progressive tax means the tax rate increases as income increases.
- b) Indirect taxes are directly paid from a person's salary.
- c) VAT is an example of an indirect tax included in the price of goods and services.

3. Open Questions

- a) Explain the difference among progressive, regressive, and proportional taxes.
- b) How do direct and indirect taxes differ in terms of who bears the tax burden in the economy?
- c) Explain the advantages and disadvantages of both direct taxes and indirect taxes in an economy, giving relevant points for each.
- d) Why do you think indirect taxes are often considered to affect consumers more than producers, even though producers are the ones who initially pay them?

9.4 Effects of taxation

Learning Activity 9.4



If a government increases taxes on goods and incomes, what possible effects might this have on consumers, businesses, and the overall economy?

Taxation has significant effects on the economy, influencing the behaviour of individuals, businesses, and government operations. It affects income distribution, consumption, investment, and overall economic development in both positive and negative ways.

Positive effects of taxation:

- 1. Provides government revenue for public services:** Taxes give the government money to run essential services like administration and security.
- 2. Funds infrastructure development:** Tax revenue is used to build and maintain roads, schools, hospitals, and other public facilities.

- 3. Promotes economic development:** Taxes help the government invest in projects that stimulate growth and improve living standards.
- 4. Reduces income inequality through redistribution:** Progressive taxation helps transfer resources from the rich to support the poor.
- 5. Controls inflation by reducing excess spending:** Higher taxes can reduce excessive demand in the economy, helping to stabilize prices.
- 6. Encourages fair sharing of national resources:** Everyone contributes to national development according to their ability.
- 7. Supports national security and administration:** Taxes fund the army, police, and government institutions.
- 8. Discourages harmful goods:** High taxes on products like alcohol and tobacco reduce their consumption and protect public health.

Negative Effects of Taxation:

- 1. Reduces people's disposable income:** Taxes lower the amount of money individuals can spend or save.
- 2. Can discourage investment and business expansion:** High taxes may reduce profits, making businesses less willing to grow.
- 3. May increase prices of goods and services** – Producers may pass tax costs to consumers through higher prices.
- 4. Can lead to tax evasion and avoidance:** Some people may hide income or avoid taxes to reduce their burden.
- 5. Reduces motivation to work harder:** High income taxes may discourage people from increasing their effort or productivity.
- 6. May create burden on low-income earners:** Poor people may struggle more when taxes increase the cost of basic goods.
- 7. Can cause unfair burden if poorly designed:** Inefficient tax systems may place heavier pressure on certain groups.
- 8. May slow economic growth if taxes are too high:** Excessive taxation can reduce consumption, investment, and production.

Application activity 9.4



1. Multiple Choice

- i. Which of the following is a negative effect of taxation?
 - A. Improvement of infrastructure
 - B. Reduction of disposable income
 - C. Provision of public services
 - D. Reduction of inequality

2. Open Questions

- a) Explain how taxation can promote economic development in a country.
- b) Why can high taxes discourage investment and production?

3. Scenario-Based Question

In your country, the government has increased taxes on goods and incomes. As a result, prices of goods have risen, some businesses have reduced their investment, and citizens are complaining about reduced purchasing power. However, the government has also been able to improve public services such as roads, schools, and healthcare using the increased tax revenue.

- a) Identify two positive and two negative effects of taxation shown in the scenario.
- b) Explain how taxation can affect both consumers and businesses in Rukara.

9.5 Problems facing taxation and policies to improve tax collection in developing countries

Learning Activity 9.5



Scenario-Based Question

In Kigali, it has been observed that some traders in busy markets and transport operators fail to issue receipts and underreport their daily earnings. This reduces government revenue and affects the delivery of public services such as healthcare, education, and infrastructure development. To address this challenge, the Government of Rwanda, through the Rwanda Revenue Authority, has strengthened the use of Electronic Billing Machines (EBMs) and increased inspection measures to improve tax compliance.

Questions

- 1. From the situation above, identify the main economic problem being illustrated.
- 2. Fill in the Blank

The illegal act of deliberately hiding income or failing to pay the required tax is called _____.

- 3. Suggest two measures the government has introduced to reduce the problem identified in question 1.

9.5.1 Problems facing taxation in developing countries

Taxation systems in developing countries often face a number of challenges that limit their efficiency and effectiveness in revenue collection. These difficulties affect the ability of governments to mobilize sufficient funds for public expenditure and development activities.

The main problems are discussed below.

- **Tax evasion:** Some individuals and businesses hide income to avoid paying taxes, reducing government revenue.
- **Tax avoidance:** Certain taxpayers use legal loopholes to reduce their tax burden, leading to less revenue collection.
- **Large informal sector:** Many businesses operate informally and are not registered, making it hard to tax them.
- **Limited tax awareness:** Some citizens lack knowledge about tax obligations, leading to low compliance.
- **High tax burden on formal sector:** Registered businesses may be heavily taxed compared to informal ones.
- **Administrative challenges:** Tax collection systems may face inefficiencies and delays in enforcement.
- **Corruption risks:** Misuse of tax revenues or dishonest practices can reduce trust in the system.
- **Smuggling and illegal trade:** Goods entering or leaving the country illegally reduce customs tax revenue.
- **Fluctuating incomes in agriculture sector:** Since many people depend on agriculture, unstable incomes make tax collection difficult.

9.5.2 Policies to improve tax collection in developing countries:

Tax collection in developing countries can be improved through various policy measures aimed at strengthening efficiency and compliance in the tax system. These measures help governments increase revenue and support public development needs. **The main policies are discussed below.**

- **Digital tax systems (e-filing and e-payment):** Introduction of online platforms reduces fraud, corruption, and improves efficiency.
- **Taxpayer education programs:** Awareness campaigns are conducted to educate citizens about tax obligations and benefits.
- **Expansion of tax base:** The government registers more businesses, especially in the informal sector, to increase compliance.

- **Strict penalties for tax evasion:** Laws impose fines and legal actions against individuals who avoid paying taxes.
- **Simplified tax regimes for SMEs:** Small and medium enterprises are given easier tax systems to encourage compliance.
- **Customs modernization and border control.**
- **Anti-corruption measures:** Strong policies are in place to reduce corruption in tax administration.
- **Investment incentives and tax exemptions:** Encourages compliance and attracts investors to formalize businesses.
- **Sensitizing sellers to use EBM.**

Application activity 9.5



1. What are the major challenges that hinder effective tax collection in developing countries?
2. What policy measures can governments implement to improve tax collection, and how those strategies help to increase compliance and strengthen revenue collection?

9.6 Debt financing versus taxation financing

Learning Activity 9.6



Government options for financing development projects:

The Government of Rwanda wants to build new classrooms in all districts. To raise money, it considers two options:

- 1) Increasing taxes collected from citizens and businesses, or
 - 2) Borrowing money from banks or other countries to finance the project and pay it back later.
1. What do you think is the difference between using taxes and borrowing money to finance government projects?
 2. If the government chooses to borrow money instead of increasing taxes, what might be the possible advantages and risks?
 3. Which method do you think is more sustainable for long-term development? Explain your answer.

Governments need adequate financial resources to support national development, provide essential public services, and maintain economic stability. In many countries, public expenditure often increases due to rising development needs, emergencies, and expanding social services. As a result, governments adopt different methods

of mobilising funds to bridge budget gaps and sustain economic activities. These methods can greatly influence investment, consumption, production, and the general welfare of the population.

Among the major approaches used by governments to raise public funds are **debt financing** and **taxation financing**. Each approach plays a significant role in supporting government programmes and economic development, although they differ in their sources, implications, and effects on the economy.

i. Debt financing

This refers to the government act of fulfilling its expenditure requirements through public borrowing either from internal or external sources or both.

Advantages of debt financing	Disadvantages of debt financing
Provides quick access to large amounts of capital	Involves regular interest payments
Does not dilute ownership or control	Increases risk of bankruptcy if not repaid
Interest payments may be tax-deductible	Repayment is required regardless of business performance
Helps finance expansion and long-term projects	Reduces future borrowing capacity
Encourages financial discipline due to repayment obligations	Lenders may impose strict conditions and restrictions

ii. Taxation financing

This refers to the government act of meeting its expenditure requirements through raising revenue from taxation. Taxation can either be direct or indirect which can either be levied on people's income or goods and services.

Advantages	Disadvantages
Provides a stable source of government revenue.	High taxes may discourage investment and business activities.
Reduces dependence on borrowing and foreign aid.	Heavy taxation can reduce people's disposable income.
Helps finance public services such as education, health, and infrastructure.	Tax evasion and avoidance reduce government revenue collection.
Promotes income redistribution and reduction of inequalities.	Some taxes place a heavier burden on low-income earners.
Encourages government accountability to taxpayers.	Tax collection and administration can be costly and difficult.

Application activity 9.6



1. Explain the difference between debt financing and taxation financing, and analyse two advantages and two disadvantages of each method.
2. Advise the Government of Rwanda on the most suitable method to finance the hospital project and justify your answer with relevant reasons.

9.7 Fiscal policy

Learning Activity 9.7



Visit the school library or the internet, do research and respond to the following questions:

1. Explain the meaning of fiscal policy.
2. Describe the tools and functions of fiscal policy.

9.7.1 Meaning and tools of fiscal policy

Fiscal policy refers to the deliberate use of taxation, borrowing and government expenditure to regulate the level of economic activities or stabilise development. In more developed countries, the role of the fiscal policy is to stabilise the economy, while in less developed countries, the fiscal policy is designed as an instrument of economic development.

The policy instruments or tools of fiscal policy include the following;

- a) Taxation:** This is the most effective instrument of the fiscal policy. Taxes can either be direct or indirect.
- b) Public borrowing:** This is either internal or external and can be either national borrowing or public borrowing.
- c) Government expenditure:** This involves ways of how the government uses the revenue got. It can either be through re current expenditure (expenditure for day-to-day activities) or capital expenditure or development expenditure (expenditure on development like infrastructure).

9.7.2 Forms of fiscal policy

- 1. Expansionary fiscal policy:** This type of fiscal policy is aimed at stimulating the economy and creating more growth. Government **increases spending and reduces taxes**. The idea is to put more money into consumers' hands, so they spend more. This boosts demand, which keeps businesses running and hopefully adds jobs.

The purpose of expansionary fiscal policy is to help the government to:

- Increase economic growth by stimulating demand in the economy
- Reduce unemployment through job creation in public and private sectors

- Increase consumer spending by reducing taxes and increasing disposable income
- Encourage investment by improving business confidence and market demand
- Prevent or recover from economic recession by boosting economic activity

2. Contractionary fiscal policy: This type mainly is used to control inflation and hence to slow economic growth. Government **reduces spending and increases taxes** to control inflation

The purpose of contractionary fiscal policy is to help the government to:

- Reduce the amount of money circulating in the economy
- Control rapid price increases (inflation)
- Decrease excessive consumer and business spending
- Stabilize economic growth when it is too fast or overheating
- Maintain overall economic balance and stability

Application activity 9.7



Scenario: A country is facing economic instability, including high unemployment, low business activity, reduced consumer spending, and rising prices of goods and services. The government responds by adjusting taxation and public expenditure to stabilize the economy.

Tasks:

1. Identify the two forms of fiscal policy applied in the different situations described.
2. Explain how reducing taxes and increasing government spending can help stimulate economic growth and reduce unemployment.
3. Explain how increasing taxes and reducing government expenditure can help control inflation.
4. Analyse two positive effects and two possible challenges of these fiscal policy measures on businesses and households.

9.8 Structure of taxation in Rwanda

Learning Activity 9.8



"Imagine you go shopping in Kigali and buy a mobile phone or clothes from a shop. When you check the receipt, you notice that the total amount paid is higher than the original selling price because some extra charges have been added. Also, when you receive your monthly salary, you notice that the amount you take home is less than your gross salary due to deductions made before payment."

Tasks:

1. Why do you think the total amount paid in a shop is sometimes higher than the original selling price of goods?
2. Where do you think the money deducted from salaries goes, and which authority is responsible for collecting it?
3. Do you think all taxes in Rwanda are the same, or are there different types of taxes? Explain your answer.

9.8.1 Meaning of tax Structure and tax structure in Rwanda

Structure of taxation refers to the **organized system showing how taxes are classified, collected, and managed in a country**, including the different types of taxes and the institutions responsible for administering them.

Structure of taxation in Rwanda is characterized by the following:

- **Composition:** Most of the tax revenue is got from indirect taxes like customs, sales among others.
- **Tax administration:** Revenue is normally collected by Rwanda Revenue Authority on behalf of the central government and the local government authorities.
- **Objective of taxation:** Major objective is to raise revenue for recurrent and development expenditure.
- **Tax base:** Tax base is narrow because system is not comprehensive as not all wealth, income and expenditure are taxed.
- **Taxable capacity:** This is low because of a large subsistence sector and high levels of poverty.
- **Taxation impact:** Most taxes especially indirect taxes are regressive and they tend to affect the poor more than the rich.
- **Taxation effort:** Direct taxes which are progressive tend to discourage the work effort of the people hence there is reduction in productivity.
- **Tax avoidance and evasion:** Rate of evasion is high especially through defaulting. Tax avoidance is low because most indirect taxes are levied on essential goods. However, this has been solved by introduction of the electronic billing machines.

- **Tax GDP ratio:** The proportion is not very much because of low taxable capacity and low tax base however the trend is increasing.

9.8.2 Effects of taxation structure in Rwanda

The tax structure in Rwanda plays an important role in influencing economic activities, government revenue, and the distribution of income. It determines how different taxes are applied to individuals and businesses, and how they affect production, consumption, and investment.

The following are the key facts about the effects of the tax structure in Rwanda.

- 1. On Income Distribution:** A progressive tax reduces inequality by taxing the rich more, while a regressive tax can burden the poor.
- 2. On Economic Behavior:** Taxes on certain goods can **discourage consumption** (e.g., tobacco tax). Low taxes on investment income can **encourage savings and business growth**.
- 3. On Government Revenue:** A well-structured tax system ensures **stable and sufficient revenue** for development projects.
- 4. On Inflation and Prices:** Indirect taxes on goods may **increase prices**, affecting cost of living.

9.9. End of unit assessment



KNOWLEDGE AND UNDERSTANDING

1. Define taxation and explain its main purpose in Rwanda.
2. Identify the classification of taxes used in Rwanda, giving at least one example for each.
3. Distinguish between the following concepts:
 - a) Tax evasion and tax avoidance
 - b) Direct taxes and indirect taxes
4. Choose the correct answer for each:
 - a) The institution responsible for tax collection in Rwanda is:
 - b) Ministry of Finance
 - c) Rwanda Revenue Authority
 - d) National Bank of Rwanda
 - e) Rwanda Development Board
 - b) A tax system where higher incomes are taxed at higher rates is called:
 - a) Progressive tax

- b) Regressive tax
- c) Proportional tax
- d) Indirect tax

SKILLS

1. Consumer buys goods worth FRW 50,000 including 18% VAT.
 - a) Calculate the VAT amount.
 - b) Explain why VAT is considered an indirect tax.
2. Analyse the effects of taxation in Rwanda on the following areas:
 - a) Income distribution
 - b) Government revenue
 - c) Prices and inflation
3. Study the scenario below and answer the questions:
 - a) In Kigali, some traders and transport operators often understate their daily income, reducing government revenue. The government has responded by introducing Electronic Billing Machines (EBMs). Identify the economic problem illustrated in the scenario.
 - b) Explain two causes of this problem.
 - c) Suggest two measures that can improve tax compliance.

VALUES AND ATTITUDES

1. Explain why honesty and transparency in tax payment are important for national development in Rwanda.
2. Discuss how responsible tax payment by citizens can promote fairness and improve public service delivery such as education and healthcare.



Key Unit Competence:

Analyse the impact of population growth on economic development

Learning objectives

Knowledge and understanding

1

- Explain basic population concepts and related terminology. (ex. population, density, dependency ratio, census, etc)
- Identify the reasons why the population census is carried out and the difficulties encountered.
- Explain population structures and the population pyramid.
- Explain the factors and impacts of population growth on the economy.
- Explain Malthusian population growth theory, its applicability and limitations

Skills

2

- Analyze population data to interpret trends, structures, and distribution patterns. (production of pyramids)
- Calculate population density and population growth rate
- Evaluate the causes and effects of population growth and demographic changes.
- Assess the economic implications of population size, structure, and dependency ratios.
- Suggest realistic strategies to manage population growth and enhance economic welfare of population in Rwanda and in Developing countries.
- Analyze the relevance of Malthusian population growth theory in Rwanda and in Developing countries.
- Use illustrations to explain the causes and effects of under population and over population.
- population growth theory in Rwanda and in Developing countries.
- Use illustrations to explain the causes and effects of under population and over population.

Attitudes and values

3

- Be aware of the impact of population growth on an economy and advocate for population control measures.
- Respect for national efforts in planning and data collection
- Develop Sense of civic duty and cooperation with government
- Demonstrate Integrity in providing accurate information.
- Show concern for sustainable resource use.
- Have an open mindedness and critical thinking about economic theories.
- Develop sense of fairness in resource allocation



Introductory activity

Scenario

In the district of Gasabo, the population has been increasing rapidly over the past 10 years. As a result, schools are overcrowded, jobs are limited, and hospitals are struggling to serve many patients. However, local business owners argue that the large population provides a ready market and labour force for economic activities.

Questions

1. From the situation in Gasabo, how can population growth be both a benefit and a challenge to economic development?
2. In your view, why is it important for a country to understand the size and structure of its population before planning for development?
3. How might changes in population (such as growth or decline) influence the production and distribution of goods and services in an economy?

Introduction

Population is an important concept in economics because it influences the availability of labour, the level of demand for goods and services, and the overall growth of an economy. It also helps in understanding how resources are allocated and how government plans for development.

10.1 Meaning of population and related terminologies

Learning Activity 10.1



1. What do you understand by the term population?
2. What challenges might a country face if it has too many young people compared to adults?

3. What do you think could happen if a country has too many people or too few people?

4. Why do governments sometimes count people? What do you think they use that information for?

1. Population refers to the number of people living in an area at a particular period of time. In economics, population refers to the total number of individuals living within a given area who participate in economic activities such as production, consumption, and distribution of goods and services, thereby influencing the level and structure of economic development in that area.

There are two major functions of population:

- » Population is the major consumption unit in an economy.
- » Source of labour which is the major factor of production.

1. Population census is the official process of counting all people living in a country or area at a specific time and collecting their social and economic information.

2. Population distribution is Population distribution is the way people are spread across different areas within a country or region. It shows how population is unevenly located, with some areas being densely populated while others are sparsely populated due to factors such as climate, resources, and employment opportunities.

3. Migration: this refers to the movement of people from one region to another.

4. Immigration: Coming to settle in the country

5. Emigration: Going to settle out of the country

6. Population growth is the increase in the number of people in a given area over a period of time due to births, deaths, and migration.

7. Doubling time is the period of time required for a population to double in size at a constant growth rate. It is commonly used to measure how fast a population is increasing and helps in planning for future needs such as food, housing, and services.

$$\text{Doubling time} = \frac{70}{\text{Population growth rate (PGR)}}$$

Example:

A country has an annual population growth rate of 2%. Calculate its doubling time.

Solution: Doubling time = $\frac{70}{2}$ 35 Years

Population density is the number of people living per unit area, usually measured per square kilometre.

$$\text{Population density} = \frac{\text{Total population}}{\text{Land Area}}$$

Example (Rwanda case):

Rwanda has a population of 13,246,394 people and a land area of 26,338 km². Calculate the population density of Rwanda.

Solution: Population Density = $\frac{13,246,394}{26,338}$

Population Density ≈ 503 people per km²

The population density of Rwanda is approximately **503 people per square kilometer**, showing that Rwanda is one of the most densely populated countries in Africa.

Population pyramid is a graphical representation that shows the age and sex structure of a population at a given time. It is used to understand population composition and to help in planning for future needs such as education, healthcare, and employment.

Dependency ratio is the proportion of dependents (young and elderly) compared to the working-age population.

$$\text{Dependency ratio} = \frac{\text{Number of dependents}}{\text{Working age population}} \times 100$$

Example:

In District A, the population is as follows:

- Children (0–14 years): 12,000
- Working-age population (15–64 years): 30,000
- Elderly (65+ years): 8,000

Required:

Calculate the dependency ratio of District A.

Solution:

Dependents = Children + Elderly

$$12,000 + 8,000 = 20,000$$

Working population = 30,000

$$\text{Dependency ratio} = \frac{20,000}{30,000} \times 100 = 66.7\%$$

District A has a population of 50,000, with a strong labour force of 30,000 people. The 20,000 dependents result in a dependency ratio of 66.7%, showing a moderate burden on the working population.

Implications of a high dependency ratio:

- Increased financial burden on the working population
- Higher government expenditure on social services
- Lower savings and investment levels
- Slower economic growth
- Increased unemployment and underemployment pressure

Population trend: This refers to the general pattern or direction in which a population is changing over time, such as whether it is increasing, decreasing, or remaining stable.

Ageing population is a population in which the proportion of elderly people is increasing due to low birth rates and higher life expectancy.

Effects of an ageing population

- Decline in labour force for the future.
- High dependence burden.
- Labour immobility i.e. the old are immobile.
- High levels of conservativeness because the old tend to obstruct new ideas.
- Increased government expenditure on retirement benefits.
- Reduction in aggregate demand for some commodities.
- More goods for the old people are now being produced e.g. hats, walking sticks, big trousers, very long dresses, big shirts, etc

Declining population is a situation where the total number of people in a population decreases over time due to low birth rates, high death rates, or out-migration.

Effects of declining population

A declining population can have both positive and negative effects on an economy and society.

Positive effects	Negative effects
Reduced pressure on resources	Reduced labour force
Lower environmental degradation	Slower economic growth
Improved living standards	Increased dependency ratio
Reduced urban congestion	Reduced government revenue

Application activity 10.1



1. Choose the correct answer:

a) Doubling time is calculated using:

- a) $100 \div \text{growth rate}$
- b) $70 \div \text{population growth rate}$
- c) $\text{Growth rate} \div 70$
- d) $\text{Population} \div \text{land area}$

b) Population census refers to:

- a) Estimating population growth using samples
- b) Counting only working-age people in a country
- c) Official counting of all people and their characteristics at a specific time
- d) Measuring population density in urban area

2. True or False questions:

- a) Population density is calculated by dividing total population by land area.
- b) An ageing population increases government expenditure on retirement benefits.
- c) A declining population always leads to an increase in government revenue.

10.2 Population census

Learning Activity 10.2



1. Why do you think governments need to know the exact number of people living in a country?
2. How would life be affected if no one knew how many people live in different areas of the country?
3. What kind of information about people do you think is important for planning schools, hospitals, and other services?

In Rwanda, a population census, which is the process of counting all people in the country, is conducted to gather accurate data on population size, distribution, and the socio-economic characteristics of the people. It is carried out every 10 years by the National Institute of Statistics of Rwanda, where trained enumerators visit households to collect information on age, sex, education, employment, and housing conditions.

10.2.1 Reasons for carrying out population census and difficulties met in population census.

A. Reasons for carrying out population census or importance of population census

A population census is a fundamental national exercise that provides comprehensive information about the size, structure, and distribution of a country's population. This information is essential for guiding government planning, policy formulation, and resource allocation. **The following are the main reasons for carrying out a population census:**

- It helps to determine the total population size and its distributions in the country.
- It is important for national planning purposes.

- It helps to determine population growth rate of the country overtime.
- The population figures are used in calculation per capita income in the country.
- It helps to determine the sex composition of the population that is the ratio of men to women.
- It helps to show the geographical distribution of the populations; this is important for regional resources allocation.
- It helps to determine the rate of internal and external migration.
- It helps to determine the ethnic and religious composition of the population which belongs to different tribes and religions.
- It is used to determine the population density (number of people per unit area of land)
- It helps to determine the population structure and composition in terms of age and level of education of the people.
- It is used to determine the occupation composition of the population. This is important for proper manpower planning

B. Difficulties met in population census

A population census is a large national exercise that often faces various challenges during implementation. The following are the main difficulties encountered in conducting a population census:

- It is expensive in terms of paying enumerators, processing data, logistics etc
- Remoteness of some areas which are inaccessible
- Wars and instability make it difficult since people are scattered every where
- Nomads: these are people who move from place to place. it is difficult to locate them on census day
- Illiterate persons are not able to provide required information.
- Some people give accurate information for economic, political and religious reason.
- Traditions and culture are also problem as some culture does not allow people to count their children with belief that after counting them, they die.
- Poor infrastructures that make some places unreachable.
- Lack of qualified persons to carry out population census.

10.2.2 NISR population census statistics data (2022)

The following population census statistics for Rwanda were compiled by the National Institute of Statistics of Rwanda (NISR) in 2022. The figures provide an important basis for understanding Rwanda's population structure, composition, distribution, and growth patterns. They also present key demographic, social, and economic characteristics of the population.

Experimental version

Such information is useful to governments, researchers, and other stakeholders in analysing population trends, planning for public services, and formulating appropriate development policies and programmes.

1. Total Population

Category	Statistics (2022)
Total population of Rwanda	13,246,394 people

2. Sex Structure

Category	Statistics (2022)	Percentage (%)
Men	~6,415,000 people	48.5%
Women	~6,831,394 people	51.5%

3. Regional Population Distribution

Category	Names / Indicators	Statistics (2022)	Percentage (%)
Most populated province	Eastern Province	~3,563,145 people	26.9%
Least populated province	Northern Province	~2,038,511 people	15.4%
Most populated district	Gasabo District	~879,505 people	~6.6%
Least populated district	Nyaruguru District	~318,564 people	~2.4%

4. Population Growth & Distribution

Category	Names / Indicators	Statistics (2022)	Percentage (%)
Population growth	Annual population growth rate	~2.3%	-
Urban population	-	~3.7 million people	~28%
Rural population	-	~9.5 million people	~72%

5. Household & Dependency Structure

Category	Names / Indicators	Statistics (2022)
Household characteristics	Average household size	~4 persons
Dependency structure	Dependency ratio	~76 dependents per 100 workers

6. Labour Force & Age Structure

Category	Indicators	Statistics (2022)	Percentage (%)
Labour force	Working-age population (15–64 years)	~7.1 million people	~54%
Age structure	Population below 20 years	Majority youthful population	~65%

7. Education, Health & Human Development

Category	Indicators	Statistics (2022)	Percentage (%)
Education indicators	Literacy rate (15+)	-	~73%
Fertility indicators	Total fertility rate	~3.6 children per woman	-
Mortality indicators	Life expectancy	~69 years	-
Health indicators	Health insurance coverage	Majority covered	~80%+

8. Resident population by religious affiliation and residence (2022)

Category	Percentage (%)
Religion / Christianity (overall)	>90%
Catholic	39.9%
ADEPR	21.3%
Protestant	14.6%
Adventist	12.2%
Muslim	2.0%
Jehovan witness	0.7%
No religion	3.0%

9. Language Literacy

Category	Percentage (%)
Kinyarwanda literacy	78.3%
English literacy	21.2%
French literacy	8.2%
Swahili literacy	4.0%

10. Economic & Social Indicators

Category	Names / Indicators	Statistics (2022)	Percentage (%)
Economic activities	Agriculture employment	Majority	~60%
Poverty indicators	Below poverty line	-	~27%
Technology access	Mobile phone ownership	Majority households	~80%+

Source: Reference: https://statistics.gov.rw/statistical-publications/population-size-and-population-characteristics?utm_source=chatgpt.com

Application activity 10.2



1. What do you understand by the term population census?
2. Explain the reasons why a government should conduct a population census.
3. Identify the challenges that can arise during the process of collecting population census data.

10.3 Population pyramid

Learning Activity 10.3



1. Why do some countries have a higher proportion of young people compared to older people in their population structure?
2. In Rwanda, when people are grouped by age (children, youth, adults, and the elderly), which group do you think is the largest?
3. Is the number of males and females in a population always equal? How can the structure of a country's population be represented in a simple diagram?
4. What challenges may arise if a country has a very high proportion of young people compared to working adults, and why is it important for governments to understand the structure of their population?

Population structure can be represented using different graphical methods that help to show the distribution of people by age and sex. One of the most important tools used in demography is the **population pyramid**, which provides a clear visual representation of how a population is composed.

10.3.1 Types of population pyramid and their implications

Types of population pyramids:

1. Expansive (Progressive) Population Pyramid

This is a type of population pyramid characterized by a wide base and a narrow top, indicating a high birth rate and a large proportion of young people, with relatively few elderly people. It reflects rapid population growth and is commonly found in **developing countries**.

Rwanda is an example of this pattern, as its population is still largely youthful, with a high proportion of children and young people compared to the elderly population.

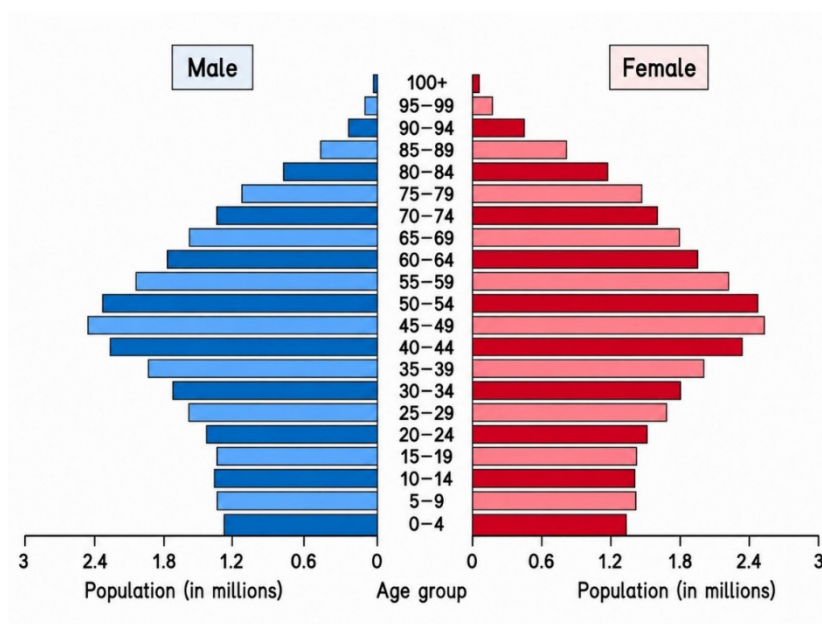


Figure 10.1.: Expansive (Progressive) Population Pyramid

Implications of expansive population pyramids:

Positive implications	Negative implications
Large future labour force due to many young people.	High dependency burden, as many children depend on few workers.
Potential for economic growth in the future if young people are well educated and employed.	High pressure on social services such as schools and hospitals.
Availability of cheap labour in the economy.	High unemployment or underemployment among young people.
Large market for goods and services due to many young consumers.	High cost of providing basic needs such as food, housing, and healthcare.
Possibility of population-driven development if well managed.	Rapid population growth can strain resources and slow down development.

2. Stationary (Stable) population pyramid

This is a type of population pyramid that has a more balanced or nearly straight shape, showing that the birth rate and death rate are almost equal. It indicates a stable population with slow or no significant growth.

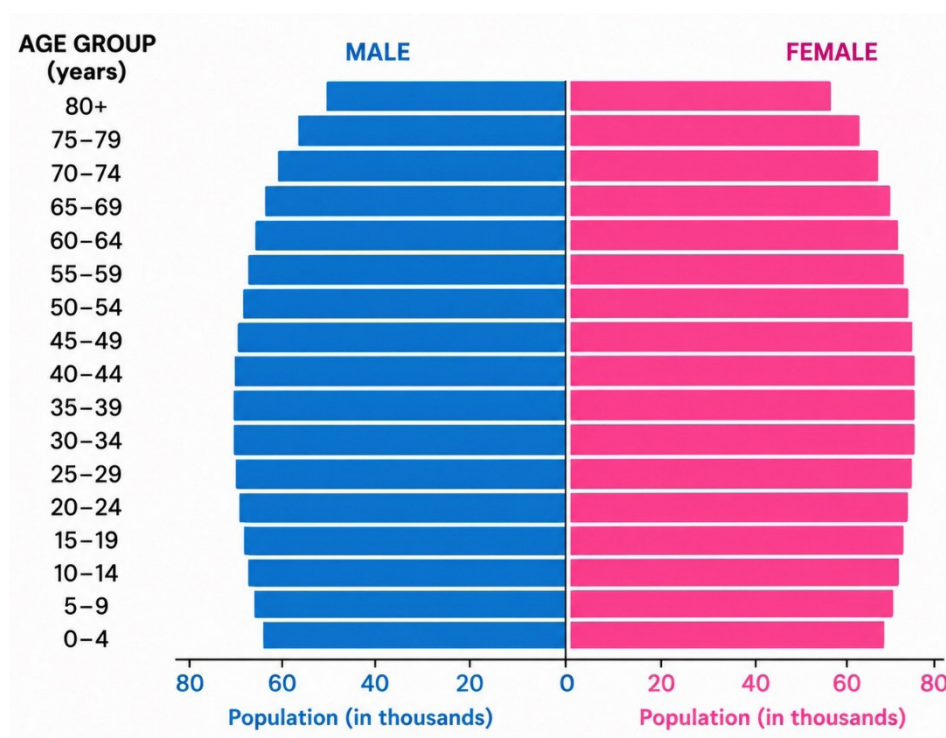


Figure 10.2.: Stationary (Stable) Population Pyramid

Implications of stationary population pyramids:

Positive implications	Negative implications
Stable population size, making planning easier for government.	Slow or no population growth may lead to a shortage of future workers.
Balanced age structure between young and old people.	Ageing population may increase the number of elderly dependents.
Less pressure on resources compared to rapidly growing populations.	Reduced labour force growth can limit economic expansion.
Better chance of providing quality services such as education and health.	Higher government spending on elderly care over time.
More sustainable use of available resources.	Possible decline in innovation and productivity if the population ages.

Constrictive (Regressive) Population Pyramid

This is a type of population pyramid with a **narrow base and a relatively wider middle or top**, showing that the **birth rate is low and the proportion of older people is higher**. It indicates a **slow or declining population growth**, commonly found in developed countries.

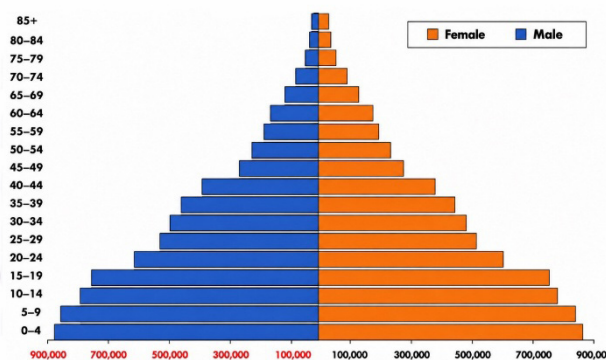


Figure 10.3.: Constrictive (Regressive) Population Pyramid

◆ Implications of constrictive population pyramids:

Positive implications	Negative implications
Lower dependency ratio due to fewer children.	Ageing population increases the number of elderly dependents.
Less pressure on schools and child-related services.	Shortage of young labour force in the economy.
Higher proportion of working-age population can support development.	High government spending on pensions and healthcare for the elderly.
Better quality of life due to smaller population size.	Reduced population growth may lead to population decline.
Easier management of resources and infrastructure.	Risk of economic slowdown due to a shrinking workforce.

Application activity 10.3



1. What is meant by a population pyramid, and what are three ways in which population structure can be represented?
2. Rwanda has a large proportion of young people compared to elderly people:
 - a) What type of population pyramid does this represent?
 - b) Give three characteristics of this type of population pyramid.
3. A country has a situation where the number of births is almost equal to the number of deaths:
 - a) Which type of population pyramid does this describe?
 - b) Is the population growing fast or slowly? Explain your answer.
4. A country has a narrow base and a wider top in its population structure:
 - a) Identify the type of population pyramid.
 - b) Suggest two challenges this country may face.
 - c) Explain two advantages and two disadvantages of having a very young population in a country like Rwanda.

10.4 Population growth

Learning Activity 10.4



1. What do you think is happening to the population of Rwanda: is it increasing, decreasing, or remaining stable? Why do you think so?
2. What factors do you think contribute to population growth in a country?
3. Why do some countries experience faster population growth than others?

Population growth is the rate at which the population of the country increases or decreases over time. It is measured by percentage change in the population. The average annual population growth rate of Rwanda (2.6%) is among the highest in Africa.

Reference: https://statistics.gov.rw/statistical-publications/population-size-and-population-characteristics?utm_source=chatgpt.com

10.4.1 Types and causes of population growth

A. Types of population growth:

- a) **Natural population growth (NPG): Natural Population Growth (NPG)** is the increase in a country's population that results from the difference between **births and deaths only**, without considering migration (immigration or emigration).

$$\text{NPGR} = \frac{\text{BR} - \text{DR}}{1000} \times 100$$

Where,

NPGR: Natural population growth rate

BR: Birth rate

DR: Death rate

- b) **Artificial population growth:** Is the growth of population that results from migration. I.e. **immigration** refers to the coming of people to stay in the country therefore it increases the number of people in the country while **emigration** refers to people moving to settle in other countries therefore it reduces number of people in the country.

$$\text{Artificial Growth Rate} = \frac{I - E}{P} \times 100$$

Where:

- I = Immigration (people entering an area)
- E = Emigration (people leaving an area)
- P = Total population of the area

c) Actual population growth: this is population growth resulting from combination of both natural and artificial growth. It is calculated as follows

$$\text{Actual population growth} = \text{BR} - \text{DR} + \text{Net migration (Immigrants - emigrants)}$$

B. Causes of population growth

Population growth is shaped by a range of interrelated factors that determine how the number of people in a given area changes over time. These factors account for variations in growth rates across different regions and are essential for understanding demographic trends. The following are the main causes of a high population growth rate:

- **High birth rate (fertility rate):** When more babies are born than before, the population increases.
- **Low death rate (mortality rate):** Improved health services, good nutrition, and sanitation reduce deaths, leading to population growth.
- **Immigration:** When people move into a country from other countries, the population increases.
- **Improved health care services:** Hospitals, vaccines, and medical treatment reduce death and increase life expectancy.
- **Better living conditions:** Availability of clean water, food, and housing improves survival rates.
- **High fertility patterns in society:** Cultural or social norms that encourage having many children increase population size.
- **Economic opportunities:** Jobs and better income attract migrants and support larger families.
- **Early marriages:** When people marry at a young age, they tend to have more children over their lifetime, increasing population growth.
- **Religious or cultural beliefs encouraging large families:** Some beliefs value having many children, which leads to higher birth rates.
- **Improved food supply and agriculture:** Better farming methods and food availability reduce starvation and increase survival rates, contributing to population growth.

Major determinants of population growth

- i. **Birth rate:** this refers to number of children who are born per 1000 of the population in the year.

$$\text{BR} = \frac{\text{Live birth in the year} \times 1000}{\text{Total population}}$$

Birth rate is affected by the following factors:

- **Ratio of women to men:** The higher the ratio of women to men, the higher the birth rate and the lower the ratio, the lower the birth rate.
 - **Level of education:** Educated people tend to produce less children compared to the uneducated.
 - **Rate of birth control:** The higher the use of control measures like family planning, the lower the birth rate and the reverse is true.
 - **Religion:** Some religions do not accept the use of family planning methods while others accept polygamy and all these increase birth rates.
 - **Culture:** Some cultures believe in polygamy while others believe in children as a source of wealth. This increases birth rates.
- ii. **Death rate:** This is the number of people who die per 1000 of the total population in a year.

$$\text{DR} = (\text{Number of deaths in one year} \div \text{Total population}) \times 1,000$$

Death rates of the country depend on the following:

- **Medical care.** If medical care is available, death rates will be low and vice versa.
 - **Poverty levels,** the higher the rates of poverty, the inability of the person to get the basics hence high death rates, lower rates of poverty will lead to low death rates.
 - **Political situation.** High rates of instabilities will cause high death rates compared to good political atmosphere.
 - **Natural disasters** like earth quakes, floods among others increase death rates.
 - **Accidents** like fire, motor etc. all increase death rates.
 - **Incidences** of diseases in some areas also influence the death rates. Etc.
- iii. **Migration:** this refers to movement of people from one country to another.

10.4.2 Effects and measures to control high population growth

A. General effects of population growth

Population growth has both positive and negative effects on a country's social, economic, and environmental conditions, and those effects are described below.

Positive effects:

- **Increased labour force:** A growing population provides more workers who can be employed in different sectors such as agriculture, industry, and services, which helps to boost production and economic development.
- **Expansion of market size:** As the population increases, the demand for goods and services also rises, creating a larger market for businesses and encouraging

investment and production.

- **Encourages economic growth:** With more people producing and consuming goods, economic activities increase, which can lead to higher national income and overall economic development.
- **Promotion of innovation and skills development:** A large population creates competition for jobs, encouraging people to acquire new skills, improve education, and develop innovative ideas to improve productivity.
- **Improved utilization of natural resources:** Population growth can lead to better use of idle or underutilized resources such as land and minerals, which contributes to national development.
- **Increased tax revenue for government:** A larger population means more taxpayers, which increases government revenue and enables it to provide better public services like schools, hospitals, and infrastructure.
- **Improved infrastructure development:** A growing population encourages governments to expand infrastructure such as roads, schools, hospitals, and electricity supply, which improves accessibility and supports economic activities.
- **Stimulates industrial development:** Population growth increases demand for manufactured goods, encouraging the growth of industries and attracting investors to produce goods locally, which creates employment opportunities.

Negative effects:

- **Unemployment:** Rapid population growth increases the number of job seekers, but job opportunities may not grow at the same rate, leading to high levels of unemployment.
- **Pressure on social services:** A large population increases demand for services such as education, health care, and housing, which can overwhelm government capacity and reduce service quality.
- **Environmental degradation:** Increased population leads to overuse of natural resources like forests, land, and water, resulting in deforestation, soil erosion, and pollution.
- **Food shortage:** When population grows faster than food production, it can lead to insufficient food supply, causing hunger and malnutrition.
- **High dependency burden:** A large number of children and non-working people increases the dependency ratio, placing financial pressure on the working population.
- **Urban overcrowding:** Population growth causes migration to urban areas, leading to overcrowding, slums, traffic congestion, and poor living conditions in towns and cities.
- **Increased government expenditure:** Governments are forced to spend more on education, health, housing, and infrastructure, which can strain national

budgets and slow development in other sectors.

- **Resource scarcity and conflict:** High population pressure can lead to competition for scarce resources such as land and water, which may result in conflicts and social tension.

B. Measures to control population growth

- **Sex education in schools:** Introducing sex education in the school curriculum helps young people understand the dangers of early sexual activity, which can lead to unwanted pregnancies and increased birth rates. It promotes responsible behavior among teenagers.
- **Family planning promotion:** Encouraging the use of family planning methods such as contraceptives helps couples plan the number and spacing of children. This directly reduces high fertility rates.
- **Promotion of girl child education:** Educating girls reduces early pregnancies and delays childbearing. Educated women are more likely to have fewer children and invest more in their wellbeing.
- **Discouraging early marriage:** Raising the legal age of marriage and enforcing laws against child marriage reduces the length of the reproductive period, leading to fewer children per woman.
- **Improving healthcare services:** Better health facilities reduce infant and child mortality rates. When parents are confident that children will survive, they tend to have smaller families.
- **Economic and social incentives for small families:** Governments can encourage small families by offering benefits such as free education, housing support, or medical care for a limited number of children, reducing the desire for large families.
- **Discouraging polygamy and harmful cultural practices:** Reducing practices like polygamy helps limit the number of children born in one household, thereby lowering overall population growth.
- **Government population control policies and awareness campaigns:** Governments and NGOs can run campaigns to educate people about the dangers of overpopulation and introduce policies such as taxes, penalties, or regulations that discourage large families.

Application activity 10.4



Read the following situation carefully and answer related questions:

In Country Y, the government has reported changes in population over the past 10 year. The country recorded 400,000 live births and 150,000 deaths. During the same period, 70,000 people entered the country for employment opportunities, while 40,000 people left the country to live abroad. The government is planning new policies to manage population growth and improve service delivery.

Questions:

1. Calculate the Natural Population Growth (NPG) of Country Y.
2. Calculate the Net Migration of Country Y.
3. Calculate the Actual Population Growth of Country Y
4. Identify five possible reasons why people are immigrating into Country Y.
5. State five effects of rapid population growth in Country Y.
6. If you were a policy maker, what advice would you give families to help manage population growth? Explain your answer.

10.5 Population density

Learning Activity 10.5



What factors do you think explain why some regions have many people while others have few inhabitants?

10.5.1 Meaning of population density

Population density refers to the number of people living in a given area, usually expressed as people per square kilometre. It is calculated by dividing the total population of a place by its land area. Population density helps to show whether an area is crowded (high density) or sparsely populated (low density), and it is useful in planning for resources, services, and infrastructure.

According to the Fifth Rwanda Population and Housing Census, Rwanda's official population density was **503 inhabitants per square kilometer** in August 2022.

10.5.2 Consequences of densely and sparsely populated areas

Differences in population distribution have important effects on economic activities, the use of resources, and the delivery of social services. Therefore, understanding these variations helps to explain the challenges and opportunities that arise in both densely and sparsely populated areas. **The following are the consequences of densely and sparsely populated areas.**

Consequences of Densely Populated Areas

A densely populated area is an area where many people live within a small piece of land, resulting in a high number of people per square kilometre. Examples in Rwanda include Kigali, Rubavu District, and Musanze District, while examples in the world include Tokyo, Mumbai, New York City, etc

Positive consequences:

- **Availability of labour force:** Businesses have access to a large number of workers, making it easier to employ people in different sectors.
- **Market expansion:** A large population creates a big market for goods and services, encouraging business growth.
- **Improved infrastructure development:** Governments are encouraged to develop roads, schools, hospitals, and other facilities due to high demand.
- **Economic growth:** High population concentration increases production, investment, and overall economic activities.
- **Better social services:** Dense areas often attract more social services such as health centres, schools, and banking services.
- **Innovation and creativity:** Interaction among many people promotes exchange of ideas, leading to innovation and new business opportunities.

Negative consequences:

- **Pressure on housing:** High demand for accommodation leads to overcrowding and informal settlements.
- **Unemployment challenges:** Many people compete for limited job opportunities.
- **Strain on social services:** Schools, hospitals, and water supply become overburdened.
- **Traffic congestion:** Increased number of people leads to transport delays and road congestion.
- **Environmental degradation:** Pollution, deforestation, and waste accumulation increase.
- **High cost of living:** Prices of goods, rent, and services tend to rise.
- **Pressure on land resources:** Limited land leads to fragmentation and reduced agricultural space.

Consequences of sparsely populated areas

A sparsely populated area is an area where few people live within a large piece of land, resulting in a low number of people per square kilometre. Examples in Rwanda include Nyagatare District, Gatsibo District, Nyaruguru district and Akagera National Park, while examples in the world include Sahara Desert, Siberia, Greenland, etc

Positive consequences:

- **Availability of land:** There is plenty of land for settlement, farming, and future expansion.
- **Low pressure on resources:** Natural resources such as water, forests, and minerals are less overexploited.
- **Less congestion:** There is reduced overcrowding, traffic, and competition for services.

- **Cleaner environment:** Less pollution of air, water, and land due to fewer human activities.
- **Peaceful living conditions:** Reduced noise, crime, and social conflicts create a calm environment.
- **Easy planning and development:** Governments find it easier to plan infrastructure like roads and schools due to less congestion.

Negative consequences:

- **Underutilization of resources:** Large areas of land and natural resources remain unused.
- **Limited labour supply:** Fewer people are available for productive economic activities.
- **Poor infrastructure development:** Roads, schools, and hospitals develop slowly due to low demand.
- **High service delivery costs:** Government spends more to provide services to scattered populations.
- **Slow economic growth:** Low population density reduces market size and investment attraction.
- **Weak markets:** Limited demand for goods and services affects business expansion.
- **Rural isolation:** People may experience limited access to social and economic opportunities.

Application activity 10.5



A Country A has a population of 20,000,000 people and a land area of 100,000 km². Country B has a population of 12,000,000 people and a land area of 60,000 km².

- Calculate the population density of each country.
- Which country is more densely populated?
- Interpret what the difference in population density might imply for economic development and resource distribution in the two countries.
- Suggest two challenges that District X and Y are likely to face.
- Suggest two measures the government can take to manage problems caused by high and low population density.

10.6 Population theories

Learning Activity 10.6



In different parts of the world, population is changing in different ways. In some countries, population is increasing very fast and people fear food shortages and unemployment. In other countries, population is growing slowly or even becoming stable, and living standards are improving. Some people believe population growth is dangerous, while others believe it is necessary for economic development.

Read the above scenario and answer the following questions:

1. Can population growth stop by itself in the future? Why do you think so?
2. Do you think population growth is always a problem, or can it also be beneficial? Explain your view.
3. Who do you think is right: those who believe population will always increase rapidly or those who believe it will stabilize? Give reasons.
4. What could explain the different views people have about population growth?

Population growth and changes in population size have been explained in different ways over time. As populations continue to change across countries and regions, different ideas have been developed to explain the causes, patterns, and effects of population growth. These ideas have resulted in the development of various population theories. **The following are the main population theories.**

10.6.1 Malthusian population theory

This was put up by **Reverend Thomas Robert Malthus** an English **Economist (1798 — 1823)** who pointed out the dangers of over population in relation to the supply of food in relation to food supply. It explains the relationship between the growth in food supply and in population. Population depends on food supply.

When food supply increases, population growth increases and vice versa. The theory states that population increases faster than food supply and if unchecked leads to **misery**. Man's biological capacity to reproduce himself exceeds his physical capacity to increase the food supply.

❖ Assumptions of the Malthusian population theory

Reverend Malthus assumed that:

- Population growth depends on food growth and that when food supply increases, population also increases.
- The supply of land is fixed thus the operation of the law of diminishing returns.
- Population grows at a geometric rate i.e. 2,4,8,16,32, etc.
- Food increases at an arithmetic rate of 2, 4,6,8,10,12, etc.

- At time 'Lo' population will be equal to food supply and this is what he calls the population trap.
- Man's capacity to reproduce himself is greater than his capacity to produce food. Malthus therefore concluded that at one time population growth will be too much to be fed by the available food and this will lead to misery, suffering and death.
- Malthus believed that any attempt to improve people's living conditions through government assistance or private charity may not provide a lasting solution to poverty. According to Thomas Robert Malthus, when people receive better food, healthcare, housing, and income support, death rates tend to fall and people are encouraged to have more children. This leads to rapid population growth.
- Beyond the population trap, people be subjected to conflicts, starvation, famine, death, accidents, wars etc. and he calls these positive checks i.e. positive checks control population growth by increasing death rates.
- He advocated for negative checks like late marriages, moral restrictions, celibacy that control population growth by reducing birth rate.

Malthus predicted that there will be a point where population growth will be equal to food production and he termed this **population trap** and beyond this point the food resources will not be enough to feed the population.

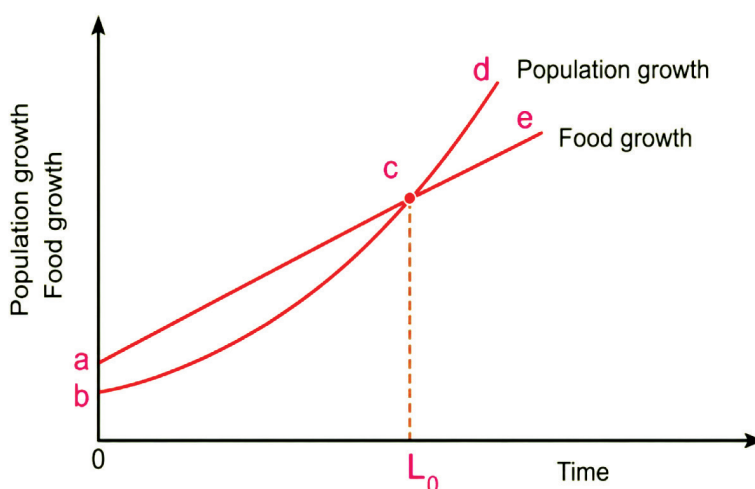


Figure 10.4.: Malthusian population trap

From the figure above, Line bcd represents the growth of population in the absence of any checks. The actual population growth rate is represented by line bce in the presence of checks. Beyond time L_0 , population growth exceeds food supply, and therefore people are likely to starve to death. Malthus indicates that always there are positive and negative checks operating.

Malthus identified **positive checks** and **negative checks** on population growth.

The positive checks check on population growth in a crude way and they include famine, death, accidents, wars etc.

The negative checks check on population growth by reducing the birth rates. They include late marriage, family planning, moral restraint etc.

Applicability/ Relevance of the Malthusian theory in developing countries:

To a small extent, the Malthusian population theory is applicable in developing economies in the following ways:

- Farming planning measures are being used in many countries as suggested by Malthus to control population growth.
- The positive checks like diseases, accidents etc. exist in many countries.
- In many parts of the world, pressure on land forces many people into disputes and migration.
- Some areas in the world face food shortages as suggested by Malthus.
- Land being a fixed factor, faces diminishing returns in that when population increases, productivity of the land decreases.

Criticisms/ limitations of Malthusian theory

To a big extent the theory is criticised because of the following:

- He did not consider the role of foreign aid in supporting a growing population.
- He did not specify the exact time when population growth would exceed food supply (population trap), which limits planning.
- He ignored other factors of population growth such as migration, education, and cultural values.
- He underestimated agricultural technological progress that has greatly increased food production.
- He did not consider international trade as a way of increasing food supply through imports.
- He failed to foresee the modernization of agriculture to boost food production.
- He assumed a mathematical relationship between food growth and population growth, which does not exist.
- He ignored international migration as a way of reducing population pressure.
- He did not consider the expansion of agricultural land through opening new areas.
- He limited preventive checks to moral restraint and late marriage, ignoring modern family planning methods.
- He overemphasized birth rates and ignored the major impact of declining death rates on population growth.

10.6.2 The theory of demographic transition

Demography is the study of population characteristics like birth rates and death rates and their effects on population changes.

The theory of demographic transition model (**DTM**) explains population growth in historical perspective in developed countries. According to this theory, population in developed countries has passed through 5 stages to reach where it is today and it is shown in the figure below.

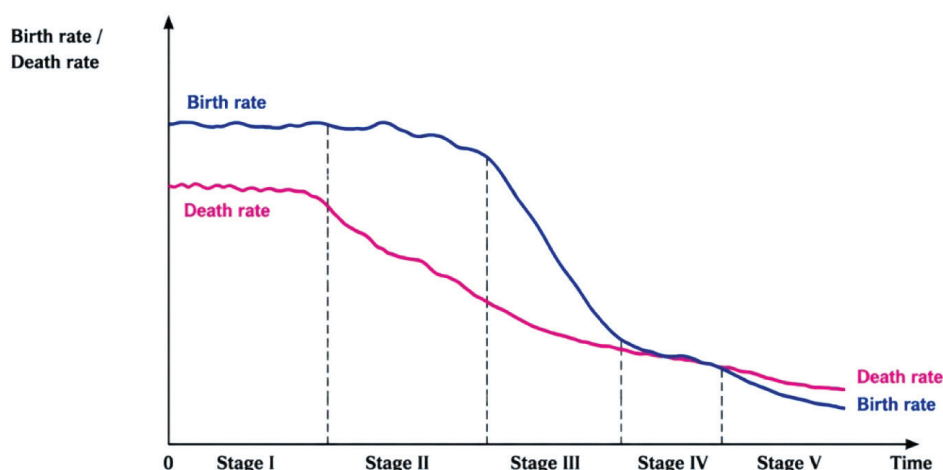


Figure 10.5.: Demographic transition

Stage 1: High Stationary

- **High birth rate:** Many children are born due to lack of family planning and children being seen as economic assets.
- **High death rate:** Many people die because of poor health services, diseases, famine, and poor sanitation.
- **Very low or no population growth:** Birth rate and death rate are almost equal, so the population remains stable or changes very slowly.
- **Low life expectancy:** People do not live long due to poor living conditions and lack of medical care.
- **Agricultural and traditional society:** The economy is mainly based on subsistence farming with little or no industrial development.

Stage 2: Early Expanding

- **High birth rate:** Birth rate remains high as in Stage 1 due to cultural values and limited access to family planning.
- **Rapidly falling death rate:** Death rate drops significantly because of improved healthcare, sanitation, and food supply.
- **Fast population growth:** The gap between birth rate and death rate becomes wide, leading to a rapid increase in population.
- **Improved living conditions:** Better medical services, clean water, and better nutrition reduce mortality.

- **Early economic development:** Some improvements in agriculture and industry begin, but the country is still largely developing.

Stage 3: Late Expanding

- **Declining birth rate:** Birth rates start to fall due to family planning, education, and changing lifestyles.
- **Low death rate:** Death rates remain low because of continued improvements in healthcare and living standards.
- **Slowing population growth:** The gap between birth rate and death rate becomes smaller, so population growth begins to slow down.
- **Urbanization increases:** More people move to towns and cities, leading to changes in family size and lifestyles.
- **Improved social and economic conditions:** Higher education levels, especially for women, and better employment opportunities reduce the desire for large families.

Stage 4: Low Stationary

- **Low birth rate:** Few children are born due to family planning, high cost of living, and preference for small families.
- **Low death rate:** Death rate remains low because of advanced healthcare, good nutrition, and high living standards.
- **Stable population growth:** Birth rate and death rate are almost equal, so the population remains stable or grows very slowly.
- **High life expectancy:** People live longer due to excellent medical services and improved quality of life.
- **Highly developed economy:** The country is industrialized with strong urbanization and a highly educated population.

Stage 5: Declining

- **Very low birth rate:** Birth rates fall below replacement level due to delayed marriage, career focus, and lifestyle choices.
- **Low but relatively higher death rate:** Death rate may slightly increase because of an aging population.
- **Negative or very slow population growth:** The population may start to decline as deaths exceed births.
- **Aging population:** A large proportion of the population is elderly, with fewer young people.
- **High dependency ratio (elderly dependents):** Fewer working-age people support a growing number of older dependents, creating economic pressure.

Problems of applying the demographic transition

- Developing countries differ in many respects from Western nations as they moved through the demographic transition.

- Population growth is higher now in many countries than before.
- Mortality declines have been much more rapid.
- Fertility levels are much higher.
- Migration does not serve as a safety valve.

10.6.3 Marxist population theory

The **Marxist population theory** was developed by Karl Marx, **argues that population problems are not caused by having too many people**, but by **the way resources are distributed in society**. Marx believed that poverty and hardship result from the capitalist system, where wealth is controlled by a small group while the majority remain poor.

According to this theory, capitalism creates a **“reserve army of labour,”** meaning a large number of unemployed or underemployed people, which helps keep wages low. As a result, poverty persists, and this **poverty often leads to high population growth**, since poorer families tend to have more children. Therefore, population growth is seen as a consequence of economic inequality rather than its cause.

Overall, the Marxist view emphasizes that population issues are **shaped by social and economic conditions**, and that **solving them requires changing the economic system to ensure a fair distribution** is the key to achieving optimum population and solving issues linked to overpopulation and underpopulation **of resources and this fair distribution can drive by moving from capitalism to socialism economic system**.

Marxist population theory is based on the following assumptions:

- Population problems are caused by unequal distribution of wealth and resources, not by overpopulation itself.
- Poverty, unemployment, and famine result from the capitalist system and exploitation of workers.
- Resources in the world are sufficient to support the population if they are distributed fairly.
- Rapid population growth is encouraged by poor living conditions and economic insecurity among the working class.
- Capitalists benefit from having a large population because it provides cheap labour.
- Improvement in social and economic conditions can reduce population problems.
- Population growth is closely related to the level of economic and social development of a society.

Criticism of the theory

- The theory puts too much blame on capitalism and ignores other causes of population problems such as culture, religion, and personal decisions.
- It assumes that resources are always sufficient, yet in some country's resources are limited and cannot adequately support a rapidly growing population.
- The theory underestimates the impact of rapid population growth on pressure over land, food supply, housing, and social services.
- It does not clearly explain population problems in socialist or non-capitalist countries where poverty and unemployment may still exist.
- The assumption that population problems disappear when wealth is fairly distributed is unrealistic because development also depends on technology, skills, and proper management of resources.
- The theory gives little attention to environmental degradation caused by high population growth and overuse of resources.
- It overlooks the importance of family planning and individual responsibility in controlling population growth.

10.6.4 The theory of optimum population

Meaning of optimum population, under population and over-population

- **Optimum population:** This is the ideal number of people in a country that allows maximum use of resources and gives the highest standard of living and economic welfare. Or is a situation where resources available are approximately equal to the total population.
- **Underpopulation:** This is a situation where the population is too small in relation to available resources, leading to underutilization of resources and low production.
- **Overpopulation:** This is a situation where the population is too large compared to available resources, leading to scarcity, unemployment, and a low standard of living.

The Optimum Population Theory explains the best population size that a country should have in relation to its available resources in order to achieve maximum economic welfare. It states that there is a balance point where the number of people is neither too high nor too low, allowing full and efficient use of resources.

At this optimum level, production is maximized, employment is high, and people enjoy a high standard of living. If the population is below this level, resources are underused (**under population**), and if it is above, resources become insufficient for the population needs (**overpopulation**), leading to low living standards.

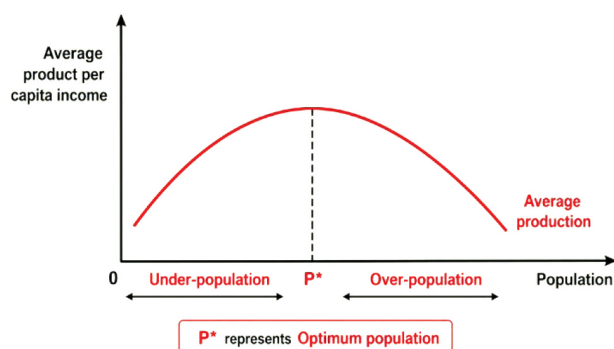


Figure 10.6.: Optimum population under-population and over-population

In the above figure before point P^* , the population is still low compared to the resources available. At point p^* there is optimum population where the population matches the available resources and after that point the population exceeds the available resources and utilisation of resources.

Implications of overpopulation

Positive Implications	Negative Implications
1. Large labor force available for production and industrial development.	1. High unemployment due to competition for limited jobs.
2. Large market for goods and services, encouraging business growth.	2. Pressure on natural resources such as land, water, and forests.
3. Increased innovation and creativity due to large population.	3. Overcrowding in urban areas leading to poor living conditions.
4. More tax revenue for government from a large population.	4. Increased poverty levels due to limited resources per person.
5. Possibility of military strength due to large manpower.	5. Strain on social services such as education, healthcare, and housing.

Implications of underpopulation

Positive Implications	Negative Implications
1. Less pressure on natural resources.	1. Shortage of labor for production and development.
2. Higher wages due to low labor supply.	2. Underutilization of resources such as land and industries.
3. Less congestion and better living conditions.	3. Small market size limits business growth.
4. More government attention per person (better services).	4. Slow economic growth due to low production levels.
5. Lower levels of pollution and environmental degradation.	5. Difficult in providing adequate defense due to small population.

10.7 Population problems and their solutions in developing countries

Learning Activity 10.7



Question 1: Explain three major **population problems in developing countries** and describe how each problem affects economic development and social services.

Question 2: For each population problem you named, propose one realistic **solution to population problems in developing countries** and justify why it would work.

Developing countries are experiencing continuous changes in population size and structure due to factors such as high birth rates and improved living conditions. These changes have a strong influence on the social and economic development of such countries. As population continues to increase, different challenges emerge, making it necessary for governments and other stakeholders to find effective ways of managing them for sustainable national development.

10.7.1 Population problems in developing countries

- **Unemployment:** This happens when many people who are willing and able to work cannot find jobs because job opportunities are limited compared to the growing labour force.
- **Poverty:** A large number of people live with very low incomes, making it difficult for them to afford basic needs such as food, clothing, education, and healthcare.
- **Food insecurity:** Food production is often not enough to meet the needs of the growing population, leading to hunger and poor nutrition in some areas.
- **Pressure on social services:** Health centres, schools, and other public services become overcrowded, reducing the quality and efficiency of service delivery.
- **Housing shortage:** The rapid increase in population leads to insufficient housing, resulting in overcrowding and the development of informal settlements (slums).
- **Environmental degradation:** High population pressure leads to overuse of natural resources, causing deforestation, soil erosion, water pollution, and loss of biodiversity.
- **High dependency ratio:** There are many dependents (children and elderly) compared to working adults, which increases the burden on the productive population.
- **Rural–urban migration problems:** Many people move from rural areas to towns, causing urban congestion, unemployment in cities, and pressure on urban infrastructure.
- **Inadequate infrastructure:** Roads, water supply, electricity, and sanitation systems become insufficient due to the rapid increase in population.
- **Health challenges:** Overcrowding and poor living conditions increase the spread of diseases and make it difficult for health services to respond effectively.

10.7.2 Solutions to population problems in developing countries

- Family planning programs: Promote use of contraceptives and birth control methods.
- Education: Especially educating girls, which helps reduce birth rates.
- Improved healthcare: Reduces death rates and encourages smaller families.
- Job creation: Develop industries to absorb the growing labour force.
- Economic development: Raise incomes and improve living standards.
- Agricultural improvement: Increase food production to match population growth.
- Urban planning: Develop better housing and infrastructure to reduce slums.
- Public awareness campaigns: Teach people about the benefits of small families.
- Empowerment of women: Give women more control over reproductive decisions.
- Migration policies: Manage rural-urban migration to reduce overcrowding

Application activity 10.7



Question 1: Choose a developing country (or a region within your country). Identify one pressing population problem there and design a short action plan with at least three steps to address it.

Question 2: Evaluate a real or hypothetical government policy aimed at reducing population pressures (e.g., family planning, education for girls, rural development). Explain how this policy addresses population problems in developing countries and list possible challenges to its success.

10.8. End of unit assessment



1. Explain the meaning of the following population terminologies:
 - a) Population census
 - b) Population density
 - c) Population pyramid
 - d) Dependency ratio
 - e) Ageing population
2. Using examples from National Institute of Statistics of Rwanda 2022 Population Census Statistics, explain five characteristics of Rwanda's population structure.
3.
 - a) Distinguish between expansive, stationary, and constrictive population pyramids.
 - b) Explain the implications of each type of population pyramid on the economy of a country.

4. a) Explain the meaning of population growth.
b) Describe the causes of high population growth in developing countries.
c) Explain four measures that can be used to control high population growth.
5. a) Explain the meaning of population density.
b) Discuss the consequences of densely populated areas and sparsely populated areas.
6. a) Explain the main ideas of the following population theories:
 - i) Thomas Robert Malthus population theory
 - ii) Demographic Transition Model
 - iii) Optimum Population Theoryb) Examine the major population problems faced by developing countries and suggest possible solutions to these problems.

UNIT 11

LABOR AND WAGES



Key Unit Competence:

Evaluate labour and wage determination and their implications for the labour market, employment levels, income distribution, labour productivity, workers' welfare, and market efficiency.

Learning objectives

Knowledge and understanding

1

- Describe the characteristics of the labor force in Developing countries using real examples.
- Explain factors influencing labor demand, supply, efficiency, and mobility, and relate them to Rwanda employment situations.
- Identify manpower problems in Developing countries.
- Explain the methods of determining wages in an economy.
- Identify the causes of wage differentials in an economy.
- Explain the forms of wage control in an economy.
- Explain the objectives, types, tools, benefits, disadvantages, and weaknesses of trade unions in Developing countries

Skills

2

- Analyze the impact of labor mobility on employment, income distribution, and economic growth.
- Analyze methods of wage determination in different economies, including Rwanda.
- Analyze how to apply wage theories, including the backward-bending supply curve to labor markets in Developing countries
- Assess the effects of wage differentials in economies particularly in Rwandan economy.
- Evaluate forms of wage control and their effects on workers, employers and economic performance.
- Analyze the operation of trade unions in an economy.
- Assess the impact of trade unions on the efficiency of labor.
- Analyze manpower problems in Developing countries.

Attitudes and values

3

- Acknowledge the role of labor mobility in solving manpower problems in Developing countries.
- Develop fairness, respect for labor, appreciation of worker rights, teamwork, and social responsibility.
- Demonstrate appreciation for the role of trade unions in promoting fairness, wage equality, and workers' rights.



Introductory activity

Scenario

In a village in Eastern Rwanda, Kankindi works on her family farm growing maize. She has never received any formal training. Her brother Eric works in a garage as a mechanic after completing vocational training. In the nearby town, Dr. Uwimana works in a hospital after many years of university education. Meanwhile, Chantal sells fruits in the local market to earn a daily income.

All these individuals are engaged in different economic activities to earn a living and support their families.

Based on the scenario:

1. Identify and classify the different categories of labour represented by Kankindi, Eric, Dr. Uwimana, and Chantal.
2. Explain two reasons why labour is important in the development of the economy.
3. Compare the workers in the scenario and explain why some may earn higher wages than others.

In every economy, workers play a key role in producing goods and services, and their performance is closely linked to how they are rewarded. Decisions about employment, earnings, and working conditions are influenced by various forces within the labour market, including policies, institutions, and economic conditions. These interactions shape not only how production takes place but also how income is distributed and how living standards change over time. The following unit examines the main aspects of labour and wages in an economy.

11.1 Meaning and categories of labour

Learning Activity 11.1



1. In your own words explain the meaning of labour.
2. Explain reasons why labour is needed in society.

Meaning of labour

As a factor of production, the term 'labour' refers to any human effort both mental and physical directed towards the process of production of goods and services. The type of job one does and the financial compensation received for it are very important in our society. In the past, the terms blue collar or white collar or pink collar employees were used to describe the type of job. Blue collar implies manual labourers, white collar for officer workers, and pink collar represents jobs associated with women like nursing, secretarial.

Categories of labour

Labour is classified according to the **level of skill, training, and education required**.

1. Skilled Labour

Skilled labour refers to work that requires **special education, professional training, and experience** to perform effectively and independently. Skilled labour is also called a **professional worker and explained as** someone with **advanced education, training, and specialized knowledge**

Examples: professionals in their fields, such as doctors who diagnose and treat patients in hospitals, engineers who design bridges, roads, and buildings, teachers who plan and deliver lessons, lawyers who defend people in court, and accountants who manage financial records and prepare financial statements.

2. Semi-skilled Labour

Semi-skilled labour refers to work that requires **some training or practical experience**, but not advanced professional education.

They include bus and truck drivers who transport goods and passengers, machine operators in factories who handle and control production equipment, laboratory assistants who support technicians in carrying out tests and experiments, and electricians who assist in wiring and installation work.

3. Unskilled Labour

Unskilled labour refers to work that requires little or no formal training or special skills. The tasks are usually simple and can be learned quickly.

They include farm workers who plant, weed, and harvest crops, street cleaners and waste collectors who maintain cleanliness in public areas, construction labourers who carry materials on building sites, and gardeners who maintain parks and compounds.

4. Professionals

These are the elite of the labour grades and include those workers who need an advanced degree to do their jobs. The three primary groups of professionals are doctors, lawyers and teachers. These are white collar workers.

Types of Labour According to Occupational Mobility

- a) **Specific labour:** This is labour that cannot be changed from one production activity to another for example doctors.

b) Non-specific labour: This is labour that can be transferred from one use to another for example unskilled labour like car washers.

Classification of labour based on its contribution to economic output:

1. Productive Labour

Productive labour refers to all human effort (physical or mental) that is used to produce goods and services that have economic value and generate income or profit. It contributes directly to the creation of wealth in an economy.

Examples:

- A farmer growing crops for sale in the market
- A factory worker producing clothes or shoes
- A doctor treating patients in a private hospital
- A carpenter making furniture for customers

2. Unproductive Labour

Unproductive labour refers to human effort that is used but does not directly produce goods or services for sale or generate profit. It may be useful to society but does not directly contribute to economic wealth creation.

Examples:

- A housewife doing unpaid household chores (cooking, cleaning)
- A volunteer working without payment in a community project
- A priest or religious leader performing spiritual duties without commercial output
- Etc.

Application activity 11.1



1. In your community, identify different types of workers you see every day. Classify them into skilled, semi-skilled, and unskilled labour, and explain your reasons for each classification.
2. Describe how both productive and unproductive labour contribute to the development of a family or society. Do you think unproductive labour is still important? Explain your answer.
3. Imagine your community suddenly loses all skilled workers for one week. Explain what problems would arise and how life would be affected in areas such as health, education, transport, and construction.

11.2 Labour force

Learning Activity 11.2



1. What do you understand by the term labour force in an economy?
2. Why is the labour force important for the development of a country?

11.2.1 Meaning and determinants of labour force

The labour force refers to all people within the working age population (usually 15–64 years) who are able and willing to work, and who are either employed or actively looking for employment. It includes both those who already have jobs and those who are unemployed but are actively seeking work.

The labour force is an important part of the economy because it provides the human effort needed to produce goods and services. A large and skilled labour force can increase productivity, promote economic growth, and improve the standard of living in a country. However, not all people within the working age are part of the labour force, as some may be students, disabled, or not actively seeking employment.

Determinants of Labour force

The size and quality of the labour force in a country are influenced by several important factors that determine how many people are available and able to work. These factors affect participation in employment, productivity levels, and the overall economic performance of a nation. They vary from one country to another depending on social, economic, and demographic conditions. The following are the main determinants of the labour force:

- 1. Size of population:** A larger population generally provides a larger labour force, while a smaller population leads to a small labour force.
- 2. Age structure of the population:** A population with many children or elderly people has a small labour force, while one with more people in the working-age group has a larger labour force.
- 3. Population growth rate:** A high population growth rate increases the future labour force, while a low growth rate reduces it.
- 4. Migration:** Emigration reduces the labour force in the home country, while immigration increases the labour force in the receiving country.
- 5. Brain drain:** The movement of skilled workers to other countries reduces both the size and quality of the labour force in the country of origin.
- 6. Retirement age:** A low retirement age reduces the labour force, while a high retirement age increases it.
- 7. Political situation:** Political stability increases labour participation, while instability and insecurity reduce the labour force due to migration and reduced economic activity.

Difference between labour force and work force

Labour Force

This includes:

- People who are currently working (employed)
- People who are not working but actively looking for work (unemployed)

Labour force = Employed + Unemployed (who available for and seeking work willing and able to work)

Workforce

This usually refers to:

People who are **actually working (Workforce = Only those who have jobs)**

Application activity 11.2



Scenario

In Country X, the government reports changes in the labour force over the past 10 years. The country has experienced rapid population growth, increased migration to neighbouring countries, political instability in some regions, and a growing number of young people entering the education system. At the same time, many skilled workers are leaving the country to work abroad, and the retirement age has recently been reduced from 70 to 64 years.

1. Define the term labour force as used in economics.
2. Identify five determinants of labour force in an economy.
3. Explain how population growth affects the size of the labour force in Country X.
4. Analyse the impact of brain drain on both the size and quality of the labour force in Country X.

11.3 Demand for labour

Learning Activity 11.3



Imagine you own a small bakery. Business is growing, and you need help. You are thinking about hiring workers.

Discuss:

How would you decide the number of workers to hire, and what considerations would guide that decision?

Demand for labour

This refers to the number of labourers needed by an employer at a given wage in a given period of time. The demand for labour is always derived. Labour is demanded not for its own sake but because there exists a demand for the goods or services which the entrepreneur may produce with the help of labour. If the demand for a certain good is increasing, the demand for labour will also increase and vice-versa.

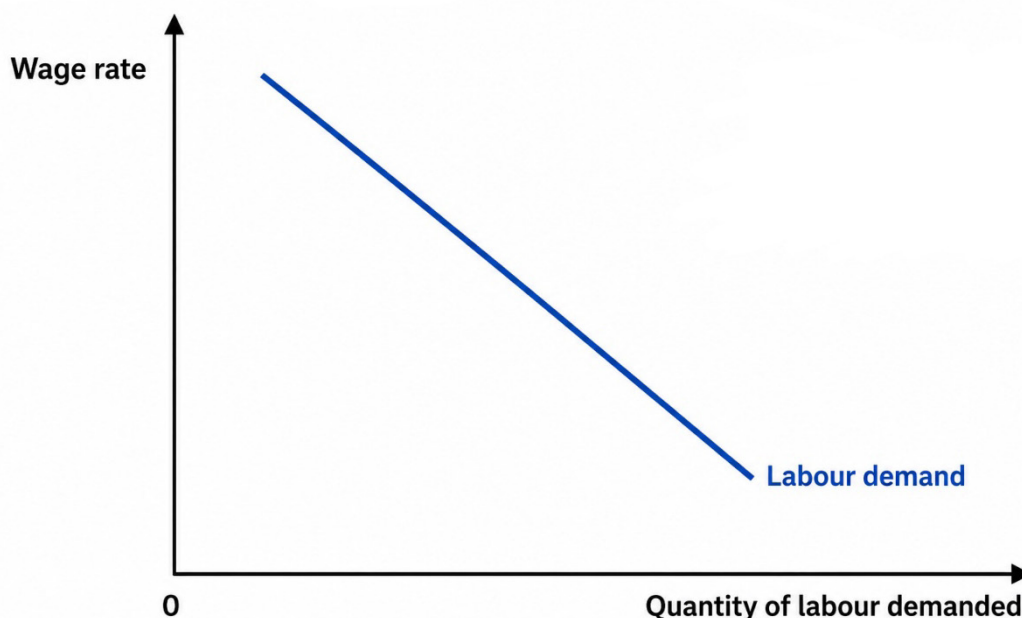


Figure 11.1.: Demand for labour

There are two main mechanisms that explain why the demand for labour curve slopes downward:

- **Substitution Effect:** If wages rise, labour becomes relatively more expensive than other inputs (e.g., machinery/capital). Firms will substitute labor for capital, reducing the quantity of labor demanded.
- **Scale Effect (Higher Costs & Output):** A wage increase raises the average and marginal costs of production. As costs rise, firms often reduce their total output (a reduction in the quantity of the product demanded), which in turn leads them to reduce the use of all inputs, including labor.

Factors affecting demand for labour

The demand for labour in an economy is influenced by several important factors that determine how many workers employers are willing and able to hire. These factors are mainly linked to the level of economic activity, production needs, and costs of hiring workers. They vary from one industry and country to another depending on market conditions, technology, and government policies. Understanding these factors helps explain why the demand for labour may increase or decrease over time. **The following are the main factors affecting the demand for labour:**

- **Level of demand for goods and services:** When the demand for goods and services is high, firms increase production and therefore hire more workers. When demand is low, labour demand decreases.

- **Wage levels:** High wages increase the cost of hiring labour, which may reduce demand for workers. Lower wages make it cheaper for firms to employ more people.
- **Productivity of labour:** When workers are highly productive, employers may hire fewer workers to produce more output. Low productivity often increases the number of workers needed.
- **Level of technology:** Advanced technology can reduce the need for labour because machines replace human work. However, in some cases, it can also create new job opportunities.
- **Cost of other factors of production:** If capital (machines) is cheaper than labour, firms may prefer machines instead of workers, reducing labour demand.
- **Government policies and regulations:** High taxes, strict labour laws, or high minimum wages can reduce labour demand, while supportive policies can increase employment.
- **Level of economic activity:** In times of economic growth, businesses expand and demand more labour. During recessions, firms reduce hiring.
- **Size and expansion of firms:** Large or expanding firms need more workers to meet production needs, while small or declining firms demand less labour.
- **Seasonal changes in production:** In some industries such as agriculture and tourism, demand for labour changes with seasons. During peak seasons, firms hire more workers, while in off-seasons, labour demand decreases.

Application activity 11.3



Read the following case study and answer the questions that follow:

In Rwanda, the government and private investors are expanding industries such as construction, manufacturing, and services. As a result, many firms are hiring more workers, especially in urban areas like Kigali. Companies such as hotels, factories, and banks are increasing their staff due to rising demand for their products and services. However, some firms use modern machines, reducing the number of workers needed in production.

Questions:

1. What do you understand by the term **demand for labour**?
2. Identify the factors that influence the demand for labour in the case study.
3. How does technology affect the demand for labour.

11.4 Labour supply

Learning Activity 11.4



Case Study

In Rwanda, last year, employers in Kigali City began offering significantly higher wages than those in Nyamagabe District. As news of these better-paying job opportunities spread, many workers in Nyamagabe started considering moving to Kigali or commuting daily in search of higher incomes. As a result, employers in Nyamagabe and nearby districts began to experience difficulties in retaining workers, particularly in the construction, hospitality, and retail sectors.

Questions

1. Explain how the situation described above is likely to affect the number of people willing to work in Nyamagabe and in Kigali City.
2. Identify and explain the main factors that determine the number of people willing to work in an economy.
3. Suggest possible measures that employers in Nyamagabe could take to reduce the loss of workers to Kigali City.

Labour supply refers to the number of hours an individual is willing to offer for work per day, week, month, or year. For an individual, the decision on how many hours to offer to work depends on the real wage and on an individual's attitude towards leisure and income.

The supply of labour is offered by households, and the traditional assumption is that more work will be offered for higher wages, implying an up sloping supply curve for labour as shown on the normal supply curve below.

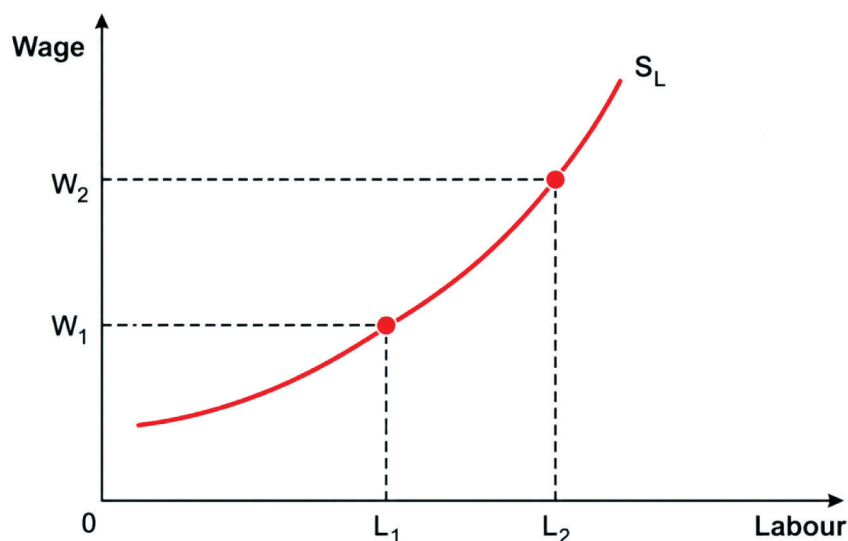


Figure 11.2.: Normal supply curve of labour

From the figure above, at a higher wage OW_2 , the supply of labour is OL_2 while at a lower wage OW_1 , the supply of labour is only OL_1

However, it has long been recognised that for some individuals or groups of workers, higher wages might lead to a reduction in hours of work offered. This may lead to a backward bending supply curve. **The shape of the curve may be due to the following:**

- The worker in effect prefers leisure at higher wages.
- Old age where even if the wages are high, labour may not increase supply.
- Presence of target workers. After achieving the target, the worker either reduces effort or stops to work despite an increase in the wage.
- Desire to live on accumulated wealth.

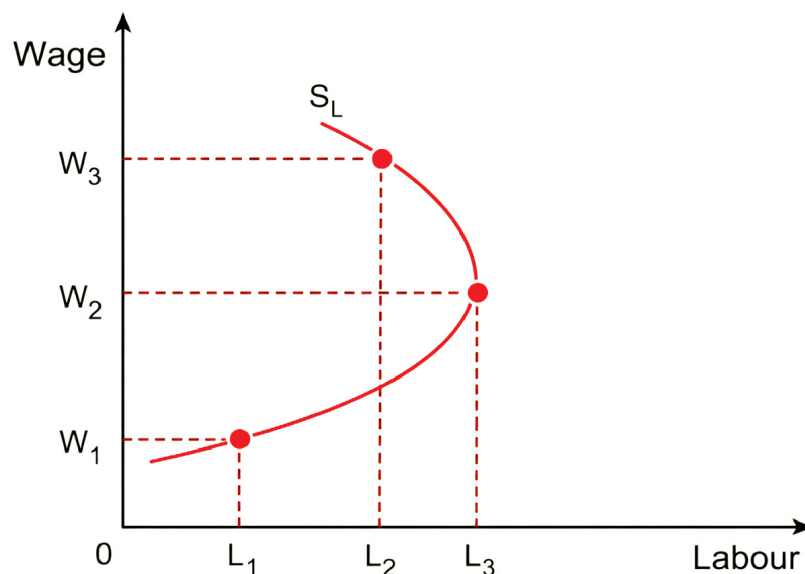


Figure 11.3.: Backward bending supply curve of labour

From the figure above, at W_1 , supply is L_1 when the wage increases to W_2 , the supply will increase to L_3 , however when the wage increases to W_3 , labour supply reduces to L_2 .

Factors determining labour supply

- **Population size:** A larger population means more people are available and willing to work, so labour supply increases.
- **Age structure of population:** If a country has more people in working-age groups (e.g. 18–60), labour supply is higher.
- **Wage level (remuneration):** Higher wages attract more people to work or to work longer hours, increasing labour supply.
- **Working conditions:** Good and safe working environments encourage more people to offer their labour.
- **Education and skills level:** Better education and skills make people more employable and increase participation in the labour market.
- **Government policies:** Policies like retirement age, minimum wage, and labour laws can increase or reduce labour supply.

- **Cultural and social factors:** Traditions, beliefs, and attitudes toward work and leisure influence how much people are willing to work.
- **Incentives offered:** If these such as wages and salaries are high, labour supply will be high and if low, labour supply will be low. Level of education: In a country where the people are educated, the supply of skilled labour will be high and vice versa.
- **Health conditions of workers:** If the conditions of workers are good, they are willing to supply more labour because they are able to work and vice versa.
- **Retirement age:** The earlier the retirement age, the higher the rate at which the people will leave the jobs and the lower the supply of labour.
- **Political situation:** If a country is politically stable, people are guaranteed security and labour supply will be high and is low in areas where people fear to risk their lives.
- **Job security:** Jobs that have high risk of retrenchment attract less labour force compared to those that have low risks of retrenchment.
- **Period of training:** The longer the period of training, the lower the supply of labour in the short-run but the shorter the period of training, the higher the supply of labour.

Application activity 11.4



1. Explain the meaning of labour supply and describe the factors that determine the number of hours an individual is willing to offer for work in an economy.
2. Distinguish between a normal upward-sloping labour supply curve and a backward-bending labour supply curve, giving reasons why each shape may occur.
3. Discuss the key factors that influence labour supply in an economy and evaluate measures that can be taken by the government to increase labour supply.

11.5 Efficiency of labour

Learning Activity 11.5



Mr Gasaro has been teaching Economics at a secondary school for the past eight years. Through consistent hard work and dedication, his students have continuously achieved the best academic performance in the district. In recognition of his outstanding performance, he received an award from the district mayor in 2014.

Basing on this information, respond to the following questions:

1. What term is used to describe Mr Gasaro's ability to produce such high academic results?
2. Explain the factors that determine Mr Gasaro's capability to perform effectively in his job.
3. Analyse the factors that may limit his capability to perform at an optimal level.

Different workers engaged in the same type of work may produce different levels of output or may require different amounts of time to complete the same quantity of work. **Efficiency of labour** refers to the ability of workers to produce a large quantity of high-quality output in the shortest possible time. Labour is said to be efficient when it can achieve maximum output in terms of both quality and quantity within a given period of time.

$$\text{Efficiency of labour} = \frac{\text{Actual output}}{\text{Expected output}} \times 100$$

Factors influencing efficiency of labour

Efficiency of labour in an economy is influenced by a wide range of economic, social, and institutional factors that determine how productively workers are able to perform their tasks. These factors affect both the quality and quantity of output produced within a given period of time. Understanding these influences is important in explaining differences in productivity among workers, firms, and countries. **The following are the key factors that influence efficiency of labour:**

- **Physical and health condition of labour:** If the health conditions of labour, levels of nutrition of the labour force is poor, its efficiency will be low but it will be high where labour is healthy and well-fed.
- **Level of technology:** Where modern equipment is provided and used such as computers, efficiency of labour tends to be high compared to situations where primitive technology is used (traditional tools).
- **Working conditions of labour:** When these are good especially job security, transport allowances etc, efficiency of labour will be high and it will be low when the conditions are poor.
- **Degree of specialisation:** If labour is highly specialized and skilled, it will become more experienced and efficient.
- **Quality and quantity of other factors of production:** If other factors of production are available and of high quality, labour efficiency will be high and it will be low if the others of production are of poor quality.
- **Political atmosphere:** Political security encourages concentration and efficiency among the labourers but where instabilities prevail, labour tends to be inefficient because it cannot settle at work.
- **Climatic conditions:** Harsh conditions such as very hot and very cold tend to make labour inefficient but if conditions are ideal or favourable, labour will be efficient.

Application activity 11.5



Case study

In Nyamagabe District, two tea processing factories, Factory A and Factory B, employ the same number of workers and use similar machines. However, Factory A produces 6,000,000 kg of processed tea per year, while Factory B produces 1,500,000 kg of processed tea per year

Questions:

1. Explain the meaning of efficiency of labour.
2. Identify the factory with higher labour efficiency and support your answer with evidence from the scenario.
3. Examine the factors that may have contributed to differences in labour efficiency between Factory A and Factory B.
4. Propose measures that Factory B can adopt to improve the efficiency of its workers.

11.6 Labour mobility

Learning Activity 11.6



In many economies, workers often change jobs or move from one place to another in search of better employment opportunities, improved income, or better working conditions. While some succeed in improving their welfare, others face challenges when adapting to new workplaces or environments. This situation raises important questions about why workers move, the effects of such movement, and the benefits it may bring to the economy.

Read the above case study and answer the following questions:

1. Why do people decide to change their jobs or move from one place to another for work?
2. How can moving from one job to another affect a worker's skills and income?
3. What challenges might workers face when they move to a new place for employment?
4. How do education and training influence a person's ability to change jobs?

Workers do not always remain in the same job, occupation, or location throughout their working lives. In many economies, people move in search of better wages, improved working conditions, career advancement, or employment opportunities. This movement of labour plays an important role in the efficient allocation of human resources and the overall performance of the economy. The following discussion focuses on the key aspects of labour mobility.

11.6.1 Meaning and forms of labour mobility

Labour mobility is the ability and willingness of workers to move from one job, place, or occupation to another in search of better opportunities.

Forms of labour mobility

a) Geographical mobility of labour

This is the ease with which labour can move from one geographical area to another in search for work. People always move from one place to another depending on the activities that may interest them. Some move for economic reasons, social reasons and political reasons among others. The movement of labour from one area to another is influenced by many factors are seen below;

Factors influencing geographical mobility of labour

- **Transport and communication:** The lower the transport cost, the easier for labour to move from one area to another and the higher the transport costs, the harder it becomes for labour to move from one place to another.
- **Political situation:** Areas which are stable tend to attract more labour than those which are unstable.
- **Nature of climate:** Areas with good climate tend to attract more labour compared to areas which have harsh climate i.e. too hot and too cold.
- **Level of wages:** Areas with high wages attract more labour than areas which give low wages.
- **Institutional barriers:** This includes things like language barrier and if this exists, immobility of labour will be low because it becomes hard to communicate.
- **Degree of ignorance:** In cases where people are ignorant about the available jobs, the lesser the rate of movement but if people are aware of the available jobs, mobility of labour will be high.
- **Degree of advertising:** If the rate of advertising is high, labour will tend to move to those areas where the jobs are but when the rate of advertising is low, labour will be immobile.
- **Cost of living:** Labour tends to move away from areas with a high cost of living to areas with low cost of living.

Factors limiting geographical mobility of labour

- **High level of ignorance about the existing jobs:** Some people do not know about the existing jobs in other areas so they end up not moving from one place.
- **High wages got at the current job:** If a worker is given a high wage at the current wage, he may not need to move to another area looking for a job.
- **Poor transport and communication:** Sometimes the transport and communication may be poor. Related to accessibility, the worker may not be able to move to another place.

- **Good climate at the current place:** People prefer to stay in good climate. When the climate in the current place is conducive to the worker, he or she may not leave that place to go to another place.
- **Low cost of living at the current place:** A cost of living in the current area may not tempt the worker to move to another area where the cost of living may be high.
- **Low level of advertising:** The low rates of advertising about the existence of other jobs in different areas may keep the labour in the same area of region.
- **High level of attachment to family members:** Some people are so much attached to their families to the extent that they cannot leave them. This closeness may hinder labour to move to another area.
- **Inadequate Institutions:** People are interested in areas that have institutions like banks, stadiums, movie halls among others. So if these are not found in a particular area, people may not move to such areas.
- **Political instability and insecurity:** Some region may be politically unstable so people may not move to them for fear of their lives.
- **Language barrier:** Internationally, people may fear to move to other regions because of the problem of language. For Rwanda, this may not be a limitation because the people all speak one language irrespective of the region of origin.
- **Old age limits movement:** Old people normally always want to stay in one place so they may not move to another area.

b) Occupational mobility of labour

This refers to the ability and willingness of workers to move from one occupation or job to another in search of better employment opportunities, income, or working conditions.

It involves two categories;

- a) **Horizontal mobility:** A situation where labour changes from one occupation to another but it does not change its status e.g. from a teacher Good Hope High School to a teacher in Corner Stone Leadership Academy.
- b) **Vertical mobility of labour:** A situation where labour changes from one occupation to another but affects or changes its status e.g. from a teacher to a headmaster. This may involve **promotion (moving to a higher position)** or **demotion (moving to a lower position)**. For example, a teacher becoming a headmaster or a headmaster being reduced to a teacher.

Factors determining occupational mobility of labour

- **Level of education and skills:** Workers with higher education and diverse skills can easily move from one occupation to another because they can meet the requirements of different jobs.
- **Training and retraining opportunities:** Availability of vocational training or on-the-job learning helps workers acquire new skills needed to shift to different occupations.

- **Wage differentials:** Differences in wages between occupations motivate workers to move to jobs that offer better pay and improved living standards.
- **Working conditions:** Jobs with safer environments, reasonable working hours, and better treatment attract workers to change occupations.
- **Personal interest and ambition:** A worker's passion, career goals, and desire for self-improvement can influence the decision to change occupations.
- **Experience and qualifications:** Some jobs require specific qualifications or experience, which can either enable or limit a worker's ability to move into them.
- **Cost of training:** High costs of acquiring new skills or education can discourage workers from changing occupations.
- **Government policies:** Policies such as free education, skill development programs, and employment regulations can encourage or restrict occupational mobility.
- **Technological changes:** Introduction of new technology can make some jobs obsolete and create new ones, forcing or encouraging workers to shift occupations.
- **Social and cultural factors:** Society's attitudes, traditions, or family expectations may either support or discourage people from entering certain occupations.

Factors limiting occupational mobility of labour

- **Low level of education and skills:** Workers with little education or limited skills find it difficult to qualify for new or specialized jobs.
- **High cost of training:** Expensive education or training programs discourage workers from acquiring new skills needed to change occupations.
- **Lack of information about job opportunities:** When workers are not aware of available jobs or required qualifications, they cannot easily shift to other occupations.
- **Occupational immobility due to specialization:** Workers highly specialized in one field may find it hard to adapt to different types of work.
- **Age factor:** Older workers may be less willing or able to learn new skills or change careers compared to younger workers.
- **Health problems and physical limitations:** Poor health can restrict a worker's ability to perform certain jobs, limiting their chances of changing occupations.
- **Social and cultural barriers:** Traditions, beliefs, or gender roles may discourage people from entering certain occupations.
- **Fear of risk and uncertainty:** Some workers prefer job security and avoid changing occupations due to fear of failure or income instability.
- **Lack of training facilities:** Limited access to institutions that provide skill development reduces opportunities for occupational change.

- **Government restrictions and regulations:** Strict licensing requirements or regulations in certain professions can make it difficult for workers to enter those fields.

Application activity 11.6



Scenario:

James is working as a cashier in a supermarket in Kigali. After gaining experience and completing a short accounting course, he is promoted to the position of junior accountant in the same company with a higher salary and increased responsibilities. Meanwhile, his friend Mugisha leaves his job as a primary school teacher in Huye and joins another school in Kigali as a secondary school teacher, performing similar duties and receiving almost the same salary.

Questions:

1. For each case, determine whether the labour mobility is **vertical or horizontal**, and give reasons for your answer.
2. Identify and explain the different forms of labour mobility shown in Mugisha's case.
3. What factors might have influenced Mugisha's decision to move and change location?
4. In what ways can labour mobility benefit both the individual worker and the economy as a whole?

11.7 Manpower problems in Developing countries

Learning Activity 11.7



Scenario:

In a certain country, there are many unemployed graduates who complain that they cannot find jobs. At the same time, many companies are importing foreign workers because they say local workers lack the required skills. Some experienced workers are also leaving the country to work abroad where salaries are higher. Despite government efforts to create jobs, the problem continues to grow.

From the scenario above answer the following questions:

1. What problems related to manpower can you identify?
2. How can a country have many unemployed people while still importing labour from other countries?
3. Why do you think some skilled workers leave their country even when jobs are available locally?
4. What do you think this lesson will help you understand about manpower problems?

Manpower problems are among the key challenges affecting the development of many economies. The way human resources are available, distributed, and utilized plays an important role in determining productivity and economic progress. In different countries, these challenges vary in nature and intensity, influencing how effectively labour contributes to national development. Understanding these problems is therefore essential for identifying their causes and finding appropriate solutions.

11.7.1 Meaning and causes of manpower problems in developing countries.

Manpower refers to the total number of people who are available and able to work in a country, organization, or economy. It includes both skilled and unskilled labour that contributes to the production of goods and services.

Manpower problems arise when a country is unable to use its labour force effectively due to imbalances in its availability, quality, and distribution. This situation occurs when workers do not fully match available job opportunities in terms of skills, numbers, or location, resulting in low efficiency in production and slowing down economic development.

In any given economy, the quantity and quality of labour as a factor of production is important to the labourers themselves, the employers, the government, and the economy as a whole. Most developing countries are faced with numerous manpower related problems, which include among others the following;

- Excessive labour supply in some sectors e.g. agriculture.
- Scarcity of labour in some sectors e.g. engineering, medical services etc.
- Low skills and competence due to low and poor education system.

Low levels of occupational mobility.

- Brain drain of highly skilled manpower to developed countries.
- Educated people are trained for white collar jobs which are not readily available.
- High rural urban migration due to an urban biased education system.
- Poor health due to poor feeding and diseases e.g. HIV/ AIDS.
- Over dependency on certain categories of labour e.g. in consultancy work.

Causes of manpower problems in developing countries

- **Low level of education and training:** Many workers lack adequate education and vocational skills, which reduces their ability to perform tasks efficiently and adapt to modern production methods.
- **Rapid population growth:** When the population grows faster than job creation, it leads to an oversupply of labour and increases manpower-related pressures in the economy.
- **Rural-urban migration:** Movement of workers from rural areas to urban centres creates labour shortages in rural areas while causing congestion and unemployment in towns.

- **Mismatch between skills and job requirements:** The skills possessed by workers often do not match the needs of employers, leading to inefficiency and underutilization of labour.
- **Limited job opportunities:** When the economy does not create enough employment, many people remain unemployed or underemployed, worsening manpower problems.
- **Poor working conditions:** Unfavourable working environments discourage productivity and can lead to low labour retention and reduced efficiency.
- **Inadequate investment in human capital:** Low spending on education, health, and training reduces the quality and productivity of the labour force over time.
- **Technological changes:** Rapid advancement in technology can make some skills obsolete, leaving many workers unfit for available modern jobs.
- **High rate of population dependency:** In many developing economies, a large proportion of the population is not in the labour force (children and elders), which increases the burden on the working population and reduces effective manpower utilization.
- **Health problems and poor nutrition:** Poor health conditions and inadequate nutrition weaken workers physically and mentally, leading to low productivity, frequent absenteeism, and reduced efficiency at work.
- **Political instability and insecurity:** Conflicts and unstable political conditions discourage investment and disrupt economic activities, limiting job creation and causing loss or displacement of skilled labour.

11.7.2 Effects and Measures to minimise the current manpower problems in developing countries

Manpower problems in developing countries have wide-ranging effects on economic and social development by limiting the efficient use of labour. These challenges also require appropriate measures to reduce their impact and improve the utilization of human resources. Therefore, it is important to understand both the effects and the possible solutions to these problems.

A. Effects of manpower problems

- **Low productivity in production activities:** When workers are poorly educated or lack skills, they are unable to perform tasks efficiently, which reduces overall output in the economy.
- **High unemployment and underemployment:** Rapid population growth and limited job opportunities result in many people remaining without jobs or working below their capacity.
- **Uneven regional development:** Rural–urban migration leads to labour shortages in rural areas and overcrowding in urban areas, causing imbalance in development between regions.

- **Inefficient use of labour resources:** When workers' skills do not match job requirements, many employees are not fully utilized, leading to wastage of human resources.
- **Low economic growth and development:** Poor manpower utilization reduces production levels and slows down the overall growth of the economy.
- **Low labour morale and high turnover:** Poor working conditions discourage workers, leading to low motivation, frequent job changes, and reduced stability in the labour market.
- **Reduced innovation and slow technological progress:** Limited investment in human capital and outdated skills make it difficult for economies to adopt and effectively use modern technology.
- **Increased dependency burden on the working population:** A high proportion of dependants reduces the proportion of active workers, increasing pressure on the labour force and lowering overall productivity.
- **Loss of skilled labour and reduced investment confidence:** Health problems, poor conditions, and political instability can cause skilled workers to leave the country, while also discouraging investors from creating jobs, weakening economic development.

B. Measures to minimize manpower problems in developing countries:

Manpower problems can be minimized through manpower planning, which helps a country to anticipate future labour needs and prepare accordingly.

Manpower planning This refers to a long-term projection of labour demand and supply aimed at attaining a balance between labour demand and supply at different levels of economic development. It is investment in education through a particular number of people/ labour in specific fields so that in the long-run there are no deficits or surpluses in the labour market. It involves a wide range of predictions and forecasting of the unskilled, semi-skilled and skilled workforce in an endeavour to develop the necessary human resource requirements of the economy.

Some of the measures include:

- Appropriate manpower planning necessary to minimise such irregularities.
- Redeployment of the workforce should be undertaken to reduce shortage of manpower in some sectors of the national economy.
 - The education system should be revised to avoid the production of more job seekers instead more job makers should be produced.
 - Retrenchment i.e. the old, the sick, and the very disabled workers should be terminated to create an effective labour force.
 - Official exchange of manpower should be enhanced between governments, to not only reduce shortages of indigenous manpower but also increase the supply of labour in all productive sectors of the economy.
 - Promotion of science and technology in all the needy fields e.g. medical, engineering etc.

Application activity 11.7



Using Rwanda NST1, NST2 and Vision 2050 answer the following questions:

1. Identify and explain any three measures taken by the Government of Rwanda under NST1 and NST2 to reduce manpower problems such as unemployment, skills mismatch, and low productivity.
2. How does Vision 2050 aim to develop a skilled and productive manpower in Rwanda?

11.8 Wages

Learning Activity 11.8



Read the following dialogue and answer the questions that follow.

Mrs Mukamana: Good morning, Eric. I was informed that you are skilled in repairing computers. Our school computer laboratory has many machines that need regular maintenance. Can you handle the work?

Eric: Yes Madam, I can. I have been repairing computers and installing software for several years, so I have enough experience.

Mrs Mukamana: That sounds good. I think it would be easier to pay you a fixed amount every month.

Eric: I would prefer to be paid according to the number of computers I repair.

Mrs Mukamana: Alright. How much do you charge per computer?

Eric: I charge 8,000 RWF for repairing one computer and 5,000 RWF for software installation.

Mrs Mukamana: Okay. You may start work next week.

Questions

- i. Who is the employer and who is the employee in the dialogue above?
- ii. If Eric repairs 6 computers and installs software in 4 computers each week for 4 weeks, calculate the total amount he will earn. What is this payment called?
- iii. What factors might Eric have considered before deciding the amount he requested for the work?

11.8.1 Meaning and forms of wages

A wage is the reward to labour as a factor of production for services rendered. A salary is a fixed reward to labour and it does not change because of time or pieces of work done. Wages may be;

Real wage refers to the purchasing power of nominal wage. The amount of goods and services that nominal wage can purchase at a certain price ruling in the market. It is the total amount of satisfaction labour receives in return for the services offered.

$$\text{Real wage} = \frac{\text{nominal wage}}{\text{CPI}} \times 100$$

Nominal wage refers to the total amount of money earned by a unit of labour as payment for services rendered during production expressed in units of a currency e.g. Francs, shillings, dollars. For instance, you employ a worker and pay him/her 200 000frw per month for the services he/she renders to you. This amount which is paid in terms of money is what is called **nominal wage**.

Forms of Wages

Wages may be expressed in three different forms, as explained below:

1. Living Wage

A living wage is the amount of money paid to a worker that is enough to cover basic needs such as food, shelter, clothing, and also allow a decent standard of living.

2. Minimum Wage

A minimum wage is the lowest amount of money that a worker is legally allowed to be paid, usually set by the government to protect workers from exploitation.

3. Fair Wage

A fair wage is a reasonable level of payment that lies between the minimum wage and the living wage, considering both the worker's needs and the employer's ability to pay.

11.8.2 Methods of wage determination

Wages are determined in different ways depending on the nature of work, the conditions of the labour market, and the agreement between employers and workers. The method used to determine wages influences workers' earnings, motivation, and productivity. Understanding these methods helps to explain how workers are compensated for the services they provide in the production process.

Some of the methods used to determine wages include:

1. Piece-Rate Earning

Piece-rate earning is a method of wage payment where workers are paid according to the quantity of work they produce or complete. The payment is directly linked to output, meaning that the more units of work a worker completes, the more they earn.

Advantages and disadvantages of piece rate earning

Advantages	Disadvantages
Better-performing workers receive higher remuneration.	Workers may overwork themselves in order to earn more income.
It eliminates the need for constant supervision.	Workers do not have strong bargaining power for higher wages.
Workers perform tasks at their own pace.	Misunderstandings may arise regarding payment.
It encourages workers to work efficiently and quickly.	Workers often demand immediate payment upon completion of work.
Workers have opportunities to work in multiple places.	Employers have limited control over workers' activities.
It reduces the need for full-time staff when work is periodic.	Employers must ensure availability of funds for timely payment.
Workers are paid immediately upon completion of a task.	It may lead to low quality of output as workers rush to finish tasks.

2. Time-rate earning

Time-rate earning is a method where workers are paid a fixed amount of money based on the amount of time they spend working, such as per hour, per day, or per month. The wage is agreed upon in advance between the employer and employee, and it does not depend on the level of output produced.

Advantages and disadvantages of time rate earning

Advantages	Disadvantages
It promotes high quality of work as workers are not rushed to complete tasks for payment.	It does not provide strong incentives for higher productivity since payment is based on time worked.
It helps to protect delicate machinery since workers operate more carefully.	It requires close supervision to ensure workers remain productive.
It is suitable for jobs where output is difficult to measure, such as in the service sector.	Workers may reduce their pace of work knowing their wages are not affected.
It facilitates easy organization of workers into shifts.	Workers are paid after a fixed period, which may require them to wait a long time for income.
It simplifies taxation since income is easily identifiable for tax purposes.	It does not reward highly productive or skilled workers fairly.
It gives employers time to prepare funds for wage payment when due.	It can discourage innovation and initiative among workers.

3. Sliding scale method

The sliding scale method is a system of wage adjustment where workers' wages change according to changes in the cost of living or economic conditions. When the cost of living (prices of goods and services) increases, wages are increased to maintain workers' purchasing power. Conversely, when the cost of living decreases, wages may also be adjusted downward.

Sliding scale is important in the following ways:

- It adjusts wages in line with changes in the cost of living, helping workers maintain their purchasing power.
- It motivates workers since higher productivity and improved economic conditions can lead to higher wages.
- It reduces labour disputes by providing a fair and flexible wage adjustment system.
- It helps maintain workers' real income during inflationary periods.
- It promotes industrial harmony by linking wages to both economic performance and living standards.

4. Minimum Wage Legislation

This is where the government fixes the wage above the equilibrium wage below which it is illegal to pay the workers. The wage is also known as wage floor.

The following are purposes of minimum wage legislation:

- Protection of workers from exploitation.
- Improvement of workers' standard of living.
- Reduction of income inequality.
- Promotion of social and economic stability.
- Encouragement of productivity and efficiency among low-paid workers.

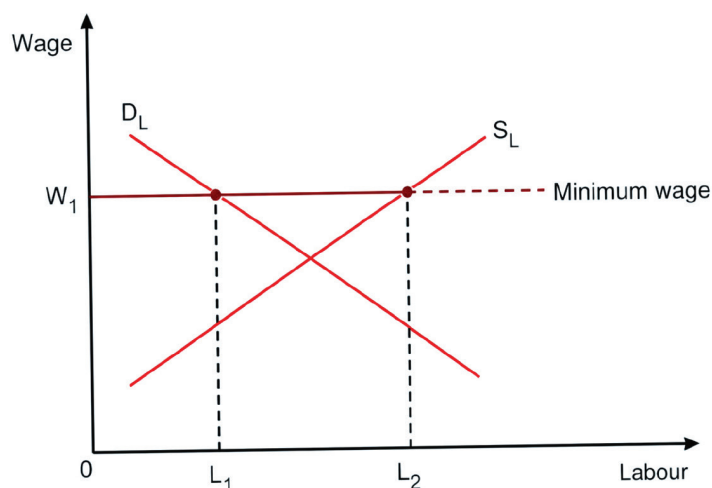


Figure 11.4.: Minimum wage

From the figure above, the minimum wage is fixed at W_1 above the equilibrium wage where demand for labour meets with labour for labour.

5. Trade Union Bargaining

Trade union bargaining is a method where workers form organisations called trade unions to collectively negotiate with employers regarding wages and working conditions. Instead of negotiating individually, workers use their union representatives to demand fair wages, better working conditions, and job security. This method strengthens workers' bargaining power, especially when dealing with large employers. It promotes fairness and protects workers' rights, although it may sometimes lead to industrial disputes if agreement is not reached.

11.8.3 Wage theories

Wages differ widely from one worker to another, even among people doing similar jobs. These differences raise important questions: why do some workers earn higher wages than others? What determines the level of wages in an economy? Is it the worker's skills, the type of job, the forces of demand and supply, or other factors? To answer these questions, economists have developed various explanations known as **wage theories**.

These theories help us understand how wages are determined and why wage differences exist in different occupations and sectors of the economy. Some of those theories are explained below:

1. Marginal productivity theory of wages

This theory states that a worker is paid according to the extra output or value they add to production. In simple terms, the more a worker contributes to production, the higher their wage should be. Employers hire workers up to the point where the cost of employing an extra worker is equal to the value of what that worker produces. It assumes perfect competition, free movement of labour, and no government interference, although in real life these conditions rarely exist.

Assumptions of the marginal productivity theory of

There is a free and complete competition among workers for jobs.

There is a free and complete competition among the employers for workers.

Labour knows its marginal product.

Labour and capital are fully employed. Labour is homogeneous

Employers are able to measure and predict in advance the marginal product of labour.

Capital and labour are perfectly mobile.

Government does not interfere in wage determination.

It is the forces of demand and supply that determine the wage.

The bargaining power of labour and management are equal.

Weaknesses of Marginal Productivity Wage Theory

- It assumes perfect competition in the labour market, which rarely exists in reality.
- It is difficult to measure the exact marginal productivity of each worker, especially in group production.
- It ignores the role of trade unions and collective bargaining in wage determination.
- It assumes all factors of production are variable and easily adjustable, which is unrealistic in the short run.
- It neglects non-economic factors such as skills, experience, and working conditions.
- It assumes workers are paid exactly equal to their marginal productivity, which is often not the case.
- It does not explain wage differences between similar workers in different industries or regions.
- It ignores government intervention such as minimum wage laws and labour regulations.

2. Wages fund theory

This theory was first introduced by Adam Smith in 1776 in his book *The Wealth of Nations* and later popularized and refined by John Stuart Mill in 1848. The theory explains that wages are paid from a fixed fund of money saved by employers for paying workers. If this fund is large, workers receive higher wages; if it is small, wages are low. Therefore, wages depend on the size of the employer's savings set aside for labour payment.

$$\text{Wage rate} = \frac{\text{Wage fund}}{\text{Number of workers}}$$

Weaknesses of the wage fund theory

- It assumes the wage fund is fixed, which is unrealistic.
- It ignores the role of labour demand in wage determination.
- It does not take into account differences in worker productivity.
- It overlooks the influence of trade unions and collective bargaining.
- It fails to explain wage differences across occupations and regions.
- It ignores government intervention such as minimum wage legislation.

3. Bargaining Theory of Wages

According to John Davidson wages are determined through negotiation or bargaining between workers and employers. Workers try to push wages up, while employers try to keep wages low. The final wage depends on the bargaining strength of each side.

4. Iron Law of Wages

This also called the subsistence wage theory was propounded by **David Ricardo (1772-1823)**. According to this theory This theory states that labour should be paid a wage which is enough to cater for their basic needs of life (Subsistence life).

It is argued that if wages are increased beyond the subsistence level, the labour will marry earlier and will produce more children. This will result in the increase in population and labour supply will exceed demand. So, the money wages will fall to the level of subsistence.

If wages remain below the subsistence level, the labour will not be able to maintain their families. Due to starvation and malnutrition, etc. the death rate will increase. The supply of labour will fall short of demand. This will lead to increase in wages.

Increase in wages will be followed by increase in population which will lead to increase in the supply of labour resulting into a reduction in wages again to the subsistence level.

Criticism of the Subsistence Theory of Wages

This theory has been criticized on the following grounds,

- The theory does not explain the differences wages in different employments. According to the theory, the wage rate tends to be equal to the subsistence level of all the workers. Then, why is it that wages differ from industry to industry, worker to worker and occupation to occupation.
- The theory does not take into account the influence that trade unions have in the determination of wage rate in the labour market. Trade union activity is vital factor in wage determination.
- There is no direct relationship between change in money incomes and marriage. It is incorrect to assume that when the money income of a person increases above the subsistence level, then he marries early and the birth rate increases. On the other hand, the trend in modern life is that that when the income increases, it is generally followed by a higher standard of living and the workers do not produce more children.
- The Subsistence theory of wages entirely ignores the demand side of the labour and emphasizes only the supply side for the determination of the wages

5. Residual claimant theory (Francis A. Walker)

This theory was developed by **Francis A. Walker (1840-1897)**. According to Walker, there are four factors of production or business activity. Those are land, labour, capital, and entrepreneurship. He views that once all other three factors are rewarded, what remains is paid as wages to workers. Thus, according to this theory, worker is the residual claimant i.e. he should be given what remains after paying all other factors of production. According to Walker: "Wages equal to whole product minus rent, interest and profit".

11.8.4 Wage differentials

Wage differentials are a common feature in the labour market where workers receive different levels of pay for different types of work. These variations are observed across occupations, industries, and regions, and they play an important role in the structure of earnings within an economy.

Meaning and causes of wage differentials

Wage differentials refer to the differences in wages or salaries earned by different workers for doing different jobs or even similar jobs. This means that not all workers are paid the same amount of money, even if they are working in the same economy or sometimes in the same sector.

Causes of wage differentials

Wage differentials arise in the labour market due to various underlying factors that influence how different types of work are rewarded. These differences reflect the conditions under which labour is supplied and demanded, as well as the characteristics of workers and jobs. Understanding the causes helps to explain why wages are not uniform across all sectors of the economy.

The following are the main causes of wage differentials:

- **Human capital:** Workers with higher levels of education, training, skills, and experience are generally more productive and efficient, so employers are willing to pay them higher wages. In contrast, unskilled workers earn lower wages because they require little training and can easily be replaced.
- **Cost of training (learning a job):** Occupations that require long periods of education and expensive training tend to offer higher wages to compensate workers for the time, effort, and money invested. Jobs that require little or no training usually pay lower wages.
- **Immobility of labour:** When workers are unwilling or unable to move from one place or occupation to another, they may remain in low-paying jobs even when better opportunities exist elsewhere. Labour mobility allows workers to access higher-paying jobs, reducing wage differences.
- **Risk and working conditions:** Occupations that involve danger, health risks, or harsh working conditions usually pay higher wages as compensation for the risk involved. Safer and more comfortable jobs tend to offer lower wages since they are more desirable to workers *ceteris paribus*.
- **Government policy and trade unions** Government intervention, such as minimum wage laws and salary regulations, can influence wage levels across different sectors.
- **Trade unions:** Strong trade unions negotiate higher wages and better working conditions for their members, leading to wage differences between unionized and non-unionized workers.
- **Natural abilities and talent:** Workers with exceptional innate abilities, talent, or aptitude are often more productive and therefore receive higher wages compared to others.

Effects of wage differentials

Wage differentials are common in the labour market and involve differences in pay among workers. They raise important issues about fairness and efficiency, and can have both positive and negative effects on individuals and the economy.

The following are the effects of wage differentials:

Positive Effects of Wage Differentials:

- **Encourages Hard Work and Productivity:** Higher wages act as an incentive for workers to put in more effort and improve their performance in order to earn better pay.
- **Promotes Skill Acquisition:** People are motivated to acquire education, training, and specialized skills so they can qualify for higher-paying jobs.
- **Efficient Allocation of Labour:** Workers are attracted to jobs or sectors where wages are higher, ensuring labour moves to where it is most needed in the economy.
- **Encourages Innovation and Career Progression:** Wage differences motivate individuals to be creative, innovative, and seek promotions or better opportunities.
- **Attracts Talent to Scarce or Risky Jobs:** Occupations that are dangerous, highly skilled, or less attractive can offer higher wages to attract qualified workers.
- **Promotes Economic Growth:** When people work harder and improve their skills, productivity increases, leading to higher output and overall economic growth.

Negative Effects of Wage Differentials:

- **Low Welfare:** Workers earning low wages struggle to meet basic needs, leading to poverty and a low standard of living.
- **Reduced Demand:** Low-income earners have limited purchasing power, which reduces demand for goods and services and discourages investment.
- **Misallocation of Resources:** High-income groups spend more on luxury goods, while low-income groups cannot afford basic necessities, leading to inefficient resource use.
- **Social Disharmony:** Wide income gaps can cause resentment, crime, insecurity, and even political instability in society.
- **Regional Imbalance:** Areas with better-paying jobs develop faster, while low-income regions remain underdeveloped, increasing inequality.
- **Capital Outflow and Low Local Investment:** High earners, especially foreigners, may send money abroad instead of investing locally, reducing funds available for domestic development.

11.8.5 Wage controls

Wage levels in an economy are not always left to market forces alone, as they can be influenced or regulated by certain policies. In some cases, governments take action to guide how wages are set in order to achieve broader economic and social goals. These interventions help shape how labour is rewarded within the economy. Therefore, governments may intervene through wage control, using different measures to regulate wages so as to protect workers and support sustainable economic development.

Wage control refers to the measures taken by the government to regulate how wages are set, increased, or reduced in an economy. The aim is to ensure fairness between workers and employers, prevent extreme wage inequalities, protect workers' living standards, and maintain economic stability.

Wage control is done through setting minimum or maxim wage:

a) Minimum wage

A minimum wage is a payment for labour that is fixed by the government above the equilibrium wage below which the employer is by law not allowed to pay the worker.

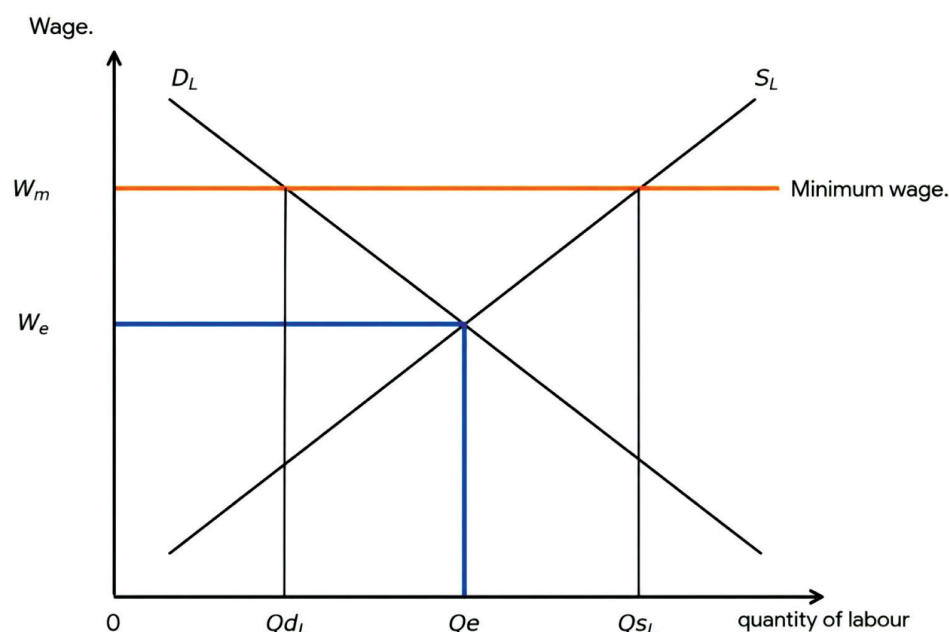


Figure 11.5.: Minimum wage

The Equilibrium (W_e , Q_e): This is where labor demand (D_L) naturally meets labor supply (S_L) without government intervention. When the government fixes the minimum wage at W_m (Minimum wage), the demand for labour reduces because labour becomes expensive. However, the supply of labour rises because the wage is high. This creates unemployment.

Arguments in favour of minimum wage legislation

- To encourage employers to use labour intensive methods, thereby providing employment to more employees.
- To discourage labour mobility especially from rural to urban areas. Paying the same wage to rural and urban dwellers discourages rural-urban migration.
- To reduce wage differentials and as a consequence, the gap between the poor and the rich is reduced.
- To satisfy the workers' needs and raise standard of living.
- To protect the workers from employers' exploitation.
- To increase the purchasing power of the workers. Consumption expenditures increase which in turn stimulates demand, investment and consequently employment opportunities are generated. The increase in the purchasing power of the people gives an incentive to producers and hence an increase in productivity.
- Minimum wages prove beneficial to the employers. This is because labour efficiency increases.
- Minimum wages tend to reduce labour unrest and maintain industrial peace. Workers are now more satisfied.
- It reduces relative geographical and industrial wage differential

Arguments against minimum wage legislation:

- Unemployment results. As wages increase, employers demand less workers but supply increases and hence unemployment.
- The purchasing power may decrease due to unemployment and consequently this discourages producers.
- Leads to rural-urban migration since they are normally set and adopted in urban areas.
- Employers may now prefer to use capital-intensive techniques of production because machines may not demand for wages.
- Cost of production increases leading to high prices.
- A minimum wage makes employers to have preference for high wage.

b) A maximum wage

This is the highest wage set by the government below equilibrium point above which it is illegal to pay workers.

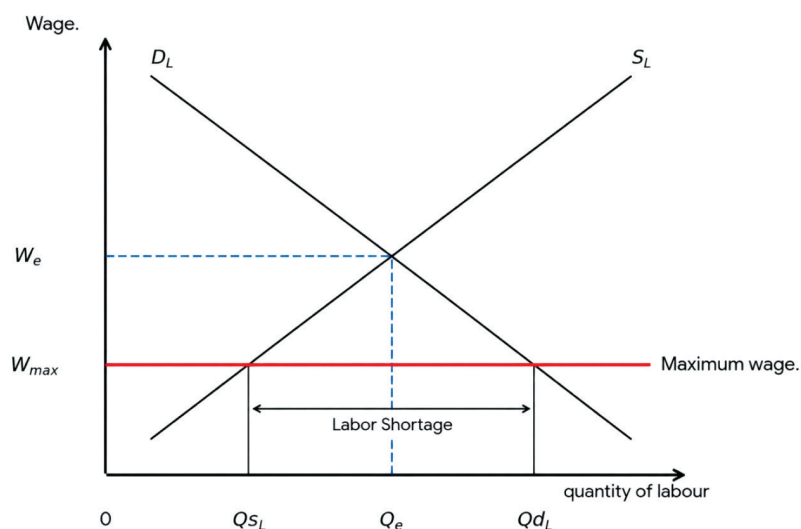


Figure 15.2.: Maximum wage

From the above graph the should be noted:

We: Equilibrium wage

Wmax: Maximum wage

Qs_L: Quantity supplied of labour

Qd_L: Qunatity demanded of labour

The government set maximum wage below equilibrium with the aim of protecting employers from paying a very high wage to their workers. However, this kind of wage legislation is not common in low developing countries. It is not common to find the government setting a wage above which people should not pay since wages are mostly determined by demand and supply of labour.

- Reduces public spending, especially for government officials and public servants.
- Prevent wage inflation in key sectors that could destabilize the economy.
- Promote fairness in income distribution within organizations or society.
- Increase funds for public use by limiting excessive pay and redirecting resources to services or development
- Reduces exploitation of employers.
- Encourages fair use of national resources: Resources saved from limiting excessive wages can be redirected towards public services, investment, or improving workers' welfare.

Arguments against maximum wage legislation:

- It may reduce workers' motivation and productivity because highly skilled or hardworking employees may feel underpaid.
- It can encourage skilled workers to migrate to countries or firms that offer better wages, leading to brain drain.

- Employers may find it difficult to attract and retain qualified workers if wages are legally restricted.
- Maximum wage legislation can discourage innovation, effort, and career development since extra performance may not lead to higher earnings.
- **Encourages informal or hidden payments** employers and employees may try to bypass the wage ceiling through bonuses, under-the-table payments, or benefits in kind, increasing corruption and reducing transparency.
- **Discourages investment and business growth:** Firms may find it harder to attract top managerial and technical talent if wages are restricted, reducing competitiveness, innovation, and foreign investment in the economy.

Application activity 11.8



1. Define the term wage differentials in your own words.
2. Why do you think different workers are paid different salaries even when they work in the same country?
3. Discuss effects of wage differentials on:
 - a) workers
 - b) the economy
4. Suggest five measures that can be used to reduce excessive wage differentials.
5. What do you understand by wage control in the Rwandan context?
6. Explain the meaning of:
 - a) Minimum wage
 - b) Maximum wage
7. In your opinion, why might a government decide to control wages instead of leaving them to employers and workers?

11.9 Trade unions

Learning Activity 11.9



1. What comes to your mind when you hear about workers joining together?
2. What do you think might make workers decide to come together and form a group?
3. If workers have a problem at work, how do you think they can solve it collectively?

11.9.1 Meaning and types of trade union

Trade union is a labour organization voluntarily created by workers with the aim of improving their standard of living through better pays and better working conditions.

They have the following objectives:

- To negotiate for fair wages, safe working conditions, social protection and constructive employer–employee relations. To advocate for good employer–employee relationships.
- To advocate for good working conditions.
- To advocate for social benefits like medicine, transport among others.
- To advocate for respect between employers and employees.
- To advocate for job security.
- To advocate for representation in company's decision making.

Types of trade unions

Workers in an economy may organize themselves in different ways depending on their occupation, skills, and sector of employment. These forms of organization vary in structure and purpose, influencing how workers represent their interests and negotiate with employers. This variation gives rise to different types of trade unions. These types are described below:

1. Craft unions

These are trade unions that bring together workers who have the same skills regardless of the industry in which they work. Members usually have similar training or expertise, which allows them to share common professional interests and challenges. Examples include engineers, teachers, doctors, lawyers, and printers.

2. Industrial unions

Industrial unions are trade unions that organize all workers within the same industry, regardless of their skills or job roles. They include skilled, semi-skilled, and unskilled workers, bringing them together under one organization. Their main focus is to represent and protect the interests of workers in a particular industry. For example, a union may represent all workers in the manufacturing or construction industry.

3. White-collar unions

These unions bring together administrative and professional workers who are usually paid salaries. Members mainly engage in office or professional work that requires formal qualifications and specialized knowledge. Examples include teachers, bankers, and legal workers.

4. Open shop (General) unions

These are trade unions that allow any worker to join, regardless of their industry or skill level. Membership is open to all workers, whether skilled, semi-skilled, or unskilled, making them more inclusive in nature.

5. Closed shop unions

These are trade unions where workers are required to join the union before being employed or in order to keep their jobs. Membership is therefore a condition for employment, meaning that only union members are allowed to work in a given firm or industry.

For instance, in some unionized industries such as certain transport or mining companies, workers may be required to belong to a designated workers' union as a condition for employment.

11.9.2 Tools / Methods used by trade unions to achieve their objectives

In any workplace, disagreements may arise between workers and employers over issues such as wages, working conditions, and job security. In **Rwanda**, as in many other countries, workers form trade unions to represent their interests and ensure their voices are heard. However, for these unions to achieve their objectives, they must use different approaches or strategies. Some of these methods are peaceful and promote cooperation, while others involve applying pressure on employers when negotiations fail. **The following are the main tools and methods used by trade unions to achieve their goals.**

A. Peaceful methods

These are **non-violent ways** used by trade unions to solve disputes through negotiation, persuasion, or legal means.

- 1. Collective Bargaining:** This is a peaceful negotiation process between workers' representatives (trade unions) and employers to agree on wages, working hours, benefits, and conditions of work. This tool helps employers and workers to reach mutual agreement without conflict. **Example:** Teachers in **Rwanda** negotiating salary improvements with the government.
- 2. Ultimatum / Grace Period:** Workers give employers a specific time limit to respond to their demands before taking further action. **Example:** A union gives a factory 10 days to improve safety conditions.
- 3. Work-to-Rule:** Workers strictly follow all rules and procedures without extra effort or flexibility, which slows down production. Workers use this tool in order to reduce productivity without breaking rules. **Example:** Employees leaving work exactly at closing time even if tasks are unfinished.
- 4. Sit-down strike:** Worker's report to work but refuse to perform any duties while remaining at the workplace to stop production and pressure employers. **Example:** Workers sitting in offices without working until wages are increased.
- 5. Demonstration:** A public protest where workers march peacefully carrying placards to express dissatisfaction. The purpose is to attract public attention and pressure employers or government. **Example:** Workers marching in **Kigali** demanding better pay.

6. Mediation / Arbitration

Mediation and arbitration are peaceful methods used in industrial relations to settle disputes between workers (through trade unions) and employers with the help of a neutral third party when direct negotiations fail or become difficult. **Example:** A government labour officer mediating between a company and its workers.

- **In mediation**, the neutral person (mediator) listens to both sides, encourages dialogue, and helps them reach a voluntary agreement. The mediator does not impose a decision.
- **In arbitration**, the neutral person (arbitrator) examines the evidence and arguments from both sides and then makes a binding decision that both workers and employers must follow.

7. Industrial court: is a formal legal institution established by the government to hear, examine, and decide on disputes between workers (often represented by trade unions) and employers using the principles of labour law. It deals with conflicts that cannot be solved through negotiation, mediation, or arbitration.

Example: Workers taking a salary dispute to court for judgment.

8. Mass media compain / Press war: is a method used by trade unions where workers communicate their demands or complaints to the public through radio, television, newspapers, and social media in order to create awareness about their working conditions and issues with employers. It involves spreading information widely so that many people can understand the workers' situation and pressure employers or authorities to respond fairly.

This method is powerful because it reaches a large audience quickly and helps to build public understanding, sympathy, and sometimes government intervention in favour of workers.

Example: Union leaders speaking on radio about unfair dismissal of workers.

9. Education and training: is a method used by trade unions where they organize structured learning programs such as seminars, workshops, and short courses to improve workers' **skills, knowledge, and competence** in their jobs and in labour relations. It helps workers understand their rights, improve their performance, and become more effective in their occupations.

Through this method, workers become more skilled and confident, which increases their efficiency at work and strengthens their ability to negotiate better wages and working conditions because employers value skilled labour more highly.

10. Restricting labour supply: is a method used by trade unions where they influence or require employers to hire only workers who are members of the union or who agree to join the union before or during employment. This limits the number of available non-union workers in the labour market for a particular job or firm.

By controlling entry into employment, the union increases its membership size and strengthens solidarity among workers, which gives it greater power to negotiate for better wages, improved working conditions, and stronger protection of workers' rights.

Example: A factory requiring all new workers to join the union.

B. Violent (Forceful) methods

These are **coercive or disruptive actions** used when peaceful methods fail. They aim to force employers to meet workers' demands.

1. Go-Slow Tactics

Go-slow tactics is a method used by trade unions where workers deliberately reduce their normal speed of work without completely stopping work. They continue reporting to work but perform their duties slowly and less efficiently than usual in order to reduce the level of production in the organization. This is done as a form of pressure on the employer to respond to workers' demands, especially when negotiations have failed.

This method disrupts normal production, increases production costs per unit, and creates delays in delivery of goods or services, which forces employers to reconsider workers' demands such as better pay or improved working conditions.

Example: Workers in a factory producing fewer goods per day than usual.

2. Picketing: is a method used by trade unions where workers position themselves at or near the entrance of a workplace to discourage, block, or prevent other workers (especially non-union members or replacement workers) from entering and carrying out work. It is usually organized during a strike or labour dispute to show strong opposition to the employer's decisions.

This method aims to put pressure on the employer by disrupting normal business activities, creating awareness of the workers' grievances, and encouraging negotiation. It is often used to strengthen the impact of a strike by making it difficult for the organization to continue operating normally. **Example:** Union members blocking entry to a factory gate.

3. Boycott of products / Sabotage: Workers refuse to buy or encourage others not to buy a company's products, or disrupt production in order to affect the employer's income.

Example: Workers asking the public not to buy goods from a certain company.

4. Boycott of personnel: Workers refuse to work under specific managers or supervisors in order to pressure **their removal**. **Example:** Employees refusing to work under a harsh supervisor.

5. Violent industrial strike: A complete stop of work by workers, sometimes involving destruction of property, to force employers to meet their demands.

Example: Workers stopping all production after failed negotiations.

11.9.3 Weaknesses of trade union in developing countries

Trade unions in developing countries face several challenges that limit their effectiveness in representing and protecting workers. These challenges weaken their bargaining power and reduce their ability to fully influence labour relations and improve working conditions. **The main weaknesses are discussed below.**

- **Limited financial resources:** Trade unions often lack sufficient funds from members, which limits their ability to organize activities, support strikes, or provide services to workers.

- **Poor leadership:** In some cases, union leaders may lack skills, experience, or commitment, leading to weak representation of workers' interests.
- **Low membership participation:** Many workers do not actively participate in union activities, which reduces unity and weakens bargaining power.
- **Government interference:** Governments may control or restrict union activities, reducing their independence and ability to negotiate freely.
- **Employers' resistance:** Some employers oppose trade unions and may refuse to recognize them or negotiate in good faith.
- **Lack of education and awareness among workers**
When workers are not well informed about their rights, it becomes difficult for unions to mobilize and represent them effectively.
- **Internal conflicts and divisions:** Disagreements among union members or leaders weaken unity and reduce the effectiveness of collective action.
- **Low bargaining power in informal sectors:** Many workers in developing countries are in informal employment, making it difficult for trade unions to organize them and negotiate effectively on their behalf.
- **Limited legal enforcement power:** Even when agreements are reached, trade unions may lack strong legal mechanisms to ensure employers fully implement the agreed terms and conditions.

Application activity 11.9



In Rwanda, workers in sectors such as education, manufacturing, and banking form trade unions to improve wages, working conditions, and job security. These unions represent workers in negotiations with employers and help to resolve workplace disputes. In some workplaces, membership in a union is voluntary, while in others it may be required as a condition of employment.

Questions

1. From the situation above, define a trade union and identify three categories of workers mentioned, showing the type of trade union that each group is likely to form.
2. Distinguish between **open shop unions** and **closed shop unions**.
3. Describe any four objectives of trade unions as reflected in the case study.
4. Explain how trade unions can improve the welfare of workers in Rwanda, giving clear economic reasons.

11.10. End of unit assessment



1. Explain the meaning of labour and labour force, and discuss the factors that determine the size of the labour force in an economy.
2. Discuss the factors that influence both the demand for labour and the supply of labour in an economy, and explain how these factors affect employment levels.
3. Examine the concept of efficiency of labour and evaluate the factors that can improve or reduce labour productivity in developing countries.
4. Analyse the meaning, forms, and factors influencing labour mobility, and assess its importance to economic development.
5. Discuss the causes and effects of manpower problems in developing countries, and evaluate the measures that can be taken to solve them.
6. Explain the meaning of wages and critically examine the methods of wage determination and wage theories in explaining wage differences in an economy.
7. Evaluate the role of trade unions in improving workers' welfare, and discuss the challenges they face in developing countries.

References

At the end of the Teacher's Guide, references should be presented in **APA style** to acknowledge the sources used in preparing the content. This ensures academic integrity, provides credibility, and allows learners and teachers to consult the original materials for deeper understanding. Below is a sample structure of how the references can be listed:

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